Elliptic Regularity

Scott Armstrong [*] Tuomo Kuusi [†] Amélie Loher [‡]

October 22, 2026

Abstract

We present the quantitative regularity theory of linear elliptic and parabolic equations in divergence form, whose coefficient field is uniformly elliptic and, in general, merely measurable and possibly nonsymmetric. Throughout, we track how the constants depend on the dimension, the ellipticity and the exponents involved. We begin with function spaces and functional inequalities. We then study the Laplace equation: estimates for harmonic functions, the monotonicity formulas of Almgren and of Alt, Caffarelli and Friedman, and Poon's frequency function for the heat equation. Next come the perturbative estimates of Schauder and of Calderón and Zygmund. The latter are proved with constants explicit enough to yield Yudovich's uniqueness theorem for the two-dimensional Euler equations. The scale-invariant estimates, which require no regularity of the coefficients, form the core of the manuscript: the hole-filling estimate, the Meyers estimate, the De Giorgi–Nash–Moser theory and the Harnack inequality, examples showing that several of the resulting exponents and constants are sharp, and two-sided bounds on the Green function. The last chapter treats parabolic equations, from the Caccioppoli inequality to the Gaussian bounds of Nash and Aronson. The results are accompanied by a machine-checked formalization in Lean 4, available at https://github.com/scottnarmstrong/EllipticRegularityLean.

Contents

0 Preface **2**

1 Function spaces and functional inequalities **4**

1.1 The isoperimetric inequality 4

1.2 The Sobolev inequality 7

1.3 Trace inequalities 24

2 The Laplace equation **29**

2.1 Harmonic functions 29

2.2 Monotonicity formulas 31

2.3 The heat equation 38

3 Perturbative estimates **40**

3.1 Schauder estimates 40

[*]CNRS and Laboratoire Jacques-Louis Lions, Sorbonne Université; Courant Institute of Mathematical Sciences, New York University. scottnarmstrong@gmail.com.

[†]Department of Mathematics and Statistics, University of Helsinki. tuomo.kuusi@helsinki.fi.

[‡]All Souls College, University of Oxford. amelie.loher@all-souls.ox.ac.uk.

3.2 Calderón-Zygmund estimates . . . 45
4 Scale-invariant estimates 55
4.1 The hole-filling estimate . . . 55
4.2 Higher integrability and the Meyers estimate . . . 60
4.3 The De Giorgi-Nash Hölder estimate . . . 68
4.4 Moser iteration . . . 76
4.5 The sharp exponent in two dimensions . . . 88
4.6 Sharpness of the Hölder exponent and of the Harnack ratio for $d \geqslant 3$ . . . 94
4.7 Green function decay estimates . . . 100
5 Parabolic equations 112
5.1 Function spaces for parabolic equations . . . 112
5.2 Parabolic Caccioppoli inequality . . . 115
5.3 The Nash-Aronson estimate . . . 118
References 124

0. Preface

This manuscript is an introduction to the regularity theory of linear, scalar, second-order elliptic and parabolic equations in divergence form,

$$-\nabla \cdot \mathbf{a}\nabla u = 0 \quad \text{and} \quad \partial_t u - \nabla \cdot \mathbf{a}\nabla u = 0\,,$$

in which the coefficient field $\mathbf{a}$ is bounded and uniformly elliptic but, in general, neither continuous nor symmetric. It is written for graduate students and for researchers in analysis and partial differential equations who want a self-contained account of the basic estimates of the theory, with complete proofs. We assume familiarity with measure theory and Lebesgue integration, with basic functional analysis in Banach and Hilbert spaces, and with the elementary theory of Sobolev spaces (weak derivatives, the spaces $W^{1,p}$ and $W^{1,p}_0$, and the density of smooth functions), as presented for instance in [Eva98]. The functional inequalities needed later, such as the Sobolev and Poincaré inequalities, are proved in Chapter 1. The few standard facts that are used without proof are stated where they are used.

Chapter 1 develops the functional inequalities used throughout: the isoperimetric inequality, proved by means of the Prékopa–Leindler and Brunn–Minkowski inequalities; the Sobolev, Poincaré, John–Nirenberg, Trudinger, Gaussian Poincaré and Nash inequalities; the Campanato characterization of Hölder spaces; and the trace theorem. Chapter 2 concerns the Laplace equation: derivative bounds and analyticity of harmonic functions, the monotonicity formulas of Almgren and of Alt, Caffarelli and Friedman, and Poon's frequency function for the heat equation. Chapter 3 presents the perturbative theory, in which the coefficients are close to constant: Schauder estimates for continuous and Hölder continuous coefficients, and the Calderón–Zygmund estimate. The dependence of the latter on the integrability exponent is made explicit, and this is what yields Yudovich's uniqueness theorem for the two-dimensional Euler equations with bounded vorticity. Chapter 4 is the core of the manuscript. It treats estimates which are invariant under scaling and require no regularity of the coefficients: the hole-filling estimate, the Meyers estimate, the De Giorgi–Nash Hölder estimate and Moser's iteration, the weak Harnack and Harnack inequalities,

examples showing that several of the exponents and constants obtained are sharp, and two-sided bounds on the Green function. Chapter 5 extends the theory to parabolic equations with coefficients depending on time and space: the energy space, the parabolic Caccioppoli inequality and the two-sided Gaussian bounds of Nash and Aronson, which are obtained together with the parabolic weak Harnack and Harnack inequalities.

The estimates are quantitative throughout. Constants are tracked explicitly in terms of the dimension d, the ellipticity Λ and the exponents involved, and in several cases the resulting dependence is shown to be optimal. The Meyers estimate holds for exponents up to $2 + c\Lambda^{-1/2}$; the constant in the Harnack inequality is $\exp(C\Lambda^{1/2})$; and, for $d \geqslant 3$, the Hölder exponent of solutions is at least $\exp(-C\Lambda^{1/2})$. Examples in Chapter 4 show that each of these orders is optimal, and that in two dimensions the Hölder exponent $\Lambda^{-1/2}$ obtained for symmetric coefficients cannot be improved. Similarly, the Calderón–Zygmund estimate in L^p requires only that the oscillation of the coefficients be of order p^{-1}, with a constant of order p, and both orders are optimal.

The results of this manuscript have been formalized in the Lean 4 proof assistant, using the Mathlib library. Every numbered theorem, proposition, lemma and corollary, as well as every definition and example, has a formal counterpart in Lean, every formal result is proved there, and the formal statements have been approved by the authors. Remarks and exercises are not formalized. The formalization is available at `https://github.com/scottnarmstrong/EllipticRegularityLean`, where the file `RESULTS.md` lists the Lean declaration corresponding to each numbered result, together with notes on individual statements. For instance, the formal version of Proposition 1.1 concerns only the inequality, since the equality case is not proved here.

Notation

- *Euclidean space.* The dimension is $d \geqslant 1$. For $x, y \in \mathbb{R}^d$, we write $x \cdot y$ for the inner product and $|x|$ for the Euclidean norm, and $e_1, \dots, e_d$ is the standard basis. The open ball is $B_r(x) := \{y \in \mathbb{R}^d : |y - x| < r\}$, and $B_r := B_r(0)$. In Chapters 1 and 4, $Q_r(x) := x + (-r, r)^d$ denotes the open cube; in Chapter 5, $Q_r(t_0, x_0) := (t_0 - r^2, t_0] \times B_r(x_0)$ denotes the parabolic cylinder (see Section 5.1).

- *Measures.* For a measurable set $E \subseteq \mathbb{R}^d$, $|E|$ is its Lebesgue measure and $\mathbf{1}_E$ its indicator function. We write $\mathcal{H}^{d-1}$ for the $(d-1)$-dimensional Hausdorff measure.

- *Averages and normalized norms.* If $0 < |E| < \infty$, then

$$(f)_E := \fint_E f := \frac{1}{|E|} \int_E f \quad \text{and} \quad \|f\|_{\underline{L}^p(E)} := \left(\fint_E |f|^p \right)^{1/p} = |E|^{-1/p} \|f\|_{L^p(E)} .$$

- *Oscillation and Hölder seminorms.* We write $\operatorname{osc}_S u := \sup_S u - \inf_S u$ and $\operatorname{ess\,osc}_S u := \operatorname{ess\,sup}_S u - \operatorname{ess\,inf}_S u$. For $\alpha \in (0, 1]$ and $U \subseteq \mathbb{R}^d$,

$$[u]_{C^{0,\alpha}(U)} := \sup_{x,y \in U,\, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha} \quad \text{and} \quad \|u\|_{C^{0,\alpha}(U)} := \sup_U |u| + [u]_{C^{0,\alpha}(U)} .$$

  The labels $C^{0,1-}$ and $C^{1,1-}$ in Chapter 3 refer to estimates in $C^{0,\alpha}$ for u, respectively for ∇u, which hold for every $\alpha \in (0, 1)$.

- *Function spaces.* For an open set $U \subseteq \mathbb{R}^d$, we use the Lebesgue spaces $L^p(U)$ and $L^p(U;\mathbb{R}^d)$, the latter also written $L^p(U)^d$, and the Sobolev spaces $W^{1,p}(U)$ and $W_0^{1,p}(U)$, the closure of $C_c^\infty(U)$ in $W^{1,p}(U)$. We write $H^1 := W^{1,2}$ and $H_0^1 := W_0^{1,2}$.
- *Matrices and coefficient fields.* We write $\mathbb{R}^{d\times d}$ for the real $d \times d$ matrices, I_d for the identity matrix and $\mathbf{a}^t$, or $\mathbf{a}^\top$, for the transpose of $\mathbf{a}$. A coefficient field is a measurable map $\mathbf{a} : U \to \mathbb{R}^{d\times d}$, which need not be symmetric. Its ellipticity is measured by (4.1), which normalizes the lower bound to one and introduces the ellipticity $\Lambda \geqslant 1$; it implies $|\mathbf{a}e| \leqslant \Lambda|e|$ and (4.2). Parabolic coefficient fields $\mathbf{a}(t,x)$ satisfy (5.1), with constants $0 < \lambda \leqslant \Lambda$.
- *Equations.* Elliptic equations are understood in the weak sense of (4.135), with $\mathbf{b} = \mathbf{a}$. Subsolutions and supersolutions satisfy the corresponding inequalities for nonnegative test functions. Parabolic solutions are defined in Definition 5.4.
- *Miscellaneous.* We write $s_+ := \max\{s, 0\}$ and $s \wedge t := \min\{s, t\}$ for $s, t \in \mathbb{R}$. The notation $C(d, \Lambda)$ indicates that the constant C depends only on d and Λ.

1. Function spaces and functional inequalities

1.1 The isoperimetric inequality

Among all sets of given volume, the ball has the smallest perimeter.

Proposition 1.1 (Isoperimetric inequality)**.** *For every bounded open set $E \subset \mathbb{R}^d$ with C^1 boundary,*

$$\mathcal{H}^{d-1}(\partial E) \geqslant d|B_1|^{\frac{1}{d}}|E|^{\frac{d-1}{d}}\,, \tag{1.1}$$

with equality if and only if E is a ball.

The three proofs below establish the inequality (1.1); the characterization of the equality case is not proved in these notes.

First proof of Proposition 1.1 by Steiner symmetrization. By translation and scaling, we may assume $|E| = |B_1|$, so that (1.1) becomes $\mathcal{H}^{d-1}(\partial E) \geqslant |\partial B_1|$. Write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{d-1}$.

Step 1. The Steiner symmetral. For each $x' \in \mathbb{R}^{d-1}$, let $\ell(x') := \mathcal{H}^1(\{x_1 \in \mathbb{R} : (x_1, x') \in E\})$ be the length of the slice. The *Steiner symmetral* of E in the x_1-direction is

$$E^\star := \left\{(x_1, x') \in \mathbb{R}^d : |x_1| < \tfrac{1}{2}\ell(x')\right\}.$$

By Fubini's theorem, $|E^\star| = |E|$.

Step 2. Perimeter does not increase. We claim that

$$\mathcal{H}^{d-1}(\partial E^\star) \leqslant \mathcal{H}^{d-1}(\partial E)\,. \tag{1.2}$$

Since E has C^1 boundary, for a.e. x' the slice $\{x_1 : (x_1, x') \in E\}$ consists of finitely many open intervals $(a_1, b_1), \dots, (a_N, b_N)$ with $a_1 < b_1 < a_2 < \cdots < b_N$, and the boundary ∂E

intersects the line $\{x'\}\times\mathbb{R}$ at $2N$ points. On each boundary sheet, ∂E is locally a graph $x_1 = f_j(x')$. Writing $\nu = (\nu_1, \nu')$ for the outer unit normal, the co-area-type formula gives

$$\mathcal{H}^{d-1}(\partial E) = \int_{\mathbb{R}^{d-1}} \sum_{j=1}^{2N} \frac{1}{|\nu_1(f_j(x'), x')|}\, dx' \,,$$

which, using $|\nu_1|^{-1} = \sqrt{1+|\nabla_{x'} f_j|^2}$, becomes

$$\mathcal{H}^{d-1}(\partial E) = \int_{\mathbb{R}^{d-1}} \sum_{j=1}^{2N} \sqrt{1+|\nabla_{x'} f_j|^2}\, dx' \,.$$

After symmetrization, $\partial E^\star$ has exactly two sheets: $x_1 = \pm\frac{1}{2}\ell(x')$, contributing

$$\mathcal{H}^{d-1}(\partial E^\star) = 2\int_{\mathbb{R}^{d-1}} \sqrt{1+\tfrac{1}{4}|\nabla_{x'}\ell|^2}\, dx' \,.$$

Since $\ell = \sum_{j=1}^N (b_j - a_j)$, we have $\nabla_{x'}\ell = \sum_{j=1}^N (\nabla_{x'} b_j - \nabla_{x'} a_j)$, and by the triangle inequality and the convexity of $t \mapsto \sqrt{1+t^2}$,

$$2\sqrt{1+\tfrac{1}{4}|\nabla_{x'}\ell|^2} \leqslant \sum_{j=1}^N \left(\sqrt{1+|\nabla_{x'} a_j|^2} + \sqrt{1+|\nabla_{x'} b_j|^2} \right) = \sum_{j=1}^{2N} \sqrt{1+|\nabla_{x'} f_j|^2} \,.$$

Integrating over x' gives (1.2).

Step 3. Convergence to a ball. Let S_k denote Steiner symmetrization in the x_k-direction, and define $E_n := (S_d \circ \cdots \circ S_1)^n(E)$. Then $|E_n| = |E| = |B_1|$ for every n, and $\mathcal{H}^{d-1}(\partial E_n)$ is non-increasing. Moreover, E_n is symmetric with respect to each coordinate hyperplane, and by compactness (in the L^1 topology on characteristic functions), a subsequence converges to some set F with $|F| = |B_1|$ and $\mathcal{H}^{d-1}(\partial F) \leqslant \lim_{n\to\infty} \mathcal{H}^{d-1}(\partial E_n)$. Since F is symmetric and Steiner-invariant in every direction, F must be a ball. Therefore $F = B_1$ and

$$\mathcal{H}^{d-1}(\partial E) \geqslant \lim_{n\to\infty} \mathcal{H}^{d-1}(\partial E_n) \geqslant \mathcal{H}^{d-1}(\partial B_1) = |\partial B_1| = d|B_1| \,.$$

This gives (1.1). □

Second proof of Proposition 1.1 by the Knothe map. We may assume $|E| = |B_1|$, so the target is the unit ball B_1. Write $x = (x_1, \dots, x_d)$.

Step 1. Construction of the Knothe map. We construct a smooth map $T : E \to B_1$ that pushes forward the uniform measure on E to the uniform measure on B_1, and whose Jacobian DT is upper triangular with positive diagonal entries. The map is built coordinate-by-coordinate using the quantile (inverse CDF) construction:

- Define $T_1(x)$ so that the x_1-marginals match: let $F_E(t) := |E|^{-1}|\{y \in E : y_1 \leqslant t\}|$ and $F_{B_1}(t) := |B_1|^{-1}|\{y \in B_1 : y_1 \leqslant t\}|$, and set $T_1(x) := F_{B_1}^{-1}(F_E(x_1))$.
- For $i = 2, \dots, d$: conditional on $(x_1, \dots, x_{i-1})$ (equivalently, on $(T_1, \dots, T_{i-1})$), match the x_i-conditional marginals by the same quantile construction.

By construction, each T_i depends only on $(x_1, \dots, x_i)$, so DT is upper triangular. The diagonal entries $\partial T_i/\partial x_i$ are positive (as derivatives of increasing quantile functions), and the change of variables formula gives $\det DT = 1$ a.e.

Step 2. AM-GM and the divergence theorem. Since DT is upper triangular with positive diagonal,

$$1 = \det DT = \prod_{i=1}^{d} \frac{\partial T_i}{\partial x_i} \leqslant \left(\frac{1}{d} \sum_{i=1}^{d} \frac{\partial T_i}{\partial x_i} \right)^d = \left(\frac{\operatorname{div} T}{d} \right)^d ,$$

where we used the AM-GM inequality. Hence $\operatorname{div} T \geqslant d$. Integrating over E and applying the divergence theorem,

$$d|E| \leqslant \int_E \operatorname{div} T = \int_{\partial E} T \cdot \nu_E \, d\mathcal{H}^{d-1} \leqslant \mathcal{H}^{d-1}(\partial E) \cdot \sup_{x \in E} |T(x)| .$$

Since T maps into B_1, we have $|T| \leqslant 1$, and the right side is $\mathcal{H}^{d-1}(\partial E)$. Therefore $\mathcal{H}^{d-1}(\partial E) \geqslant d|E| = d|B_1|$, which is (1.1). □

The third proof deduces the isoperimetric inequality from the Brunn–Minkowski inequality, which we now state and prove.

Theorem 1.2 (Prékopa–Leindler inequality)**.** *Let $\lambda \in (0,1)$ and $f, g, h : \mathbb{R}^d \to [0, \infty)$ be measurable functions satisfying*

$$h\big(\lambda x + (1-\lambda)y\big) \geqslant f(x)^\lambda \, g(y)^{1-\lambda} \quad \forall \, x, y \in \mathbb{R}^d . \tag{1.3}$$

Then

$$\int_{\mathbb{R}^d} h \geqslant \left(\int_{\mathbb{R}^d} f \right)^\lambda \left(\int_{\mathbb{R}^d} g \right)^{1-\lambda} . \tag{1.4}$$

Proof. We first treat the case $d = 1$. We may assume $\int f = \int g = 1$ (otherwise rescale f and g). Let $F(t) := \int_{-\infty}^t f(s)\, ds$ and $G(t) := \int_{-\infty}^t g(s)\, ds$ be the cumulative distribution functions, and define the quantile functions $\alpha(p) := F^{-1}(p)$ and $\beta(p) := G^{-1}(p)$ for $p \in (0,1)$. Then $f(\alpha(p))\, \alpha'(p) = 1$ and $g(\beta(p))\, \beta'(p) = 1$ for a.e. p. By the change of variables $t = \lambda\alpha(p) + (1-\lambda)\beta(p)$,

$$\int_{\mathbb{R}} h \geqslant \int_0^1 h\big(\lambda\alpha(p) + (1-\lambda)\beta(p)\big) \, \big(\lambda\alpha'(p) + (1-\lambda)\beta'(p)\big) \, dp .$$

By (1.3) the first factor is $\geqslant f(\alpha)^\lambda \, g(\beta)^{1-\lambda} = \alpha'(p)^{-\lambda} \, \beta'(p)^{-(1-\lambda)}$, while the AM-GM inequality gives $\lambda\alpha' + (1-\lambda)\beta' \geqslant (\alpha')^\lambda (\beta')^{1-\lambda}$. Multiplying, the integrand is $\geqslant 1$, so $\int h \geqslant 1$.

For general d, we proceed by induction. Write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{d-1}$. For each $x_1 \in \mathbb{R}$, define $f_{x_1}(x') := f(x_1, x')$ and likewise for g, h. The hypothesis (1.3) gives

$$h_{\lambda x_1 + (1-\lambda) y_1}\big(\lambda x' + (1-\lambda) y'\big) \geqslant f_{x_1}(x')^\lambda \, g_{y_1}(y')^{1-\lambda} .$$

By the $(d-1)$-dimensional Prékopa–Leindler inequality (induction hypothesis),

$$H\big(\lambda x_1 + (1-\lambda) y_1\big) \geqslant F(x_1)^\lambda \, G(y_1)^{1-\lambda} ,$$

where $F(x_1) := \int_{\mathbb{R}^{d-1}} f_{x_1}$, $G(y_1) := \int_{\mathbb{R}^{d-1}} g_{y_1}$ and $H(t) := \int_{\mathbb{R}^{d-1}} h_t$. By the one-dimensional case, $\int_{\mathbb{R}} H \geqslant (\int_{\mathbb{R}} F)^\lambda (\int_{\mathbb{R}} G)^{1-\lambda}$, which is (1.4). □

Theorem 1.3 (Brunn–Minkowski inequality)**.** *For every pair of nonempty compact sets $A, B \subset \mathbb{R}^d$,*

$$|A+B|^{\frac{1}{d}} \geqslant |A|^{\frac{1}{d}} + |B|^{\frac{1}{d}}. \tag{1.5}$$

Proof. Let $c := |A|^{1/d} + |B|^{1/d}$ and $\lambda := |A|^{1/d}/c$. Set $X := \lambda^{-1}A$ and $Y := (1-\lambda)^{-1}B$, so that $A + B = \lambda X + (1-\lambda)Y$. Observe that $|X| = \lambda^{-d}|A| = c^d$ and $|Y| = (1-\lambda)^{-d}|B| = c^d$. Applying Theorem 1.2 with $f = \mathbf{1}_X$, $g = \mathbf{1}_Y$, $h = \mathbf{1}_{\lambda X + (1-\lambda)Y}$ (the hypothesis (1.3) holds since $\lambda X + (1-\lambda)Y$ is convex-combination-closed),

$$|A+B| = |\lambda X + (1-\lambda)Y| \geqslant |X|^{\lambda}\,|Y|^{1-\lambda} = c^d = \left(|A|^{\frac{1}{d}} + |B|^{\frac{1}{d}}\right)^d. \qquad \square$$

Third proof of Proposition 1.1 by the Brunn–Minkowski inequality. For $\varepsilon > 0$, the ε-neighbourhood of E satisfies $E + \varepsilon B_1 \supseteq E$ and, by the Brunn–Minkowski inequality (Theorem 1.3),

$$|E + \varepsilon B_1|^{\frac{1}{d}} \geqslant |E|^{\frac{1}{d}} + \varepsilon\,|B_1|^{\frac{1}{d}}.$$

Since E has C^1 boundary, the Minkowski content equals the perimeter:

$$\mathcal{H}^{d-1}(\partial E) = \lim_{\varepsilon \to 0} \frac{|E + \varepsilon B_1| - |E|}{\varepsilon}.$$

Computing the right-derivative of $\varepsilon \mapsto (|E|^{1/d} + \varepsilon |B_1|^{1/d})^d$ at $\varepsilon = 0$,

$$\mathcal{H}^{d-1}(\partial E) \geqslant d\,|E|^{\frac{d-1}{d}}\,|B_1|^{\frac{1}{d}}.$$

This is (1.1). $\square$

1.2 The Sobolev inequality

The Sobolev inequality is equivalent to the isoperimetric inequality on general Riemannian manifolds.

Proposition 1.4 (Sobolev inequality)**.** *For every $u \in W^{1,1}(\mathbb{R}^d)$,*

$$\|u\|_{L^{\frac{d}{d-1}}(\mathbb{R}^d)} \leqslant d^{-1}|B_1|^{-\frac{1}{d}}\|\nabla u\|_{L^1(\mathbb{R}^d)}. \tag{1.6}$$

Proof. By considering the positive and negative parts of u separately, we may assume that $u \geqslant 0$. By density, we may also assume that $u \in C_c^\infty(\mathbb{R}^d)$. By the layer-cake formula,

$$\int_{\mathbb{R}^d} |u|^{\frac{d}{d-1}} = \int_0^\infty \left|\left\{|u|^{\frac{d}{d-1}} > t\right\}\right| dt = \frac{d}{d-1}\int_0^\infty \left|\{|u| > s\}\right| s^{\frac{1}{d-1}}\, ds.$$

By the co-area formula,

$$\int_{\mathbb{R}^d} |\nabla u| = \int_0^\infty |\{u = s\}|\, ds$$

By the isoperimetric inequality, for every $s > 0$,

$$\left|\{u = s\}\right| \geqslant \frac{|\partial B_1|}{|B_1|^{\frac{d-1}{d}}}\left|\{u > s\}\right|^{\frac{d-1}{d}} = d|B_1|^{\frac{1}{d}}\left|\{u > s\}\right|^{\frac{d-1}{d}}.$$

Therefore it suffices to show that

$$\int_0^\infty \big|\{|u|>s\}\big|\, s^{\frac{1}{d-1}}\, ds \leqslant \frac{d-1}{d}\left(\int_0^\infty |\{u>s\}|^{\frac{d-1}{d}}\, ds\right)^{\frac{d}{d-1}}. \tag{1.7}$$

To prove (1.7), we define

$$f(t) := \int_0^t \big|\{|u|>s\}\big|\, s^{\frac{1}{d-1}}\, ds \quad \text{and} \quad g(t) := \frac{d-1}{d}\left(\int_0^t |\{u>s\}|^{\frac{d-1}{d}}\, ds\right)^{\frac{d}{d-1}},$$

observe that $f(0) = 0 = g(0)$ and

$$\begin{aligned} f'(t) = \big|\{|u|>t\}\big|\, t^{\frac{1}{d-1}} &= \left(\int_0^t |\{u>t\}|^{\frac{d-1}{d}}\, ds\right)^{\frac{1}{d-1}} |\{u>t\}|^{\frac{d-1}{d}} \\ &\leqslant \left(\int_0^t |\{u>s\}|^{\frac{d-1}{d}}\, ds\right)^{\frac{1}{d-1}} |\{u>t\}|^{\frac{d-1}{d}} = g'(t)\,. \end{aligned}$$

Therefore by integration we obtain $f(t) \leqslant g(t)$ for every $t>0$. Sending $t\to\infty$ yields (1.7) and completes the proof. □

The sharp constant in the previous proposition relies on the isoperimetric inequality. For most purposes a dimensional constant suffices, and the case $p=1$ with such a constant follows from the elementary argument of Gagliardo and Nirenberg (see, e.g., [Eva98, Section 5.6.1]), which avoids isoperimetry. Throughout, we write $p^* := \frac{dp}{d-p}$ for $p\in[1,d)$ and

$$\gamma_p := \frac{p(d-1)}{d-p}\,, \tag{1.8}$$

so that $\gamma_1 = 1$ and $\gamma_p \frac{d}{d-1} = (\gamma_p - 1)\frac{p}{p-1} = p^*$ for $p>1$.

Proposition 1.5 (Sobolev inequality for $1\leqslant p<d$)**.** *Let $d\geqslant 2$. There exists $C(d)<\infty$ such that, for every $p\in[1,d)$ and $u\in W^{1,p}(\mathbb{R}^d)$,*

$$\|u\|_{L^{p^*}(\mathbb{R}^d)} \leqslant C\gamma_p \|\nabla u\|_{L^p(\mathbb{R}^d)}. \tag{1.9}$$

Moreover, if $U\subseteq\mathbb{R}^d$ is open and $u\in W^{1,p}_0(U)$, then $\|u\|_{L^{p^*}(U)} \leqslant C\gamma_p\|\nabla u\|_{L^p(U)}$.

Proof. Let C be the constant of the Gagliardo–Nirenberg inequality

$$\|v\|_{L^{d/(d-1)}(\mathbb{R}^d)} \leqslant C\|\nabla v\|_{L^1(\mathbb{R}^d)}, \qquad v\in C^1_c(\mathbb{R}^d).$$

This is (1.9) for $p=1$ and $u\in C^1_c(\mathbb{R}^d)$. Let $p\in(1,d)$ and $u\in C^1_c(\mathbb{R}^d)$. Since $\gamma_p>1$, the function $v := |u|^{\gamma_p}$ belongs to $C^1_c(\mathbb{R}^d)$ and $|\nabla v| = \gamma_p |u|^{\gamma_p-1}|\nabla u|$. By Hölder's inequality and the identities following (1.8),

$$\|u\|^{\gamma_p}_{L^{p^*}(\mathbb{R}^d)} = \|v\|_{L^{d/(d-1)}(\mathbb{R}^d)} \leqslant C\gamma_p \int_{\mathbb{R}^d} |u|^{\gamma_p-1}|\nabla u| \leqslant C\gamma_p \|u\|^{\gamma_p-1}_{L^{p^*}(\mathbb{R}^d)} \|\nabla u\|_{L^p(\mathbb{R}^d)}.$$

If $u=0$ there is nothing to prove. Otherwise, dividing by $\|u\|^{\gamma_p-1}_{L^{p^*}(\mathbb{R}^d)}$, which is finite and positive, gives (1.9) for $u\in C^1_c(\mathbb{R}^d)$. For $u\in W^{1,p}(\mathbb{R}^d)$ we take $u_n\in C^\infty_c(\mathbb{R}^d)$ with $u_n\to u$ in $W^{1,p}(\mathbb{R}^d)$ and almost everywhere, and apply Fatou's lemma. If $u\in W^{1,p}_0(U)$, we apply the same argument to approximants $u_n\in C^\infty_c(U)$, extended by zero. □

We next prove the Sobolev–Poincaré inequality on bounded convex domains. The constant depends only on d, p and the diameter, and not on the eccentricity of the domain. The first step is to put the domain in a normalized position by an affine map. Recall that normalized norms are invariant under affine changes of variables.

Lemma 1.6 (Affine normalization of convex domains)**.** *Let $V \subseteq \mathbb{R}^d$ be a nonempty bounded open convex set with $\operatorname{diam} V \leqslant D$, and set $R_d := \sqrt{d}\,(1 + d + \sqrt{d})$. There is an invertible affine map $T(x) = Ax + b$ such that $B_1 \subseteq T(V) \subseteq B_{R_d}$ and $|A^{-1}e| \leqslant \frac{D}{2}|e|$ for every $e \in \mathbb{R}^d$.*

Proof. The closure $K := \overline{V}$ is compact. The function $(x_0, \dots, x_d) \mapsto |\det(x_1 - x_0, \dots, x_d - x_0)|$ attains its maximum on K^{d+1} at some $(v_0, \dots, v_d)$. The maximum is positive, since V contains a small simplex. Let M be the matrix with columns $v_i - v_0$, and let $T_0(x) := M^{-1}(x - v_0)$, so that $T_0(v_0) = 0$ and $T_0(v_i) = e_i$. Fix $x \in K$ and put $y := T_0(x)$. Replacing v_i by x for $i \geqslant 1$ replaces the ith column of M by My, which multiplies the determinant by y_i. By maximality, $|y_i| \leqslant 1$ for every i, so $T_0(K) \subseteq [-1, 1]^d \subseteq \overline{B}_{\sqrt{d}}$. Moreover, K is convex and contains $v_0, \dots, v_d$, so $T_0(K)$ contains the standard simplex $\operatorname{conv}\{0, e_1, \dots, e_d\}$. With $r := (d + \sqrt{d})^{-1}$, the point $r\mathbf{1}$ has distance r to each face of this simplex, so the simplex contains $B_r(r\mathbf{1})$. Since V is open and convex, the interior of $T_0(K)$ is $T_0(V)$, and therefore $B_r(r\mathbf{1}) \subseteq T_0(V)$. Set $T := r^{-1}(T_0 - r\mathbf{1})$. Then $B_1 \subseteq T(V)$, and $T(V) \subseteq r^{-1}(\overline{B}_{\sqrt{d}} - r\mathbf{1}) \subseteq \overline{B}_{R_d}$ because $\sqrt{d}(1 + r)/r = R_d$. As $T(V)$ is open, it lies in the interior B_{R_d} of this closed ball. Finally, for $|e| = 1$ and $t \in (0, 1)$, the points $T^{-1}(\pm te)$ lie in V, so $2t|A^{-1}e| \leqslant D$. Letting $t \to 1$ gives $|A^{-1}e| \leqslant \frac{D}{2}$. □

The next lemma is the classical representation of the oscillation of a function on a convex domain by a Riesz potential [GT01, Lemma 7.16]. We write $I_1 f(x) := \int_{\mathbb{R}^d} |x - y|^{1-d} f(y)\, dy$.

Lemma 1.7 (Riesz potential bound on convex domains)**.** *Let $V \subseteq \mathbb{R}^d$ be a bounded open convex set with $\operatorname{diam} V \leqslant D$, let $u \in W^{1,1}(V)$, and let $S \subseteq V$ be measurable with $|S| > 0$. Then, for almost every $x \in V$,*

$$|u(x) - (u)_S| \leqslant \frac{D^d}{d|S|}\, I_1\big(|\nabla u|\mathbf{1}_V\big)(x). \tag{1.10}$$

Proof. Suppose first that $u \in C^1(V) \cap W^{1,1}(V)$. For $x, y \in V$ the segment $[x, y]$ lies in V, and with $\omega := (y - x)/|y - x|$,

$$u(x) - u(y) = -\int_0^{|x-y|} \omega \cdot \nabla u(x + r\omega)\, dr.$$

Let $g := |\nabla u|\mathbf{1}_V$. Integrating over $y \in S \subseteq B_D(x)$ and using polar coordinates centered at x,

$$|S||u(x) - (u)_S| \leqslant \int_{B_D(x)} \int_0^\infty g(x + r\omega)\, dr\, dy = \frac{D^d}{d} \int_{\partial B_1} \int_0^\infty g(x + r\omega)\, dr\, d\omega = \frac{D^d}{d}\, I_1 g(x).$$

For general $u \in W^{1,1}(V)$, take $u_n \in C^\infty(V) \cap W^{1,1}(V)$ with $u_n \to u$ in $W^{1,1}(V)$, by the Meyers–Serrin theorem. Then $(u_n)_S \to (u)_S$, and $I_1(|\nabla u_n|\mathbf{1}_V) \to I_1(|\nabla u|\mathbf{1}_V)$ in $L^1(V)$, since $\int_V |x - y|^{1-d}\, dx \leqslant d|B_1|D$ for every $y \in V$. Passing to an almost everywhere convergent subsequence gives (1.10). □

Lemma 1.8 (Weak-type bound for the Riesz potential)**.** *Let $d \geqslant 2$ and $d' := \frac{d}{d-1}$. For every nonnegative $f \in L^1(\mathbb{R}^d)$ and $\lambda > 0$,*

$$|\{I_1 f \geqslant \lambda\}| \leqslant 2^{d'} d|B_1| \left(\frac{\|f\|_{L^1(\mathbb{R}^d)}}{\lambda} \right)^{d'}.$$

Proof. If $\|f\|_{L^1} = 0$, then $I_1 f = 0$ and there is nothing to prove. Otherwise, let $\delta := (2\|f\|_{L^1}/\lambda)^{1/(d-1)}$, and split $I_1 f = N + F$, where N is the contribution of $|x-y| < \delta$. Then $F \leqslant \delta^{1-d}\|f\|_{L^1} = \lambda/2$, so $\{I_1 f \geqslant \lambda\} \subseteq \{N \geqslant \lambda/2\}$. By Fubini's theorem $\|N\|_{L^1} = d|B_1|\delta\|f\|_{L^1}$, and Chebyshev's inequality gives $|\{N \geqslant \lambda/2\}| \leqslant 2\lambda^{-1} d|B_1|\delta\|f\|_{L^1}$, which is the claim. □

Theorem 1.9 (Sobolev–Poincaré inequality on convex domains)**.** *Let $d \geqslant 2$. There exists $C(d) < \infty$ such that, for every nonempty bounded open convex set $V \subseteq \mathbb{R}^d$ with $\operatorname{diam} V \leqslant D$, every $p \in [1, d)$ and every $u \in W^{1,p}(V)$,*

$$\|u - (u)_V\|_{\underline{L}^{p^*}(V)} \leqslant C\gamma_p D \|\nabla u\|_{\underline{L}^p(V)}. \tag{1.11}$$

The factor γ_p has order $(d-p)^{-1}$ as $p \to d$, while the optimal order is $(d-p)^{-(d-1)/d}$. Since γ_p is bounded for p away from d, this loss does not affect the applications below.

Proof. Step 1. Functions vanishing on half of a normalized domain, $p = 1$. Let W be a bounded open convex set with $B_1 \subseteq W \subseteq B_{R_d}$, and let $d' := \frac{d}{d-1}$. We show that there is $K_d < \infty$ such that, for every nonnegative $v \in W^{1,1}(W)$ whose zero set $S := \{v = 0\}$ satisfies $|S| \geqslant \frac{1}{2}|W|$,

$$\|v\|_{L^{d'}(W)} \leqslant K_d \|\nabla v\|_{L^1(W)}. \tag{1.12}$$

For $k \in \mathbb{Z}$, let $v_k := \min\{(v - 2^k)_+, 2^k\}$. Then $v_k \in W^{1,1}(W)$, $\nabla v_k = \mathbf{1}_{\{2^k < v < 2^{k+1}\}} \nabla v$ almost everywhere, and $v_k = 0$ on S, so $(v_k)_S = 0$. Since $\operatorname{diam} W \leqslant 2R_d$ and $|S| \geqslant \frac{1}{2}|B_1|$, Lemma 1.7 gives $v_k \leqslant M_d I_1 f_k$ almost everywhere in W, where $f_k := |\nabla v_k| \mathbf{1}_W$ and $M_d := 2(2R_d)^d/(d|B_1|)$. Since $v_k = 2^k$ on $\{v \geqslant 2^{k+1}\}$, Lemma 1.8 yields

$$\left|\{v \geqslant 2^{k+1}\}\right| \leqslant \left|\{I_1 f_k \geqslant 2^k/M_d\}\right| \leqslant 2^{d'} d|B_1| \big(M_d 2^{-k}\|f_k\|_{L^1}\big)^{d'}.$$

Summing over the dyadic ranges of v, and using that the sets $\{2^k < v < 2^{k+1}\}$ are disjoint,

$$\int_W v^{d'} \leqslant \sum_{k \in \mathbb{Z}} 2^{(k+2)d'} \left|\{v \geqslant 2^{k+1}\}\right| \leqslant 2^{d'} d|B_1| (4M_d)^{d'} \sum_{k \in \mathbb{Z}} \|f_k\|_{L^1}^{d'} \leqslant 2^{d'} d|B_1| (4M_d)^{d'} \|\nabla v\|_{L^1(W)}^{d'}.$$

This is (1.12).

Step 2. Functions vanishing on half of a normalized domain, $p > 1$. Let $p \in (1, d)$, and let $v \in W^{1,p}(W)$ be nonnegative and bounded with $|\{v = 0\}| \geqslant \frac{1}{2}|W|$. Then $v^{\gamma_p} \in W^{1,1}(W)$ has the same zero set and $|\nabla v^{\gamma_p}| = \gamma_p v^{\gamma_p - 1}|\nabla v|$. As in the proof of Proposition 1.5, (1.12) and Hölder's inequality (dividing by the finite quantity $\|v\|_{L^{p^*}(W)}^{\gamma_p - 1}$ when it is positive) give $\|v\|_{L^{p^*}(W)} \leqslant K_d \gamma_p \|\nabla v\|_{L^p(W)}$. Applying this to $\min\{v, n\}$ and letting $n \to \infty$ removes the boundedness assumption. By Step 1, the same bound holds for $p = 1$.

Step 3. Normalized domains. Let $u \in W^{1,p}(W)$, and let m be a median of u on W, so that $|\{u > m\}| \leqslant \frac{1}{2}|W|$ and $|\{u < m\}| \leqslant \frac{1}{2}|W|$. The functions $(u-m)_\pm$ vanish on sets of measure at least $\frac{1}{2}|W|$, so Step 2 gives $\|u-m\|_{L^{p^*}(W)} \leqslant 2K_d\gamma_p\|\nabla u\|_{L^p(W)}$. By Hölder's inequality, $|(u)_W - m| \leqslant |W|^{-1/p^*}\|u-m\|_{L^{p^*}(W)}$, and hence

$$\|u-(u)_W\|_{L^{p^*}(W)} \leqslant 2\|u-m\|_{L^{p^*}(W)} \leqslant 4K_d\gamma_p\|\nabla u\|_{L^p(W)}.$$

Since $\frac{1}{p} - \frac{1}{p^*} = \frac{1}{d}$ and $|W| \leqslant |B_{R_d}|$, the normalized form reads $\|u-(u)_W\|_{\underline{L}^{p^*}(W)} \leqslant 4K_d|B_{R_d}|^{1/d}\gamma_p\|\nabla u\|_{\underline{L}^p(W)}$.

Step 4. General convex domains. Let $T(x) = Ax+b$ be as in Lemma 1.6, let $W := T(V)$ and $w := u \circ T^{-1} \in W^{1,p}(W)$. Then $\nabla w = A^{-t}(\nabla u)\circ T^{-1}$, and $|A^{-t}e| \leqslant \frac{D}{2}|e|$ because A^{-t} and A^{-1} have the same operator norm. Normalized norms and means are invariant under T, so Step 3 gives

$$\begin{aligned}\|u-(u)_V\|_{\underline{L}^{p^*}(V)} = \|w-(w)_W\|_{\underline{L}^{p^*}(W)} &\leqslant 4K_d|B_{R_d}|^{1/d}\gamma_p\|\nabla w\|_{\underline{L}^p(W)}\\ &\leqslant 2K_d|B_{R_d}|^{1/d}\gamma_p D\|\nabla u\|_{\underline{L}^p(V)}.\end{aligned}$$

This is (1.11) with $C := 2K_d|B_{R_d}|^{1/d}$. □

Since $\operatorname{diam} B_r = 2r$ and $\operatorname{diam} Q_r = 2\sqrt{d}\,r$ for the cubes $Q_r(x) := x + (-r,r)^d$, Theorem 1.9 gives, for every $p \in [1,d)$ and $u \in W^{1,p}$ of the ball or cube,

$$\begin{aligned}\|u-(u)_{B_r(x)}\|_{\underline{L}^{p^*}(B_r(x))} &\leqslant 2C\gamma_p r\|\nabla u\|_{\underline{L}^p(B_r(x))},\\ \|u-(u)_{Q_r(x)}\|_{\underline{L}^{p^*}(Q_r(x))} &\leqslant 2\sqrt{d}\,C\gamma_p r\|\nabla u\|_{\underline{L}^p(Q_r(x))}.\end{aligned} \tag{1.13}$$

The Poincaré inequality for mean-zero functions is still valid if we allow the mean to be defined with respect to any subset of positive measure.

Proposition 1.10 (Poincaré inequality)**.** *Let $d \geqslant 2$. There exists $C(d) < \infty$ such that, for every $\theta \in (0,1)$, $u \in H^1(B_1)$ and measurable $E \subseteq B_1$ satisfying $|E| \geqslant \theta|B_1|$, we have that*

$$\|u-(u)_E\|_{L^2(B_1)} \leqslant C\|\nabla u\|_{L^2(B_1)} \cdot \begin{cases}(1+|\log\theta|)^{\frac{1}{2}} & \text{if } d=2,\\ \theta^{\frac{2-d}{2d}} & \text{if } d>2.\end{cases} \tag{1.14}$$

Moreover, the difference of the means satisfies

$$\big|(u)_{B_1} - (u)_E\big| \leqslant C\|\nabla u\|_{L^2(B_1)} \cdot \begin{cases}|\log\theta|^{\frac{1}{2}} & \text{if } d=2,\\ \theta^{\frac{2-d}{2d}} & \text{if } d>2.\end{cases} \tag{1.15}$$

Proof. Write $\omega_d := |B_1|$, $w := u-(u)_{B_1}$ and $g := \|\nabla u\|_{L^2(B_1)}$. We may assume $g > 0$: if $g = 0$ then u is a.e. equal to a constant on the connected set B_1, and both sides of the estimates vanish.

Step 1. Poincaré inequalities for w. We claim that, for $C(d) < \infty$,

$$\|w\|_{L^2(B_1)} \leqslant Cg, \qquad \|w\|_{L^{2^*}(B_1)} \leqslant Cg \quad \text{if } d>2, \quad 2^* := \tfrac{2d}{d-2}. \tag{1.16}$$

If $d > 2$, apply (1.13) with $p = 2 < d$, $r = 1$, so that $p^* = 2^*$. It gives $\|w\|_{\underline{L}^{2^*}(B_1)} \leqslant C\|\nabla u\|_{\underline{L}^2(B_1)}$, and since $\|f\|_{L^s(B_1)} = \omega_d^{1/s}\|f\|_{\underline{L}^s(B_1)}$, the second bound in (1.16) holds with a constant depending on d. The first follows by Hölder's inequality on the ball. If $d = 2$, apply (1.13) with $p = 1 < d$, for which $p^* = 2$; together with Hölder's inequality, $\|w\|_{\underline{L}^2(B_1)} \leqslant C\|\nabla u\|_{\underline{L}^1(B_1)} \leqslant C\|\nabla u\|_{\underline{L}^2(B_1)}$, which is the first bound.

Step 2. Reduction. Since $\|u - (u)_E\|_{L^2(B_1)} \leqslant \|w\|_{L^2(B_1)} + \omega_d^{1/2}|(u)_{B_1} - (u)_E|$, by Step 1 the estimate (1.14) follows from (1.15), because for $\theta \in (0,1)$ we have $1 \leqslant \theta^{\frac{2-d}{2d}}$ if $d > 2$, and $1 \leqslant (1 + |\log\theta|)^{1/2}$ and $|\log\theta|^{1/2} \leqslant (1 + |\log\theta|)^{1/2}$ if $d = 2$. (The 1 in the case $d = 2$ cannot be dropped in (1.14): the right side of (1.15) tends to zero as $\theta \to 1$, while $\|u - (u)_E\|_{L^2(B_1)}$ does not.) It remains to prove (1.15). Note that, since $|E| \geqslant \theta\omega_d > 0$ and $\int_{B_1} w = 0$,

$$(u)_E - (u)_{B_1} = |E|^{-1}\int_E w. \tag{1.17}$$

Step 3. The case $d > 2$. Let $\phi := |E|^{-1}\mathbf{1}_E - \omega_d^{-1}\mathbf{1}_{B_1}$, so that $\int_{B_1}\phi\, w = |E|^{-1}\int_E w$ since $\int_{B_1} w = 0$. Set $s := \frac{2d}{d+2}$, so that $\frac{1}{s} + \frac{1}{2^*} = 1$ and $1 - \frac{1}{s} = \frac{d-2}{2d}$. By the triangle inequality, $|E| \leqslant \omega_d$ and $|E| \geqslant \theta\omega_d$,

$$\|\phi\|_{L^s(B_1)} \leqslant |E|^{-1+\frac{1}{s}} + \omega_d^{-1+\frac{1}{s}} \leqslant 2|E|^{-\frac{d-2}{2d}} \leqslant 2\omega_d^{-\frac{d-2}{2d}}\,\theta^{\frac{2-d}{2d}}.$$

By (1.17), Hölder's inequality and (1.16),

$$|(u)_E - (u)_{B_1}| = \left|\int_{B_1}\phi\, w\right| \leqslant \|\phi\|_{L^s(B_1)}\|w\|_{L^{2^*}(B_1)} \leqslant C\theta^{\frac{2-d}{2d}}\, g.$$

Step 4. The case $d = 2$ and $\theta \geqslant \frac{1}{2}$. Since $\int_{B_1} w = 0$, we have $\int_E w = -\int_{B_1\setminus E} w$, and $|B_1\setminus E| \leqslant (1-\theta)\omega_2$. By (1.17), the Cauchy-Schwarz inequality, $|E| \geqslant \frac{1}{2}\omega_2$, Step 1 and $1 - \theta \leqslant -\log\theta$,

$$|(u)_E - (u)_{B_1}| \leqslant |E|^{-1}|B_1\setminus E|^{\frac{1}{2}}\|w\|_{L^2(B_1)} \leqslant \frac{2}{\omega_2}\big((1-\theta)\omega_2\big)^{\frac{1}{2}}\, Cg \leqslant C|\log\theta|^{\frac{1}{2}}\, g.$$

Step 5. The case $d = 2$ and $\theta < \frac{1}{2}$. Proposition 1.18 with $d = 2$ (so $\frac{d}{d-1} = 2$) gives, for $C_T(2) < \infty$ and every $t > 0$, $|\{|w| > t\}| \leqslant C_T\exp(-t^2/a^2)$ with $a := C_T^{1/2}g$. Let $t_0 := a|\log\theta|^{1/2}$. By the layer cake formula, using $|E \cap \{|w| > t\}| \leqslant |E|$ for $t < t_0$,

$$\int_E |w| = \int_0^\infty |E \cap \{|w| > t\}|\, dt \leqslant |E|\, t_0 + C_T\int_{t_0}^\infty e^{-t^2/a^2}\, dt.$$

Writing $t = t_0 + ar$ and using $(t_0 + ar)^2 \geqslant t_0^2 + a^2r^2$, and $e^{-t_0^2/a^2} = \theta$, we get $\int_{t_0}^\infty e^{-t^2/a^2}dt \leqslant a\theta\int_0^\infty e^{-r^2}dr = \frac{\sqrt{\pi}}{2}a\theta$. Dividing by $|E| \geqslant \theta\omega_2$ and using (1.17),

$$|(u)_E - (u)_{B_1}| \leqslant |E|^{-1}\int_E |w| \leqslant t_0 + \frac{\sqrt{\pi}\, C_T}{2\omega_2}\, a = a\Big(|\log\theta|^{\frac{1}{2}} + C\Big) \leqslant Cg\,|\log\theta|^{\frac{1}{2}},$$

where in the last step we used $1 \leqslant |\log\theta|^{1/2}/(\log 2)^{1/2}$ since $\theta < \frac{1}{2}$.

Steps 3, 4 and 5 give (1.15), and the proof is complete. □

Remark 1.11. The proof shows that in the case $d = 2$ the Moser-Trudinger route (Step 5) is needed only for $\theta < \frac{1}{2}$; for θ near 1 the elementary Step 4 gives the correct (vanishing) dependence.

The Bombieri lemma is an application of the following abstract iteration, in the form of Bombieri and Giusti (see also Saloff-Coste, *Aspects of Sobolev-type inequalities*, Lemma 2.2.6). We include the proof, which uses only Hölder's and Chebyshev's inequalities.

Lemma 1.12 (Bombieri-Giusti lemma)**.** *Let $\delta \in (0,1)$, $\alpha_0 \in (0,\infty)$, $c \geqslant 1$, $m > 0$ and $\alpha_1 := \min\{1, \frac{\alpha_0}{2}\}$. Let ν be a probability measure on a set $U(1)$, let $\{U(\sigma)\}_{\sigma\in[\delta,1]}$ be ν-measurable subsets which are increasing in σ, and write $\|f\|_{\beta,\sigma} := (\int_{U(\sigma)} |f|^\beta \, d\nu)^{1/\beta}$. Let $f > 0$ be measurable with $f \in L^{\alpha_0}(U(1),\nu)$, and assume that*

$$\nu\big(U(1) \cap \{\log f > \lambda\}\big) \leqslant \frac{c}{\lambda} \quad \text{for every } \lambda > 0, \tag{1.18}$$

and that, for every $\delta \leqslant \sigma' < \sigma \leqslant 1$ and $\alpha \in (0, \alpha_1]$,

$$\|f\|_{\alpha_0,\sigma'} \leqslant \big(c(\sigma - \sigma')^{-m}\big)^{\frac{1}{\alpha}-\frac{1}{\alpha_0}} \|f\|_{\alpha,\sigma}. \tag{1.19}$$

Then $\|f\|_{\alpha_0,\delta} \leqslant C$ for some $C(\delta, c, m, \alpha_0) < \infty$.

Proof. Let $L(\sigma) := \log \|f\|_{\alpha_0,\sigma} \in [-\infty, \infty)$ and $h := \max\{L, 0\}$. Then h is nondecreasing and $h \leqslant h(1) < \infty$.

Step 1. A one-step estimate. Fix $\delta \leqslant \sigma' < \sigma \leqslant 1$, and set $\kappa := c(\sigma - \sigma')^{-m} \geqslant 1$, $L := L(\sigma)$ and $L_\kappa := 2c(2\kappa)^4$. We claim that

$$h(\sigma') \leqslant \tfrac{3}{4} h(\sigma) + L_\kappa + \alpha_1^{-1} \log(2\kappa). \tag{1.20}$$

If $L \leqslant L_\kappa$, then $h(\sigma') \leqslant h(\sigma) \leqslant \max\{L_\kappa, 0\} = L_\kappa$ and (1.20) holds. Suppose $L > L_\kappa$ and let $\lambda := L/2$, so that $\lambda > c(2\kappa)^4 \geqslant c$. For $\alpha \in (0, \alpha_1]$, split $U(\sigma)$ into $\{f \leqslant e^\lambda\}$ and $\{f > e^\lambda\}$ and apply Hölder's inequality with exponents $\frac{\alpha_0}{\alpha}$ and its conjugate on the second set, together with (1.18):

$$\int_{U(\sigma)} f^\alpha \, d\nu \leqslant e^{\alpha\lambda} + \|f\|_{\alpha_0,\sigma}^{\alpha} \, \nu(U(1) \cap \{f > e^\lambda\})^{1-\frac{\alpha}{\alpha_0}} \leqslant e^{\frac{\alpha L}{2}} + e^{\alpha L} \Big(\frac{c}{\lambda}\Big)^{\frac{1}{2}},$$

where we used $\frac{\alpha}{\alpha_0} \leqslant \frac{1}{2}$ and $c/\lambda < 1$. Now choose $\alpha := \min\{\alpha_1, \ L^{-1}\log(\lambda/c)\} \in (0, \alpha_1]$. Since $\alpha L \leqslant \log(\lambda/c)$, we have $e^{\alpha L}(c/\lambda)^{1/2} \leqslant e^{\alpha L/2}$, hence $\|f\|_{\alpha,\sigma} \leqslant 2^{1/\alpha} e^{L/2}$. By (1.19) and $\kappa \geqslant 1$,

$$L(\sigma') \leqslant \Big(\frac{1}{\alpha} - \frac{1}{\alpha_0}\Big) \log \kappa + \frac{\log 2}{\alpha} + \frac{L}{2} \leqslant \frac{\log(2\kappa)}{\alpha} + \frac{L}{2}.$$

If $\alpha = \alpha_1$, then $\alpha^{-1}\log(2\kappa) = \alpha_1^{-1}\log(2\kappa)$. Otherwise $\alpha = L^{-1}\log(\lambda/c)$ and, since $L > 2c(2\kappa)^4$, $\log(\lambda/c) = \log \frac{L}{2c} > 4\log(2\kappa)$, so $\alpha^{-1}\log(2\kappa) < L/4$. In both cases $L(\sigma') \leqslant \frac{3}{4}L + \alpha_1^{-1}\log(2\kappa)$, and $L = h(\sigma) > 0$; this gives (1.20).

Step 2. Iteration. Since $\log(2\kappa) \leqslant 2\kappa \leqslant 2\kappa^4$ and $L_\kappa = 32c\kappa^4 = 32c^5(\sigma - \sigma')^{-4m}$, (1.20) yields, with $K := 32c^5 + 2\alpha_1^{-1}c^4$, $\vartheta := \frac{3}{4}$ and $\beta := 4m$,

$$h(\sigma') \leqslant \vartheta h(\sigma) + K(\sigma - \sigma')^{-\beta}, \qquad \delta \leqslant \sigma' < \sigma \leqslant 1.$$

Choose $\tau := \frac{1}{2}(1+\vartheta^{1/\beta}) \in (0,1)$, so that $\vartheta\tau^{-\beta} < 1$, and let $\sigma_i := 1-(1-\delta)\tau^i$, so that $\sigma_0 = \delta$, $\sigma_{i+1}-\sigma_i = (1-\delta)(1-\tau)\tau^i$ and $\sigma_i \leqslant 1$. Iterating the previous display k times,

$$h(\delta) \leqslant \vartheta^k h(\sigma_k) + K\sum_{i=0}^{k-1} \vartheta^i(\sigma_{i+1}-\sigma_i)^{-\beta} \leqslant \vartheta^k h(1) + \frac{K}{((1-\delta)(1-\tau))^\beta}\sum_{i=0}^{k-1}(\vartheta\tau^{-\beta})^i.$$

Letting $k\to\infty$ gives $h(\delta) \leqslant K((1-\delta)(1-\tau))^{-\beta}(1-\vartheta\tau^{-\beta})^{-1}$, and $\|f\|_{\alpha_0,\delta} \leqslant e^{h(\delta)}$. □

Lemma 1.13 (Bombieri lemma)**.** *Assume $q,k\in(0,\infty)$, $A>0$ and $v\in L^q(B_1)$ satisfy $v>0$ a.e.,*

$$\|\log v\|_{L^1(B_1)} \leqslant A, \tag{1.21}$$

and, for every $\frac{1}{2}\leqslant r<s\leqslant 1$ and $p\in(0,\min\{1,\frac{q}{2}\}]$,

$$\|v\|_{L^q(B_r)} \leqslant \left(\frac{A}{(s-r)^k}\right)^{\frac{1}{p}} \|v\|_{\underline{L}^p(B_s)}. \tag{1.22}$$

There exists a constant $C(q,k,A,d)<\infty$ such that

$$\|v\|_{L^q(B_{1/2})} \leqslant C. \tag{1.23}$$

Proof. Let $\omega_d := |B_1|$ and apply Lemma 1.12 with $U(\sigma) := B_\sigma$, $\delta := \frac{1}{2}$, $\nu := \omega_d^{-1}dx$ on B_1, $f := v$, $\alpha_0 := q$ (so $\alpha_1 = \min\{1,\frac{q}{2}\}$), $m := 2k$ and

$$c := \max\{1,\omega_d^{-1}\}\max\{1,(2^dA)^2\} \geqslant 1.$$

Step 1. Hypothesis (1.18). For $\lambda>0$, Chebyshev's inequality and (1.21) give $|B_1\cap\{\log v>\lambda\}| \leqslant \lambda^{-1}\int_{B_1}(\log v)_+ \leqslant A/\lambda$, that is, $\nu(\{\log v>\lambda\}) \leqslant \frac{A}{\omega_d\lambda} \leqslant \frac{c}{\lambda}$, since $\frac{A}{\omega_d} \leqslant \max\{1,\omega_d^{-1}\}\max\{1,A\} \leqslant c$.

Step 2. Hypothesis (1.19). Let $\frac{1}{2}\leqslant r<s\leqslant 1$ and $p\in(0,\alpha_1]$, so $p\leqslant\frac{q}{2}$ and $\frac{1}{q}\leqslant\frac{1}{p}-\frac{1}{q}$. Since $\|f\|_{L^\beta(B_\sigma)} = \omega_d^{1/\beta}\|f\|_{\beta,\sigma}$ and $\|v\|_{\underline{L}^p(B_s)} = |B_s|^{-1/p}\|v\|_{L^p(B_s)} = s^{-d/p}\|v\|_{p,s}$, (1.22) becomes

$$\|v\|_{q,r} \leqslant \omega_d^{-\frac{1}{q}} Y^{\frac{1}{p}} \|v\|_{p,s}, \qquad Y := \frac{A}{(s-r)^k s^d} \leqslant \frac{2^dA}{(s-r)^k}.$$

Since $s-r\leqslant\frac{1}{2}$, we have $\max\{1,Y^2\} \leqslant \max\{1,(2^dA)^2\}(s-r)^{-2k}$. Using $\omega_d^{-1/q} \leqslant \max\{1,\omega_d^{-1}\}^{\frac{1}{p}-\frac{1}{q}}$, and $Y^{1/p} \leqslant \max\{1,Y^2\}^{\frac{1}{2p}} \leqslant \max\{1,Y^2\}^{\frac{1}{p}-\frac{1}{q}}$ (as $\frac{1}{2p}\leqslant\frac{1}{p}-\frac{1}{q}$, and $\max\{1,Y^2\}\geqslant 1$), we conclude

$$\|v\|_{q,r} \leqslant \left(c(s-r)^{-2k}\right)^{\frac{1}{p}-\frac{1}{q}} \|v\|_{p,s},$$

which is (1.19) with $m=2k$ (for $k=2$ this is $m=4$).

Step 3. Conclusion. Since $v\in L^q(B_1)$, Lemma 1.12 applies and gives $\|v\|_{L^q(B_{1/2})} = \omega_d^{1/q}\|v\|_{q,1/2} \leqslant C(q,c,2k,d)$, and c depends only on A and d. □

For any $u \in L^1_{\mathrm{loc}}(U)$, we define the norm

$$\|u\|_{\mathrm{BMO}(U)} := \sup\Big\{ \fint_B |u - (u)_B| \,:\, B \text{ is a ball such that } \overline{B} \subseteq U \Big\}. \tag{1.24}$$

We define the space of *bounded mean oscillation*, denoted by $\mathrm{BMO}(U)$, by

$$\mathrm{BMO}(U) := \Big\{ u \in L^1_{\mathrm{loc}}(U) \,:\, \|u\|_{\mathrm{BMO}(U)} < \infty \Big\}. \tag{1.25}$$

If U is connected, then $\mathrm{BMO}(U)$ is a Banach space after we identify functions which differ by a constant.

The Poincaré inequality implies that $W^{1,d}(\mathbb{R}^d) \subseteq \mathrm{BMO}(\mathbb{R}^d)$ and, for a constant $C(d) < \infty$,

$$\|u\|_{\mathrm{BMO}(\mathbb{R}^d)} \leqslant C \|\nabla u\|_{L^d(\mathbb{R}^d)}. \tag{1.26}$$

Indeed, let B be a ball of radius r. For $d \geqslant 2$, Theorem 1.9 with $p = 1$ and $D = 2r$ gives $\fint_B |u-(u)_B| \leqslant \|u-(u)_B\|_{L^{d/(d-1)}(B)} \leqslant Cr \fint_B |\nabla u|$, and for $d = 1$ this is the fundamental theorem of calculus. By Hölder's inequality,

$$\fint_B |u - (u)_B| \leqslant Cr \fint_B |\nabla u| \leqslant Cr|B|^{-\frac{1}{d}} \left(\int_B |\nabla u|^d \right)^{\frac{1}{d}} = C(d) \|\nabla u\|_{L^d(B)}. \tag{1.27}$$

The John–Nirenberg inequality states that BMO functions are locally exponentially integrable. We prove it below by a stopping-time argument on dyadic cubes. The Vitali covering lemma, which we prove first, is used for the maximal function estimate in the proof of Trudinger's inequality.

Lemma 1.14 (Vitali covering lemma)**.** *Let $\mathcal{C} = \{B^\alpha : \alpha \in \mathcal{I}\}$ be a nonempty collection of open balls of positive radius in $\mathbb{R}^d$ with $R := \sup_{\alpha \in \mathcal{I}} \operatorname{rad}(B^\alpha) < \infty$. Then there exists a countable (possibly finite), pairwise disjoint subcollection $\{B^{\alpha_i}\}_i \subseteq \mathcal{C}$ such that, for every $\alpha \in \mathcal{I}$, there exists an index i with*

$$B^\alpha \cap B^{\alpha_i} \neq \varnothing \quad \text{and} \quad \operatorname{rad}(B^\alpha) < 2 \operatorname{rad}(B^{\alpha_i}). \tag{1.28}$$

In particular, $B^\alpha \subseteq 5B^{\alpha_i}$ for this i, and

$$\bigcup_{\alpha \in \mathcal{I}} B^\alpha \subseteq \bigcup_i 5B^{\alpha_i}. \tag{1.29}$$

Here $5B$ denotes the ball with the same center as B and five times the radius.

Proof. For $j \geqslant 0$, let $\mathcal{F}_j := \{\alpha \in \mathcal{I} : 2^{-j-1}R < \operatorname{rad}(B^\alpha) \leqslant 2^{-j}R\}$; these classes partition $\mathcal{I}$. We define subsets $\mathcal{G}_j \subseteq \mathcal{F}_j$ inductively. Given $\mathcal{G}_0, \dots, \mathcal{G}_{j-1}$, let $\mathcal{F}'_j$ be the set of $\alpha \in \mathcal{F}_j$ such that B^α is disjoint from every ball B^β with $\beta \in \mathcal{G}_0 \cup \cdots \cup \mathcal{G}_{j-1}$. By Zorn's lemma, $\mathcal{F}'_j$ has a maximal subset $\mathcal{G}_j$ of pairwise disjoint balls. The union of the $\mathcal{G}_j$ is a pairwise disjoint subcollection by construction, and it is countable since every ball contains a point of $\mathbb{Q}^d$.

Let $\alpha \in \mathcal{F}_j$. If $\alpha \in \mathcal{F}'_j$, then either $\alpha \in \mathcal{G}_j$ or, by maximality, B^α meets some ball of $\mathcal{G}_j$. If $\alpha \notin \mathcal{F}'_j$, then B^α meets some ball of $\mathcal{G}_{j'}$ with $j' < j$. In either case, B^α meets some B^β with $\beta \in \mathcal{G}_{j'}$ for some $j' \leqslant j$, and then $\operatorname{rad}(B^\beta) > 2^{-j'-1}R \geqslant 2^{-j-1}R \geqslant \frac{1}{2}\operatorname{rad}(B^\alpha)$. This is (1.28). Finally, let x_β and x_α be the centers, $\rho := \operatorname{rad}(B^\alpha)$ and $z \in B^\alpha \cap B^\beta$. Every $y \in B^\alpha$ satisfies $|y - x_\beta| < \rho + \rho + \operatorname{rad}(B^\beta) < 5 \operatorname{rad}(B^\beta)$, since $|y - x_\alpha| < \rho$, $|x_\alpha - z| < \rho$ and $|z - x_\beta| < \operatorname{rad}(B^\beta)$. □

The following consequence is the weak $(1,1)$ estimate for the centered maximal function. For $g \in L^1(\mathbb{R}^d)$ we set

$$\mathsf{M}g(x) := \sup_{\rho>0} \fint_{B_\rho(x)} |g|,$$

which is lower semicontinuous, since each $x \mapsto \fint_{B_\rho(x)} |g|$ is continuous.

Lemma 1.15 (Weak-type maximal inequality)**.** *For every $g \in L^1(\mathbb{R}^d)$ and $\lambda > 0$,*

$$|\{\mathsf{M}g > \lambda\}| \leqslant \frac{5^d}{\lambda} \|g\|_{L^1(\mathbb{R}^d)}.$$

Proof. If $\{\mathsf{M}g > \lambda\} = \varnothing$ there is nothing to prove. Otherwise, for every x with $\mathsf{M}g(x) > \lambda$ there is a ball $B^x = B_{\rho_x}(x)$ with $\int_{B^x} |g| > \lambda |B^x|$; in particular $|B^x| < \|g\|_{L^1}/\lambda$, so the radii are bounded. By Lemma 1.14 there is a countable disjoint subfamily $\{B_i\}$ with $\{\mathsf{M}g > \lambda\} \subseteq \bigcup_i 5B_i$. Then $|\{\mathsf{M}g > \lambda\}| \leqslant 5^d \sum_i |B_i| \leqslant 5^d \lambda^{-1} \sum_i \int_{B_i} |g| \leqslant 5^d \lambda^{-1} \|g\|_{L^1}$. □

A *cube* is a closed axis-parallel cube $Q \subseteq \mathbb{R}^d$, with side length $\ell(Q)$; its *dyadic subcubes* are obtained by repeatedly bisecting all sides.

Lemma 1.16 (John–Nirenberg inequality on cubes)**.** *Let Q be a cube, $v \in L^1(Q)$ and $N > 0$ be such that $\fint_P |v - (v)_P| \leqslant N$ for every dyadic subcube P of Q (including Q). Then, for every $t > 0$,*

$$|\{x \in Q \,:\, |v(x) - (v)_Q| > t\}| \leqslant e^2 |Q| \exp\left(-\frac{t}{2^d e N}\right). \tag{1.30}$$

Proof. Replacing v by v/N and t by t/N, we may assume $N = 1$.

Step 1. One generation. Let P be a dyadic subcube of Q. Let $\mathcal{S}(P)$ be the family of maximal proper dyadic subcubes $P' \subsetneq P$ with $\fint_{P'} |v - (v)_P| > e$. (Every chain of dyadic ancestors is finite, so every such cube lies in a maximal one.) Then:

(i) the cubes of $\mathcal{S}(P)$ have pairwise disjoint interiors (and boundaries of measure zero), so $\sum_{P' \in \mathcal{S}(P)} |P'| \leqslant e^{-1} \sum_{P'} \int_{P'} |v - (v)_P| \leqslant e^{-1} \int_P |v - (v)_P| \leqslant e^{-1} |P|$;

(ii) $|v - (v)_P| \leqslant e$ almost everywhere in $P \setminus \bigcup \mathcal{S}(P)$: if x is in no $P' \in \mathcal{S}(P)$, every proper dyadic subcube containing x has average of $|v - (v)_P|$ at most e, and the Lebesgue differentiation theorem along these shrinking cubes gives the claim for almost every such x;

(iii) $|(v)_{P'} - (v)_P| \leqslant 2^d e$ for every $P' \in \mathcal{S}(P)$: the dyadic parent $\widehat{P}$ of P' satisfies $\fint_{\widehat{P}} |v - (v)_P| \leqslant e$, either since $\widehat{P} \neq P$ is not selected by maximality, or since $\widehat{P} = P$ and $\fint_P |v - (v)_P| \leqslant 1$. Hence $|(v)_{P'} - (v)_P| \leqslant \fint_{P'} |v - (v)_P| \leqslant 2^d \fint_{\widehat{P}} |v - (v)_P| \leqslant 2^d e$.

Step 2. Iteration. Let $\mathcal{G}_0 := \{Q\}$ and $\mathcal{G}_{m+1} := \bigcup_{P \in \mathcal{G}_m} \mathcal{S}(P)$, and let G_m be the union of $\mathcal{G}_m$. By (i) (the cubes of $\mathcal{G}_m$ have pairwise disjoint interiors), $|G_m| \leqslant e^{-m} |Q|$. Fix $m \geqslant 0$ and let $x \in Q \setminus G_m$ outside a null set. Then $x \in G_i \setminus G_{i+1}$ for some $i < m$, so x belongs to a chain $Q = P^{(0)} \supset P^{(1)} \supset \cdots \supset P^{(i)}$ with $P^{(q+1)} \in \mathcal{S}(P^{(q)})$ and $x \notin \bigcup \mathcal{S}(P^{(i)})$. By (ii) and (iii),

$$|v(x) - (v)_Q| \leqslant |v(x) - (v)_{P^{(i)}}| + \sum_{q<i} |(v)_{P^{(q+1)}} - (v)_{P^{(q)}}| \leqslant e + i 2^d e \leqslant e(1 + m 2^d).$$

Therefore $\{|v-(v)_Q| > e(1+m2^d)\} \subseteq G_m$ up to a null set, and its measure is at most $e^{-m}|Q|$.

Step 3. Conclusion. If $t \geqslant e$, let $m := \lfloor (t/e-1)/2^d \rfloor \geqslant 0$. Then $t \geqslant e(1+m2^d)$ and $m > (t/e-1)2^{-d} - 1$, so the left side of (1.30) is at most $e^{-m}|Q| \leqslant e^{1+2^{-d}} \exp(-t/(2^d e))|Q|$. If $t < e$, the right side of (1.30) is at least $e^2 e^{-1}|Q| \geqslant |Q|$. □

We now deduce the ball version. In the statement, balls are open, and the right side is interpreted as 0 if $\|u\|_{\mathrm{BMO}(B)} = 0$.

Proposition 1.17 (John–Nirenberg inequality)**.** *There exists $C(d) < \infty$ such that, for every ball $B \subseteq \mathbb{R}^d$, $u \in \mathrm{BMO}(B)$ and $t > 0$,*

$$\frac{\left|\{x \in B \,:\, |u(x) - (u)_B| > t\}\right|}{|B|} \leqslant C \exp\left(- \frac{t}{C\|u\|_{\mathrm{BMO}(B)}} \right). \tag{1.31}$$

In particular, $u \in L^1(B)$, so that $(u)_B$ is defined.

Proof. Both sides of (1.31) are invariant under translations and dilations of B, so we may assume $B = B_1$. Fix $N > \|u\|_{\mathrm{BMO}(B_1)}$; it suffices to prove (1.31) with N in place of $\|u\|_{\mathrm{BMO}(B_1)}$ and $C(d)$ independent of N, and then let $N \downarrow \|u\|_{\mathrm{BMO}(B_1)}$. For a point $x \in B_1$ write $s(x) := 1 - |x| \in (0,1]$. Two elementary facts are used repeatedly: if $E \subseteq G$ are balls with closures in B_1, then

$$|(u)_E - (u)_G| \leqslant \fint_E |u - (u)_G| \leqslant \frac{|G|}{|E|} \fint_G |u - (u)_G| \leqslant \frac{|G|}{|E|} N, \tag{1.32}$$

and every cube contained in a ball G with $\overline{G} \subseteq B_1$ is compact in B_1, so that u is integrable on it.

Step 1. Oscillation on cubes. Let Q be a cube with center c such that the ball $B_Q := B_{\sqrt{d}\ell(Q)/2}(c)$ satisfies $\overline{B_Q} \subseteq B_1$. Let $Q' \subseteq Q$ be any cube, with center c', and $B' := B_{\sqrt{d}\ell(Q')/2}(c') \supseteq Q'$. Since $c' - c$ has all coordinates of size at most $(\ell(Q) - \ell(Q'))/2$, we have $|c'-c| + \sqrt{d}\ell(Q')/2 \leqslant \sqrt{d}\ell(Q)/2$, so $\overline{B'} \subseteq \overline{B_Q} \subseteq B_1$. With $\omega := |B_1|(\sqrt{d}/2)^d = |B'|/|Q'|$,

$$\fint_{Q'} |u - (u)_{Q'}| \leqslant 2 \fint_{Q'} |u - (u)_{B'}| \leqslant 2\omega \fint_{B'} |u - (u)_{B'}| \leqslant 2\omega N. \tag{1.33}$$

Lemma 1.16 applied to $v = u$ and $2\omega N$ in place of N gives, for every $t > 0$,

$$|\{x \in Q \,:\, |u(x) - (u)_Q| > t\}| \leqslant e^2 |Q| \exp\left(- \frac{t}{2^{d+1} e \omega N} \right). \tag{1.34}$$

Step 2. A comparison of averages. Let $H := B_{3/4}$. Let $k \geqslant 0$ be an integer and let Q be a cube of side $\ell_k := 2^{-k}/(4\sqrt{d})$ whose center c satisfies $s(c) \in [\frac{3}{8}2^{-k}, \frac{9}{8}2^{-k}]$ (this is the situation in Step 3). Then $B_Q = B_{2^{-k}/8}(c)$, $\overline{B_Q} \subseteq B_{s(c)/2}(c)$, and $\overline{B_{s(c)/2}(c)} \subseteq B_1$, so Step 1 applies, and

$$|(u)_Q - (u)_H| \leqslant K(k+1)N, \qquad K = K(d) := \omega + 5^d + 3^{d+1}. \tag{1.35}$$

Write $s := s(c)$ and $E_0 := B_{s/2}(c)$, whose closure lies in B_1 since $|c| + s/2 = 1 - s/2$. We have $|E_0|/|B_Q| \leqslant (s2^{k+2})^d \leqslant 5^d$, hence by (1.32)

$$|(u)_Q - (u)_{E_0}| \leqslant |(u)_Q - (u)_{B_Q}| + |(u)_{B_Q} - (u)_{E_0}| \leqslant (\omega + 5^d)N.$$

(The first term is bounded by $\fint_Q |u - (u)_{B_Q}| \leqslant \omega N$.) Now let $n := c/|c|$ (any unit vector if $c = 0$), $s_j := 2^j s$, $p_j := (1 - s_j)n$ and $E_j := B_{s_j/2}(p_j)$ for $0 \leqslant j \leqslant J$, where $J \geqslant 0$ is the least integer with $s_J \geqslant \frac{1}{2}$. If $J = 0$ then $J \leqslant k + 1$ trivially. If $J \geqslant 1$, then $s_{J-1} < \frac{1}{2}$, so $2^J s < 1$, and $s \geqslant \frac{3}{8}2^{-k}$ gives $2^J < \frac{8}{3}2^k < 2^{k+2}$, hence $J \leqslant k + 1$. For $j < J$ let $G_j := B_{3s_j/2}(p_{j+1})$. Since $|p_j - p_{j+1}| = s_j$, we have $E_j \subseteq G_j$ and $E_{j+1} \subseteq G_j$, and $\overline{G_j} \subseteq B_1$ because $|p_{j+1}| + \frac{3}{2}s_j = 1 - \frac{1}{2}s_j$. By (1.32), since $|G_j|/|E_j| = 3^d$ and $|G_j|/|E_{j+1}| = (3/2)^d$,

$$|(u)_{E_j} - (u)_{E_{j+1}}| \leqslant (3^d + (3/2)^d)N \leqslant 2 \cdot 3^d N.$$

Finally $s_J \in [\frac{1}{2}, 1)$ when $J \geqslant 1$ and $s_J = s \geqslant \frac{1}{2}$ if $J = 0$, so $E_J \subseteq H$ (as $|p_J| + s_J/2 = 1 - s_J/2 \leqslant \frac{3}{4}$) and $|H|/|E_J| \leqslant 3^d$, so $|(u)_{E_J} - (u)_H| \leqslant 3^d N$. Adding up, $|(u)_Q - (u)_H| \leqslant (\omega + 5^d + 2(k+1)3^d + 3^d)N$, which is (1.35).

Step 3. A covering of B_1 by cubes. Let $A_0 := \{s \geqslant \frac{1}{2}\}$ and, for $k \geqslant 1$, $A_k := \{2^{-k-1} \leqslant s < 2^{-k}\}$, so that $B_1 = \bigcup_{k \geqslant 0} A_k$. Let $\ell_k := 2^{-k}/(4\sqrt{d})$ and let $\mathcal{Q}_k$ be the set of cubes of the grid $\ell_k \mathbb{Z}^d$ (side ℓ_k) which meet A_k. If $Q \in \mathcal{Q}_k$ has center c, then $|c - x_0| \leqslant \sqrt{d}\ell_k/2 = 2^{-k}/8$ for some $x_0 \in A_k$, so $s(c) \in [\frac{3}{8}2^{-k}, \frac{9}{8}2^{-k}]$ (the upper bound is clear for $k = 0$ as well). In particular $B_Q = B_{2^{-k}/8}(c)$ has closure in $B_{s(c)/2}(c)$, which has closure in B_1, and the hypotheses of Steps 1 and 2 hold. The cubes of $\mathcal{Q}_k$ have disjoint interiors and lie in $\{x \in B_1 : s(x) \leqslant \frac{5}{4}2^{-k}\}$, a set of measure at most $\frac{5}{4}d2^{-k}|B_1|$. Thus

$$\sum_{Q \in \mathcal{Q}_k} |Q| \leqslant 2d\, 2^{-k}|B_1|. \tag{1.36}$$

Step 4. Integrability. By Step 1 with $Q' = Q$, and (1.35), for $Q \in \mathcal{Q}_k$ we have $\int_Q |u - (u)_H| \leqslant |Q|(2\omega + K(k+1))N$. Summing over $Q \in \mathcal{Q}_k$ and k and using (1.36),

$$\int_{B_1} |u - (u)_H| \leqslant 2d|B_1|N \sum_{k \geqslant 0} 2^{-k}(2\omega + K(k+1)) = K_1|B_1|N,$$

with $K_1 = K_1(d) < \infty$. Hence $u \in L^1(B_1)$ and $|(u)_{B_1} - (u)_H| \leqslant K_1 N$.

Step 5. Distribution function. Let $a := (2^{d+1}e\omega)^{-1}$ and $\tau := t/N$. For $Q \in \mathcal{Q}_k$, (1.35) and (1.34) give

$$|\{x \in Q \,:\, |u - (u)_H| > t\}| \leqslant |\{x \in Q : |u - (u)_Q| > t - K(k+1)N\}| \leqslant |Q|\, m_k,$$

where $m_k := \min\{1, e^2 \exp(-a(\tau - K(k+1)))\}$. Let k_0 be the least integer $k \geqslant 0$ with $K(k+1) > \tau/2$. Since $m_k \leqslant e^2 e^{-a\tau/2}$ for $k < k_0$ and $m_k \leqslant 1$ otherwise, (1.36) and the covering property of Step 3 yield

$$\frac{|\{x \in B_1 : |u - (u)_H| > t\}|}{|B_1|} \leqslant 2d \sum_{k < k_0} 2^{-k} e^2 e^{-a\tau/2} + 2d \sum_{k \geqslant k_0} 2^{-k} \leqslant 4de^2 e^{-a\tau/2} + 4d\, 2^{-k_0}.$$

As $k_0 \geqslant \tau/(2K) - 1$, the right side is at most $C \exp(-\tau/C)$ for a constant $C(d)$. Finally, if $\tau \geqslant 2K_1$, then $\{|u - (u)_{B_1}| > t\} \subseteq \{|u - (u)_H| > t/2\}$ by Step 4, and if $\tau < 2K_1$ the inequality is trivial after enlarging C. This proves (1.31) with N in place of $\|u\|_{\mathrm{BMO}(B_1)}$. □

The John–Nirenberg inequality is sharp. Indeed, $u(x) = \log|x|$ belongs to $\mathrm{BMO}(B_1)$ (see below) and $(u)_{B_1} = -\frac{1}{d}$, while $|\{x \in B_1 : u(x) < -t\}| = |B_1|e^{-dt}$, so the reverse of (1.31) holds in $B = B_1$ with a small constant $c(d) > 0$ in place of C. To see that $u \in \mathrm{BMO}(B_1)$, let $B_\rho(z) \subseteq B_1$. If $|z| \geqslant 2\rho$, then u is Lipschitz on $B_\rho(z)$ with constant $2/|z|$, so its oscillation is at most $2\rho \cdot 2/|z| \leqslant 2$. If $|z| < 2\rho$, then $B_\rho(z) \subseteq B_{3\rho}$ and $\fint_{B_\rho(z)} |u - \log(3\rho)| \leqslant 3^d \fint_{B_{3\rho}} |\log(|x|/3\rho)|\,dx = 3^d \fint_{B_1} |\log|x||\,dx$; the average of $|u-(u)_{B_\rho(z)}|$ is at most twice this.

The combination of (1.26) and (1.31) yields exponential integrability for functions in $W^{1,d}$. However, this is *not* a sharp result. Indeed, by a straightforward computation one can check that, for $p > 0$,

$$u(x) = \log^p\bigl(2 + |x|^{-1}\bigr) \in W^{1,d}(B_1) \quad \text{if and only if } p < \tfrac{d-1}{d}, \tag{1.37}$$

since $|\nabla u| \approx \log^{p-1}(1/|x|)\,|x|^{-1}$ near the origin, so that $\int_{B_1} |\nabla u|^d$ is comparable to $\int_0 \log^{d(p-1)}(1/r)\,\frac{dr}{r}$, which is finite if and only if $d(p-1) < -1$. (For $p \leqslant 0$ the function u is bounded and the same computation shows that $u \in W^{1,d}(B_1)$.) Therefore there is a gap between the integrability of the "worst" BMO function and the worst $W^{1,d}$ function. For this reason, contrary to the usual practice, we should not think of the embedding $W^{1,d}(\mathbb{R}^d) \hookrightarrow \mathrm{BMO}(\mathbb{R}^d)$ as the "critical Sobolev embedding," at least when it comes to the best estimate on the distribution function of u. This title should be bestowed rather on the following result, which can be guessed from (1.37).

Proposition 1.18 (Trudinger inequality)**.** *Let $d \geqslant 2$. There exists $C(d) < \infty$ such that, for every $u \in W^{1,d}(B_1)$ and $t > 0$,*

$$\Bigl|\bigl\{x \in B_1 \,:\, |u(x) - (u)_{B_1}| > t\bigr\}\Bigr| \leqslant C \exp\left(-\frac{1}{C}\left(\frac{t}{\|\nabla u\|_{L^d(B_1)}}\right)^{\frac{d}{d-1}}\right), \tag{1.38}$$

where the right side is interpreted as 0 if $\nabla u = 0$.

Proof. If $\nabla u = 0$ almost everywhere, then u is constant on the connected set B_1 and there is nothing to prove. Otherwise, by dividing u by $\|\nabla u\|_{L^d(B_1)}$ and subtracting $(u)_{B_1}$, we may assume $\|\nabla u\|_{L^d(B_1)} = 1$ and $(u)_{B_1} = 0$. Let $g := |\nabla u|\mathbf{1}_{B_1}$, extended by zero, and let $d' := \frac{d}{d-1}$. Then $g \in L^1(\mathbb{R}^d)$ with $\|g\|_{L^1} \leqslant |B_1|^{1/d'}\|g\|_{L^d} = |B_1|^{1/d'}$. Write $\mathsf{M}g$ for the maximal function of Lemma 1.15. Since g is not zero almost everywhere, $\mathsf{M}g(x) > 0$ for every x.

Step 1. Pointwise bound. Lemma 1.7 with $V = S = B_1$ and $D = 2$ gives, for almost every $x \in B_1$,

$$|u(x)| \leqslant c_1 I_1 g(x) = c_1 \int_{B_1} \frac{|\nabla u(y)|}{|x-y|^{d-1}}\,dy, \qquad c_1 := \frac{2^d}{d|B_1|}. \tag{1.39}$$

Fix such an x with $m := \mathsf{M}g(x) < \infty$ (if $m = \infty$, the bound (1.40) below is trivial), and fix $\varepsilon > 0$. We split the integral at $|x-y| = \varepsilon$.

Step 2. The near part. Decomposing into the annuli $\{2^{-k-1}\varepsilon \leqslant |y-x| < 2^{-k}\varepsilon\}$, $k \geqslant 0$,

$$\int_{\{|y-x|<\varepsilon\}} \frac{g(y)}{|x-y|^{d-1}}\,dy \leqslant \sum_{k\geqslant 0} (2^{-k-1}\varepsilon)^{1-d} \int_{B_{2^{-k}\varepsilon}(x)} g \leqslant \sum_{k\geqslant 0} (2^{-k-1}\varepsilon)^{1-d} |B_1| (2^{-k}\varepsilon)^d m = 2^d |B_1|\,\varepsilon\, m.$$

Step 3. The far part. Since $|x-y|<2$ for $y\in B_1$, Hölder's inequality gives, if $\varepsilon<2$,

$$\int_{B_1\cap\{|y-x|\geqslant\varepsilon\}}\frac{g(y)}{|x-y|^{d-1}}\,dy\leqslant\|g\|_{L^d}\left(\int_{\{\varepsilon\leqslant|z|<2\}}|z|^{-d}\,dz\right)^{\frac{d-1}{d}}=\left(d|B_1|\log\frac{2}{\varepsilon}\right)^{\frac{d-1}{d}},$$

and the left side vanishes if $\varepsilon\geqslant 2$.

Step 4. Choice of ε. Let $L:=\log(2+m)\geqslant\log 2$ and take $\varepsilon:=L^{\frac{d-1}{d}}/m$, which is well defined since $m>0$. The near part is at most $2^d|B_1|L^{\frac{d-1}{d}}$. If $\varepsilon<2$, then $\log\frac{2}{\varepsilon}\leqslant 2L$. Indeed, $\log\frac{2}{\varepsilon}=\log 2+\log m+\frac{d-1}{d}\log\frac{1}{L}$ with $\log m<L$. If $L\geqslant 1$, then $\log\frac{1}{L}\leqslant 0$ and $\log\frac{2}{\varepsilon}\leqslant\log 2+L\leqslant 2L$. If $L<1$, then $m<e-2<1$, so $\log m<0$, and $L\geqslant\log 2$ gives $\log\frac{1}{L}\leqslant\log\frac{1}{\log 2}$, hence $\log\frac{2}{\varepsilon}<\log\frac{2}{\log 2}<\log 4=2\log 2\leqslant 2L$. Thus the far part is at most $(2d|B_1|)^{\frac{d-1}{d}}L^{\frac{d-1}{d}}$. We conclude that, for almost every $x\in B_1$,

$$|u(x)|\leqslant C_1\log\bigl(2+\mathsf{M}g(x)\bigr)^{\frac{d-1}{d}},\qquad C_1:=c_1\bigl(2^d|B_1|+(2d|B_1|)^{\frac{d-1}{d}}\bigr).\tag{1.40}$$

Step 5. Distribution function. Let $\lambda\geqslant\lambda_0:=C_1(\log 4)^{1/d'}$. Then $e^{(\lambda/C_1)^{d'}}\geqslant 4$, so $e^{(\lambda/C_1)^{d'}}-2\geqslant\frac12 e^{(\lambda/C_1)^{d'}}$. By (1.40) and Lemma 1.15,

$$\begin{aligned}|\{x\in B_1:|u(x)|>\lambda\}|&\leqslant\left|\left\{\bigl(\log(2+\mathsf{M}g)\bigr)^{\frac{d-1}{d}}>\frac{\lambda}{C_1}\right\}\right|=\left|\left\{\mathsf{M}g>e^{(\lambda/C_1)^{d'}}-2\right\}\right|\\&\leqslant\left|\left\{\mathsf{M}g>\tfrac12 e^{(\lambda/C_1)^{d'}}\right\}\right|\leqslant 2\cdot 5^d|B_1|^{1/d'}\exp\bigl(-(\lambda/C_1)^{d'}\bigr).\end{aligned}$$

If $\lambda<\lambda_0$, then $|\{|u|>\lambda\}|\leqslant|B_1|\leqslant 4|B_1|\exp(-(\lambda/C_1)^{d'})$, since $(\lambda_0/C_1)^{d'}=\log 4$. Together, the left side is at most $C\exp(-(\lambda/C_1)^{d'})$ for every $\lambda>0$ with $C:=\max\{2\cdot 5^d|B_1|^{1/d'},4|B_1|\}$, which is (1.38) after enlarging C so that $C\geqslant C_1^{d'}$. □

Lemma 1.19 (Campanato characterization of Hölder spaces)**.** *There exists a constant $C(d)<\infty$ such that, for every $\alpha\in(0,1]$ and $u\in L^1(B_1)$,*

$$C^{-1}\alpha[u]_{C^{0,\alpha}(B_{1/2})}\leqslant\sup_{x\in B_{1/2}}\sup_{r\in(0,1/4)}r^{-\alpha}\bigl\|u-(u)_{B_r(x)}\bigr\|_{\underline{L}^1(B_r(x))}\leqslant C[u]_{C^{0,\alpha}(B_1)}.\tag{1.41}$$

Proof. The only interesting inequality is the first one, as the second is obvious. Denote

$$A:=\sup_{x\in B_{1/2}}\sup_{r\in(0,1/4)}r^{-\alpha}\bigl\|u-(u)_{B_r(x)}\bigr\|_{\underline{L}^1(B_r(x))}$$

and set $u_r:=u*\eta_r$, where $\{\eta_r\}_{r>0}$ is the standard mollifier. Observe that

$$\|\nabla u_r\|_{L^\infty(B_{1/2})}\leqslant\|u-(u)_{B_r}\|_{L^1(B_r)}\|\nabla\eta_r\|_{L^\infty(B_r)}\leqslant CAr^{\alpha-1}$$

and, for every $x\in B_{1/2}$ and $r<1/8$,

$$\begin{aligned}|(u_r-u_{2r})(x)|&=\left|\int_{B_{2r}(x)}\bigl(u(y)-(u)_{B_{2r}}\bigr)(\eta_r(y-x)-\eta_{2r}(y-x))\,dy\right|\\&\leqslant C\|u-(u)_{B_{2r}}\|_{L^1(B_{2r})}\Bigl(\|\eta_r\|_{L^\infty(B_{2r})}+\|\eta_{2r}\|_{L^\infty(B_{2r})}\Bigr)\leqslant CAr^\alpha.\end{aligned}$$

By summing the latter inequality over dyadic scales and applying the Lebesgue differentiation theorem, we get

$$\|u - u_r\|_{L^\infty(B_{1/2})} \leqslant \frac{C}{\alpha} A r^\alpha.$$

By the triangle inequality, we deduce that, for every $x, y \in B_{1/2}$ with $|x-y| < 1/4$ and every $r \in (0, 1/4)$,

$$|u(x) - u(y)| \leqslant |u_r(x) - u_r(y)| + 2\|u - u_r\|_{L^\infty(B_{1/2})} \leqslant CAr^{\alpha-1}|x-y| + \frac{C}{\alpha} A r^\alpha.$$

We may optimize by taking $r = \alpha|x-y|$ to obtain

$$|u(x) - u(y)| \leqslant CA\alpha^{\alpha-1}|x-y|^\alpha \leqslant CA\alpha^{-1}|x-y|^\alpha.$$

This proves the bound for $|x-y| < 1/4$. If $x, y \in B_{1/2}$ and $|x-y| \geqslant 1/4$, choose the points $x = z_0, z_1, \dots, z_5 = y$ on the segment $[x,y] \subseteq B_{1/2}$ with $|z_i - z_{i-1}| = |x-y|/5 < 1/5$. Then

$$|u(x) - u(y)| \leqslant \sum_{i=1}^{5} |u(z_i) - u(z_{i-1})| \leqslant 5CA\alpha^{-1}\left(\tfrac{1}{5}\right)^\alpha \leqslant 5CA\alpha^{-1} \leqslant 20CA\alpha^{-1}|x-y|^\alpha,$$

since $|x-y|^\alpha \geqslant 4^{-\alpha} \geqslant 1/4$. Finally, $u_r \to u$ almost everywhere as $r \to 0$, and the estimate above shows that u_r converges uniformly on $B_{1/2}$; hence u agrees a.e. with a continuous function, and the estimates hold for it. This completes the proof. □

The factor α on the left side of (1.41) cannot be removed. It comes from summing the oscillations of the mean over dyadic scales. For instance, the function $|x|^\alpha$ has Hölder seminorm of order one, while the middle quantity in (1.41) is of order α. If the mean oscillation is replaced by the essential oscillation $\operatorname{ess\,osc}_S u := \operatorname{ess\,sup}_S u - \operatorname{ess\,inf}_S u$, then no summation is needed and the loss disappears. We state this for cubes $Q_r(x) := x + (-r, r)^d$.

Lemma 1.20 (Hölder continuity from oscillation decay)**.** *Let $\alpha \in (0,1]$, $K \geqslant 0$, $R > 0$ and $x_0 \in \mathbb{R}^d$, and let $u \in L^1(Q_R(x_0))$ satisfy*

$$\operatorname*{ess\,osc}_{Q_r(x)} u \leqslant K r^\alpha \quad \text{for every cube } Q_r(x) \subseteq Q_R(x_0). \tag{1.42}$$

Then there is a function u^* on $\overline{Q}_{R/8}(x_0)$, equal to u almost everywhere there, such that

$$|u^*(x) - u^*(y)| \leqslant 2^\alpha K |x-y|^\alpha, \qquad x, y \in \overline{Q}_{R/8}(x_0). \tag{1.43}$$

Proof. Let $x \in \overline{Q}_{R/8}(x_0)$. For $r \in (0, R/2]$ we have $Q_r(x) \subseteq Q_R(x_0)$. Hence the quantities $M_r(x) := \operatorname{ess\,sup}_{Q_r(x)} u$ and $m_r(x) := \operatorname{ess\,inf}_{Q_r(x)} u$ are finite, by (1.42). As $r \downarrow 0$, $M_r(x)$ is nonincreasing, $m_r(x)$ is nondecreasing, and $M_r(x) - m_r(x) \leqslant K r^\alpha \to 0$. We define $u^*(x)$ as their common limit, so that

$$m_r(x) \leqslant u^*(x) \leqslant M_r(x), \qquad r \in (0, R/2]. \tag{1.44}$$

At every Lebesgue point x of u, the mean $(u)_{Q_r(x)}$ lies in $[m_r(x), M_r(x)]$ and converges to $u(x)$. Hence $u^* = u$ almost everywhere.

Let $x \neq y$ in $\overline{Q}_{R/8}(x_0)$, and put $\rho := 2|x-y|_\infty \leqslant R/2$, where $|z|_\infty := \max_i |z_i|$. Then $y \in Q_\rho(x)$, and $Q_r(y) \subseteq Q_\rho(x)$ for all small $r > 0$. Applying (1.44) at y with such r, and at x with $r = \rho$, we see that both $u^*(x)$ and $u^*(y)$ lie in $[m_\rho(x), M_\rho(x)]$. Therefore

$$|u^*(x) - u^*(y)| \leqslant M_\rho(x) - m_\rho(x) \leqslant K\rho^\alpha \leqslant 2^\alpha K |x-y|^\alpha. \qquad \square$$

Denote the standard heat kernel on $\mathbb{R}^d$ by

$$\Phi(t,x) := (4\pi t)^{-d/2} \exp\!\left(-\frac{|x|^2}{4t}\right).$$

Proposition 1.21 (Gaussian Poincaré inequality)**.** *For every $t > 0$ and every real-valued $u \in C_c^\infty(\mathbb{R}^d)$,*

$$\int_{\mathbb{R}^d} |u(x)|^2 \Phi(t,x)\,dx - \left(\int_{\mathbb{R}^d} u(x)\Phi(t,x)\,dx\right)^2 \leqslant 2t \int_{\mathbb{R}^d} |\nabla u(x)|^2 \Phi(t,x)\,dx.$$

Proof. For complex-valued u one applies the statement to $\operatorname{Re} u$ and $\operatorname{Im} u$ and adds, after replacing the square of the mean by its modulus squared. Fix $t > 0$ and denote the left side by $\operatorname{Var}_{\Phi(t)}(u)$. Define $w(s,x) := (u * \Phi(s,\cdot))(x)$ for $s > 0$ and set $w(0,\cdot) := u$.

Step 1: regularity. Since $w(s,x) = \int_{\mathbb{R}^d} u\big(x - \sqrt{2s}\,z\big)\Phi(1/2, z)\,dz$, dominated convergence shows that w is continuous on $[0,\infty) \times \mathbb{R}^d$ with $|w| \leqslant \sup|u|$. For each multi-index α, we have $D^\alpha w(s) = (D^\alpha u) * \Phi(s)$, so w is smooth in x with $|D^\alpha w| \leqslant \sup|D^\alpha u|$, and $\partial_s w = \Delta w = (\Delta u) * \Phi(s)$ is continuous on $[0,\infty)\times\mathbb{R}^d$ and bounded. Moreover, for every $\varepsilon \in (0,t)$ there is $C_\varepsilon < \infty$ such that $|\Phi| + |\nabla\Phi| + |\Delta\Phi| + |\partial_s\Phi| \leqslant C_\varepsilon e^{-|x|^2/(8t)}$ at the point $(t-s,x)$ for every $s \in [0, t-\varepsilon]$ and $x \in \mathbb{R}^d$. Consequently, on $[0, t-\varepsilon]$ we may differentiate under the integral sign in the functions $s \mapsto \int f(s,x)\Phi(t-s,x)\,dx$ below (with $f = w^2$ or $f = w$), and integrate by parts in x without boundary terms.

Step 2: the derivative of the variance along the flow. Write $\Phi = \Phi(t-s,x)$ and recall that $\partial_s[\Phi(t-s,x)] = -\Delta\Phi(t-s,x)$. For $0 \leqslant s < t$, integrating by parts twice for the linear term and once for the quadratic one,

$$\begin{aligned}
\frac{d}{ds}\int_{\mathbb{R}^d} w\,\Phi\,dx &= \int_{\mathbb{R}^d} \big(\Delta w\,\Phi - w\,\Delta\Phi\big)\,dx = 0\,,\\
\frac{d}{ds}\int_{\mathbb{R}^d} |w|^2\Phi\,dx &= \int_{\mathbb{R}^d} \big(2w\,\Delta w\,\Phi - |w|^2\,\Delta\Phi\big)dx = \int_{\mathbb{R}^d} \big(-2|\nabla w|^2\Phi - 2w\,\nabla w\cdot\nabla\Phi + 2w\,\nabla w\cdot\nabla\Phi\big)\,dx\\
&= -2\int_{\mathbb{R}^d} |\nabla w(s,x)|^2\,\Phi(t-s,x)\,dx\,.
\end{aligned}$$

Here $-\int |w|^2\Delta\Phi\,dx = 2\int w\,\nabla w\cdot\nabla\Phi\,dx$ produces the second cross term, which cancels the first one coming from $\int 2w\,\Delta w\,\Phi\,dx = -\int(2|\nabla w|^2\Phi + 2w\nabla w\cdot\nabla\Phi)\,dx$. Set

$$F(s) := \int_{\mathbb{R}^d} |w(s,x)|^2\Phi(t-s,x)\,dx - \left(\int_{\mathbb{R}^d} w(s,x)\,\Phi(t-s,x)\,dx\right)^2, \qquad 0 \leqslant s < t\,.$$

Then $\int w(s)\Phi(t-s)\,dx$ is constant in s, equal to its value $\int u\,\Phi(t)\,dx$ at $s = 0$, and $F(0) = \operatorname{Var}_{\Phi(t)}(u)$ and $F' = -2\int|\nabla w|^2\Phi(t-s)\,dx \leqslant 0$.

Step 3: the endpoint $s = t$. Since $\Phi(t-s,\cdot)$ is a probability density with variance $2(t-s)$ per coordinate, and w is bounded and continuous on $[0,t]\times\mathbb{R}^d$, we have $\int |w(s)|^2\Phi(t-s)\,dx \to |w(t,0)|^2$ as $s\uparrow t$; likewise, by the constancy just observed and the evenness of $\Phi(t)$, $\int w(s)\Phi(t-s)\,dx = \int u\,\Phi(t)\,dx = w(t,0)$. Therefore $F(s)\to 0$ as $s\uparrow t$. Integrating F' over $[0,t-\varepsilon]$ gives $\mathrm{Var}_{\Phi(t)}(u) = F(0) = F(t-\varepsilon) + 2\int_0^{t-\varepsilon}\!\int_{\mathbb{R}^d} |\nabla w|^2\Phi(t-s)\,dx\,ds$, and letting $\varepsilon\to 0$ (monotone convergence),

$$\mathrm{Var}_{\Phi(t)}(u) = 2\int_0^t\!\int_{\mathbb{R}^d} |\nabla w(s,x)|^2\,\Phi(t-s,x)\,dx\,ds\,.$$

Step 4: Jensen. Since $\nabla w(s) = (\nabla u)*\Phi(s)$ and $\Phi(s)$ is a probability density, Jensen's inequality yields

$$|\nabla w(s,x)|^2 = \left|\int_{\mathbb{R}^d} \nabla u(y)\,\Phi(s,x-y)\,dy\right|^2 \leqslant \int_{\mathbb{R}^d} |\nabla u(y)|^2\,\Phi(s,x-y)\,dy\,.$$

Integrating against $\Phi(t-s,x)$ in x, using Tonelli's theorem, the evenness of Φ and the semigroup property $\Phi(s)*\Phi(t-s)=\Phi(t)$, we obtain

$$\int_{\mathbb{R}^d} |\nabla w(s,x)|^2\,\Phi(t-s,x)\,dx \leqslant \int_{\mathbb{R}^d} |\nabla u(y)|^2\,\Phi(t,y)\,dy\,.$$

Inserting this into the identity of Step 3 and integrating over $s\in[0,t]$ produces the factor $2t$ and completes the proof. □

Proposition 1.22 (Nash inequality)**.** *There exists a constant $C(d)<\infty$ such that, for every $u\in H^1(\mathbb{R}^d)\cap L^1(\mathbb{R}^d)$,*

$$\int_{\mathbb{R}^d} |u(x)|^2\,dx \leqslant C\left(\int_{\mathbb{R}^d} |\nabla u(x)|^2\,dx\right)^{\frac{d}{d+2}}\left(\int_{\mathbb{R}^d} |u(x)|\,dx\right)^{\frac{4}{d+2}}$$

Proof. Step 1: reductions. Both sides are continuous with respect to the norm $\|u\|_{H^1}+\|u\|_{L^1}$, and $C_c^\infty(\mathbb{R}^d)$ is dense in $H^1(\mathbb{R}^d)\cap L^1(\mathbb{R}^d)$ for this norm: if $\chi\in C_c^\infty(\mathbb{R}^d)$ satisfies $\chi=1$ near 0 and $\chi_R := \chi(\cdot/R)$, then $\chi_R u\to u$ in L^1 and L^2 by dominated convergence, and $\|\nabla(\chi_R u)-\nabla u\|_{L^2}\leqslant \|(1-\chi_R)\nabla u\|_{L^2} + R^{-1}\|\nabla\chi\|_{L^\infty}\|u\|_{L^2}\to 0$; a compactly supported element of H^1 lies in L^1, and its mollifications converge to it in H^1 and L^1. We may therefore assume that $u\in C_c^\infty(\mathbb{R}^d)$ and $u\not\equiv 0$. The inequality is invariant under translations and under $u\mapsto u(\lambda\,\cdot)$: the latter multiplies $\|u\|_{L^2}^2$ by λ^{-d}, $\|\nabla u\|_{L^2}^2$ by λ^{2-d} and $\|u\|_{L^1}$ by λ^{-d}, so the right side is multiplied by $\lambda^{\frac{(2-d)d-4d}{d+2}} = \lambda^{-d}$. Hence we may assume that $\operatorname{supp} u\subseteq Q := (0,1)^d$, and all norms below may be computed over Q. Write $A := \|\nabla u\|_{L^2(Q)}$ and $B := \|u\|_{L^1(Q)}$; both are positive.

Step 2: two Poincaré-type inequalities on cubes. Let $R = a+(0,\ell)^d$ be a cube and $v\in C^1(\overline{R})$. For $x,y\in R$ let $p_i := (y_1,\dots,y_i,x_{i+1},\dots,x_d)$, so that $p_0=x$, $p_d=y$, and p_{i-1},p_i differ only in the ith coordinate. By Cauchy–Schwarz, $|v(p_{i-1})-v(p_i)|^2\leqslant \ell\int_{a_i}^{a_i+\ell}|\partial_i v|^2(y_1,\dots,y_{i-1},r,x_{i+1},\dots,x$ and $|v(x)-v(y)|^2\leqslant d\sum_{i=1}^d |v(p_{i-1})-v(p_i)|^2$. Integrating over $(x,y)\in R\times R$, the integrand of the ith term does not depend on $x_1,\dots,x_i,y_i,\dots,y_d$, which contribute ℓ^{d+1}, so

that $\iint_{R\times R}|v(x)-v(y)|^2\,dx\,dy \leqslant d\,\ell^{d+2}\int_R|\nabla v|^2\,dx$. Since $\int_R|v-(v)_R|^2\,dx = \frac{1}{2|R|}\iint_{R\times R}|v(x)-v(y)|^2\,dx\,dy$, we get

$$\int_R |v-(v)_R|^2\,dx \leqslant \tfrac{d}{2}\,\ell^2\int_R|\nabla v|^2\,dx\,. \tag{1.45}$$

This elementary argument is valid for every $d\geqslant 1$. If moreover $v\in C_c^\infty(Q)$, then $|v(x)|\leqslant\int_0^{x_1}|\partial_1 v(r,x_2,\dots,x_d)|\,dr$, so $|v(x)|^2\leqslant\int_0^1|\partial_1 v(r,x')|^2\,dr$ and integrating gives the Friedrichs inequality $\|v\|_{L^2(Q)}\leqslant\|\nabla v\|_{L^2(Q)}$. Applied to u, and combined with $|Q|=1$ and Cauchy–Schwarz, this yields $B\leqslant\|u\|_{L^2(Q)}\leqslant A$.

Step 3: decomposition into small cubes. Let $h\in\mathbb{N}\cup\{0\}$ and let $\mathcal{Q}_h$ be the collection of the 3^{dh} cubes $z+(0,3^{-h})^d$ with $z\in 3^{-h}\{0,\dots,3^h-1\}^d$; they are disjoint, have volume 3^{-dh}, and their closures cover $\overline{Q}$. For $R\in\mathcal{Q}_h$, we have $\int_R|u|^2\,dx=\int_R|u-(u)_R|^2\,dx+|R|\,|(u)_R|^2$, and $|R|\,|(u)_R|^2\leqslant|R|^{-1}\big(\int_R|u|\,dx\big)^2$. Summing over $R\in\mathcal{Q}_h$, using (1.45) with $\ell=3^{-h}$ and the inequality $\sum a_R^2\leqslant(\sum a_R)^2$ for $a_R\geqslant 0$, we get

$$\int_Q|u|^2\,dx\leqslant\tfrac{d}{2}\,3^{-2h}\sum_{R\in\mathcal{Q}_h}\int_R|\nabla u|^2\,dx+3^{dh}\Big(\sum_{R\in\mathcal{Q}_h}\int_R|u|\,dx\Big)^2=\tfrac{d}{2}\,3^{-2h}A^2+3^{dh}B^2\,.$$

Step 4: choice of the scale. By Step 2, $A/B\geqslant 1$, so that

$$h:=\left\lceil\frac{2}{d+2}\log_3\frac{A}{B}\right\rceil$$

is a nonnegative integer, and it satisfies $\frac{A}{B}\leqslant 3^{\frac{d+2}{2}h}<3^{\frac{d+2}{2}}\frac{A}{B}$. Therefore $3^{-2h}A^2\leqslant A^2(A/B)^{-\frac{4}{d+2}}=A^{\frac{2d}{d+2}}B^{\frac{4}{d+2}}$ and $3^{dh}B^2\leqslant 3^d(A/B)^{\frac{2d}{d+2}}B^2=3^dA^{\frac{2d}{d+2}}B^{\frac{4}{d+2}}$. Inserting these bounds into the previous display gives

$$\|u\|_{L^2(\mathbb{R}^d)}^2\leqslant\big(\tfrac{d}{2}+3^d\big)\|\nabla u\|_{L^2(\mathbb{R}^d)}^{\frac{2d}{d+2}}\|u\|_{L^1(\mathbb{R}^d)}^{\frac{4}{d+2}}\,,$$

which is the claim for $u\in C_c^\infty(Q)$, and hence, by Step 1, for every $u\in H^1(\mathbb{R}^d)\cap L^1(\mathbb{R}^d)$ with $C=\frac{d}{2}+3^d$. □

1.3 Trace inequalities

Throughout this subsection, $\Omega\subset\mathbb{R}^d$ denotes a bounded domain with C^1 boundary. We write $d\sigma$ for the surface measure on $\partial\Omega$. For $s\in(0,1)$ and $p\in[1,\infty)$, we define the *Gagliardo seminorm* of a measurable function $f:\partial\Omega\to\mathbb{R}$ by

$$[f]_{B^s_{p,p}(\partial\Omega)}:=\left(\int_{\partial\Omega}\int_{\partial\Omega}\frac{|f(x)-f(y)|^p}{|x-y|^{d-1+sp}}\,d\sigma(x)\,d\sigma(y)\right)^{1/p} \tag{1.46}$$

and the *Besov norm*

$$\|f\|_{B^s_{p,p}(\partial\Omega)}:=\|f\|_{L^p(\partial\Omega)}+[f]_{B^s_{p,p}(\partial\Omega)}\,.$$

When $p=2$, the space $B^s_{2,2}(\partial\Omega)$ coincides with the fractional Sobolev space $H^s(\partial\Omega)$.

For $p\in[1,\infty)$ and $u\in W^{1,p}(\Omega)$, a function $g\in L^p(\partial\Omega,d\sigma)$ is called the *trace* of u, and we write $Tu=u|_{\partial\Omega}:=g$, if there exist $\varphi_n\in C^1(\overline{\Omega})$ such that $\varphi_n\to u$ in $W^{1,p}(\Omega)$

and $\varphi_n|_{\partial\Omega} \to g$ in $L^p(\partial\Omega)$. The following theorem shows that, for $p \in (1,\infty)$, the trace of every $u \in W^{1,p}(\Omega)$ exists, is unique, and satisfies the bound (1.47); uniqueness follows from the L^p part of that inequality.

Theorem 1.23 (Trace inequality)**.** *For every $p \in (1,\infty)$ there exists $C(d,p,\Omega) < \infty$ such that every $u \in W^{1,p}(\Omega)$ has a unique trace $Tu = u|_{\partial\Omega}$, and*

$$\|Tu\|_{B^{1-1/p}_{p,p}(\partial\Omega)} \leqslant C\|u\|_{W^{1,p}(\Omega)}\,. \tag{1.47}$$

Proof. Write $s := 1 - \frac{1}{p}$, so that $sp = p-1$. The proof has two parts. In the first (Steps 1–3) we prove the estimate for functions in $C^1_c(\overline{\mathbb{R}^d_+})$ on the half-space $\mathbb{R}^d_+ := \{(x', x_d) \in \mathbb{R}^d : x_d > 0\}$, whose boundary we identify with $\mathbb{R}^{d-1}$. In the second (Steps 4–5) we transfer it to $u \in C^1(\overline{\Omega})$ by a partition of unity and C^1 charts, and then extend it to $W^{1,p}(\Omega)$ by density.

Step 1: the L^p bound on the half-space. Let $v \in C^1_c(\overline{\mathbb{R}^d_+})$ and write $w := v(\cdot,0)$. Since $p > 1$, the function $|v|^p$ is C^1 with $\partial_t|v|^p = p|v|^{p-2}v\,\partial_t v$, and v has compact support, so

$$|w(x')|^p = -\int_0^\infty \partial_t\big(|v(x',t)|^p\big)\,dt \leqslant p\int_0^\infty |v(x',t)|^{p-1}|\partial_d v(x',t)|\,dt\,.$$

Integrating in x' and applying Hölder's inequality in (x',t) with exponents $\frac{p}{p-1}$ and p gives

$$\|w\|^p_{L^p(\mathbb{R}^{d-1})} \leqslant p\,\|v\|^{p-1}_{L^p(\mathbb{R}^d_+)}\,\|\partial_d v\|_{L^p(\mathbb{R}^d_+)}\,. \tag{1.48}$$

Step 2: a Hardy inequality. For every measurable $g : (0,\infty) \to [0,\infty)$,

$$\int_0^\infty \left(\frac{1}{r}\int_0^r g(t)\,dt\right)^p dr \leqslant \left(\frac{p}{p-1}\right)^p \int_0^\infty g(t)^p\,dt\,. \tag{1.49}$$

Indeed, the substitution $t = \theta r$ gives $\frac{1}{r}\int_0^r g(t)\,dt = \int_0^1 g(\theta r)\,d\theta$. By Minkowski's integral inequality, the $L^p(0,\infty)$ norm in r of the left side is at most $\int_0^1 \|g(\theta\,\cdot)\|_{L^p(0,\infty)}\,d\theta = \|g\|_{L^p(0,\infty)}\int_0^1 \theta^{-1/p}\,d\theta$, and $\int_0^1 \theta^{-1/p}\,d\theta = \frac{p}{p-1}$.

Step 3: the Gagliardo seminorm on the half-space. Let v and w be as in Step 1. We show that

$$[w]^p_{B^s_{p,p}(\mathbb{R}^{d-1})} = \int_{\mathbb{R}^{d-1}}\int_{\mathbb{R}^{d-1}} \frac{|w(x') - w(y')|^p}{|x'-y'|^{d-1+sp}}\,dx'\,dy' \leqslant C(d,p)\,\|\nabla v\|^p_{L^p(\mathbb{R}^d_+)}\,. \tag{1.50}$$

Fix $x' \neq y'$, and set $k := y' - x'$ and $h := |k|$. Comparing $w(x')$ and $w(y')$ through the values of v at height h gives

$$w(x') - w(y') = \big(v(x',0) - v(x',h)\big) + \big(v(x',h) - v(y',h)\big) + \big(v(y',h) - v(y',0)\big)\,,$$

and by the fundamental theorem of calculus,

$$|w(x') - w(y')| \leqslant \int_0^h |\partial_d v(x',t)|\,dt + h\int_0^1 \big|\nabla' v(x'+\theta k, h)\big|\,d\theta + \int_0^h |\partial_d v(y',t)|\,dt\,,$$

where ∇' denotes the gradient in x'. Raising to the power p, using $(a+b+c)^p \leqslant 3^{p-1}(a^p + b^p + c^p)$ and Jensen's inequality for the middle term, we get

$$|w(x')-w(y')|^p \leqslant 3^{p-1}\Big(\Big(\int_0^h |\partial_d v(x',t)|\,dt\Big)^p + h^p \int_0^1 \big|\nabla' v(x'+\theta k,h)\big|^p\,d\theta + \Big(\int_0^h |\partial_d v(y',t)|\,dt\Big)^p\Big)\,.$$

We multiply by $|x'-y'|^{-(d-1+sp)} = h^{-(d-1+sp)}$ and integrate in (x',y'). The first and third terms give the same integral by symmetry, so that

$$[w]^p_{B^s_{p,p}} \leqslant 3^{p-1}(2I_V + I_H)\,,$$

where

$$I_V := \int_{\mathbb{R}^{d-1}}\int_{\mathbb{R}^{d-1}} \frac{\big(\int_0^{|k|} |\partial_d v(x',t)|\,dt\big)^p}{|k|^{d-1+sp}}\,dk\,dx'\,, \quad I_H := \int_{\mathbb{R}^{d-1}}\int_{\mathbb{R}^{d-1}} \frac{\int_0^1 |\nabla' v(x'+\theta k,|k|)|^p\,d\theta}{|k|^{d-1+sp-p}}\,dk\,dx'\,.$$

Let κ denote the surface measure of the unit sphere of $\mathbb{R}^{d-1}$ (so $\kappa = 2$ if $d=2$), and recall that $dk = \kappa r^{d-2}\,dr$ in polar coordinates, for functions of $r = |k|$.

For I_V, fix x' and let $g(t) := |\partial_d v(x',t)|$. Since $d-2-(d-1+sp) = -1-sp = -p$, polar coordinates in k and (1.49) give

$$\int_{\mathbb{R}^{d-1}} \frac{\big(\int_0^{|k|} g\,dt\big)^p}{|k|^{d-1+sp}}\,dk = \kappa\int_0^\infty \Big(\frac{1}{r}\int_0^r g\,dt\Big)^p dr \leqslant \kappa\Big(\frac{p}{p-1}\Big)^p \int_0^\infty |\partial_d v(x',t)|^p\,dt\,.$$

Integrating in x' yields $I_V \leqslant \kappa\,(\frac{p}{p-1})^p\,\|\partial_d v\|^p_{L^p(\mathbb{R}^d_+)}$.

For I_H, we exchange the order of integration and, for fixed k and θ, translate $x' \mapsto x'-\theta k$. Since $d-1+sp-p = d-2$,

$$I_H = \int_{\mathbb{R}^{d-1}} |k|^{2-d}\int_{\mathbb{R}^{d-1}} |\nabla' v(z',|k|)|^p\,dz'\,dk = \kappa\int_0^\infty\int_{\mathbb{R}^{d-1}} |\nabla' v(z',r)|^p\,dz'\,dr = \kappa\,\|\nabla' v\|^p_{L^p(\mathbb{R}^d_+)}\,,$$

because $r^{2-d}r^{d-2} = 1$. Combining the bounds on I_V and I_H yields (1.50) with $C(d,p) = 3^{p-1}\kappa\big(2(\frac{p}{p-1})^p + 1\big)$.

Step 4: transfer to Ω for $u \in C^1(\overline{\Omega})$. Since $\partial\Omega$ is compact and of class C^1, it is covered by finitely many open cylinders $U_1,\dots,U_N$ with the following property. After a rigid motion, which does not affect any of the quantities involved, we have $U_j = \{|x'| < r_j,\ |x_d| < a_j\}$, $\Omega\cap U_j = \{x\in U_j : x_d > \gamma_j(x')\}$ and $\partial\Omega\cap U_j = \{x\in U_j : x_d = \gamma_j(x')\}$, for some $\gamma_j \in C^1(\overline{B_{r_j}})$ with $|\gamma_j| < a_j$. Let $L \geqslant 1$ bound $1 + \|\nabla\gamma_j\|_{L^\infty}$ for all j. We fix once and for all $\zeta_j \in C^\infty_c(U_j)$ with $\sum_j \zeta_j = 1$ on $\partial\Omega$. All constants below depend only on d, p and these choices, hence on d, p and Ω.

Let $u \in C^1(\overline{\Omega})$. The map $\Phi_j(x) := (x', x_d - \gamma_j(x'))$ is a C^1 diffeomorphism of U_j onto its image, with Jacobian determinant 1, and it maps $\Omega\cap U_j$ into $\mathbb{R}^d_+$ and $\partial\Omega\cap U_j$ into $\mathbb{R}^{d-1}$. Define $v_j := (\zeta_j u)\circ\Phi_j^{-1}$ on $\Phi_j(U_j\cap\Omega)$ and $v_j := 0$ elsewhere in $\overline{\mathbb{R}^d_+}$. Since $\operatorname{supp}\zeta_j$ is a compact subset of U_j, we have $v_j \in C^1_c(\overline{\mathbb{R}^d_+})$, and, by the change of variables formula, $\|v_j\|_{W^{1,p}(\mathbb{R}^d_+)} \leqslant C\|\zeta_j u\|_{W^{1,p}(\Omega)} \leqslant C\|u\|_{W^{1,p}(\Omega)}$. Set $f_j := (\zeta_j u)|_{\partial\Omega}$, extended by zero outside U_j, and $w_j := v_j(\cdot,0) \in C^1_c(\mathbb{R}^{d-1})$, so that $f_j(x',\gamma_j(x')) = w_j(x')$ and $u|_{\partial\Omega} = \sum_j f_j$.

By (1.48) and Young's inequality, $\|w_j\|^p_{L^p(\mathbb{R}^{d-1})} \leqslant C\|v_j\|^p_{W^{1,p}(\mathbb{R}^d_+)}$. On $\partial\Omega\cap U_j$ the surface measure is $d\sigma = \sqrt{1+|\nabla\gamma_j|^2}\,dx' \leqslant L\,dx'$, so $\|f_j\|^p_{L^p(\partial\Omega)} \leqslant L\|w_j\|^p_{L^p(\mathbb{R}^{d-1})}$. For the seminorm, split $\partial\Omega\times\partial\Omega$ into the pairs lying in $U_j\times U_j$ and the remaining ones. For $x=(x',\gamma_j(x'))$ and $y=(y',\gamma_j(y'))$ in U_j we have $|x-y|\geqslant|x'-y'|$, so by (1.50),

$$\iint\limits_{(\partial\Omega\cap U_j)^2} \frac{|f_j(x)-f_j(y)|^p}{|x-y|^{d-1+sp}}\,d\sigma\,d\sigma \leqslant L^2\int_{\mathbb{R}^{d-1}}\int_{\mathbb{R}^{d-1}} \frac{|w_j(x')-w_j(y')|^p}{|x'-y'|^{d-1+sp}}\,dx'\,dy' \leqslant C\|\nabla v_j\|^p_{L^p(\mathbb{R}^d_+)}\,.$$

A remaining pair has $f_j(x)\neq 0$ and $y\notin U_j$, or the reverse. Then $x\in K_j := \operatorname{supp}\zeta_j\cap\partial\Omega$ and $|x-y|\geqslant\delta_j := \operatorname{dist}(K_j,\mathbb{R}^d\setminus U_j)>0$, so these pairs contribute at most $2\sigma(\partial\Omega)\,\delta_j^{-(d-1+sp)}\|f_j\|^p_{L^p(\partial\Omega)}$. Altogether $\|f_j\|_{L^p(\partial\Omega)}+[f_j]_{B^s_{p,p}(\partial\Omega)}\leqslant C\|u\|_{W^{1,p}(\Omega)}$ for each j. Since $[\,\cdot\,]_{B^s_{p,p}(\partial\Omega)}$ is a seminorm and $u|_{\partial\Omega}=\sum_j f_j$, we obtain

$$\|u|_{\partial\Omega}\|_{B^s_{p,p}(\partial\Omega)} \leqslant C\|u\|_{W^{1,p}(\Omega)} \qquad \text{for every } u\in C^1(\overline{\Omega}). \tag{1.51}$$

Step 5: general u. Let $u\in W^{1,p}(\Omega)$. Since Ω is bounded with C^1 boundary, $C^1(\overline{\Omega})$ is dense in $W^{1,p}(\Omega)$ (see, e.g., [Eva98, Section 5.3]), so there are $\varphi_n\in C^1(\overline{\Omega})$ with $\varphi_n\to u$ in $W^{1,p}(\Omega)$. Applying the L^p part of (1.51) to $\varphi_n-\varphi_m$ shows that $(\varphi_n|_{\partial\Omega})$ is Cauchy in $L^p(\partial\Omega)$; let g be its limit. Then g is a trace of u, which proves existence. If $g'\in L^p(\partial\Omega)$ is any trace of u, that is, $\varphi_n'\in C^1(\overline{\Omega})$ satisfy $\varphi_n'\to u$ in $W^{1,p}(\Omega)$ and $\varphi_n'|_{\partial\Omega}\to g'$ in $L^p(\partial\Omega)$, then $\varphi_n-\varphi_n'\to 0$ in $W^{1,p}(\Omega)$, and the L^p part of (1.51) gives $\varphi_n|_{\partial\Omega}-\varphi_n'|_{\partial\Omega}\to 0$ in $L^p(\partial\Omega)$, so $g=g'$. Finally, passing to a subsequence, $\varphi_n|_{\partial\Omega}\to g$ σ-almost everywhere. Fatou's lemma applied to the integral defining $[\,\cdot\,]_{B^s_{p,p}(\partial\Omega)}$ gives $[g]_{B^s_{p,p}(\partial\Omega)}\leqslant \liminf_n [\varphi_n|_{\partial\Omega}]_{B^s_{p,p}(\partial\Omega)}$, while $\|g\|_{L^p(\partial\Omega)}=\lim_n\|\varphi_n|_{\partial\Omega}\|_{L^p(\partial\Omega)}$. Together with (1.51) for φ_n and $\|\varphi_n\|_{W^{1,p}(\Omega)}\to\|u\|_{W^{1,p}(\Omega)}$, this yields (1.47) for u. □

Remark 1.24. Theorem 1.23 extends to bounded Lipschitz domains: the partition of unity and chart argument in the proof works because $\partial\Omega$ is locally the graph of a Lipschitz function, and the Gagliardo seminorm (1.46) is invariant under bi-Lipschitz changes of variable (up to a constant depending on the Lipschitz constant). The constant C in (1.47) then depends on the Lipschitz constant of $\partial\Omega$ in addition to d and p.

We next show that the trace operator is surjective: every function in $B^{1-1/p}_{p,p}(\partial\Omega)$ is the trace of some $W^{1,p}(\Omega)$ function, and that the extension can be chosen linearly.

Theorem 1.25 (Right inverse for the trace)**.** *For every $p\in(1,\infty)$ there exist $C(d,p,\Omega)<\infty$ and a bounded linear operator $E : B^{1-1/p}_{p,p}(\partial\Omega)\to W^{1,p}(\Omega)$ such that, for every $g\in B^{1-1/p}_{p,p}(\partial\Omega)$, $T(Eg)=g$ and*

$$\|Eg\|_{W^{1,p}(\Omega)} \leqslant C\|g\|_{B^{1-1/p}_{p,p}(\partial\Omega)}\,. \tag{1.52}$$

Proof. As in the proof of Theorem 1.23, we reduce to the half-space $\mathbb{R}^d_+$ with boundary $\mathbb{R}^{d-1}$. Write $s:=1-\frac{1}{p}$. Let $\varphi\in C^\infty_c(\mathbb{R}^{d-1})$ be a mollifier with $\int_{\mathbb{R}^{d-1}}\varphi=1$ and $\varphi\geqslant 0$, and define $\varphi_t(z'):=t^{-(d-1)}\varphi(z'/t)$ for $t>0$. For $g\in B^s_{p,p}(\mathbb{R}^{d-1})$, define the extension

$$Eg(x',t) := (\varphi_t * g)(x') = \int_{\mathbb{R}^{d-1}} \varphi_t(x'-y')\,g(y')\,dy'\,, \quad (x',t)\in\mathbb{R}^d_+\,. \tag{1.53}$$

Since $\varphi_t \to \delta_0$ as $t \to 0^+$, we have $Eg(\cdot,0) = g$ in the trace sense. The map $g \mapsto Eg$ is linear, since it is a convolution in x' for each fixed t. The operator on Ω is obtained from this one by composing with the charts and multiplying by a fixed cutoff function depending only on the distance to $\partial\Omega$ and a partition of unity, and each of these operations is linear in g; thus the resulting E is linear. Throughout the proof, C denotes a constant depending only on d, p and Ω. It remains to show (1.52).

Step 1. The L^p bound: $\|Eg\|_{L^p(\mathbb{R}^d_+)} \leqslant C\|g\|_{L^p(\mathbb{R}^{d-1})}$. By Young's convolution inequality, $\|Eg(\cdot,t)\|_{L^p(\mathbb{R}^{d-1})} \leqslant \|g\|_{L^p(\mathbb{R}^{d-1})}$ for every $t > 0$. Since Eg will be multiplied by a cutoff supported near $t = 0$, this term is bounded.

Step 2. The tangential gradient: $\|\nabla' Eg\|_{L^p(\mathbb{R}^d_+)} \leqslant C\|g\|_{B^s_{p,p}}$. We have $\nabla' Eg(\cdot,t) = (\nabla'\varphi_t) * g$. Since $\int \nabla'\varphi_t = 0$:

$$\nabla' Eg(x',t) = \int_{\mathbb{R}^{d-1}} \nabla'\varphi_t(x'-y')\big(g(y') - g(x')\big)\,dy'\,.$$

By Minkowski's inequality and the scaling $\|\nabla'\varphi_t\|_{L^1} = t^{-1}\|\nabla'\varphi\|_{L^1}$, a direct estimate gives $|\nabla' Eg(x',t)| \leqslant Ct^{-1}\int_{|z'|\leqslant Ct} |g(x'-z') - g(x')|\frac{dz'}{t^{d-1}}$. Using Hölder and integrating in t:

$$\int_0^\infty \int_{\mathbb{R}^{d-1}} |\nabla' Eg|^p\,dx'\,dt \leqslant C\int_0^\infty t^{-p}\int_{\mathbb{R}^{d-1}} \left(\fint_{B'_t} |g(x'-z') - g(x')|\,dz'\right)^p dx'\,dt\,.$$

By the substitution $t = |z'|$ and Fubini, this is bounded by $C[g]^p_{B^s_{p,p}}$.

Step 3. The normal derivative: $\|\partial_t Eg\|_{L^p(\mathbb{R}^d_+)} \leqslant C[g]_{B^s_{p,p}}$. This is the key estimate. We compute

$$\partial_t Eg(x',t) = \int_{\mathbb{R}^{d-1}} \partial_t\big[\varphi_t(x'-y')\big]\,g(y')\,dy'\,.$$

Since $\int \partial_t[\varphi_t] = \partial_t \int \varphi_t = 0$, we may replace $g(y')$ by $g(y') - g(x')$:

$$\partial_t Eg(x',t) = \int_{\mathbb{R}^{d-1}} \partial_t\big[\varphi_t(x'-y')\big]\,\big(g(y') - g(x')\big)\,dy'\,.$$

The kernel satisfies $|\partial_t[\varphi_t(z')]| \leqslant Ct^{-d}$ and is supported in $|z'| \leqslant Ct$, so

$$|\partial_t Eg(x',t)| \leqslant \frac{C}{t^d}\int_{|z'|\leqslant Ct} |g(x'-z') - g(x')|\,dz'\,.$$

Raising to the power p, integrating over $x' \in \mathbb{R}^{d-1}$ and $t \in (0,\infty)$, and applying Hölder's inequality inside the ball B'_{Ct}:

$$\begin{aligned}\int_0^\infty \int_{\mathbb{R}^{d-1}} |\partial_t Eg|^p\,dx'\,dt &\leqslant C\int_0^\infty t^{-dp}\int_{\mathbb{R}^{d-1}} \left(\int_{|z'|\leqslant Ct} |g(x'-z') - g(x')|\,dz'\right)^p dx'\,dt \\ &\leqslant C\int_0^\infty t^{-dp} t^{(d-1)(p-1)} \int_{\mathbb{R}^{d-1}}\int_{|z'|\leqslant Ct} |g(x'-z') - g(x')|^p\,dz'\,dx'\,dt\,.\end{aligned}$$

The exponent of t is $-dp+(d-1)(p-1) = -dp+dp-d-p+1 = -d-p+1$. Substituting $r = |z'|$ and exchanging the order of integration (Fubini with $t \geqslant r/C$):

$$\leqslant C \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \frac{|g(x') - g(y')|^p}{|x' - y'|^{d-1+sp}} \, dx' \, dy' = C[g]^p_{B^s_{p,p}} ,$$

where the last step uses $\int_{r/C}^{\infty} t^{-d-p+1} \, dt = Cr^{-d-p+2} = C|z'|^{-(d-2+p)}$ and $(d-2+p)+(d-2) = d-1+sp$ (since $sp = p-1$, so $d-1+sp = d+p-2$). Combining Steps 1–3 yields (1.52). □

Corollary 1.26 (Characterization of the trace space)**.** *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain with C^1 boundary and $p \in (1, \infty)$. The trace operator $T : W^{1,p}(\Omega) \to B^{1-1/p}_{p,p}(\partial\Omega)$ is bounded and surjective, with a bounded linear right inverse E. In particular, $B^{1-1/p}_{p,p}(\partial\Omega)$ is the trace space of $W^{1,p}(\Omega)$.*

Proof. Boundedness of T is Theorem 1.23, and Theorem 1.25 gives a bounded linear operator E with $TE = \mathrm{Id}$, so T is surjective. □

Exercise 1.27. Prove the $p = 2$ case of Theorem 1.23 directly using the Fourier transform on the half-space: show that the trace maps $H^1(\mathbb{R}^d_+)$ onto $H^{1/2}(\mathbb{R}^{d-1})$.

Exercise 1.28. Show that the trace of $W^{1,1}(\Omega)$ is $L^1(\partial\Omega)$, not a Besov space with positive smoothness. *Hint:* Approximate a Dirac mass on $\partial\Omega$ by traces of $W^{1,1}$ functions.

Exercise 1.29. Show that $u \in W^{1,p}(\Omega)$ satisfies $u|_{\partial\Omega} = 0$ in the trace sense if and only if $u \in W^{1,p}_0(\Omega)$.

2. The Laplace equation

2.1 Harmonic functions

Proposition 2.1 (Pointwise derivative bounds for harmonic functions)**.** *There exists a constant $C(d) < \infty$ such that, if $r > 0$ and $u \in H^1(B_r)$ is harmonic in B_r, then u has a representative that is real analytic in B_r and satisfies, for every integer $m \geqslant 1$,*

$$|\nabla^m u(0)| \leqslant \left(\frac{Cm}{r}\right)^m \|u - (u)_{B_r}\|_{\underline{L}^2(B_r)} . \tag{2.1}$$

Here $|\nabla^m u(0)|$ is the Euclidean norm of the tensor of mth partial derivatives: $|\nabla^m u(0)|^2$ is the sum of $|\partial_{i_1} \cdots \partial_{i_m} u(0)|^2$ over all $i_1, \ldots, i_m \in \{1, \ldots, d\}$.

The restriction $m \geqslant 1$ is necessary: the right side of (2.1) vanishes for constant functions.

Proof. The argument uses only the mean value formula on balls and convolution with a radial kernel; no boundary integrals are needed. Recall that, for $d \geqslant 2$, if $v \in C^\infty(B_R(x))$ is harmonic then

$$v(x) = \fint_{B_s(x)} v \quad \text{for every } s \in (0, R) . \tag{2.2}$$

If $d = 1$, then harmonic functions are affine and (2.2) is immediate.

Step 1. Reproduction by radial kernels. Fix a radial function $\varphi \in C_c^\infty(B_1)$ with $\int_{\mathbb{R}^d} \varphi = 1$, and write $\varphi(y) = g(|y|)$ with $g \in C^1([0,\infty))$ vanishing on $[1,\infty)$. Set $\varphi_s(y) := s^{-d}\varphi(y/s)$ and

$$K := \max_{1\leqslant i\leqslant d} \|\partial_i \varphi\|_{L^1(\mathbb{R}^d)}\,,$$

which depends only on d once φ is fixed; note that $\|\partial_i \varphi_s\|_{L^1(\mathbb{R}^d)} = s^{-1}\|\partial_i\varphi\|_{L^1(\mathbb{R}^d)}$. We claim that, if $v \in C^\infty(B_R(x))$ is harmonic and $0 < s < R$, then

$$\int_{\mathbb{R}^d} \varphi_s(y) v(x-y)\,dy = v(x)\,. \tag{2.3}$$

Writing $g_s(t) := s^{-d} g(t/s)$, we have $\varphi_s(y) = \int_0^s (-g_s'(t))\mathbf{1}_{\{|y|<t\}}\,dt$ for $y \neq 0$. By Fubini's theorem and (2.2),

$$\int_{\mathbb{R}^d} \varphi_s(y) v(x-y)\,dy = \int_0^s (-g_s'(t))|B_t| \fint_{B_t(x)} v\,dt = v(x)\int_0^s (-g_s'(t))|B_t|\,dt\,,$$

and the same computation with $v \equiv 1$ shows that the last integral equals $\int_{\mathbb{R}^d}\varphi_s = 1$.

Step 2. Harmonic functions in H^1 are smooth. For $\delta \in (0,r)$, the mollification $u_\delta := u * \varphi_\delta$ is smooth in $B_{r-\delta}$, and it is harmonic there: for $x \in B_{r-\delta}$, testing the equation of u with $\varphi_\delta(x-\cdot) \in C_c^\infty(B_r)$ gives $\Delta u_\delta(x) = 0$. Let $\delta, \varepsilon > 0$ with $\delta + \varepsilon < r$ and $x \in B_{r-\delta-\varepsilon}$. Applying (2.3) to u_δ and to u_ε, and using the commutativity of convolution (which involves only the values of u in $B_{\delta+\varepsilon}(x) \subseteq B_r$), we find

$$u_\delta(x) = (u_\delta * \varphi_\varepsilon)(x) = (u * \varphi_\delta * \varphi_\varepsilon)(x) = (u_\varepsilon * \varphi_\delta)(x) = u_\varepsilon(x)\,.$$

Consequently $\widetilde{u}(x) := u_\delta(x)$, for any $\delta < \frac{1}{2}(r - |x|)$, is a well-defined smooth harmonic function in B_r. For each $\rho < r$, we have $u_\delta \to u$ in $L^2(B_\rho)$ as $\delta \to 0$ and $u_\delta = \widetilde{u}$ in B_ρ for $\delta < \frac{1}{2}(r-\rho)$. Therefore $\widetilde{u} = u$ almost everywhere in B_r. From now on we identify u with $\widetilde{u}$.

Step 3. The first derivative estimate. Let $v \in C^\infty(B_R(x))$ be harmonic and $0 < s < R$. Since (2.3) holds at every point of $B_{R-s}(x)$, we may differentiate it at x to obtain $\partial_i v(x) = \int_{\mathbb{R}^d} \partial_i \varphi_s(y)\, v(x-y)\,dy$. Hence

$$|\partial_i v(x)| \leqslant \frac{K}{s} \sup_{B_s(x)} |v|\,, \quad i \in \{1,\dots,d\}\,. \tag{2.4}$$

Step 4. Iteration. Let $v \in C^\infty(B_R(x))$ be harmonic, let $0 < \rho < R$, and set $M := \sup_{\overline{B}_\rho(x)} |v|$. We claim that, for every $m \geqslant 1$ and $i_1,\dots,i_m \in \{1,\dots,d\}$,

$$|\partial_{i_1}\cdots\partial_{i_m} v(x)| \leqslant \Big(\frac{Km}{\rho}\Big)^m M\,. \tag{2.5}$$

Since partial derivatives of v are harmonic, each $w_k := \partial_{i_{m-k+1}}\cdots\partial_{i_m} v$ is harmonic in $B_R(x)$. We show by induction on $k \in \{0,\dots,m\}$ that

$$|w_k(y)| \leqslant \Big(\frac{Km}{\rho}\Big)^k M \quad \text{for every } y \in \overline{B}_{\rho(1-k/m)}(x)\,.$$

The case $k = 0$ is the definition of M. If it holds for $k < m$ and $y \in \overline{B}_{\rho(1-(k+1)/m)}(x)$, then $B_{\rho/m}(y) \subseteq \overline{B}_{\rho(1-k/m)}(x)$ and w_k is harmonic in $B_{R-|y-x|}(y)$, whose radius exceeds ρ/m. Thus (2.4) with $s = \rho/m$ gives $|w_{k+1}(y)| \leqslant (Km/\rho) \sup_{B_{\rho/m}(y)} |w_k| \leqslant (Km/\rho)^{k+1} M$. The case $k = m$, $y = x$ is (2.5).

Step 5. From L^∞ to averaged L^2. Let $v \in C^\infty(B_r)$ be harmonic and $y \in \overline{B}_{r/2}$. Then v is harmonic in $B_{r-|y|}(y)$ and $r - |y| \geqslant r/2 > r/4$, so (2.2) and the Cauchy-Schwarz inequality give

$$|v(y)| = \left| \fint_{B_{r/4}(y)} v \right| \leqslant \left(\frac{|B_r|}{|B_{r/4}|} \fint_{B_r} v^2 \right)^{1/2} = 2^d \|v\|_{\underline{L}^2(B_r)} \,. \tag{2.6}$$

Step 6. Conclusion. The function $v := u - (u)_{B_r}$ is smooth and harmonic in B_r and has the same derivatives of order $m \geqslant 1$ as u. By (2.5) with $x = 0$, $R = r$ and $\rho = r/2$, followed by (2.6),

$$|\partial_{i_1} \cdots \partial_{i_m} u(0)| \leqslant \left(\frac{2Km}{r} \right)^m 2^d \|u - (u)_{B_r}\|_{\underline{L}^2(B_r)} \,.$$

Summing the squares of the d^m partial derivatives and using $2^d \leqslant 2^{dm}$ yields (2.1) with $C := 2^{d+1} d^{1/2} K$.

Step 7. Analyticity. Fix $x_0 \in B_r$, put $\delta := \frac{1}{4}(r - |x_0|)$ and $M := \sup_{\overline{B}_{3\delta}(x_0)} |u| < \infty$. For $y \in B_\delta(x_0)$ we have $\overline{B}_{2\delta}(y) \subseteq \overline{B}_{3\delta}(x_0)$ and u is harmonic in $B_{r-|y|}(y)$ with $r - |y| > 3\delta$, so (2.5) with $\rho = 2\delta$ gives $|\partial_{i_1} \cdots \partial_{i_m} u(y)| \leqslant (Km/(2\delta))^m M$. For $h \in \mathbb{R}^d$ with $|h| < \delta$, the function $\gamma(t) := u(x_0 + th)$ satisfies $\gamma^{(m)}(t) = \sum_{i_1,\dots,i_m} \partial_{i_1} \cdots \partial_{i_m} u(x_0 + th)\, h_{i_1} \cdots h_{i_m}$, hence $\sup_{[0,1]} |\gamma^{(m)}| \leqslant (d^{1/2}|h| Km/(2\delta))^m M$. By Taylor's theorem with Lagrange remainder and $m^m \leqslant e^m m!$,

$$\left| u(x_0 + h) - \sum_{k=0}^{m-1} \frac{\gamma^{(k)}(0)}{k!} \right| \leqslant \frac{1}{m!} \sup_{[0,1]} |\gamma^{(m)}| \leqslant \left(\frac{ed^{1/2} K |h|}{2\delta} \right)^m M \,,$$

which tends to zero as $m \to \infty$ whenever $|h| < \min\{1, 2(ed^{1/2}K)^{-1}\}\delta$. At $y = x_0$, the same bounds control the k-linear form $\frac{1}{k!}\nabla^k u(x_0)$: each entry of $\nabla^k u(x_0)$ is at most $(Kk/(2\delta))^k M$ and $\sum_i |h_i| \leqslant d^{1/2}|h|$, so its operator norm is at most $(ed^{1/2}K/(2\delta))^k M$. Hence the Taylor series of u at x_0, grouped by homogeneous degree, has a positive radius of convergence and converges absolutely for such h. Thus the Taylor series of u at x_0 converges to u in a neighborhood of x_0, and u is real analytic in B_r. □

2.2 Monotonicity formulas

Given $u \in H^1(B_R)$, we define the *Almgren frequency function of* u by

$$N_u(r) := \frac{r \int_{B_r} |\nabla u|^2}{\int_{\partial B_r} u^2}, \quad r \in (0, R). \tag{2.7}$$

The Almgren monotonicity formula states that, if u is a harmonic function, then $r \mapsto N_u(r)$ is monotone increasing.

Theorem 2.2 (Almgren monotonicity formula)**.** *Let $R > 0$ and $u \in H^1(B_R)$ be harmonic. Then*

$$r \mapsto N_u(r) \quad \text{is monotone increasing} \tag{2.8}$$

and, for every $r \in (0, R)$, we have that $N_u'(r) = 0$ if and only if u is a homogeneous harmonic polynomial.

First proof of Theorem 2.2. Write

$$N_u(r) = \frac{rD_u(r)}{H_u(r)} \quad \text{where} \quad D_u(r) := \int_{B_r} |\nabla u|^2 \quad \text{and} \quad H_u(r) := \int_{\partial B_r} u^2.$$

Observe that

$$H_u(r) = \frac{1}{r} \int_{B_r} \nabla \cdot \left(u^2 x\right) dx$$

and, by Green's formula,

$$D_u(r) = \frac{1}{r} \int_{\partial B_r} u(x \cdot \nabla u). \tag{2.9}$$

Compute

$$\begin{aligned} H_u'(r) &= -\frac{1}{r^2} \int_{B_r} \nabla \cdot \left(u^2 x\right) dx + \frac{1}{r} \int_{\partial B_r} \nabla \cdot \left(u^2 x\right) dx \\ &= -\frac{1}{r} H_u(r) + \frac{1}{r} \int_{\partial B_r} \left(du^2 + 2u(\nabla u \cdot x)\right) dx \\ &= \frac{d-1}{r} H_u(r) + 2D_u(r) \end{aligned} \tag{2.10}$$

and

$$\begin{aligned} D_u'(r) = \int_{\partial B_r} |\nabla u|^2 &= \frac{1}{r} \int_{B_r} \nabla \cdot \left(|\nabla u|^2 x\right) dx \\ &= \frac{1}{r} \int_{B_r} \left(d|\nabla u|^2 + 2x \cdot \nabla^2 u \nabla u\right) \\ &= \frac{d-2}{r} \int_{B_r} |\nabla u|^2 + \frac{2}{r^2} \int_{\partial B_r} (x \cdot \nabla u)^2 \\ &= \frac{d-2}{r} D_u(r) + \frac{2}{r^2} \int_{\partial B_r} (x \cdot \nabla u)^2. \end{aligned}$$

Therefore, using (2.9) and the Hölder inequality, we obtain

$$\begin{aligned} N_u'(r) &= \frac{1}{H_u(r)^2} \left(D_u(r) H_u(r) + rD_u'(r) H_u(r) - rD_u(r) H_u'(r)\right) \\ &= \frac{2}{rH_u(r)^2} \left(H_u(r) \int_{\partial B_r} (x \cdot \nabla u)^2 - r^2 D_u(r)^2\right) \\ &= \frac{2}{rH_u(r)^2} \left(\int_{\partial B_r} u^2 \int_{\partial B_r} (x \cdot \nabla u)^2 - \left(\int_{\partial B_r} u(x \cdot \nabla u)\right)^2\right) \geqslant 0. \end{aligned}$$

Notice that equality holds if and only if u is proportional to $x \cdot \nabla u$ on ∂B_r. In this case, let $\alpha \in \mathbb{R}$ be such that $v := x \cdot \nabla u - \alpha u = 0$ on ∂B_r. Since v is also harmonic, it follows that $v = 0$ in B_r. This implies that $\alpha \geqslant 0$ and u is positively homogeneous of degree α. It follows from regularity that $\alpha \in \mathbb{N}$ and that u is a harmonic polynomial. □

Second proof of Theorem 2.2. This proof uses the series expansion of u into homogeneous harmonic polynomials. Let $\mathbb{H}_k$ denote the set of homogeneous harmonic polynomials of degree k. Observe by Green's formula that, for every $p \in \mathbb{H}_k$,

$$r \int_{B_r} |\nabla p|^2 = \int_{\partial B_r} p(x \cdot \nabla p) = k \int_{\partial B_r} p^2$$

and thus

$$N_p \equiv k, \quad \forall p \in \mathbb{H}_k. \tag{2.11}$$

Now let u be a harmonic function and expand it into a series of homogeneous harmonic polynomials:

$$u = \sum_{k=0}^{\infty} p_k, \quad p_k \in \mathbb{H}_k. \tag{2.12}$$

By orthogonality and the fact that $N_{p_k} \equiv k$, we find that

$$N_u(r) = \frac{r \sum_{k=0}^{\infty} \int_{B_r} |\nabla p_k|^2}{\sum_{k=0}^{\infty} \int_{\partial B_r} p_k^2} = \frac{\sum_{k=0}^{\infty} k r^{2k} \int_{\partial B_1} p_k^2}{\sum_{k=0}^{\infty} r^{2k} \int_{\partial B_1} p_k^2}. \tag{2.13}$$

Differentiating, we find that

$$\begin{aligned} N_u'(r) &= \left(\sum_{k=0}^{\infty} r^{2k} \int_{\partial B_1} p_k^2 \right)^{-2} \left(\sum_{j,k=0}^{\infty} 2k(k-j) r^{2k+2j-1} \int_{\partial B_1} p_k^2 \int_{\partial B_1} p_j^2 \right) \\ &= \left(\sum_{k=0}^{\infty} r^{2k} \int_{\partial B_1} p_k^2 \right)^{-2} \left(\sum_{j,k=0}^{\infty} (k-j)^2 r^{2k+2j-1} \int_{\partial B_1} p_k^2 \int_{\partial B_1} p_j^2 \right) \geqslant 0. \end{aligned}$$

We see that equality holds only if there exists at most $k \in \mathbb{N}$ such that $p_k \not\equiv 0$, in other words, if there exists $k \in \mathbb{N}$ such that $u = p_k$. This completes the proof. □

Corollary 2.3 (Three sphere theorem)**.** *Let u be a harmonic function on B_{r_0}. Then*

$$t \mapsto \log \left(\fint_{\partial B_{\exp(t)}} u^2 \right) \quad \text{is convex on} \quad (-\infty, \log r_0)\,. \tag{2.14}$$

Proof. We observe by (2.10) that

$$\frac{d}{dr} \left(\log \left(r^{1-d} H_u(r) \right) \right) = \frac{H_u'(r) - (d-1) r^{-1} H_u(r)}{H_u(r)} = \frac{2 D_u(r)}{H_u(r)} = \frac{2}{r} N_u(r). \tag{2.15}$$

By Theorem 2.2, we deduce that

$$\frac{d^2}{dr^2}\left(\log\left(\fint_{\partial B_r} u^2\right)\right) + \frac{1}{r}\frac{d}{dr}\left(\log\left(\fint_{\partial B_r} u^2\right)\right) = \frac{d}{dr}\left(r\frac{d}{dr}\left(\log\left(r^{1-d}H_u(r)\right)\right)\right) = 2N_u'(r) \geqslant 0\,.$$

Combining this with (2.15) yields (2.14). □

Let us now turn to the issue of *doubling conditions*. Notice (2.14) implies that, for every $\lambda > 1$,

$$r \mapsto \frac{\int_{B_{\lambda r}} u^2}{\int_{B_r} u^2} \quad \text{is monotone increasing.} \tag{2.16}$$

Sometimes (2.16) with $\lambda = 2$ is called a *doubling inequality*. Doubling inequalities play an important role in unique continuation results.

Exercise 2.4 (Unique continuation)**.** Let u be a harmonic function on a connected open subset $U \subseteq \mathbb{R}^d$ which vanishes on an open ball $B_r(x_0) \subseteq U$. Using (2.16), show that u must vanish on U.

Exercise 2.5. Prove a variant of Theorem 2.2 by showing that

$$r \mapsto M_u(r) := \frac{r^2\int_{B_r}|\nabla u|^2}{\int_{B_r} u^2} \quad \text{is nondecreasing.} \tag{2.17}$$

Exercise 2.6. Let u be a non-constant harmonic function on B_{r_0} with $u(0) = 0$. Show that

$$N_u(r) \geqslant 1\,, \quad \forall r \in (0, r_0)\,.$$

Exercise 2.7. Deduce that u satisfies

$$\frac{d}{dr}\fint_{\partial B_r} u^2 \geqslant \frac{2}{r}\fint_{\partial B_r} u^2$$

and, consequently,

$$r \mapsto r^{-2}\fint_{\partial B_r} u^2 \quad \text{is nondecreasing}\,.$$

Hint: use the identity

$$\frac{d}{dr}\fint_{\partial B_r} u^2 = \fint_{\partial B_r} 2u\left(\nabla u \cdot \frac{x}{r}\right) = \frac{2}{|\partial B_r|}\int_{B_r}|\nabla u|^2 = \frac{2N_u(r)}{r}\fint_{\partial B_r} u^2\,.$$

Lemma 2.8. *Let u be a non-constant harmonic function on B_{r_0} with $u(0) = 0$. Then*

$$\frac{d}{dr}\fint_{B_r} u^2 \geqslant \frac{2}{r}\fint_{B_r} u^2\,. \tag{2.18}$$

Proof. Compute, using that $|\partial B_1| = d|B_1|$,

$$\frac{d}{dt}\Big(\fint_{B_t} u^2\Big) = \frac{1}{|B_1|}\frac{d}{dt}\Big(t^{-d}\int_{B_t} u^2\Big) = \frac{1}{|B_1|t^d}\Big(\int_{\partial B_t} u^2 - \frac{d}{t}\int_{B_t} u^2\Big) = \frac{d}{t}\Big(\fint_{\partial B_t} u^2 - \fint_{B_t} u^2\Big).$$

Using Exercise 2.7, we find that

$$\begin{aligned}\fint_{\partial B_t} u^2 - \fint_{B_t} u^2 = \fint_{\partial B_t} u^2 - \frac{1}{|B_1|}t^{-d}\int_0^t\int_{\partial B_h} u^2\,dh &= dt^{-d}\int_0^t h^{d-1}\Big(\fint_{\partial B_t} u^2 - \fint_{\partial B_h} u^2\Big)\,dh \\ &\geqslant dt^{-d}\int_0^t h^{d-1}\Big(1-\frac{h^2}{t^2}\Big)\fint_{\partial B_t} u^2\,dh \\ &= \frac{2}{d+2}\fint_{\partial B_t} u^2\,.\end{aligned}$$

Thus

$$\fint_{\partial B_t} u^2 \geqslant \frac{d+2}{d}\fint_{B_t} u^2$$

and therefore

$$\fint_{B_r} u^2 - \fint_{B_s} u^2 = d\int_s^r t^{-1}\Big(\fint_{\partial B_t} u^2 - \fint_{B_t} u^2\Big)\,dt \geqslant 2\int_s^r t^{-1}\fint_{B_t} u^2\,dt \geqslant 2(r-s)r^{-1}\fint_{B_s} u^2\,.$$

Dividing by $r-s$ and sending $s \uparrow r$ yields (2.18). □

The following corollary makes explicit the connection between the Almgren frequency and polynomial growth.

Corollary 2.9 (Growth estimate from frequency)**.** *Let u be a harmonic function on B_R with $u(0)=0$. For every $0<s<r<R$,*

$$\Big(\frac{s}{r}\Big)^{2N_u(r)}\fint_{\partial B_r} u^2 \leqslant \fint_{\partial B_s} u^2 \leqslant \Big(\frac{s}{r}\Big)^{2N_u(s)}\fint_{\partial B_r} u^2\,. \tag{2.19}$$

If, in addition, u is harmonic on all of $\mathbb{R}^d$ and $N_u(r) \leqslant N$ for every $r>0$, then u is a harmonic polynomial of degree at most $\lfloor N\rfloor$.

Proof. By (2.15),

$$\log\frac{\fint_{\partial B_r} u^2}{\fint_{\partial B_s} u^2} = 2\int_s^r \frac{N_u(t)}{t}\,dt\,.$$

Since N_u is nondecreasing (Theorem 2.2), the integrand satisfies $N_u(s)/t \leqslant N_u(t)/t \leqslant N_u(r)/t$ for $t\in[s,r]$, and integrating yields (2.19). For the second claim, expand $u=\sum_k p_k$ with $p_k\in\mathbb{H}_k$, which converges on all of $\mathbb{R}^d$. Write $a_k := \fint_{\partial B_1} p_k^2$, so that $\fint_{\partial B_r} p_k^2 = a_k r^{2k+d-1}$ and (2.13) reads $N_u(r) = \sum_k k\,a_k r^{2k}/\sum_k a_k r^{2k}$. If $u\not\equiv 0$, the bound $N_u(r)\leqslant N$ is therefore equivalent to

$$\sum_{k>N}(k-N)\,a_k r^{2k} \leqslant \sum_{k\leqslant N}(N-k)\,a_k r^{2k}\,.$$

The right side is a polynomial in r of degree at most $2\lfloor N\rfloor$. If $a_m>0$ for some $m>N$, then the left side is at least $(m-N)a_m r^{2m}$ with $m\geqslant\lfloor N\rfloor+1$, which is impossible for large r.

Hence $p_k \equiv 0$ for all $k > \lfloor N \rfloor$. The restriction to entire u is necessary: for $d = 2$, the harmonic polynomial $u = r\cos\theta + r^3\cos 3\theta$ has $N_u(r) = (1+3r^4)/(1+r^4) \leqslant 2$ for all $r \leqslant 1$ but degree 3. □

We next discuss the Alt-Caffarelli-Friedman monotonicity formula, which concerns pairs of non-negative subharmonic functions with disjoint supports. We first observe that the Almgren monotonicity extends to subharmonic functions.

Remark 2.10 (Almgren for subharmonic functions)**.** The conclusion of Theorem 2.2 remains valid for non-negative continuous subharmonic functions u (i.e., $\Delta u \geqslant 0$). Indeed, integration by parts gives $D_u(r) = \int_{B_r} |\nabla u|^2 \leqslant \frac{1}{r}\int_{\partial B_r} u(x\cdot\nabla u)$ (since the additional term $\int_{B_r} u\,\Delta u \geqslant 0$), and the Cauchy-Schwarz step in the first proof of Theorem 2.2 yields $N_u'(r) \geqslant 0$.

The key ingredient for the Alt-Caffarelli-Friedman formula is the following eigenvalue inequality on the sphere.

Lemma 2.11 (Friedland-Hayman inequality)**.** *Let Ω_+, Ω_- be complementary open subsets of ∂B_1 (i.e., $\Omega_+ \cap \Omega_- = \varnothing$ and $\overline{\Omega_+} \cup \overline{\Omega_-} = \partial B_1$). Let $\lambda_\pm$ denote the first Dirichlet eigenvalue of $-\Delta_{\partial B_1}$ on $\Omega_\pm$, and define $\alpha_\pm \geqslant 0$ by*

$$\alpha_\pm(\alpha_\pm + d - 2) = \lambda_\pm\,. \tag{2.20}$$

Then

$$\alpha_+ + \alpha_- \geqslant 2\,. \tag{2.21}$$

Proof. We reduce to caps by spherical symmetrization and then invoke the sharp inequality for caps from [FH76], which is not reproved here. Replace Ω_+ by the spherical cap C_+ of the same surface measure. By the spherical Faber-Krahn inequality (Sperner), the first Dirichlet eigenvalue does not increase, and so α_+ does not increase. Since $\Omega_- \subseteq \partial B_1 \backslash \Omega_+$ has measure at most that of $\partial B_1 \backslash C_+$, the same argument yields a cap $C_- \subseteq \partial B_1 \backslash C_+$ (of equal measure to Ω_-) with $\alpha_- \geqslant \alpha(C_-)$, and monotonicity of the eigenvalue under inclusion gives $\alpha(C_-) \geqslant \alpha(\partial B_1 \backslash C_+)$. It therefore suffices to prove (2.21) for complementary caps of opening angles θ and $\pi - \theta$. Writing $\alpha(\theta)$ for the exponent of a cap of opening angle $\theta \in (0,\pi)$ (the degree of the least homogeneous harmonic function in the cone over the cap vanishing on its boundary), the required inequality is

$$\alpha(\theta) + \alpha(\pi - \theta) \geqslant 2\,, \qquad \theta \in (0,\pi)\,.$$

For $\theta = \pi/2$ we have $\alpha = 1$, since x_1 restricted to the hemisphere is a positive eigenfunction with eigenvalue $d-1$. That $\alpha(\theta)$ is decreasing in θ does not by itself yield the inequality for $\theta \neq \pi/2$; the remaining step is the analysis of the cap exponent $\alpha(\theta)$ carried out in [FH76], and we take it from there. This is the form in which the inequality is used in [ACF84]. □

Remark 2.12. Equality in (2.21) holds for complementary hemispheres, but it is not true that equality characterizes them under the hypotheses of Lemma 2.11. For $d \geqslant 3$, take Ω_+ to be an open hemisphere with one point removed and Ω_- the opposite open hemisphere. These are disjoint open sets whose closures cover ∂B_1, and a point has zero capacity on ∂B_1, so $\lambda_+ = \lambda_- = d-1$ and $\alpha_+ + \alpha_- = 2$. The equality case is therefore only meaningful modulo sets of zero capacity, and we do not use it.

The connection to monotonicity formulas is as follows. For a non-negative subharmonic function w vanishing at the origin, the Almgren frequency $N_w(r)$ controls the growth of $\int_{\partial B_r} w^2$. The quantity $\alpha_\pm$ in (2.20) controls the *angular* component of this growth: the eigenvalue bound says that w cannot grow slowly in the radial direction unless it is spread over a large portion of the sphere. When two such functions w_+, w_- have disjoint supports, the Friedland-Hayman inequality forces the sum of their growth rates to be at least 2.

Theorem 2.13 (Alt-Caffarelli-Friedman monotonicity formula)**.** *Let $d \geqslant 3$, $R > 0$, and let $u_+, u_- \in H^1(B_R)$ be non-negative, continuous, and subharmonic in B_R, with $u_+(0) = u_-(0) = 0$ and $u_+ \cdot u_- = 0$. Then*

$$r \mapsto \Phi(r) := \frac{1}{r^4} \prod_{\pm} \int_{B_r} \frac{|\nabla u_\pm|^2}{|x|^{d-2}}\, dx \quad \text{is nondecreasing for } r \in (0, R)\,. \tag{2.22}$$

Proof. Define $J_\pm(r) := \int_{B_r} |\nabla u_\pm|^2 |x|^{2-d}\, dx$, so that $\Phi(r) = r^{-4} J_+(r) J_-(r)$, and compute

$$J'_\pm(r) = r^{2-d} \int_{\partial B_r} |\nabla u_\pm|^2\,.$$

Taking the logarithmic derivative,

$$\frac{d}{dr} \log \Phi(r) = \sum_{\pm} \frac{J'_\pm(r)}{J_\pm(r)} - \frac{4}{r}\,, \tag{2.23}$$

so it suffices to show that $\sum_\pm r J'_\pm(r)/J_\pm(r) \geqslant 4$.

Decompose the gradient on ∂B_r into its radial and tangential components:

$$\int_{\partial B_r} |\nabla u_\pm|^2 = \int_{\partial B_r} \Big(\frac{\partial u_\pm}{\partial r}\Big)^2 + \frac{1}{r^2} \int_{\partial B_r} |\nabla_\theta u_\pm|^2\,, \tag{2.24}$$

where ∇_θ denotes the tangential gradient on ∂B_r. By the variational characterization of the first Dirichlet eigenvalue on the support $\Omega_\pm(r) := \{x/|x| : u_\pm(x) > 0,\ |x| = r\} \subseteq \partial B_1$ and the fact that $u_\pm = 0$ on $\partial B_1 \backslash \Omega_\pm(r)$,

$$\int_{\partial B_r} |\nabla_\theta u_\pm|^2 \geqslant \lambda_\pm(r) \int_{\partial B_r} u_\pm^2\,, \tag{2.25}$$

where $\lambda_\pm(r)$ is the first Dirichlet eigenvalue of $-\Delta_{\partial B_1}$ on $\Omega_\pm(r)$.

Define $\alpha_\pm(r) \geqslant 0$ by $\alpha_\pm(r)(\alpha_\pm(r) + d - 2) = \lambda_\pm(r)$. Since $u_+ u_- = 0$ implies that $\Omega_+(r)$ and $\Omega_-(r)$ are complementary, the Friedland-Hayman inequality (Lemma 2.11) gives

$$\alpha_+(r) + \alpha_-(r) \geqslant 2\,.$$

Combining (2.24) and (2.25), and writing $H_\pm(r) := \int_{\partial B_r} u_\pm^2$ and $D_\pm(r) := \int_{B_r} |\nabla u_\pm|^2$, we obtain

$$r^{2-d} \int_{\partial B_r} |\nabla u_\pm|^2 \geqslant r^{2-d} \int_{\partial B_r} \Big(\frac{\partial u_\pm}{\partial r}\Big)^2 + \frac{\alpha_\pm(r)(\alpha_\pm(r) + d - 2)}{r^d}\, H_\pm(r)\,.$$

We claim that for each sign, the quotient $rJ'_\pm(r)/J_\pm(r) \geqslant 2\alpha_\pm(r)$. To see this, observe that since $u_\pm$ is subharmonic and non-negative with $u_\pm(0) = 0$, the function $t \mapsto t^{-2\alpha_\pm(t)} \int_{\partial B_t} u_\pm^2$ is nondecreasing for $t \leqslant r$ (by the Almgren monotonicity and the Rayleigh bound). Thus $H_\pm(t) \lesssim (t/r)^{2\alpha_\pm(r)+d-1} H_\pm(r)$, and therefore

$$J_\pm(r) = \int_0^r t^{2-d} D'_\pm(t)\, dt \leqslant \int_0^r t^{2-d} \int_{\partial B_t} |\nabla u_\pm|^2\, dt \leqslant C(\alpha_\pm, d)\, r^{3-d} \int_{\partial B_r} |\nabla u_\pm|^2 = C(\alpha_\pm, d)\, r\, J'_\pm(r)\,,$$

where the constant $C(\alpha_\pm, d)$ satisfies $1/C \geqslant 2\alpha_\pm(r)$ (this can be verified by a homogeneity computation). It follows that

$$\sum_\pm \frac{rJ'_\pm(r)}{J_\pm(r)} \geqslant 2\alpha_+(r) + 2\alpha_-(r) \geqslant 4\,,$$

and (2.23) gives $\Phi'(r) \geqslant 0$. □

Remark 2.14. In dimension $d = 2$, the weight $|x|^{2-d} = 1$ and the functional becomes $\Phi(r) = r^{-4} \prod_\pm \int_{B_r} |\nabla u_\pm|^2$. The result remains valid with the same proof.

2.3 The heat equation

We now present the parabolic analogue of the Almgren frequency function, due to Poon. Let $\Phi(s, x) = (4\pi s)^{-d/2} \exp(-|x|^2/(4s))$ be the heat kernel on $\mathbb{R}^d$ as defined above.

Theorem 2.15 (Poon's parabolic frequency function)**.** *Let u be a smooth solution of $\partial_t u = \Delta u$ on $\mathbb{R}^d \times (-T, 0)$ with $|u(x,t)| \leqslant C \exp(A|x|^2)$ for some constants $A, C < \infty$. Define, for $s \in (0, T)$,*

$$H(s) := \int_{\mathbb{R}^d} u(-s,x)^2\, \Phi(s,x)\, dx\,, \quad D(s) := \int_{\mathbb{R}^d} |\nabla u(-s,x)|^2\, \Phi(s,x)\, dx\,, \tag{2.26}$$

and the parabolic frequency

$$\mathcal{N}(s) := \frac{s\, D(s)}{H(s)}\,. \tag{2.27}$$

Then $H(s)$ and $D(s)$ are finite for every $s \in (0, T)$ with $8As < 1$, and $s \mapsto \mathcal{N}(s)$ is nondecreasing on $\{s \in (0,T) : 8As < 1\}$. (The threshold $8As < 1$ is not claimed to be sharp.)

Proof. Write $v(s, x) := u(-s, x)$, so that $\partial_s v = -\Delta v$. All integrals below are over $\mathbb{R}^d$.

Step 0. We show that H and D are finite on $I := \{s \in (0,T) : 8As < 1\}$, an interval, and that the integrations by parts and differentiations below are justified there. For $s \in I$ we have $2A < 1/(4s)$, so $v^2\Phi \leqslant C^2(4\pi s)^{-d/2} \exp\big((2A - \frac{1}{4s})|x|^2\big)$ is integrable, locally uniformly in $s \in I$ with a Gaussian majorant. Interior parabolic estimates on parabolic cylinders of a radius $\rho = \rho(K) \in (0, 1]$ with $\rho^2 < T - s$, for s in a compact set $K \subset I$, give $|\nabla v| + |\nabla^2 v| + |\partial_s v| + |\nabla \partial_s v| \leqslant C' e^{A'|x|^2}$ for every $A' > A$ close to A, locally uniformly in time, and since $8As < 1$ is a strict inequality we may choose A' so that $8A's < 1$ as well.

Hence all integrands in Steps 1–4 are dominated by a Gaussian times a function integrable in x, locally uniformly in $s \in I$, which justifies differentiation under the integral sign and the integrations by parts (with the boundary terms on ∂B_ρ tending to zero as $\rho \to \infty$). Without a restriction on s the claim fails: for $A_0 > 0$ and $0 < 4A_0T < 1$, the function $u(t,x) = (1-4A_0(t+T))^{-d/2} \exp\bigl(A_0|x|^2/(1-4A_0(t+T))\bigr)$ is a solution in the growth class with $A = A_0/(1-4A_0T)$, but $H(s) = D(s) = \infty$ for $s \in [1/(4A), T)$ whenever $T > 1/(4A)$. We use the identities $\partial_s \Phi = \Delta\Phi$ and $\nabla\Phi = -\frac{x}{2s}\Phi$.

Step 1. We show $H'(s) = 2D(s)$. Differentiating,

$$H'(s) = \int \bigl(-2v\,\Delta v \cdot \Phi + v^2\,\Delta\Phi\bigr)\,.$$

Integration by parts on the first term gives $-2\int v\,\Delta v\,\Phi = 2\int |\nabla v|^2\Phi + 2\int v\,\nabla v \cdot \nabla\Phi$. On the second, $\int v^2\Delta\Phi = -2\int v\,\nabla v \cdot \nabla\Phi$. The cross terms cancel, yielding $H'(s) = 2D(s)$.

Step 2. Define the *parabolic scaling* of v:

$$w := x \cdot \nabla v + 2s\,\partial_s v = x \cdot \nabla v - 2s\,\Delta v\,.$$

We claim that

$$\int v\,w\,\Phi = 2s\,D(s)\,. \tag{2.28}$$

Indeed, $\int v\,w\,\Phi = \int v(x\cdot\nabla v)\,\Phi + 2s\int v\,\partial_s v\,\Phi$. Since $H'(s) = 2\int v\,\partial_s v\,\Phi + \int v^2\partial_s\Phi$ and $\int v^2\partial_s\Phi = \int v^2\Delta\Phi = -2\int v\,\nabla v\cdot\nabla\Phi = \frac{1}{s}\int v(x\cdot\nabla v)\,\Phi$, we get $\int v(x\cdot\nabla v)\,\Phi + 2s\int v\,\partial_s v\,\Phi = sH'(s) = 2sD$.

Step 3. We show the parabolic Rellich identity

$$sD'(s) + D(s) = \frac{1}{2s}\int w^2\,\Phi\,. \tag{2.29}$$

Using $\partial_s v = -\Delta v$ and $\partial_s\Phi = \Delta\Phi$,

$$sD'(s) = s\int \bigl(-2\nabla\Delta v \cdot \nabla v\,\Phi + |\nabla v|^2\,\Delta\Phi\bigr)\,.$$

For the first integral, integration by parts gives

$$-2s\int \nabla\Delta v\cdot\nabla v\,\Phi = 2s\int (\Delta v)^2\,\Phi + 2s\int \Delta v\,\nabla v\cdot\nabla\Phi = 2s\int (\partial_s v)^2\,\Phi + \int \partial_s v\,(x\cdot\nabla v)\,\Phi\,.$$

For the second, using $\nabla\Phi = -\frac{x}{2s}\Phi$ and the identity $x\cdot\nabla(|\nabla v|^2) = 2\nabla(x\cdot\nabla v)\cdot\nabla v - 2|\nabla v|^2$,

$$s\int |\nabla v|^2\,\Delta\Phi = \frac{1}{2}\int x\cdot\nabla(|\nabla v|^2)\,\Phi = \int \nabla(x\cdot\nabla v)\cdot\nabla v\,\Phi - D(s)\,.$$

Integrating by parts and using $\nabla\Phi = -\frac{x}{2s}\Phi$,

$$\int \nabla(x\cdot\nabla v)\cdot\nabla v\,\Phi = -\int (x\cdot\nabla v)\,\Delta v\,\Phi - \int (x\cdot\nabla v)\,\nabla v\cdot\nabla\Phi = \int (x\cdot\nabla v)\,\partial_s v\,\Phi + \frac{1}{2s}\int (x\cdot\nabla v)^2\,\Phi\,.$$

Combining the two integrals,

$$sD'(s) = 2s\int(\partial_s v)^2\,\Phi + 2\int \partial_s v\,(x\cdot\nabla v)\,\Phi + \frac{1}{2s}\int (x\cdot\nabla v)^2\,\Phi - D(s)$$
$$= \frac{1}{2s}\int \big(x\cdot\nabla v + 2s\,\partial_s v\big)^2\,\Phi - D(s) = \frac{1}{2s}\int w^2\,\Phi - D(s)\,.$$

Step 4. Using $H' = 2D$, (2.28), and (2.29),

$$\mathcal{N}'(s) = \frac{1}{H^2}\big(H(sD'+D) - 2sD^2\big) = \frac{1}{2sH^2}\left(H\int w^2\,\Phi - \left(\int v\,w\,\Phi\right)^2\right) \geqslant 0$$

by the Cauchy-Schwarz inequality in $L^2(\Phi\,dx)$. □

Remark 2.16. The parabolic frequency satisfies $\mathcal{N}(s)\equiv k$ if and only if $v(s,x) = s^k p(x/\sqrt{s})$ for some caloric polynomial p of parabolic degree $2k$ (that is, $w = 2kv$). For a caloric function with $u(0,0) = 0$, the Gaussian Poincaré inequality (Proposition 1.21), extended by a standard density argument to functions $f\in C^1(\mathbb{R}^d)$ with $f, \nabla f\in L^2(\Phi(s,\cdot)\,dx)$ (cut off by $\chi(\cdot/R)f$ and let $R\to\infty$), gives $\mathcal{N}(s)\geqslant\frac12$ at every s for which $u(-s,\cdot)$ and $\nabla u(-s,\cdot)$ belong to $L^2(\Phi(s,\cdot)\,dx)$ (that is, whenever $H(s), D(s)<\infty$), applied to $f = u(-s,\cdot)$, whose mean against $\Phi(s)$ is $u(0,0) = 0$.

Notes and references

3. Perturbative estimates

3.1 Schauder estimates

Proposition 3.1 ($C^{0,1-}$ Schauder estimate)**.** *There exists $C(d)<\infty$ such that, for every $\alpha\in(0,1)$, and coefficient field $\mathbf{a}: B_1\to\mathbb{R}^{d\times d}$ satisfying*

$$\|\mathbf{a} - \mathrm{I}_d\|_{L^\infty(B_1)} \leqslant 2^{-3-d/2}(1-\alpha)\,, \tag{3.1}$$

and functions $u\in H^1(B_1)$, $\mathbf{f}\in L^p(B_1)^d$, with $p = d(1-\alpha)^{-1}$, satisfying the equation

$$-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f}\quad \text{in } B_1,$$

we have that $u\in C^{0,\alpha}(B_{1/2})$ and the estimate

$$[u]_{C^{0,\alpha}(B_{1/2})} \leqslant \frac{C}{\alpha}\big(\|\nabla u\|_{L^2(B_1)} + (1-\alpha)^{-1}\|\mathbf{f}\|_{L^p(B_1)}\big)\,.$$

Proof. In dimension $d=1$, the weak equation implies that $\mathbf{a}u' + \mathbf{f} = c$ is constant on $B_1 = (-1,1)$. Since $1/2\leqslant\mathbf{a}\leqslant 3/2$, integration and Hölder's inequality give

$$|c|\leqslant C\big(\|u'\|_{L^2(B_1)} + \|\mathbf{f}\|_{L^p(B_1)}\big),\qquad |u(x)-u(y)|\leqslant 2|c||x-y| + 2\|\mathbf{f}\|_{L^p(B_1)}|x-y|^{1-1/p}.$$

Here C is independent of $p>1$. As $1-1/p=\alpha$ and $|x-y|\leqslant 1$ for $x,y\in B_{1/2}$, this proves the claimed estimate. We henceforth assume $d\geqslant 2$, so that $p\geqslant 2$.

Let $\delta := \|\mathbf{a} - \mathrm{I}_d\|_{L^\infty(B_1)}$. For each $r \in (0,1)$, let $h_r \in H^1(B_r)$ be the solution of

$$\begin{cases} -\Delta h_r = 0 & \text{in } B_r, \\ h_r = u & \text{on } \partial B_r. \end{cases}$$

Observe that the difference $u - h_r$ satisfies

$$\begin{cases} -\nabla \cdot \mathbf{a}\nabla(u - h_r) = \nabla \cdot \big(\mathbf{f} + (\mathbf{a} - I_d)\nabla h_r\big) & \text{in } B_r, \\ u - h_r = 0 & \text{on } \partial B_r. \end{cases} \tag{3.2}$$

Testing (3.2) with $u - h_r$ and applying the Cauchy-Schwarz inequality yields

$$\begin{aligned} (1-\delta)\|\nabla(u-h_r)\|^2_{\underline{L}^2(B_r)} &\leqslant \fint_{B_r} \nabla(u-h_r)\cdot \mathbf{a}\nabla(u-h_r) \\ &= -\fint_{B_r} \nabla(u-h_r)\cdot\big(\mathbf{f} + (\mathbf{a}-I_d)\nabla h_r\big) \\ &\leqslant \|\nabla(u-h_r)\|_{\underline{L}^2(B_r)}\big(\delta\|\nabla h_r\|_{\underline{L}^2(B_r)} + \|\mathbf{f}\|_{\underline{L}^2(B_r)}\big)\,. \end{aligned} \tag{3.3}$$

Since $\delta < 1/2$, dividing by $\|\nabla(u-h_r)\|_{\underline{L}^2(B_r)}$ gives

$$\begin{aligned} \|\nabla(u-h_r)\|_{\underline{L}^2(B_r)} &\leqslant \frac{\delta}{1-\delta}\|\nabla h_r\|_{\underline{L}^2(B_r)} + \frac{1}{1-\delta}\|\mathbf{f}\|_{\underline{L}^2(B_r)} \\ &\leqslant 2\delta\|\nabla h_r\|_{\underline{L}^2(B_r)} + 2\|\mathbf{f}\|_{\underline{L}^2(B_r)}\,. \end{aligned}$$

Since $|\nabla h_r|^2$ is subharmonic, the submean-value inequality gives $\|\nabla h_r\|_{\underline{L}^2(B_{\theta r})} \leqslant \|\nabla h_r\|_{\underline{L}^2(B_r)}$. Combined with the Dirichlet principle, this yields

$$\|\nabla h_r\|_{\underline{L}^2(B_{\theta r})} \leqslant \|\nabla h_r\|_{\underline{L}^2(B_r)} \leqslant \|\nabla u\|_{\underline{L}^2(B_r)}\,.$$

By the triangle inequality, we obtain, for every $\theta \in (0, 1/2]$,

$$\begin{aligned} \|\nabla u\|_{\underline{L}^2(B_{\theta r})} &\leqslant \|\nabla h_r\|_{\underline{L}^2(B_{\theta r})} + \|\nabla(u-h_r)\|_{\underline{L}^2(B_{\theta r})} \\ &\leqslant \|\nabla h_r\|_{\underline{L}^2(B_r)} + 2\delta\theta^{-d/2}\|\nabla h_r\|_{\underline{L}^2(B_r)} + 2\theta^{-d/2}\|\mathbf{f}\|_{\underline{L}^2(B_r)} \\ &\leqslant \big(1 + 2\delta\theta^{-d/2}\big)\|\nabla u\|_{\underline{L}^2(B_r)} + 2|B_1|^{-1/p}\theta^{-d/2}r^{-d/p}\|\mathbf{f}\|_{L^p(B_1)}\,. \end{aligned}$$

Multiplying by $(\theta r)^{1-\alpha} = (\theta r)^{d/p}$, we obtain

$$(\theta r)^{1-\alpha}\|\nabla u\|_{\underline{L}^2(B_{\theta r})} \leqslant \big(1 + 2\delta\theta^{-d/2}\big)\theta^{1-\alpha} r^{1-\alpha}\|\nabla u\|_{\underline{L}^2(B_r)} + 2|B_1|^{-1/p}\theta^{d/p-d/2}\|\mathbf{f}\|_{L^p(B_1)}\,. \tag{3.4}$$

We next select $\theta := 1/2$ and $\delta_0 > 0$ so small that

$$\big(1 + 2\delta_0\theta^{-d/2}\big)\theta^{1-\alpha} \leqslant 1 - \frac{1}{8}(1-\alpha)\,. \tag{3.5}$$

The constraint (3.5) is equivalent to $1 + 2^{1+d/2}\delta_0 \leqslant \big(1 - \frac{1}{8}(1-\alpha)\big)2^{1-\alpha}$. In view of the inequality

$$1 + \frac{1}{4}x \leqslant \Big(1 - \frac{1}{8}x\Big)2^x\,, \quad \forall x \in [0,1]\,,$$

it suffices to select

$$\delta_0 := 2^{-3-d/2}(1-\alpha)\,.$$

Inserting these choices into (3.4), we obtain

$$(r/2)^{1-\alpha}\|\nabla u\|_{\underline{L}^2(B_{r/2})} \leqslant \Big(1-\frac{1}{8}(1-\alpha)\Big) r^{1-\alpha}\|\nabla u\|_{\underline{L}^2(B_r)} + 2^{1+d/2-d/p}|B_1|^{-1/p}\|\mathbf{f}\|_{L^p(B_1)}\,. \tag{3.6}$$

An iteration of (3.6) yields

$$\begin{aligned}\sup_{s\in(0,1)} s^{1-\alpha}\|\nabla u\|_{\underline{L}^2(B_s)} &\leqslant 2^{d/2}\sup_{k\in\mathbb{N}} 2^{-k(1-\alpha)}\|\nabla u\|_{\underline{L}^2(B_{2^{-k}})}\\ &\leqslant 2^{d/2}\big(\|\nabla u\|_{\underline{L}^2(B_1)} + 2^{4+d/2-d/p}|B_1|^{-1/p}(1-\alpha)^{-1}\|\mathbf{f}\|_{L^p(B_1)}\big).\end{aligned}$$

The normalization factors $|B_1|^{-1/2}$ and $|B_1|^{-1/p}$ are bounded in terms of d alone, since $p \geqslant 2$. By the Poincaré inequality, we deduce that, for a constant $C(d) < \infty$,

$$\sup_{s\in(0,1/2)} s^{-\alpha}\|u-(u)_{B_s}\|_{\underline{L}^2(B_s)} \leqslant C\big(\|\nabla u\|_{L^2(B_1)} + (1-\alpha)^{-1}\|\mathbf{f}\|_{L^p(B_1)}\big)\,.$$

By repeating the argument (or performing a translation and dilation applying the result already proved) we obtain

$$\sup_{x\in B_{1/2}}\sup_{s\in(0,1/4)} s^{-\alpha}\|u-(u)_{B_s}\|_{\underline{L}^2(B_s(x))} \leqslant C\big(\|\nabla u\|_{L^2(B_1)} + (1-\alpha)^{-1}\|\mathbf{f}\|_{L^p(B_1)}\big)\,.$$

Applying Lemma 1.19, after bounding the normalized L^1 oscillation by the normalized L^2 oscillation, gives the claimed estimate with the additional factor α^{-1}. This factor remains bounded when $\alpha \geqslant 1/2$, so the estimate has the stated order $(1-\alpha)^{-1}$ as $\alpha \uparrow 1$. □

Corollary 3.2 ($C^{0,1-}$ estimate for continuous coefficients)**.** *Suppose that $\mathbf{a} : \overline{B}_1 \to \mathbb{R}^{d\times d}$ is continuous and uniformly elliptic with constant Λ. Then, for every $\alpha \in (0,1)$, there exists a constant $C < \infty$, which depends on (d,Λ,α) and the modulus of continuity for $\mathbf{a}(\cdot)$ such that, for every pair $u \in H^1(B_1)$ and $\mathbf{f} \in L^p(B_1)^d$, with $p = d(1-\alpha)^{-1}$, satisfying the equation*

$$-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f} \quad \text{in } B_1,$$

we have that $u \in C^{0,\alpha}(B_{1/2})$ and the estimate

$$[u]_{C^{0,\alpha}(B_{1/2})} \leqslant \frac{C}{\alpha}\|\nabla u\|_{L^2(B_1)} + \frac{C}{\alpha(1-\alpha)}\|\mathbf{f}\|_{L^p(B_1)}\,.$$

Proof. By performing an affine change of variables, we may formulate a version of Proposition 3.1 in which the left side of (3.1) is replaced by $\operatorname{osc}_{B_1}\mathbf{a}$ and the constant C depends additionally on Λ. Moreover, the assumptions of the proposition are invariant under dilation and translation, so the proposition may be applied in balls of any size and centered at any point. We may then use the assumption that $\mathbf{a}$ is continuous to find a scale $r > 0$ such that $\operatorname{osc}_{B_r(x)}\mathbf{a}$ is sufficiently small, for each $x \in B_{1/2}$, that we may apply the proposition in each such ball $B_r(x)$. Since a finite number of balls of the form $B_{r/2}(x)$ covers $B_{1/2}$, we obtain the corollary. The scale r depends on α through the smallness threshold (3.1); its dependence on the modulus of continuity and the parameters in the statement enters the constant C when we estimate

$$\|\nabla u\|_{\underline{L}^2(B_r(x))} + \|\mathbf{f}\|_{\underline{L}^p(B_r(x))} \leqslant Cr^{-d/2}\|\nabla u\|_{L^2(B_1)} + Cr^{-d/p}\|\mathbf{f}\|_{L^p(B_1)}\,.$$ □

Proposition 3.3 ($C^{1,1-}$ Schauder estimate)**.** *There exists $C(d,\Lambda) < \infty$ such that, for every $\alpha \in (0,1)$, coefficient field $\mathbf{a} \in C^{0,\alpha}(B_1;\mathbb{R}^{d\times d})$ and vector field $\mathbf{f} \in C^{0,\alpha}(B_1;\mathbb{R}^d)$, with*

$$e\cdot\mathbf{a}(x)e \geqslant |e|^2 \quad \text{and} \quad e\cdot\mathbf{a}(x)^{-1}e \geqslant \Lambda^{-1}|e|^2\,, \quad \forall x \in B_1,\ e\in\mathbb{R}^d,$$

and every $r_0 \in (0, 1/4]$ satisfying

$$r_0^\alpha [\mathbf{a}]_{C^{0,\alpha}(B_1)} \leqslant C^{-1}\alpha(1-\alpha)$$

and every solution $u \in H^1(B_1)$ of

$$-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f} \quad \text{in } B_1,$$

we have the estimate

$$\|\nabla u\|_{L^\infty(B_{1/2})} + r_0^\alpha [\nabla u]_{C^{0,\alpha}(B_{1/2})} \leqslant \frac{C}{\alpha^2}\Big(r_0^{-d/2}\|\nabla u\|_{\underline{L}^2(B_1)} + \frac{r_0^\alpha}{1-\alpha}[\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,. \tag{3.7}$$

Proof. By the Lebesgue differentiation theorem and Lemma 1.19, it suffices to show the following estimate for each $x \in B_{1/2}$. The first term controls the L^∞ norm via Lebesgue differentiation, while Lemma 1.19 applied to the second term contributes an additional factor α^{-1} from (1.41). Bounding the initial local norm by $Cr_0^{-d/2}\|\nabla u\|_{\underline{L}^2(B_1)}$ then gives (3.7). The required local estimate is

$$\sup_{r\in(0,r_0)}\Big(\|\nabla u\|_{\underline{L}^2(B_r(x))} + \Big(\frac{r_0}{r}\Big)^\alpha \|\nabla u - (\nabla u)_{B_r(x)}\|_{\underline{L}^2(B_r(x))}\Big) \leqslant \frac{C}{\alpha}\Big(\|\nabla u\|_{\underline{L}^2(B_{r_0}(x))} + \frac{r_0^\alpha}{1-\alpha}[\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,. \tag{3.8}$$

Translate the center to 0. Write $\mathbf{a}(0) = S + K$, where S is symmetric and K is skew-symmetric. The constant skew part contributes zero to the scalar weak operator, so we may replace $\mathbf{a}$ by $\mathbf{a} - K$. The change of variables $y = S^{-1/2}x$ transforms the coefficient to $S^{-1/2}(\mathbf{a}-K)S^{-1/2}$, whose value at the center is I_d. Since $I_d \leqslant S \leqslant \Lambda I_d$, the transformed image of B_{r_0} contains $B_{r_0/\sqrt{\Lambda}}$ and is contained in B_{r_0}. Apply the argument below on the contained ball. The transformed Hölder seminorm is at most $\Lambda^{\alpha/2}[\mathbf{a}]_{C^{0,\alpha}}$, so its product with the transformed radius to the power α is at most $r_0^\alpha[\mathbf{a}]_{C^{0,\alpha}}$. Norm and volume comparisons introduce only constants depending on (d,Λ). Radii between $r_0/\sqrt{\Lambda}$ and r_0 are controlled directly by the initial local energy. We therefore give the remaining argument with $\mathbf{a}(0) = I_d$. Finally, subtracting a constant from $\mathbf{f}$ preserves the equation and its Hölder seminorm, so we may suppose that $\mathbf{f}(0) = 0$.

We follow the reasoning in the proof of Proposition 3.1 to derive (3.3), and then use the stronger hypotheses to estimate the right side differently: we get

$$\begin{aligned}\|\nabla(u-h_r)\|_{\underline{L}^2(B_r)} &\leqslant 2\|\mathbf{a}-I_d\|_{L^\infty(B_r)}\|\nabla h_r\|_{\underline{L}^2(B_r)} + 2\|\mathbf{f}\|_{\underline{L}^2(B_r)}\\ &\leqslant 2r^\alpha\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\|\nabla h_r\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\\ &\leqslant 2r^\alpha\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,.\end{aligned} \tag{3.9}$$

Here comes the main point in the proof, where we use the regularity of harmonic functions. We use that ∇h_r is harmonic and apply Lemma 2.8 to obtain that, for every $0 < s < r < 1$,

$$\|\nabla h_r - \nabla h_r(0)\|_{\underline{L}^2(B_s)} \leqslant \frac{s}{r}\|\nabla h_r - \nabla h_r(0)\|_{\underline{L}^2(B_r)} = \frac{s}{r}\|\nabla h_r - (\nabla h_r)_{B_r}\|_{\underline{L}^2(B_r)}\,.$$

We use the triangle inequality to write this inequality with u in place of h_r, paying the price of the harmonic approximation error. We obtain, for every $0 < s < r < 1$,

$$\begin{aligned}\|\nabla u - (\nabla u)_{B_s}\|_{\underline{L}^2(B_s)} &\leqslant \|\nabla u - \nabla h_r(0)\|_{\underline{L}^2(B_s)}\\ &\leqslant \|\nabla h_r - \nabla h_r(0)\|_{\underline{L}^2(B_s)} + \|\nabla u - \nabla h_r\|_{\underline{L}^2(B_s)}\\ &\leqslant \frac{s}{r}\|\nabla h_r - (\nabla h_r)_{B_r}\|_{\underline{L}^2(B_r)} + \Big(\frac{r}{s}\Big)^{d/2}\|\nabla(u-h_r)\|_{\underline{L}^2(B_r)}\\ &\leqslant \frac{s}{r}\|\nabla u - (\nabla u)_{B_r}\|_{\underline{L}^2(B_r)} + \Big(1+\Big(\frac{r}{s}\Big)^{d/2}\Big)\|\nabla(u-h_r)\|_{\underline{L}^2(B_r)}\,.\end{aligned}$$

Multiplying by $s^{-\alpha}$ and using (3.9), we obtain

$$\begin{aligned}s^{-\alpha}\|\nabla u - (\nabla u)_{B_s}\|_{\underline{L}^2(B_s)} \leqslant \Big(\frac{s}{r}\Big)^{1-\alpha} r^{-\alpha}\|\nabla u - (\nabla u)_{B_r}\|_{\underline{L}^2(B_r)}\\ + 2\Big(\frac{r}{s}\Big)^{\alpha}\Big(1+\Big(\frac{r}{s}\Big)^{d/2}\Big)\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,.\end{aligned}$$

Taking $s = r/2$, we obtain, for every $r \in (0, r_0)$,

$$\begin{aligned}\big(r/2\big)^{-\alpha}\|\nabla u - (\nabla u)_{B_{r/2}}\|_{\underline{L}^2(B_{r/2})} \leqslant \Big(1-\frac{1}{2}(1-\alpha)\Big)r^{-\alpha}\|\nabla u - (\nabla u)_{B_r}\|_{\underline{L}^2(B_r)}\\ + 2^{3+d/2}\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,.\end{aligned}$$

Iterating this inequality yields a constant $C(d) < \infty$ such that, for every $r_0 \in (0,1)$,

$$\begin{aligned}&\sup_{r\in(0,r_0)} r^{-\alpha}\|\nabla u - (\nabla u)_{B_r}\|_{\underline{L}^2(B_r)}\\ &\quad\leqslant Cr_0^{-\alpha}\|\nabla u\|_{\underline{L}^2(B_{r_0})} + \frac{C}{1-\alpha}\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,.\end{aligned} \tag{3.10}$$

Applying this inequality with r and $r/2$ and using the triangle inequality, we obtain, for every $r \in (0, r_0)$,

$$\begin{aligned}&\big|(\nabla u)_{B_r} - (\nabla u)_{B_{r/2}}\big|\\ &\quad\leqslant C\Big(\frac{r}{r_0}\Big)^{\alpha}\|\nabla u\|_{\underline{L}^2(B_{r_0})} + \frac{Cr^{\alpha}}{1-\alpha}\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,.\end{aligned}$$

Summing this inequality over dyadic scales (the geometric series $\sum_{k\geqslant 0} 2^{-k\alpha} = (1-2^{-\alpha})^{-1} \leqslant C\alpha^{-1}$ accounts for the factor of α^{-1}) yields

$$\sup_{r\in(0,r_0)}\big|(\nabla u)_{B_r}\big| \leqslant \frac{C}{\alpha}\|\nabla u\|_{\underline{L}^2(B_{r_0})} + \frac{Cr_0^{\alpha}}{\alpha(1-\alpha)}\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} + [\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big)\,. \tag{3.11}$$

By (3.10), (3.11) and the triangle inequality, we obtain

$$\begin{aligned}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} &\leqslant r_0^\alpha \sup_{r\in(0,r_0)} r^{-\alpha}\|\nabla u-(\nabla u)_{B_r}\|_{\underline{L}^2(B_r)}+\sup_{r\in(0,r_0)}\left|(\nabla u)_{B_r}\right|\\ &\leqslant \frac{C}{\alpha}\|\nabla u\|_{\underline{L}^2(B_{r_0})}+\frac{Cr_0^\alpha}{\alpha(1-\alpha)}\Big([\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)}+[\mathbf{f}]_{C^{0,\alpha}(B_1)}\Big).\end{aligned}$$

To justify absorption before knowing that the supremum is finite, first perform the iteration on the finite set of dyadic radii $r_0, r_0/2, \dots, r_0/2^N$. Every corresponding local energy is finite. The oscillation recurrence and the finite telescoping sum of averages give the preceding estimate with these truncated suprema and a constant independent of N. If $r_0 \in (0,1)$ is small enough that $C(\alpha(1-\alpha))^{-1}r_0^\alpha[\mathbf{a}]_{C^{0,\alpha}(B_1)} < \frac{1}{2}$, we may reabsorb the term with $C(\alpha(1-\alpha))^{-1}r_0^\alpha[\mathbf{a}]_{C^{0,\alpha}(B_1)}\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)}$ appearing on the right side, to obtain

$$\sup_{r\in(0,r_0)}\|\nabla u\|_{\underline{L}^2(B_r)} \leqslant \frac{C}{\alpha}\left(\|\nabla u\|_{\underline{L}^2(B_{r_0})}+\frac{r_0^\alpha}{1-\alpha}[\mathbf{f}]_{C^{0,\alpha}(B_1)}\right).$$

The absorbed bounds are uniform in N. Letting $N\to\infty$ gives control of every dyadic radius, and comparison with a neighboring dyadic radius gives the displayed bound at every radius. Inserting this estimate back into (3.10) and using again that $C(\alpha(1-\alpha))^{-1}r_0^\alpha[\mathbf{a}]_{C^{0,\alpha}(B_1)} < \frac{1}{2}$, we obtain (3.8) for $x=0$. □

3.2 Calderón-Zygmund estimates

The estimates in this subsection improve the integrability of the gradient when the coefficients are close to a constant matrix. The dependence on the exponent is important: an oscillation of order p^{-1} suffices for an L^p estimate, and the coefficient of the forcing norm is of order p. We obtain these bounds by comparing the gradient with a harmonic gradient and decomposing its oscillations on dyadic cubes. The decay of the harmonic oscillations allows us to sum the contribution of the initial energy before taking the L^p norm.

Throughout this subsection $d \geqslant 1$ and matrix norms are Euclidean operator norms.

Proposition 3.4 (Calderón-Zygmund estimate)**.** *For every $\Lambda \geqslant 1$, there exists $C(d,\Lambda) < \infty$ such that, for every $p \in [2,\infty)$, measurable coefficient field $\mathbf{a} : B_1 \to \mathbb{R}^{d\times d}$ satisfying*

$$e\cdot\mathbf{a}(x)e \geqslant |e|^2 \quad \text{and} \quad e\cdot\mathbf{a}(x)^{-1}e \geqslant \Lambda^{-1}|e|^2, \quad \text{for a.e. } x\in B_1 \text{ and every } e\in\mathbb{R}^d,$$

and

$$\|\mathbf{a}-(\mathbf{a})_{B_1}\|_{L^\infty(B_1)} \leqslant \frac{1}{Cp}, \tag{3.12}$$

and every $u \in H^1(B_1)$ and $\mathbf{f}\in L^p(B_1)^d$ satisfying

$$-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f} \quad \text{in } B_1,$$

we have $u \in W^{1,p}(B_{1/2})$ and

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \leqslant C\Big(\|\nabla u\|_{\underline{L}^2(B_1)}+p\|\mathbf{f}\|_{\underline{L}^p(B_1)}\Big). \tag{3.13}$$

We first prove the estimate for coefficients close to I_d. A change of variables using the symmetric part of the average coefficient will then give Proposition 3.4. The harmonic comparison is measured by the gradient excess

$$\mathcal{E}(g;B) := \left(\fint_B |g-(g)_B|^2 \right)^{1/2}.$$

If h is harmonic in $B_R(z)$, the exact decay estimate proved in the preceding subsection gives

$$\mathcal{E}(\nabla h; B_r(z)) \leqslant \frac{r}{R}\mathcal{E}(\nabla h; B_R(z)), \quad 0<r\leqslant R. \tag{3.14}$$

In dimension one the gradient of a harmonic function is constant, so this estimate holds there as well.

For a locally integrable function b on $\mathbb{R}^d$, let $\mathcal{M}b(x)$ be the supremum of $\fint_B |b|$ over all Euclidean balls containing x. For a vector field G, write $\mathcal{M}_2 G := (\mathcal{M}(|G|^2))^{1/2}$. Fields defined on a bounded domain are extended by zero when these operators are used.

Lemma 3.5 (Harmonic comparison)**.** *Suppose $-\Delta u = \nabla\cdot G$ in $B_R(z)$, with $u\in H^1(B_R(z))$ and $G\in L^2(B_R(z))^d$. For every $0<r\leqslant R$ and $x\in B_r(z)$, we have*

$$\mathcal{E}(\nabla u; B_r(z)) \leqslant \frac{r}{R}\mathcal{E}(\nabla u; B_R(z)) + \kappa_d \mathcal{M}_2(G\mathbf{1}_{B_R(z)})(x), \tag{3.15}$$

where $\kappa_d := \frac{d+2}{2}(1+\frac{2}{d})^{d/2}$. The zero extension on the right may be replaced by any extension agreeing with G on $B_R(z)$.

Proof. Let h be the harmonic function in $B_R(z)$ with the boundary values of u, and set $w := u-h\in H^1_0(B_R(z))$. Testing the equation for w gives

$$\|\nabla w\|_{\underline{L}^2(B_R(z))} \leqslant \|G\|_{\underline{L}^2(B_R(z))}.$$

Since $\int_{B_R(z)}\nabla w = 0$ and $\int_{B_R(z)}\nabla h\cdot\nabla w = 0$, the centered energies are orthogonal, and thus $\mathcal{E}(\nabla h; B_R(z)) \leqslant \mathcal{E}(\nabla u; B_R(z))$. Combining this with (3.14) and the minimizing property of the average yields, for every $\theta\in(0,1)$,

$$\mathcal{E}(\nabla u; B_{\theta R}(z)) \leqslant \theta\mathcal{E}(\nabla u; B_R(z)) + \theta^{-d/2}\|G\|_{\underline{L}^2(B_R(z))}. \tag{3.16}$$

Starting at r, enlarge the radius by successive factors θ^{-1} until reaching R; the last factor, if necessary, is smaller. The product of the coefficients on the final excess is exactly r/R. Every ball used in the comparison contains x, and the forcing terms are bounded by the geometric sum $\theta^{-d/2}(1-\theta)^{-1}\mathcal{M}_2 G(x)$. Taking $\theta = d/d+2$ gives (3.15). □

The factor r/R in (3.15) will be summed along a descending chain of cubes. We next give the decomposition which identifies those chains. A family $\mathcal{S}$ of dyadic subcubes of a cube is called $3/4$-sparse if there are pairwise disjoint measurable sets $E_Q\subseteq Q$, indexed by $Q\in\mathcal{S}$, with $|E_Q|\geqslant 3/4|Q|$.

Lemma 3.6 (Oscillation decomposition)**.** *Let Q_0 be a cube in $\mathbb{R}^d$, let $g \in L^2(Q_0)^m$, and write*

$$e_Q := \left(\fint_Q |g - (g)_Q|^2 \right)^{1/2} .$$

There exists a $3/4$-sparse family $\mathcal{S}$ of dyadic subcubes, including Q_0, such that, almost everywhere in Q_0,

$$|g - (g)_{Q_0}| \leqslant 2^{1+d/2} \sum_{Q \in \mathcal{S}} e_Q \mathbf{1}_Q . \tag{3.17}$$

For every $1 < p < \infty$ and nonnegative $b \in L^p(Q_0)$,

$$\left\| \sum_{Q \in \mathcal{S}} (\operatorname*{ess\,inf}_Q b) \mathbf{1}_Q \right\|_{L^p(Q_0)} \leqslant \frac{4}{3} p \|b\|_{L^p(Q_0)} . \tag{3.18}$$

The latter estimate also holds for a forest of such families whose root cubes have disjoint interiors.

Proof. At a selected cube Q, select the maximal strict dyadic subcubes P for which

$$\fint_P |g - (g)_Q|^2 > 4 e_Q^2 .$$

If $e_Q = 0$, there are no such cubes and $g = (g)_Q$ almost everywhere in Q. Otherwise, summing the stopping inequality over the disjoint selected subcubes shows that their union has measure at most $|Q|/4$. Outside that union, dyadic differentiation gives $|g - (g)_Q| \leqslant 2e_Q$. The dyadic parent $\widehat{P}$ of a stopping cube was not stopped and has volume $2^d|P|$. Consequently,

$$|(g)_P - (g)_Q| \leqslant \left(\fint_P |g - (g)_Q|^2 \right)^{1/2} \leqslant \left(2^d \fint_{\widehat{P}} |g - (g)_Q|^2 \right)^{1/2} \leqslant 2^{1+d/2} e_Q .$$

Iterate the selection, starting at Q_0. The complements of the stopping children give the disjoint sets E_Q required for sparsity. The union of the cubes at the nth stopping depth has measure at most $4^{-n}|Q_0|$, so almost every point lies on a finite stopping chain. Telescoping the means along that chain proves (3.17).

To prove (3.18), let $\varphi \geqslant 0$ belong to $L^{p'}$, where $p' = p/p-1$, and let $M_{\mathcal{D}}$ be the dyadic maximal operator on the root cube, or on each root separately in a forest. Sparsity and the definition of the maximal operator give

$$\begin{aligned} \sum_{Q \in \mathcal{S}} (\operatorname*{ess\,inf}_Q b) \int_Q \varphi &\leqslant \frac{4}{3} \sum_{Q \in \mathcal{S}} \int_{E_Q} b M_{\mathcal{D}} \varphi \\ &\leqslant \frac{4}{3} \|b\|_{L^p} \|M_{\mathcal{D}} \varphi\|_{L^{p'}} \leqslant \frac{4}{3} p \|b\|_{L^p} \|\varphi\|_{L^{p'}} . \end{aligned}$$

Here we used the dyadic maximal inequality $\|M_{\mathcal{D}}\|_{L^q \to L^q} \leqslant q/q-1$. For a finite number of generations, the maximal-cube decomposition gives

$$t|\{M_{\mathcal{D}} \varphi > t\}| \leqslant \int_{\{M_{\mathcal{D}} \varphi > t\}} \varphi .$$

Multiplying by qt^{q-2}, integrating in t, and using Hölder's inequality yields $\|M_{\mathcal{D}}\varphi\|_{L^q} \leqslant {}^{q}\!/_{q-1}\|\varphi\|_{L^q}$. Monotone convergence proves the same bound for all generations. Duality, first for finite sparse sums and then by monotone convergence, gives (3.18) without assuming that the sum belongs to L^p. □

We also need the ball maximal estimate. For every $q > 1$,

$$\|\mathcal{M}b\|_{L^q(\mathbb{R}^d)} \leqslant 2\left(3^d \frac{q}{q-1}\right)^{1/q} \|b\|_{L^q(\mathbb{R}^d)}\,. \tag{3.19}$$

Indeed, for $b \in L^1$ and $t > 0$, cover a compact subset of $\{\mathcal{M}b > t\}$ by finitely many balls on which the average of $|b|$ exceeds t. Selecting the largest remaining ball and discarding balls which meet it gives disjoint balls whose triple dilates cover the compact set. Consequently its measure is at most $3^d t^{-1}\|b\|_{L^1}$; inner regularity gives the same bound for the full level set. For $b \in L^q$, apply this weak estimate to $b\mathbf{1}_{\{|b|>t/2\}}$, which belongs to L^1, to obtain

$$|\{\mathcal{M}b > t\}| \leqslant \frac{2\cdot 3^d}{t} \int_{\{|b|>t/2\}} |b|\,.$$

Integrating against qt^{q-1} proves (3.19). In particular, for every $p \geqslant 4$,

$$\|\mathcal{M}_2 G\|_{L^p(\mathbb{R}^d)} \leqslant \sqrt{2}\left(3^d \frac{p}{p-2}\right)^{1/p} \|G\|_{L^p(\mathbb{R}^d)} \leqslant 2\cdot 3^{d/4}\|G\|_{L^p(\mathbb{R}^d)}\,. \tag{3.20}$$

Lemma 3.7 (Local Laplacian estimate)**.** *There exist $A_d, K_d < \infty$, depending only on d, such that, for every $p \geqslant 4$, $0 < s < t \leqslant 1$, $u \in H^1(B_t)$ and $G \in L^p(B_t)^d$ satisfying $-\Delta u = \nabla\cdot G$ in B_t, we have $\nabla u \in L^p(B_s)^d$ and*

$$N_p(\nabla u; B_s) \leqslant A_d(t-s)^{-d/2} N_2(\nabla u; B_t) + K_d p N_p(G; B_t)\,, \tag{3.21}$$

where $N_q(g; D) := \left(|B_1|^{-1}\int_D |g|^q\right)^{1/q}$.

Proof. First consider a cube Q_0 with side length ℓ, center z_0, and circumscribed radius $r_0 = \sqrt{d}\ell/2$, such that $B_{2r_0}(z_0) \subseteq B_t$. Set $g := \nabla u$, $U_0 := \|g\|_{\underline{L}^2(B_{2r_0}(z_0))}$ and $b := \mathcal{M}_2(G\mathbf{1}_{B_t})$. If Q is a generation-k dyadic subcube of Q_0, its circumscribed ball has radius $2^{-k}r_0$ and center z_Q, and $B_{r_0}(z_Q) \subseteq B_{2r_0}(z_0)$. The minimizing property of the average, comparison of the ball and cube volumes, and Lemma 3.5 give

$$e_Q \leqslant V_d 2^{-k} U_0 + 2^{-d/2} V_d \kappa_d \operatorname*{ess\,inf}_Q b\,, \qquad V_d := \left(|B_1| d^{d/2}\right)^{1/2}. \tag{3.22}$$

The square root of the ratio between the ball and cube volumes is $2^{-d/2}V_d$, and the excess on $B_{r_0}(z_Q)$ is at most $2^{d/2}U_0$. The average on the root satisfies $|(g)_{Q_0}| \leqslant V_d U_0$.

Apply Lemma 3.6. At a point of Q_0, there is at most one cube of each dyadic generation, so $\sum_{Q\ni x} 2^{-k(Q)} \leqslant 2$. Therefore (3.22) gives the pointwise bound

$$|g|\mathbf{1}_{Q_0} \leqslant V_d(1 + 2^{2+d/2})U_0\mathbf{1}_{Q_0} + 2V_d\kappa_d \sum_{Q\in\mathcal{S}} (\operatorname*{ess\,inf}_Q b)\mathbf{1}_Q\,. \tag{3.23}$$

The energy contribution has been summed pointwise, and thus will not acquire the factor p in (3.18).

Partition space into cubes of side $\ell = {}^{t-s}\!/_{2\sqrt{d}}$, and retain those which meet B_s. Their interiors are disjoint. For every retained cube, we have $r_0 = {}^{t-s}\!/_4$, $B_{2r_0}(z_0) \subset B_t$, and

$$U_0 \leqslant \left(\frac{2}{t-s}\right)^{d/2} N_2(g; B_t)\,.$$

Use (3.23) on all these roots. Restricting the energy term to B_s costs no further volume factor, since $|B_s| \leqslant |B_1|$. For the forcing term, use the forest estimate (3.18), followed by (3.20). This proves (3.21) with

$$A_d := 2^{d/2} V_d (1 + 2^{2+d/2}) \quad \text{and} \quad K_d := \frac{16}{3}\, 3^{d/4} V_d \kappa_d\,.$$

All sums may first be truncated at a finite stopping depth and then passed to the limit. In particular, the argument establishes the asserted L^p integrability without assuming it in advance. □

Proposition 3.8 (Coefficients close to the identity)**.** *There exists $C(d) < \infty$ such that, for every $p \in [2, \infty)$, measurable $\mathbf{a} : B_1 \to \mathbb{R}^{d\times d}$ satisfying*

$$\|\mathbf{a} - \mathrm{I}_d\|_{L^\infty(B_1)} \leqslant \frac{1}{Cp}\,,$$

and every $u \in H^1(B_1)$ and $\mathbf{f} \in L^p(B_1)^d$ satisfying $-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f}$ in B_1, we have $u \in W^{1,p}(B_{1/2})$ and

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \leqslant C\big(\|\nabla u\|_{\underline{L}^2(B_1)} + p\|\mathbf{f}\|_{\underline{L}^p(B_1)}\big)\,. \tag{3.24}$$

Proof. Write $\delta := \|\mathbf{a} - \mathrm{I}_d\|_{L^\infty(B_1)}$. We choose $C \geqslant 1$, so that the hypothesis implies $\delta \leqslant {}^1\!/_2$.

Step 1: The estimate for $p \geqslant 4$. We first prove the estimate under the condition $\delta \leqslant (2K_d p)^{-1}$ and the additional assumption that ∇u is bounded on compact subsets of B_1. Setting

$$U := \|\nabla u\|_{\underline{L}^2(B_1)}, \quad F := \|\mathbf{f}\|_{\underline{L}^p(B_1)}, \quad \Phi(s) := N_p(\nabla u; B_s), \quad \eta := K_d p \delta,$$

and writing $-\Delta u = \nabla\cdot G$ with $G = \mathbf{f} + (\mathbf{a} - \mathrm{I}_d)\nabla u$, Lemma 3.7 gives

$$\Phi(s) \leqslant A_d (t-s)^{-d/2} U + K_d p F + \eta\Phi(t)\,, \quad \frac{1}{2} \leqslant s < t < 1. \tag{3.25}$$

Let $\sigma_d := 2^{-1/d}$ and $r_j := {}^3\!/_4 - {}^1\!/_4\sigma_d^j$. Iterating (3.25) a finite number of times gives

$$\Phi({}^1\!/_2) \leqslant A_d \left(\frac{4}{1-\sigma_d}\right)^{d/2} \sum_{j=0}^{n-1} (\eta\sqrt{2})^j U + K_d p \sum_{j=0}^{n-1} \eta^j F + \eta^n \Phi(r_n)\,.$$

Since $\eta \leqslant {}^1\!/_2$ and Φ is bounded on $[{}^1\!/_2, {}^3\!/_4]$, the remainder tends to zero. The two geometric sums are bounded independently of p. Passing from $N_p(\nabla u; B_{1/2})$ to the normalized norm costs $2^{d/p} \leqslant 2^{d/4}$. Thus there are constants L_d, J_d, depending only on d, such that

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \leqslant L_d U + J_d p F\,, \quad p \geqslant 4, \tag{3.26}$$

whenever $\delta \leqslant (2K_d p)^{-1}$.

Step 2: Approximation. Extend $\mathbf{a} - \mathrm{I}_d$ and $\mathbf{f}$ by zero outside B_1, and mollify them, setting $\mathbf{a}_\varepsilon := \mathrm{I}_d + \rho_\varepsilon * ((\mathbf{a} - \mathrm{I}_d)\mathbf{1}_{B_1})$ and $\mathbf{f}_\varepsilon := \rho_\varepsilon * (\mathbf{f}\mathbf{1}_{B_1})$. Then $\|\mathbf{a}_\varepsilon - \mathrm{I}_d\|_{L^\infty} \leqslant \delta$, so the symmetric part of $\mathbf{a}_\varepsilon$ is bounded below by $(1-\delta)\mathrm{I}_d$. By Lax–Milgram there is $u_\varepsilon \in u + H^1_0(B_1)$ satisfying the equation with coefficients $\mathbf{a}_\varepsilon$ and forcing $\mathbf{f}_\varepsilon$. Apply Proposition 3.3 on interior balls to the coefficient $\mathbf{a}_\varepsilon/(1-\delta)$ and forcing $\mathbf{f}_\varepsilon/(1-\delta)$. These smooth coefficients have lower coercivity 1 and inverse coercivity at least $1/9$, since $\delta \leqslant 1/2$. The proposition therefore shows that ∇u_ε is bounded on every compact subball, and Step 1 applies to u_ε.

Subtract the weak equations and test by $u_\varepsilon - u$. Since $\delta < 1$, this gives

$$\|\nabla(u_\varepsilon - u)\|_{L^2(B_1)} \leqslant \frac{\|\mathbf{f}_\varepsilon - \mathbf{f}\|_{L^2(B_1)} + \|(\mathbf{a}_\varepsilon - \mathbf{a})\nabla u\|_{L^2(B_1)}}{1-\delta} \longrightarrow 0\,.$$

The forcing converges strongly in L^p and L^2. The coefficient term converges by almost everywhere convergence of the mollifications and dominated convergence. A subsequence of the gradients therefore converges almost everywhere, and Fatou's lemma passes (3.26) to u. To obtain the L^p integrability of u, set $T_k u := \max\{-k, \min\{u, k\}\}$. These bounded truncations satisfy $|\nabla T_k u| \leqslant |\nabla u|$ almost everywhere, and $|(T_k u)_{B_{1/2}}| \leqslant \|u\|_{\underline{L}^2(B_{1/2})}$. The mean-zero Poincaré inequality thus gives a bound on $\|T_k u\|_{L^p(B_{1/2})}$ independent of k. Fatou's lemma yields $u \in L^p(B_{1/2})$. We have proved (3.26) and the Sobolev membership without the extra boundedness assumption.

Step 3: The interval $2 \leqslant p < 4$. Decrease the dimensional smallness constant so that $\delta \leqslant (4K_d p)^{-1}$. This ensures $\delta \leqslant (8K_d)^{-1}$, and hence the established estimate at exponent 4 applies. For the fixed field $\mathbf{a}$, let $v \in H^1_0(B_1)$ solve $-\nabla\cdot\mathbf{a}\nabla v = \nabla\cdot\mathbf{f}$. The energy estimate gives

$$\|\nabla v\|_{\underline{L}^2(B_1)} \leqslant 2\|\mathbf{f}\|_{\underline{L}^2(B_1)}\,.$$

The linear map $T : \mathbf{f} \mapsto \nabla v|_{B_{1/2}}$, with normalized Lebesgue measures on its input and output, has endpoint bounds

$$\|T\|_{L^2\to L^2} \leqslant 2^{1+d/2} \quad \text{and} \quad \|T\|_{L^4\to L^4} \leqslant 2L_d + 4J_d\,.$$

Enlarge L_d, J_d if necessary so that the second bound dominates the first. Interpolation yields $\|T\|_{L^p\to L^p} \leqslant 2L_d + 4J_d$ for $2 \leqslant p \leqslant 4$. The $\mathbf{a}$-harmonic function $h := u - v$ satisfies

$$\|\nabla h\|_{\underline{L}^p(B_{1/2})} \leqslant \|\nabla h\|_{\underline{L}^4(B_{1/2})} \leqslant L_d\big(U + 2\|\mathbf{f}\|_{\underline{L}^p(B_1)}\big)\,.$$

Combining the estimates for v and h gives

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \leqslant L_d U + 4(L_d + J_d)\|\mathbf{f}\|_{\underline{L}^p(B_1)} \leqslant 2(L_d + J_d)(U + pF)\,.$$

Choose C large enough to include this bound, the high-exponent bound, and the smallness conditions. This proves (3.24). The truncation argument in Step 2 again gives $u \in W^{1,p}(B_{1/2})$. □

Proof of Proposition 3.4. Let $\mathbf{a}_0 := (\mathbf{a})_{B_1}$ and $S := \operatorname{sym}\mathbf{a}_0$. The ellipticity assumptions give $\mathrm{I}_d \leqslant S \leqslant \Lambda\mathrm{I}_d$. The constant skew part $\mathbf{a}_0 - S$ contributes zero to the scalar divergence-form operator. Under the change of variables $x = S^{1/2}y$, the coefficient and forcing become

$$\tilde{\mathbf{a}}(y) = \mathrm{I}_d + S^{-1/2}(\mathbf{a}(S^{1/2}y) - \mathbf{a}_0)S^{-1/2} \quad \text{and} \quad \tilde{\mathbf{f}}(y) = S^{-1/2}\mathbf{f}(S^{1/2}y)\,.$$

In particular, $\|\widetilde{\mathbf{a}} - \mathrm{I}_d\|_{L^\infty} \leqslant \|\mathbf{a} - \mathbf{a}_0\|_{L^\infty}$, and $\nabla_y \widetilde{u} = S^{1/2}\nabla_x u$.

Set $\rho := (4\sqrt{\Lambda})^{-1}$ and $E_z := z + S^{1/2}B_\rho$ for $z \in B_{1/2}$. Then $B_\rho(z) \subseteq E_z \subseteq B_{1/4}(z) \subseteq B_1$. Rescaling Proposition 3.8 and changing variables back gives

$$\|\nabla u\|_{\underline{L}^p(1/2 E_z)} \leqslant C(d,\Lambda)\big(\|\nabla u\|_{\underline{L}^2(E_z)} + p\|\mathbf{f}\|_{\underline{L}^p(E_z)}\big)\,.$$

Here $1/2 E_z$ denotes $z + S^{1/2}B_{\rho/2}$. Cover $B_{1/2}$ by finitely many such half ellipsoids, with their number bounded in terms of d, Λ. Their volumes are bounded below by a positive constant depending only on d, Λ. Summing the pth powers, and using $p \geqslant 2$, bounds the energy contribution by $C(d,\Lambda)\|\nabla u\|_{\underline{L}^2(B_1)}$ and the forcing contribution by $C(d,\Lambda)p\|\mathbf{f}\|_{\underline{L}^p(B_1)}$. Taking the constant in (3.12) large enough to imply the identity smallness condition proves (3.13) and the asserted local Sobolev membership. □

Remark 3.9 (Dependence on p)**.** For constant coefficients, the oscillation condition is automatic and (3.13) holds for every finite $p \geqslant 2$. The factor p and the smallness order p^{-1} are both optimal in dimensions $d \geqslant 2$, as the following examples show. Dimension one is exceptional: the equation can be integrated directly.

Example 3.10 (The oscillation threshold)**.** In dimension two, write e_r, e_θ for the polar unit vectors and set

$$\mathbf{a}_\delta = (1+\delta)e_r \otimes e_r + (1-\delta)e_\theta \otimes e_\theta, \qquad u(r,\theta) = r^\beta \cos\theta, \qquad \beta = \sqrt{\frac{1-\delta}{1+\delta}}\,,$$

where $0 < \delta < 1$. A computation gives

$$\nabla \cdot \mathbf{a}_\delta \nabla u = \big((1+\delta)\beta^2 - (1-\delta)\big) r^{\beta-2}\cos\theta = 0\,.$$

The function belongs to $H^1(B_1)$, and the equation holds weakly through the origin: the boundary term on ∂B_ε tends to zero like ε^β. However, $|\nabla u|$ is comparable to $r^{\beta-1}$, so $\nabla u \notin L^p$ when $p(1-\beta) \geqslant 2$. For a given $p > 2$, take $\beta = 1 - 2/p$. Failure occurs at

$$\delta = \frac{1-\beta^2}{1+\beta^2} = \frac{2(p-1)}{p^2 - 2p + 2} \sim \frac{2}{p}\,.$$

Moreover, $(\mathbf{a}_\delta)_{B_1} = \mathrm{I}_2$ and $\|\mathbf{a}_\delta - \mathrm{I}_2\|_{L^\infty} = \delta$. Dividing $\mathbf{a}_\delta$ by $1-\delta$ puts it in the normalized ellipticity convention of Proposition 3.4, with $\Lambda = (1+\delta)/(1-\delta)$. Thus this example also tests smallness relative to the average coefficient. Extending u constantly in the remaining variables and taking the identity coefficient in those directions gives the same obstruction in every $d \geqslant 2$.

Example 3.11 (The forcing constant)**.** In dimension two, set $u(x) := x_1 \log|x|$ and $\mathbf{f}(x) := -2x_1 x/|x|^2$, assigning arbitrary values at the origin. Then $u \in H^1(B_1)$, $\|\mathbf{f}\|_{L^\infty} \leqslant 2$, and, distributionally,

$$-\Delta u = \nabla \cdot \mathbf{f} = -2\frac{x_1}{|x|^2} \quad \text{and} \quad \nabla u = e_1 \log|x| + \frac{x_1 x}{|x|^2}\,.$$

There is no point mass at the origin, since the boundary flux terms tend to zero. Radial integration gives $\|\nabla u\|_{\underline{L}^2(B_1)} = 2^{-1/2}$, while the triangle inequality and $-\log r \geqslant -\log(2r)$ for $r < 1/2$ give

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \geqslant \frac{1}{2}\Gamma(p+1)^{1/p} - 1 \sim \frac{p}{2e}\,.$$

Thus the coefficient of the forcing norm cannot be $o(p)$ when the coefficient of the energy norm is independent of p. The same function shows that a bound with L^∞ forcing does not yield a bounded gradient. Extension in the unused variables gives the same order in higher dimensions.

Finally, equip $W^{1,p'}_0(B_1)$ with its gradient norm. Hahn–Banach and L^p duality represent every functional in its dual as $\nabla \cdot \mathbf{f}$ for some $\mathbf{f} \in L^p(B_1)^d$; the dual norm is the infimum of the norms of all such representations. Consequently (3.13) is an interior $W^{-1,p}$ to $W^{1,p}$ estimate. The additive energy term accounts for the unrestricted boundary values of u.

We next present Yudovich's uniqueness result for the two-dimensional incompressible Euler system with bounded vorticity, which is an application of the Calderón-Zygmund estimate. The key point is that the proof uses the dependence in p of the constant as $p \to \infty$. We work on the two-dimensional torus $\mathbb{T}^2$ to avoid boundary considerations. The incompressible Euler system is

$$\begin{cases} \partial_t \mathbf{v} + \mathbf{v}\cdot\nabla\mathbf{v} + \nabla p = 0, \\ \nabla\cdot\mathbf{v} = 0. \end{cases} \tag{3.27}$$

Let $\mathbf{v}_0 \in L^2(\mathbb{T}^2;\mathbb{R}^2)$ with $\nabla\cdot\mathbf{v}_0 = 0$. We say that $\mathbf{v} \in L^\infty([0,T];L^2(\mathbb{T}^2;\mathbb{R}^2))$ with $\nabla\cdot\mathbf{v} = 0$ is a *weak solution* of (3.27) with initial datum $\mathbf{v}_0$ if, for every divergence-free test function $\boldsymbol{\varphi} \in C^\infty_c([0,T)\times\mathbb{T}^2;\mathbb{R}^2)$,

$$\int_0^T\!\!\int_{\mathbb{T}^2} \Big(\mathbf{v}\cdot\partial_t\boldsymbol{\varphi} + (\mathbf{v}\otimes\mathbf{v}) : \nabla\boldsymbol{\varphi}\Big) + \int_{\mathbb{T}^2} \mathbf{v}_0\cdot\boldsymbol{\varphi}(0,\cdot) = 0\,. \tag{3.28}$$

Here $(\mathbf{v}\otimes\mathbf{v}) : \nabla\boldsymbol{\varphi} = v_i v_j \partial_j \varphi_i$, and the pressure does not appear because the test fields are divergence-free. Test fields supported in $(0,T)\times\mathbb{T}^2$ give the weak formulation without datum; the initial term is what makes the datum meaningful.

Proposition 3.12 (Yudovich weak-strong uniqueness for 2d Euler)**.** *Let $\mathbf{v}_0 \in L^2(\mathbb{T}^2;\mathbb{R}^2)$ be divergence-free and let $\mathbf{v}_1, \mathbf{v}_2 \in L^\infty([0,T];L^2 \cap L^\infty(\mathbb{T}^2;\mathbb{R}^2))$ be weak solutions of (3.27) with initial datum $\mathbf{v}_0$. Suppose that $\mathbf{v}_1$ satisfies the energy inequality*

$$\|\mathbf{v}_1(t)\|_{L^2(\mathbb{T}^2)} \leqslant \|\mathbf{v}_0\|_{L^2(\mathbb{T}^2)} \qquad \text{for a.e. } t \in [0,T], \tag{3.29}$$

and that $\mathbf{v}_2$ has bounded vorticity $\omega_2 = \partial_1 v_{2,2} - \partial_2 v_{2,1}$ (in the sense of distributions):

$$\|\omega_2\|_{L^\infty([0,T]\times\mathbb{T}^2)} \leqslant M.$$

Then $\mathbf{v}_1 = \mathbf{v}_2$ almost everywhere in $[0,T]\times\mathbb{T}^2$.

The proof shows that $\mathbf{v}_2$ automatically satisfies the energy equality $\|\mathbf{v}_2(t)\|_{L^2} = \|\mathbf{v}_0\|_{L^2}$ for every t, so no energy hypothesis on $\mathbf{v}_2$ is needed.

Remark 3.13 (The energy inequality cannot be dropped)**.** Without (3.29) the statement fails. De Lellis and Székelyhidi [DLS09] constructed, in every dimension $n \geqslant 2$, nonzero bounded weak solutions of the Euler equations on $\mathbb{R}^n \times \mathbb{R}$ with compact support in space and time. Euler scaling $\mathbf{v}(x,t) \mapsto \mathbf{v}(\lambda x, \lambda t)$ and a translation in time place the support inside $(0,T) \times (0,1)^2$; extending periodically gives a bounded weak solution $\mathbf{v}_1 \neq 0$ on $\mathbb{T}^2$ with initial datum $\mathbf{v}_0 = 0$ in the sense of (3.28). The solution $\mathbf{v}_2 = 0$ has zero vorticity and the same datum, yet $\mathbf{v}_1 \neq \mathbf{v}_2$. Here (3.29) fails for $\mathbf{v}_1$, which has nonzero energy at some times but zero initial energy. The proof below uses (3.29) exactly once, to compare $\|\mathbf{v}_1(t)\|_{L^2}^2$ with $\|\mathbf{v}_0\|_{L^2}^2$.

The proof uses the following periodic form of the Calderón-Zygmund estimate. We identify $\mathbb{T}^2$ with $[0,1)^2$ and normalise Lebesgue measure on $\mathbb{T}^2$ to have total mass one.

Lemma 3.14 (Periodic Calderón-Zygmund estimate)**.** *There is an absolute constant $K < \infty$ such that, for every mean-zero $\omega \in L^\infty(\mathbb{T}^2)$, the mean-zero solution $\psi \in H^2(\mathbb{T}^2)$ of $-\Delta\psi = \omega$ satisfies, for every $p \in [2,\infty)$,*

$$\left\|\nabla^2 \psi\right\|_{L^p(\mathbb{T}^2)} \leqslant K p \|\omega\|_{L^\infty(\mathbb{T}^2)}\,.$$

Proof. Extend ψ, ω periodically to $\mathbb{R}^2$. Fourier series give $\|\nabla^2\psi\|_{L^2(\mathbb{T}^2)} = \|\omega\|_{L^2(\mathbb{T}^2)} \leqslant \|\omega\|_{L^\infty}$. Fix $i \in \{1,2\}$. Then $u := \partial_i \psi \in H^1_{\mathrm{loc}}(\mathbb{R}^2)$ solves $-\Delta u = \nabla \cdot \mathbf{f}$ with $\mathbf{f} := \omega e_i$, and $|\mathbf{f}| \leqslant \|\omega\|_{L^\infty}$. Apply Proposition 3.4 with $d = 2$, $\mathbf{a} = \mathrm{I}_d$ (so $\Lambda = 1$ and the oscillation condition (3.12) holds trivially, as recorded in Remark 3.9), on each ball $B_1(z)$ with $z \in \{1/4, 3/4\}^2$. Then

$$\|\nabla u\|_{\underline{L}^p(B_{1/2}(z))} \leqslant C\Big(\|\nabla u\|_{\underline{L}^2(B_1(z))} + p\|\mathbf{f}\|_{\underline{L}^p(B_1(z))}\Big).$$

The four balls $B_{1/2}(z)$ cover $[0,1]^2$, since every point of the square lies within distance $\sqrt{2}/4 < 1/2$ of some z. The ball $B_1(z)$ is contained in nine translates of the unit square, so by periodicity $\int_{B_1(z)} |\nabla u|^2 \leqslant 9\|\nabla u\|^2_{L^2(\mathbb{T}^2)} \leqslant 9\|\omega\|^2_{L^\infty}$, hence $\|\nabla u\|_{\underline{L}^2(B_1(z))} \leqslant 3(\pi)^{-1/2}\|\omega\|_{L^\infty}$, and $\|\mathbf{f}\|_{\underline{L}^p(B_1(z))} \leqslant \|\omega\|_{L^\infty}$. Summing the pth powers of the left side over the four balls, and using that $|B_{1/2}| = \pi/4$ and $p \geqslant 2$, gives

$$\|\nabla \partial_i \psi\|_{L^p(\mathbb{T}^2)} \leqslant 4^{1/p}\big(\tfrac{\pi}{4}\big)^{1/p} \cdot C(3\pi^{-1/2} + p)\|\omega\|_{L^\infty} \leqslant C' p \|\omega\|_{L^\infty}$$

with C' absolute. Taking $i = 1, 2$ completes the proof. □

Proof of Proposition 3.12. We write $\|\cdot\|_p$ for the norm of $L^p(\mathbb{T}^2)$ and $\mathbf{u}^\varepsilon := \rho_\varepsilon * \mathbf{u}$ for the spatial mollification of a field $\mathbf{u}$ on $\mathbb{T}^2$ by a standard even mollifier.

Step 1: Weak continuity in time. Let $\mathbf{u}$ be a weak solution with datum $\mathbf{v}_0$ and let $\boldsymbol{\varphi} \in C^\infty(\mathbb{T}^2;\mathbb{R}^2)$ be divergence-free. Testing (3.28) with $\chi(t)\boldsymbol{\varphi}(x)$, $\chi \in C_c^\infty([0,T))$, shows that $g(t) := \int \mathbf{u}(t) \cdot \boldsymbol{\varphi}$ has distributional derivative $\int (\mathbf{u} \otimes \mathbf{u}) : \nabla\boldsymbol{\varphi} \in L^\infty(0,T)$ and initial value $\int \mathbf{v}_0 \cdot \boldsymbol{\varphi}$. Thus g agrees a.e. with a Lipschitz function of t with $g(0) = \int \mathbf{v}_0 \cdot \boldsymbol{\varphi}$. Smooth divergence-free fields are dense in the space of divergence-free fields in L^2, and $\mathbf{u} \in L^\infty_t L^2_x$. We conclude that, after modification on a null set of times, $\mathbf{u} : [0,T] \to L^2$ is weakly continuous, divergence-free, and $\mathbf{u}(0) = \mathbf{v}_0$. We use these representatives from now on. By weak lower semicontinuity of the norm, (3.29) then holds for *every* $t \in [0,T]$. Moreover, by the same argument on Lipschitz-in-time test fields (mollification in t), for every

divergence-free $\boldsymbol{\varphi}$ that is smooth in x and Lipschitz in t with values in $C^k(\mathbb{T}^2)$ for all k, and every $0 \leqslant s \leqslant t \leqslant T$,

$$\int \mathbf{u}(t)\cdot\boldsymbol{\varphi}(t) - \int \mathbf{u}(s)\cdot\boldsymbol{\varphi}(s) = \int_s^t\int \Big(\mathbf{u}\cdot\partial_\tau\boldsymbol{\varphi} + (\mathbf{u}\otimes\mathbf{u}) : \nabla\boldsymbol{\varphi}\Big)\,. \tag{3.30}$$

Step 2: Regularity of $\mathbf{v}_2$. For a.e. τ, $\omega_2(\tau)$ is the curl of $\mathbf{v}_2(\tau)$, bounded by M, and has zero mean (test the curl with the constant 1). Let $\psi_2(\tau)$ be the mean-zero solution of $-\Delta\psi_2 = \omega_2(\tau)$ and $\nabla^\perp\psi := (\partial_2\psi, -\partial_1\psi)$. Then $\nabla^\perp\psi_2$ and $\mathbf{v}_2(\tau) - \int \mathbf{v}_2(\tau)$ are divergence-free, mean-zero L^2 fields with the same curl; their difference is both divergence-free and curl-free with zero mean, hence zero by Fourier series. So $\nabla\mathbf{v}_2 = \nabla^2\psi_2$ up to the order of indices, and Lemma 3.14 gives, for a.e. τ and all $p \in [2,\infty)$,

$$\|\nabla\mathbf{v}_2(\tau)\|_p \leqslant KpM. \tag{3.31}$$

In particular $\mathbf{v}_2\cdot\nabla\mathbf{v}_2 \in L^\infty_t L^2_x$ and $\mathbf{v}_2\cdot\nabla\mathbf{v}_2 = \nabla\cdot(\mathbf{v}_2\otimes\mathbf{v}_2)$.

Step 3: A cross-term identity. Let $\mathbf{u}$ be any weak solution with datum $\mathbf{v}_0$ in $L^\infty_{t,x}$. We claim that, for $0 \leqslant s \leqslant t \leqslant T$,

$$\int \mathbf{u}(t)\cdot\mathbf{v}_2(t) - \int \mathbf{u}(s)\cdot\mathbf{v}_2(s) = \int_s^t\int \big((\mathbf{u}-\mathbf{v}_2)\cdot\nabla\mathbf{v}_2\big)\cdot\mathbf{u}\,, \tag{3.32}$$

where $(\mathbf{a}\cdot\nabla\mathbf{b})\cdot\mathbf{c} = a_j c_i \partial_j b_i$. Test (3.30) with $\boldsymbol{\varphi} = \mathbf{v}_2^\varepsilon$, which is divergence-free and smooth in x. It is Lipschitz in time, because for any smooth $\boldsymbol{\zeta}$, $\int \mathbf{v}_2\cdot\rho_\varepsilon * \boldsymbol{\zeta} = \int \mathbf{v}_2\cdot P(\rho_\varepsilon * \boldsymbol{\zeta})$, where P is the Leray projection, and (3.30) for $\mathbf{v}_2$ applies. This also gives $\partial_\tau\mathbf{v}_2^\varepsilon = -\rho_\varepsilon * P(\mathbf{v}_2\cdot\nabla\mathbf{v}_2)$. Since $\mathbf{u}^\varepsilon$ is divergence-free, P is invisible when paired with it, and

$$\int \mathbf{u}\cdot\partial_\tau\mathbf{v}_2^\varepsilon = -\int \mathbf{u}^\varepsilon\cdot(\mathbf{v}_2\cdot\nabla\mathbf{v}_2)\,.$$

Hence

$$\int \mathbf{u}(t)\cdot\mathbf{v}_2^\varepsilon(t) - \int \mathbf{u}(s)\cdot\mathbf{v}_2^\varepsilon(s) = \int_s^t\int \Big(-\mathbf{u}^\varepsilon\cdot(\mathbf{v}_2\cdot\nabla\mathbf{v}_2) + (\mathbf{u}\otimes\mathbf{u}) : \nabla\mathbf{v}_2^\varepsilon\Big)\,.$$

Let $\varepsilon \to 0$. Since $\mathbf{v}_2(t) \in L^2$ for every t, $\mathbf{v}_2^\varepsilon(t) \to \mathbf{v}_2(t)$ in L^2 and the left side converges. On the right, $\mathbf{u}^\varepsilon \to \mathbf{u}$ in L^2, $\mathbf{v}_2\cdot\nabla\mathbf{v}_2 \in L^2$, and $\nabla\mathbf{v}_2^\varepsilon \to \nabla\mathbf{v}_2$ in L^2 with $\mathbf{u}$ bounded. The limit is (3.32).

Step 4: Energy equality for $\mathbf{v}_2$ *and relative energy inequality.* Take $\mathbf{u} = \mathbf{v}_2$ in (3.32). The right side vanishes, so $\|\mathbf{v}_2(t)\|_2^2 = \|\mathbf{v}_2(s)\|_2^2$ for all s,t, and thus $\|\mathbf{v}_2(t)\|_2 = \|\mathbf{v}_0\|_2$ for every t, by $\mathbf{v}_2(0) = \mathbf{v}_0$. Now set $\mathbf{w} := \mathbf{v}_1 - \mathbf{v}_2$, $y(t) := \|\mathbf{w}(t)\|_2^2$ and $B := \|\mathbf{v}_1\|_{L^\infty_{t,x}} + \|\mathbf{v}_2\|_{L^\infty_{t,x}}$. In (3.32) with $\mathbf{u} = \mathbf{v}_1$ write $\mathbf{v}_1 = \mathbf{w} + \mathbf{v}_2$. Since $(\mathbf{w}\cdot\nabla\mathbf{v}_2)\cdot\mathbf{v}_2 = \mathbf{w}\cdot\nabla\frac{1}{2}|\mathbf{v}_2|^2$ and $\nabla\cdot\mathbf{w} = 0$, with $|\mathbf{v}_2|^2 \in W^{1,2}$, this term integrates to zero, and

$$\int \mathbf{v}_1(t)\cdot\mathbf{v}_2(t) - \int \mathbf{v}_1(s)\cdot\mathbf{v}_2(s) = \int_s^t\int (\mathbf{w}\cdot\nabla\mathbf{v}_2)\cdot\mathbf{w}\,.$$

Suppose that $y(s) = 0$ for some $s \in [0,T]$. Then $\int \mathbf{v}_1(s)\cdot\mathbf{v}_2(s) = \|\mathbf{v}_2(s)\|_2^2 = \|\mathbf{v}_0\|_2^2$. Combined with $\|\mathbf{v}_1(t)\|_2 \leqslant \|\mathbf{v}_0\|_2$ from (3.29) and $\|\mathbf{v}_2(t)\|_2 = \|\mathbf{v}_0\|_2$, we obtain for every $t \geqslant s$

$$y(t) = \|\mathbf{v}_1(t)\|_2^2 + \|\mathbf{v}_2(t)\|_2^2 - 2\int \mathbf{v}_1(t)\cdot\mathbf{v}_2(t) \leqslant -2\int_s^t\int (\mathbf{w}\cdot\nabla\mathbf{v}_2)\cdot\mathbf{w} \leqslant 2\int_s^t\int |\mathbf{w}|^2|\nabla\mathbf{v}_2|\,. \tag{3.33}$$

Note that $y(0) = 0$ because $\mathbf{v}_1(0) = \mathbf{v}_2(0) = \mathbf{v}_0$.

Step 5: The Yudovich argument. Fix s with $y(s) = 0$ and $p \in [2, \infty)$. By Hölder's inequality with exponents p and $p/p-1$, (3.31) and the interpolation inequality $\|\mathbf{w}\|^2_{2p/p-1} \leqslant \|\mathbf{w}\|^{2/p}_\infty \|\mathbf{w}\|^{2-2/p}_2 \leqslant B^{2/p} y^{1-1/p}$, the inequality (3.33) implies

$$y(t) \leqslant a \int_s^t y(\tau)^{1-1/p}\, d\tau\,, \qquad a := 2KpMB^{2/p}, \qquad t \in [s, T].$$

If $M = 0$, then $y = 0$ on $[s, T]$. Otherwise denote the right side by $z(t)$. It is Lipschitz, and since y is bounded, $z' = ay^{1-1/p} \leqslant az^{1-1/p}$ almost everywhere. For $\delta > 0$ the function $(z+\delta)^{1/p}$ is Lipschitz with derivative at most a/p almost everywhere, so $(z(t) + \delta)^{1/p} \leqslant \delta^{1/p} + a(t-s)/p$. Letting $\delta \to 0$,

$$y(t) \leqslant z(t) \leqslant \big(2KM(t-s)\big)^p B^2\,.$$

For $t - s \leqslant (4KM)^{-1}$ the right side is at most $2^{-p}B^2$, and sending $p \to \infty$ gives $y(t) = 0$.

Step 6: Conclusion. Let $t_* := \sup\{t \in [0, T] : y = 0 \text{ on } [0, t]\}$. We have $y(t_*) = 0$: this is clear if $t_* = 0$, and otherwise $\mathbf{w}(t_*)$ is a weak limit of the vanishing fields $\mathbf{w}(t)$, $t < t_*$. If $t_* < T$, Step 5 with $s = t_*$ gives $y = 0$ on $[t_*, t_* + (4KM)^{-1}] \cap [0, T]$ (for $M = 0$ on $[t_*, T]$), contradicting the definition of t_*. Thus $t_* = T$, $\mathbf{v}_1(t) = \mathbf{v}_2(t)$ for all t, and $\mathbf{v}_1 = \mathbf{v}_2$ almost everywhere. □

Notes and references

4. Scale-invariant estimates

4.1 The hole-filling estimate

Throughout we assume that $1 \leqslant \Lambda < \infty$ and $\mathbf{a} \in L^\infty(B_1; \mathbb{R}^{d\times d})$ satisfy

$$e \cdot \mathbf{a}(x)e \geqslant |e|^2 \quad \text{and} \quad e \cdot \mathbf{a}^{-1}(x)e \geqslant \Lambda^{-1}|e|^2\,, \quad \forall e \in \mathbb{R}^d. \tag{4.1}$$

The constant Λ is called the *ellipticity*. Note that the second inequality of (4.1) implies that

$$|\mathbf{a}e| \leqslant \Lambda^{1/2}(e \cdot \mathbf{a}(x)e)^{1/2}, \tag{4.2}$$

which is how we will often use it. Indeed, applying the second inequality to $f := \mathbf{a}(x)e$ and using $f \cdot \mathbf{a}^{-1}(x)f = e \cdot \mathbf{a}(x)e$ gives $\Lambda^{-1}|\mathbf{a}(x)e|^2 \leqslant e \cdot \mathbf{a}(x)e$. Consequently $|e|^2 \leqslant e \cdot \mathbf{a}(x)e \leqslant |e||\mathbf{a}(x)e| \leqslant \Lambda^{1/2}|e|(e \cdot \mathbf{a}(x)e)^{1/2}$, so that $e \cdot \mathbf{a}(x)e \leqslant \Lambda|e|^2$, $|\mathbf{a}(x)e| \leqslant \Lambda|e|$ and $\Lambda \geqslant 1$. The matrix $\mathbf{a}(\cdot)$ may not be symmetric.

The regularity theory for divergence-form elliptic equations is based almost entirely on the following very simple energy estimate we call the *Caccioppoli inequality*.

Lemma 4.1 (Caccioppoli inequality)**.** *For every $u \in H^1(B_1)$ satisfying*

$$-\nabla \cdot \mathbf{a}\nabla u \leqslant 0 \quad \text{in } B_1, \tag{4.3}$$

and cutoff function $\varphi \in W^{1,\infty}_0(B_1)$, we have the estimate

$$\int_{B_1} \varphi^2 \nabla u_+ \cdot \mathbf{a}\nabla u_+ \leqslant 4\Lambda \int_{B_1} u_+^2 |\nabla \varphi|^2. \tag{4.4}$$

Proof. Testing the differential inequality with $\varphi^2 u_+ \in H^1_0(B_1)$ yields

$$\int_{B_1} \nabla\big(\varphi^2 u_+\big) \cdot \mathbf{a}\nabla u \leqslant 0. \tag{4.5}$$

Rearranging this and using (4.2), we obtain

$$\begin{aligned} \int_{B_1} \varphi^2 \nabla u_+ \cdot \mathbf{a}\nabla u_+ &\leqslant -\int_{B_1} 2\varphi u \nabla\varphi \cdot \mathbf{a}\nabla u_+ \\ &\leqslant \frac{1}{2}\int_{B_1} \varphi^2 \nabla u_+ \cdot \mathbf{a}\nabla u_+ + 2\Lambda \int_{B_1} |\nabla\varphi|^2 u_+^2 \,. \end{aligned} \tag{4.6}$$

This implies (4.4). □

The first clue that a solution of an elliptic equation with general measurable coefficients should be more regular than a typical H^1 function comes from the so-called *hole filling argument*, which is a simple iteration of the Caccioppoli inequality. It states that the energy density of a solution of an elliptic equation cannot be concentrated on sets of small measure.

Lemma 4.2 (Hole-filling estimate)**.** *Let $d \geqslant 2$. There exists $C(d) < \infty$ such that, for every $u \in H^1(B_1)$ satisfying*

$$-\nabla \cdot \mathbf{a}\nabla u = 0 \quad \text{in } B_1, \tag{4.7}$$

we have, with $\alpha := (C\Lambda)^{-1/2}$ and for every $r \in (0, 1/2]$,

$$\|\nabla u\|_{L^2(B_r)} \leqslant C\Lambda^{\frac{1}{2}} r^{\alpha} \|u\|_{L^2(B_1)}. \tag{4.8}$$

Proof. We work with the cubes $Q_r(x)$ of Lemma 1.20, whose shells are finite unions of boxes, and we write $E(S) := \int_S \nabla u \cdot \mathbf{a}\nabla u$ for the energy of u in a measurable set $S \subseteq B_1$. Since $e \cdot \mathbf{a}e \geqslant |e|^2$, we have $\int_S |\nabla u|^2 \leqslant E(S)$.

Step 1. Caccioppoli inequality on the transition shell. In order to obtain the sharpest dependence on the ellipticity, we use Cauchy's inequality differently than in (4.6): we keep the energy on the support of $\nabla\varphi$ instead of reabsorbing it. Let $k \in \mathbb{R}$ and let $\varphi \in W^{1,\infty}_0(B_1)$ satisfy $0 \leqslant \varphi \leqslant 1$. Since u is a solution, we may test the equation with $\varphi^2(u-k) \in H^1_0(B_1)$. Using (4.2) and $\varphi \leqslant 1$, we obtain

$$\begin{aligned} \int_{B_1} \varphi^2 \nabla u \cdot \mathbf{a}\nabla u &= -2\int_{B_1} \varphi(u-k)\nabla\varphi \cdot \mathbf{a}\nabla u \\ &\leqslant 2\Lambda^{1/2} \int_{\{\nabla\varphi \neq 0\}} |\nabla\varphi||u-k|(\nabla u \cdot \mathbf{a}\nabla u)^{1/2} \\ &\leqslant \Lambda^{1/2} \int_{\{\nabla\varphi\neq 0\}} \nabla u \cdot \mathbf{a}\nabla u + \Lambda^{1/2} \int_{B_1} |\nabla\varphi|^2 (u-k)^2 \,. \end{aligned} \tag{4.9}$$

Let $Q_{2r}(x) \subseteq B_1$ and apply (4.9) with

$$\varphi(y) := \left(2 - \frac{|y-x|_\infty}{r}\right)_+ \wedge 1,$$

with $|z|_\infty = \max_i |z_i|$ as before. Then $\varphi = 1$ in $Q_r(x)$, $\varphi = 0$ outside $Q_{2r}(x)$, and $|\nabla\varphi| \leqslant r^{-1}\mathbf{1}_{S_r(x)}$ almost everywhere, where $S_r(x) := Q_{2r}(x)\backslash\overline{Q}_r(x)$ is the cube shell. We obtain, for every $k \in \mathbb{R}$,

$$E(Q_r(x)) \leqslant \Lambda^{1/2} E(S_r(x)) + \Lambda^{1/2} r^{-2} \int_{S_r(x)} (u-k)^2 . \tag{4.10}$$

Step 2. Poincaré inequality in the shell. There exists $C_P(d) < \infty$ such that, for every $v \in H^1(S_1(0))$,

$$\int_{S_1(0)} \left|v - (v)_{S_1(0)}\right|^2 \leqslant C_P \int_{S_1(0)} |\nabla v|^2. \tag{4.11}$$

Indeed, $S_1(0)$ is the union of the $2d$ boxes $P_i^\pm := \{y \in Q_2(0) : \pm y_i > 1\}$. Each box is convex, with diameter at most $4\sqrt{d}$ and volume 4^{d-1}, so the Poincaré inequality for convex domains applies in each box with a constant depending only on d. For $i \neq j$, the intersection $P_i^\pm \cap P_j^\pm$ is a box of volume 4^{d-2}. Since $d \geqslant 2$, any two of the boxes either coincide, intersect, or both intersect a third box. Comparing the means of v on overlapping boxes through the means on their intersections, and then comparing each box mean with the mean of v on a fixed box, yields (4.11). By scaling and translation, (4.11) gives

$$\int_{S_r(x)} \left(u - (u)_{S_r(x)}\right)^2 \leqslant C_P r^2 \int_{S_r(x)} |\nabla u|^2 \leqslant C_P r^2 E(S_r(x)).$$

The hypothesis $d \geqslant 2$ is essential here: when $d = 1$, the shell is disconnected.

Step 3. Filling the hole. Taking $k := (u)_{S_r(x)}$ in (4.10) and using Step 2, we obtain

$$E(Q_r(x)) \leqslant K\, E(S_r(x)), \qquad K := (1 + C_P)\Lambda^{1/2} \geqslant 1.$$

Since the boundary of a cube is a null set, $E(S_r(x)) = E(Q_{2r}(x)) - E(Q_r(x))$. We "fill the hole" by adding $KE(Q_r(x))$ to both sides to obtain

$$E(Q_r(x)) \leqslant \theta E(Q_{2r}(x)), \qquad \theta := \frac{K}{K+1}. \tag{4.12}$$

Step 4. Iteration. Define $\beta := \log_2(1 + K^{-1})$, so that $\theta = 2^{-\beta}$. Since $K \geqslant 1$ and $\log_2(1+t) \geqslant t$ for $t \in [0,1]$, we have

$$(1 + C_P)^{-1}\Lambda^{-1/2} = K^{-1} \leqslant \beta \leqslant 1.$$

Let $Q_R(x) \subseteq B_1$ and $\rho \in (0, R]$, and choose $n \in \mathbb{N}$ with $2^{-n-1}R < \rho \leqslant 2^{-n}R$. Iterating (4.12) n times gives

$$E(Q_\rho(x)) \leqslant \theta^n E(Q_R(x)) = 2^{-n\beta} E(Q_R(x)) \leqslant 2\Big(\frac{\rho}{R}\Big)^\beta E(Q_R(x)). \tag{4.13}$$

Step 5. Conclusion. Applying Lemma 4.1 to u and to $-u$, which are both subsolutions, with $\varphi(y) := (2 - 2|y|)_+ \wedge 1$, we get

$$E(B_{1/2}) \leqslant 16\Lambda \int_{B_1} u^2.$$

Let $R_0 := (2\sqrt{d})^{-1}$, so that $Q_{R_0}(0) \subseteq B_{1/2}$. If $r \leqslant R_0$, then $B_r \subseteq Q_r(0)$, and (4.13) yields

$$E(B_r) \leqslant E(Q_r(0)) \leqslant 2(2\sqrt{d}\,r)^{\beta} E(Q_{R_0}(0)) \leqslant 64\sqrt{d}\,\Lambda\, r^{\beta} \int_{B_1} u^2.$$

If $r \in (R_0, 1/2]$, then $2\sqrt{d}\,r \geqslant 1$ and $E(B_r) \leqslant E(B_{1/2}) \leqslant 16\Lambda(2\sqrt{d}\,r)^{\beta} \int_{B_1} u^2$, which gives the same bound. Therefore

$$\|\nabla u\|_{L^2(B_r)} \leqslant E(B_r)^{1/2} \leqslant 8d^{1/4}\Lambda^{1/2} r^{\beta/2} \|u\|_{L^2(B_1)}\,, \qquad r \in (0, 1/2].$$

Let $C := \max\{4(1+C_P)^2, 8d^{1/4}\}$. Then $\alpha = (C\Lambda)^{-1/2} \leqslant \frac{1}{2}(1+C_P)^{-1}\Lambda^{-1/2} \leqslant \beta/2$, so $r^{\beta/2} \leqslant r^{\alpha}$ for $r \leqslant 1$. This gives (4.8). □

Proposition 4.3 (Hölder regularity in $d = 2$)**.** *Let $d = 2$. There exists $C < \infty$ such that, with $\alpha := (C\Lambda)^{-1/2}$, every $u \in H^1(B_1)$ satisfying $-\nabla \cdot \mathbf{a}\nabla u = 0$ in B_1 has a representative in $C^{0,\alpha}(B_{1/2})$ satisfying*

$$\|u\|_{C^{0,\alpha}(B_{1/2})} \leqslant C\Lambda^{1/2} \|u\|_{L^2(B_1)}. \tag{4.14}$$

The exponent α is sharp up to the constant C; see Proposition 4.27. The prefactor $\Lambda^{1/2}$ also has the correct order. It is tempting to deduce (4.14) from the decay (4.13) of the energy and the Morrey or Campanato characterization of Hölder spaces. However, that route sums a geometric series of oscillations and loses a factor $\alpha^{-1} \simeq \Lambda^{1/2}$, giving the weaker prefactor Λ. The proof below avoids this loss. In two dimensions, the energy in a shell controls the oscillation on a well-chosen square, and the maximum principle carries this bound to the interior. The resulting decay of the essential oscillation gives the Hölder estimate through Lemma 1.20, which has no factor α^{-1}.

Proof of Proposition 4.3. We use the notation of the proof of Lemma 4.2 with $d = 2$. In particular, the constant C is chosen at least as large as there, so that $\alpha \leqslant \beta/2 \leqslant \frac{1}{2}$.

Step 1. The oscillation in a square is controlled by the energy in its shell. We show that, whenever $Q_{2s}(x) \subseteq B_1$,

$$\operatorname*{ess\,osc}_{Q_s(x)} u \leqslant 3\left(\int_{S_s(x)} |\nabla u|^2\right)^{1/2}. \tag{4.15}$$

Choose $u_n \in C^{\infty}(\overline{Q_{2s}(x)})$ with $\sum_n \|u_n - u\|_{H^1(Q_{2s}(x))} < \infty$. Every point of the shell $S_s(x)$ lies on exactly one square $\partial Q_t(x)$ with $t \in (s, 2s)$, and $\int_s^{2s} \int_{\partial Q_t(x)} g\, d\mathcal{H}^1\, dt = \int_{S_s(x)} g$ for every integrable g, because $|\nabla|y - x|_{\infty}| = 1$ almost everywhere. By Fubini's theorem, for almost every $t \in (s, 2s)$ the following hold:

- the restrictions of u_n to the closed curve $\partial Q_t(x)$, of length $8t$, converge uniformly to a continuous function u_t, which is absolutely continuous with tangential derivative $\partial_\tau u_t = \nabla u \cdot \tau$ almost everywhere;
- on almost every segment in $Q_t(x)$ orthogonal to a side of $\partial Q_t(x)$, the restriction of u has an absolutely continuous representative whose value at the endpoint on $\partial Q_t(x)$ is given by u_t.

Averaging over $t \in (s, 2s)$, we may also choose such a t with

$$\int_{\partial Q_t(x)} |\partial_\tau u_t|^2 \, d\mathcal{H}^1 \leqslant \frac{1}{s} \int_{S_s(x)} |\nabla u|^2.$$

Any two points of $\partial Q_t(x)$ are joined by an arc of $\partial Q_t(x)$ of length at most $4t \leqslant 8s$. By the Cauchy–Schwarz inequality, $M := \max_{\partial Q_t(x)} u_t$ and $m := \min_{\partial Q_t(x)} u_t$ satisfy

$$M - m \leqslant (8s)^{1/2} \Big(\int_{\partial Q_t(x)} |\partial_\tau u_t|^2 \, d\mathcal{H}^1 \Big)^{1/2} \leqslant \sqrt{8} \Big(\int_{S_s(x)} |\nabla u|^2 \Big)^{1/2}.$$

We next show that $w := (u - M)_+ \in H^1_0(Q_t(x))$. For $\delta \in (0, t/2)$, let $\eta_\delta(y) := \min\{1, (\delta^{-1}(t - |y - x|_\infty) - 1)_+\}$. Then $\eta_\delta w \in H^1(Q_t(x))$ vanishes in the collar $C_\delta := Q_t(x) \setminus \overline{Q}_{t-\delta}(x)$ of width δ, so $\eta_\delta w \in H^1_0(Q_t(x))$. Moreover $\eta_\delta = 1$ outside $C_{2\delta}$ and $|\nabla \eta_\delta| \leqslant \delta^{-1}$. On almost every segment orthogonal to a side, w is absolutely continuous and vanishes at the endpoint on $\partial Q_t(x)$, since $u_t \leqslant M$. The one-dimensional inequality $\int_0^h f^2 \leqslant \frac{1}{2} h^2 \int_0^h |f'|^2$, valid for absolutely continuous f with $f(0) = 0$ and applied with $h = 2\delta$, and Fubini's theorem on the four overlapping strips of $C_{2\delta}$ give

$$\int_{Q_t(x)} |\nabla \eta_\delta|^2 w^2 \leqslant \delta^{-2} \int_{C_{2\delta}} w^2 \leqslant 4 \int_{C_{2\delta}} |\nabla u|^2 \xrightarrow[\delta \to 0]{} 0.$$

Consequently $\eta_\delta w \to w$ in $H^1(Q_t(x))$, and $w \in H^1_0(Q_t(x))$. We may therefore test the equation with w, extended by zero. Since $\nabla w = \mathbf{1}_{\{u > M\}} \nabla u$, we get

$$\int_{Q_t(x)} |\nabla w|^2 \leqslant \int_{Q_t(x)} \nabla w \cdot \mathbf{a} \nabla w = \int_{Q_t(x)} \nabla w \cdot \mathbf{a} \nabla u = 0,$$

so $w = 0$, that is, $u \leqslant M$ almost everywhere in $Q_t(x)$. The same argument applied to $-u$ gives $u \geqslant m$. Since $Q_s(x) \subseteq Q_t(x)$, this proves (4.15).

Step 2. Decay of the oscillation. Let $x \in B_{3/4}$ and $R := (8\sqrt{2})^{-1}$, so that $Q_R(x) \subseteq B_{1/8}(x)$ and $B_{1/4}(x) \subseteq B_1$. Lemma 4.1, applied to $\pm u$ with $\varphi(y) := (2 - 8|y - x|)_+ \wedge 1$, gives

$$E(Q_R(x)) \leqslant E(B_{1/8}(x)) \leqslant 256 \Lambda \int_{B_1} u^2.$$

For $0 < s \leqslant s_0 := R/2$, we combine (4.15), $\int_{S_s(x)} |\nabla u|^2 \leqslant E(Q_{2s}(x))$ and (4.13). Using $\beta \leqslant 1$, $(16\sqrt{2})^{\beta/2} \leqslant 5$ and $s^{\beta/2} \leqslant s^\alpha$, we get

$$\operatorname*{ess\,osc}_{Q_s(x)} u \leqslant 3 \Big(2 \Big(\frac{2s}{R} \Big)^\beta 256 \Lambda \int_{B_1} u^2 \Big)^{1/2} \leqslant C_1 \Lambda^{1/2} s^\alpha \, \|u\|_{L^2(B_1)}, \qquad C_1 := 240\sqrt{2}. \tag{4.16}$$

Step 3. The Hölder representative. Let $z \in B_{1/2}$. Every cube $Q_r(x) \subseteq Q_{s_0}(z)$ has $r \leqslant s_0$ and $x \in B_{\sqrt{2} s_0}(z) \subseteq B_{3/4}$. By (4.16), the hypothesis of Lemma 1.20 holds in $Q_{s_0}(z)$ with $K := C_1 \Lambda^{1/2} \|u\|_{L^2(B_1)}$. Hence u has a representative on $\overline{Q}_{s_0/8}(z)$ satisfying

$$|u^*(x) - u^*(y)| \leqslant 2 C_1 \Lambda^{1/2} |x - y|^\alpha \|u\|_{L^2(B_1)}, \qquad x, y \in \overline{Q}_{s_0/8}(z).$$

These representatives are continuous. Continuous functions that agree almost everywhere on an open set agree everywhere on it, so the representatives on overlapping cubes coincide and define one continuous representative u^* on $B_{1/2}$. If $x, y \in B_{1/2}$ and $|x-y| \leqslant s_0/8$, then $y \in \overline{Q}_{s_0/8}(x)$, and the display above bounds $|u^*(x) - u^*(y)|$. Moreover, $u^*(x)$ is the limit of the means of u on $Q_r(x)$ as $r \to 0$, so it lies between the essential infimum and supremum of u on $Q_{s_0}(x) \subseteq B_1$. Therefore

$$|u^*(x)| \leqslant \big|(u)_{Q_{s_0}(x)}\big| + \operatorname*{ess\,osc}_{Q_{s_0}(x)} u \leqslant \big((2s_0)^{-1} + C_1\Lambda^{1/2}\big)\|u\|_{L^2(B_1)}.$$

If $|x-y| > s_0/8$, then $|u^*(x) - u^*(y)| \leqslant 2\sup_{B_{1/2}}|u^*| \leqslant 2(s_0/8)^{-1/2}|x-y|^{\alpha}\sup_{B_{1/2}}|u^*|$. Since $\Lambda \geqslant 1$, combining these bounds yields (4.14). □

The optimal exponent in $d = 2$ turns out to be $\alpha = \Lambda^{-\frac{1}{2}}$, as we will see below in Proposition 4.27. The fact that we can obtain Hölder regularity in two dimensions relatively easily, compared with higher dimensions, reflects the fact that H^1 "almost" embeds in L^∞ when $d = 2$. In the proof above, this is visible in Step 1: in two dimensions the energy in a shell controls the oscillation on a curve, and the maximum principle turns this into an interior bound.

4.2 Higher integrability and the Meyers estimate

Throughout this subsection we assume (4.1), so in particular $\Lambda \geqslant 1$ and $\mathbf{a}$ need not be symmetric. For a solution u of an equation in divergence form we denote the square root of the energy density by

$$w := (\nabla u \cdot \mathbf{a}\nabla u)^{1/2}.$$

Using (4.2) and $e \cdot \mathbf{a}e \leqslant |e||\mathbf{a}e|$, we have

$$|\nabla u| \leqslant w, \qquad |\mathbf{a}\nabla u| \leqslant \Lambda^{1/2} w, \qquad w \leqslant \Lambda^{1/2}|\nabla u|. \tag{4.17}$$

Our goal is the following estimate of Meyers [Mey63], in which we keep track of the dependence of the range of exponents on the ellipticity Λ.

Theorem 4.4 (Meyers estimate)**.** *Let $d \geqslant 2$. There exist constants $c(d) > 0$ and $C(d) < \infty$ such that the following holds for every $\Lambda \geqslant 1$ and every measurable $\mathbf{a}$ satisfying (4.1). Set $p_\Lambda := 2 + c\Lambda^{-1/2}$.*

(i) Interior estimate. *If $p \in [2, p_\Lambda]$, $\mathbf{f} \in L^p(B_1;\mathbb{R}^d)$ and $u \in H^1(B_1)$ satisfies $-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f}$ in B_1, then*

$$\|w\|_{\underline{L}^p(B_{1/2})} \leqslant C\Big(\|w\|_{\underline{L}^2(B_1)} + \|\mathbf{f}\|_{\underline{L}^p(B_1)}\Big). \tag{4.18}$$

In particular, if $\mathbf{f} = 0$, then $\|w\|_{\underline{L}^p(B_{1/2})} \leqslant C\|w\|_{\underline{L}^2(B_1)}$, and

$$\|\nabla u\|_{\underline{L}^p(B_{1/2})} \leqslant C\big(\Lambda^{1/2}\|\nabla u\|_{\underline{L}^2(B_1)} + \|\mathbf{f}\|_{\underline{L}^p(B_1)}\big).$$

(ii) Dirichlet estimate. *If* $p \in [p'_\Lambda, p_\Lambda]$, $\mathbf{f} \in L^2(B_1;\mathbb{R}^d) \cap L^p(B_1;\mathbb{R}^d)$ *and* $u \in H^1_0(B_1)$ *satisfies* $-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f}$ *in* B_1, *then*

$$\|\nabla u\|_{L^p(B_1)} \leqslant C\|\mathbf{f}\|_{L^p(B_1)}. \tag{4.19}$$

The constant C *does not depend on* Λ *or* p.

By density, (ii) shows that, for p in this interval, the map $\mathbf{f} \mapsto u$ extends to a bounded linear map from $L^p(B_1;\mathbb{R}^d)$ to $W^{1,p}_0(B_1)$, and the resulting u solves the equation in the sense of distributions. The interval $[p'_\Lambda, p_\Lambda]$ shrinks to $\{2\}$ as $\Lambda \to \infty$, and this cannot be improved in general: the Meyers example shows that the length $c\Lambda^{-1/2}$ of the range $(2, p_\Lambda]$ has the optimal order in Λ, see Proposition 4.7 below. The proof uses only the Sobolev–Poincaré inequality, a Whitney covering, and a layer-cake integration of truncated moments. Its essential feature is that we work with the energy density w^2 rather than with $|\nabla u|^2$, and pair the flux and the gradient in the same exponent. This costs only one factor of $\Lambda^{1/2}$ (rather than Λ) in the reverse Hölder inequality, and it is what gives the order $\Lambda^{-1/2}$, which is optimal by Proposition 4.7.

We fix, throughout,

$$q := \frac{2d}{d+1} \in (1,2), \qquad \theta := \frac{q}{2} = \frac{d}{d+1}, \qquad q' = q^* = \frac{2d}{d-1}. \tag{4.20}$$

The exponent q' is the Sobolev conjugate of q, since $\frac{1}{q} - \frac{1}{d} = \frac{d+1}{2d} - \frac{1}{d} = \frac{d-1}{2d}$. By a cube we always mean a cube with sides parallel to the axes. For a cube Q we write $\ell(Q)$ for its side length and σQ for the concentric cube with side length $\sigma\ell(Q)$. By Theorem 1.9, applied with $p = q$ (so that $\gamma_q = 2$) and $D = \sqrt{d}\,\ell(Q)$, there exists $S(d) < \infty$ such that, for every cube Q and $v \in W^{1,q}(Q)$,

$$\|v - (v)_Q\|_{\underline{L}^{q'}(Q)} \leqslant S\,\ell(Q)\|\nabla v\|_{\underline{L}^q(Q)}. \tag{4.21}$$

Lemma 4.5 (Reverse Hölder inequality with contrast dependence $\Lambda^{1/2}$)**.** *There exists a constant* $C_0(d) \geqslant 1$ *such that the following holds. Let* $u \in H^1(B_1)$ *and* $\mathbf{f} \in L^2(B_1;\mathbb{R}^d)$ *satisfy*

$$-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f} \quad \text{in } B_1,$$

and set $h := |\mathbf{f}|^2$.

(i) For every cube Q *with* $2Q \subseteq B_1$,

$$\int_Q w^2 \leqslant C_0\Lambda^{1/2}|Q|\left(\fint_{2Q} w^{2\theta}\right)^{1/\theta} + C_0\int_{2Q} h. \tag{4.22}$$

(ii) If $u \in H^1_0(B_1)$, *then, with* w *and* h *extended by zero outside* B_1, *for every cube* $Q \subseteq \mathbb{R}^d$,

$$\int_Q w^2 \leqslant C_0\Lambda^{1/2}|Q|\left(\fint_{4Q} w^{2\theta}\right)^{1/\theta} + C_0\int_{4Q} h. \tag{4.23}$$

Proof. Step 1. The basic identity. Let η be a Lipschitz function with compact support and let $c \in \mathbb{R}$. Suppose that $\eta(u-c) \in H^1_0(B_1)$. Since $\mathbf{a}\nabla u + \mathbf{f} \in L^2(B_1)$, we may test the equation with $\eta(u-c)$ by density, which gives

$$\int \eta w^2 = -\int (u-c)(\mathbf{a}\nabla u + \mathbf{f}) \cdot \nabla \eta - \int \eta \mathbf{f} \cdot \nabla u. \tag{4.24}$$

The last term is bounded by $\frac{1}{2}\int \eta w^2 + \frac{1}{2}\int \eta h$, by (4.17) and Young's inequality, and it can be absorbed since $\int \eta w^2 < \infty$.

Step 2. The interior case. Let η be a Lipschitz function with $0 \leqslant \eta \leqslant 1$, $\eta = 1$ on Q, $\operatorname{supp}\eta \subseteq \overline{\frac{3}{2}Q} \subseteq 2Q$ and $|\nabla \eta| \leqslant 4/\ell(Q)$, and take $c := (u)_{2Q}$. If $2Q \subseteq B_1$, then $\eta(u-c) \in H^1_0(B_1)$. Set

$$m_w := \left(\fint_{2Q} w^{2\theta} \right)^{1/\theta}, \qquad m_h := \left(\fint_{2Q} h^{\theta} \right)^{1/\theta}.$$

By (4.17), $\|\nabla u\|_{\underline{L}^q(2Q)} \leqslant m_w^{1/2}$, $\|\mathbf{a}\nabla u + \mathbf{f}\|_{\underline{L}^q(2Q)} \leqslant \Lambda^{1/2} m_w^{1/2} + m_h^{1/2}$, because $q = 2\theta$. Hölder's inequality with the exponents q' and q, and (4.21) on the cube $2Q$, bound the first term on the right side of (4.24) by

$$\frac{4}{\ell(Q)} |2Q| \|u-c\|_{\underline{L}^{q'}(2Q)} \|\mathbf{a}\nabla u + \mathbf{f}\|_{\underline{L}^q(2Q)} \leqslant 8S|2Q|\big(\Lambda^{1/2} m_w + \sqrt{m_w m_h}\big).$$

Combining this with Step 1, and using $2\sqrt{m_w m_h} \leqslant m_w + m_h$ and $\Lambda \geqslant 1$, we get

$$\int_Q w^2 \leqslant \int \eta w^2 \leqslant 16S|2Q|\big(\tfrac{3}{2}\Lambda^{1/2} m_w + \tfrac{1}{2} m_h\big) + \int_{2Q} h.$$

Since $\theta < 1$, Jensen's inequality gives $|2Q| m_h \leqslant \int_{2Q} h$, and $|2Q| = 2^d |Q|$. This proves (4.22) with any $C_0 \geqslant \max\{24 \cdot 2^d S, 1 + 8S\}$. We stress that no higher integrability of u was used: since $\theta < 1$, $w^{2\theta} \in L^1$, and the calculation uses neither a bound on ∇u nor an absorbed Caccioppoli inequality.

Step 3. The boundary case. Now let $u \in H^1_0(B_1)$ and extend u, w, h by zero outside B_1. Then $u \in H^1(\mathbb{R}^d)$ with $\nabla u = 0$ almost everywhere outside B_1. Let Q be a cube and let η be as in Step 2. If $2Q \subseteq B_1$, then take $c := (u)_{4Q}$. The function $\eta(u-c)$ has compact support in B_1. Since $\mathbf{a}\nabla u$ and $\mathbf{f}$ are extended by zero outside B_1, the bounds (4.17) continue to hold on $4Q$, and (4.21) on $4Q$ gives

$$\|u-c\|_{\underline{L}^{q'}(4Q)} \leqslant 4S\ell(Q) \|\nabla u\|_{\underline{L}^q(4Q)}.$$

If $2Q \nsubseteq B_1$, take $c := 0$. Then $\eta u \in H^1_0(B_1)$, since u has zero trace. In this case we claim that $Z := \{u = 0\} \cap 4Q$ has measure at least $4^{-d}|4Q|$. Let x_Q be the center of Q, and let $z \in 2Q$ with $|z| \geqslant 1$. Put $\nu := z/|z|$. The half space $\{x : \nu \cdot x \geqslant 1\}$ lies outside B_1. Let $s_i := 1$ if $\nu_i \geqslant 0$ and $s_i := -1$ otherwise, and consider the subcube Q' of $4Q$ of side length $\ell(Q)$ consisting of the points x with $s_i(x_i - x_{Q,i}) \in [\ell(Q), 2\ell(Q)]$ for each i. For every $x \in Q'$,

$$\nu \cdot (x - x_Q) = \sum_i |\nu_i| s_i (x_i - x_{Q,i}) \geqslant \ell(Q)|\nu|_1 \geqslant \nu \cdot (z - x_Q),$$

because $|z_i - x_{Q,i}| \leqslant \ell(Q)$. Hence $\nu \cdot x \geqslant \nu \cdot z = |z| \geqslant 1$, so $Q' \subseteq Z$, and the claim follows. Since $u = 0$ on Z,

$$|(u)_{4Q}| \leqslant \left(\frac{|4Q|}{|Z|}\right)^{1/q'} \|u - (u)_{4Q}\|_{\underline{L}^{q'}(4Q)} \leqslant 2^{d-1}\|u - (u)_{4Q}\|_{\underline{L}^{q'}(4Q)}.$$

Consequently, by (4.21), $\|u\|_{\underline{L}^{q'}(4Q)} \leqslant 4(1 + 2^{d-1})S\ell(Q)\|\nabla u\|_{\underline{L}^{q}(4Q)}$. In both cases we have $\|u - c\|_{\underline{L}^{q'}(4Q)} \leqslant 4(1 + 2^{d-1})S\ell(Q)\|\nabla u\|_{\underline{L}^{q}(4Q)}$, and (4.24) holds. Proceeding as in Step 2, with the means now over $4Q$, we obtain (4.23), with a constant that depends only on d. Taking for C_0 the largest of the constants required in Steps 2 and 3 completes the proof. □

We now give a quantitative version of Gehring's lemma [Geh73]. We work with the constants

$$H := \left(\tfrac{9}{2}\right)^{d+1}, \qquad D_d := 2 \cdot 3^d H, \qquad G := 4(d+1)3^d H, \qquad N_\sigma := 3(\sigma+1)^d. \tag{4.25}$$

The absorption argument below takes place on truncated moments, so that we never have to know a priori that $g \in L^{1+\varepsilon}$.

Lemma 4.6 (Quantitative Gehring lemma)**.** *Let $d \geqslant 2$. There exists a constant $C_1(d) < \infty$ such that the following holds. Let $\sigma \in \{2, 4\}$, $A \geqslant 1$, $B \geqslant 0$, $\theta = \frac{d}{d+1}$, and*

$$0 < \varepsilon \leqslant \frac{1}{C_1 A}. \tag{4.26}$$

Let $g, h \geqslant 0$ be functions such that, for every cube Q in the region indicated below,

$$\int_Q g \leqslant A|Q|\left(\fint_{\sigma Q} g^\theta\right)^{1/\theta} + B\int_{\sigma Q} h. \tag{4.27}$$

(a) If $g \in L^1(\mathbb{R}^d)$ has compact support, $h \in L^{1+\varepsilon}(\mathbb{R}^d)$, and (4.27) holds for every cube $Q \subseteq \mathbb{R}^d$, then

$$\|g\|_{L^{1+\varepsilon}(\mathbb{R}^d)} \leqslant C_1 B\|h\|_{L^{1+\varepsilon}(\mathbb{R}^d)}. \tag{4.28}$$

(b) If $g \in L^1(B_1)$, $h \in L^{1+\varepsilon}(B_1)$ and (4.27) holds for every cube Q with $\sigma Q \subseteq B_1$, then

$$\|g\|_{\underline{L}^{1+\varepsilon}(B_{1/2})} \leqslant C_1\left(\fint_{B_1} g + B\|h\|_{\underline{L}^{1+\varepsilon}(B_1)}\right). \tag{4.29}$$

Proof. We prove the lemma with $C_1 := \max\{8G, 2^{d+4}N_4\}$. Since $N_\sigma \leqslant N_4$ for $\sigma \in \{2, 4\}$, the proof yields the constants $8BN_\sigma$ in (a) and 2^{d+4} in (b), which are at most $C_1 B$ and C_1 respectively, and it uses (4.26) only through its consequence $\varepsilon \leqslant \frac{1}{8GA}$.

Step 1. The maximal function. Let $\mathcal{M}$ denote the uncentered maximal operator over closed cubes in $\mathbb{R}^d$. The Vitali covering argument, applied with dilation factor three,

gives $|\{\mathcal{M}k > s\}| \leqslant 3^d s^{-1} \int |k|$. Applying this to $k\mathbf{1}_{\{k>s/2\}}$ and noting that $\mathcal{M}(k\mathbf{1}_{\{k\leqslant s/2\}}) \leqslant \frac{s}{2}$, we get, for every $k \geqslant 0$ and $s > 0$,

$$|\{\mathcal{M}k > s\}| \leqslant \frac{2 \cdot 3^d}{s} \int_{\{k>s/2\}} k. \tag{4.30}$$

Step 2. A Whitney covering. In the setting of (a), let $V := (\mathcal{M}(g^\theta))^{1/\theta}$ and, for $t > 0$, $E_t := \{V > t\}$. Since $g^\theta \in L^1$ has compact support, E_t is open and bounded. Let $\{Q\}$ be the maximal dyadic cubes such that $\overline{2\sigma Q} \subseteq E_t$. They have disjoint interiors and cover E_t up to a null set. The dyadic parent of Q is not admissible, so there is $y \in \overline{(4\sigma+1)Q} \backslash E_t$. The cube $(4\sigma+1)Q$ contains both σQ and y, and $V(y) \leqslant t$, so that

$$\left(\fint_{\sigma Q} g^\theta \right)^{1/\theta} \leqslant (4 + \tfrac{1}{\sigma})^{d/\theta} t \leqslant Ht. \tag{4.31}$$

The dilated cubes $\sigma Q \subseteq E_t$ have bounded overlap: $\sum_Q \mathbf{1}_{\sigma Q} \leqslant N_\sigma$. Indeed, if $x \in \sigma Q$, then the distance of x to E_t^c in the maximum norm lies between $\frac{\sigma}{2}\ell(Q)$ and $\frac{5\sigma+1}{2}\ell(Q)$. So two cubes whose dilates contain x have side lengths with ratio at most $5 + \frac{1}{\sigma} < 8$. There are therefore at most three dyadic scales, and at most $(\sigma+1)^d$ cubes of each scale, since their centers lie in a cube of side $\sigma\ell$ and form a lattice of spacing ℓ.

Since $g \leqslant V$ almost everywhere, $\{g > t\} \subseteq E_t$ up to a null set. We sum (4.27) over the cubes Q, use (4.31), $\sum_Q |Q| = |E_t|$, the bounded overlap, and (4.30) applied to $k = g^\theta$ and $s = t^\theta$. This gives

$$\int_{\{g>t\}} g \leqslant D_d A\, t^{1-\theta} \int_{\{g>2^{-1/\theta}t\}} g^\theta + BN_\sigma \int_{E_t} h. \tag{4.32}$$

Step 3. Truncated absorption. For $L > 0$ let

$$I_L := \int g \min(g, L)^\varepsilon, \qquad V_L := \min(V, L).$$

Then $I_L \leqslant L^\varepsilon \int g < \infty$. We multiply (4.32) by $\varepsilon t^{\varepsilon-1}$ and integrate over $t \in (0, L)$, using Fubini's theorem. The left side becomes I_L and the last term on the right becomes $BN_\sigma \int h V_L^\varepsilon$. For the first term on the right, set $c := 2^{-1/\theta}$ and compute

$$\varepsilon D_d A \int g^\theta \int_0^{\min(L, g/c)} t^{\varepsilon-\theta}\, dt\, dx = \frac{\varepsilon D_d A}{1-\theta+\varepsilon} \int g^\theta \min(L, g/c)^{1-\theta+\varepsilon} \leqslant \beta I_L,$$

where we used $\min(L, g/c) \leqslant c^{-1} \min(g, L)$ and $g^\theta \min(g, L)^{1-\theta} \leqslant g$, and, since $\varepsilon \leqslant \frac{1}{8GA}$,

$$\beta := \frac{\varepsilon D_d A}{1-\theta+\varepsilon} 2^{(1-\theta+\varepsilon)/\theta} \leqslant GA\varepsilon \leqslant \frac{1}{8}.$$

The first bound on β holds because $\varepsilon \leqslant \frac{1}{8G}$ implies $\varepsilon \leqslant 1 - \theta = \frac{1}{d+1}$, so that $1 - \theta + \varepsilon \geqslant \frac{1}{d+1}$ and $2^{(1-\theta+\varepsilon)/\theta} \leqslant 2^{2/d} \leqslant 2$. We conclude that

$$I_L \leqslant \beta I_L + BN_\sigma \int h V_L^\varepsilon. \tag{4.33}$$

To estimate the last term, let $s := (1+\varepsilon)/\theta > 1$. By the layer-cake formula and (4.30) for $k = g^\theta$, and the same elementary inequalities as before,

$$\int V_L^{1+\varepsilon} \leqslant 2^s 3^d \frac{s}{s-1} I_L \leqslant 2^s 3^d (d+1) I_L.$$

Hölder's inequality with exponents $1+\varepsilon$ and $\frac{1+\varepsilon}{\varepsilon}$ therefore gives

$$\int h V_L^{\varepsilon} \leqslant \|h\|_{L^{1+\varepsilon}} \Big(2^s 3^d (d+1) I_L\Big)^{\varepsilon/(1+\varepsilon)} \leqslant 6 \|h\|_{L^{1+\varepsilon}} I_L^{\varepsilon/(1+\varepsilon)}.$$

Here we used $2^{\varepsilon/\theta} \leqslant 2^{1/d}$ and $[3^d(d+1)]^{\varepsilon/(1+\varepsilon)} \leqslant [3^d(d+1)]^{1/(d+2)}$, which follow from $\varepsilon \leqslant \frac{1}{d+1}$. Inserting this in (4.33) and using $I_L < \infty$, $\beta \leqslant \frac18$, we obtain $I_L^{1/(1+\varepsilon)} \leqslant \frac{6BN_\sigma}{1-\beta}\|h\|_{L^{1+\varepsilon}} \leqslant 8BN_\sigma \|h\|_{L^{1+\varepsilon}}$. Sending $L \to \infty$ by monotone convergence gives (4.28).

Step 4. Localization. We now prove (b). Let $\frac12 \leqslant \rho < R \leqslant \frac34$ and let $m := \fint_{B_1} g$; we may assume $m > 0$. Let g and h be replaced by their extensions by zero outside B_R, and define V, E_t as above. Set

$$T_{\rho,R} := \left(\frac{3\sqrt{d}}{R-\rho}\right)^{d+1} m.$$

By the weak-type estimate and Jensen's inequality, $|E_t| \leqslant 3^d \omega_d (m/t)^\theta$, where $\omega_d := |B_1| \leqslant 2^d$. So, for every $t \geqslant T_{\rho,R}$, $|E_t| \leqslant (2(R-\rho)/\sqrt{d})^d$. If a selected cube Q meets B_ρ, then $|2\sigma Q| \leqslant |E_t|$ gives $\ell(Q) \leqslant (R-\rho)/(\sigma\sqrt{d})$. Therefore every point of σQ lies within Euclidean distance $\frac{\sqrt{d}(\sigma+1)}{2}\ell(Q) \leqslant \frac34(R-\rho)$ of a point of $Q \cap B_\rho$, and $\sigma Q \subseteq B_R \subseteq B_1$. Thus (4.27) is available for every selected cube meeting B_ρ. Summing over these cubes as in Step 2 gives (4.32), for every $t \geqslant T_{\rho,R}$, with the left side restricted to B_ρ. Integrating $\varepsilon t^{\varepsilon-1}$ over $t \in (T_{\rho,R}, L)$ and bounding the lower levels by $T_{\rho,R}^{\varepsilon} \int_{B_1} g$, we get, for $I_L(D) := \int_D g \min(g,L)^\varepsilon$,

$$I_L(B_\rho) \leqslant T_{\rho,R}^{\varepsilon} \int_{B_1} g + \beta I_L(B_R) + 6BN_\sigma \|h\|_{L^{1+\varepsilon}(B_R)} I_L(B_R)^{\varepsilon/(1+\varepsilon)}.$$

By Young's inequality, the last term is at most $\frac18 I_L(B_R) + 8^\varepsilon (6BN_\sigma)^{1+\varepsilon} \int_{B_R} h^{1+\varepsilon}$. The coefficient of $I_L(B_R)$ is therefore at most $\beta + \frac18 \leqslant \frac14$. We iterate with $\rho_j := \frac34 - \frac{2^{-j}}{4}$, whose successive gaps are $2^{-j}/8$, so that $T_{\rho_j,\rho_{j+1}} = (24\sqrt{d})^{d+1} 2^{j(d+1)} m$. Since $\varepsilon \leqslant \frac{1}{8GA} \leqslant \frac{1}{8G}$ gives $(d+1)\varepsilon \leqslant \frac{1}{32\cdot 3^d H}$, we have $(24\sqrt{d})^{(d+1)\varepsilon} \leqslant 2$ and $\frac14 2^{(d+1)\varepsilon} \leqslant \frac12$, and the series converges. The remainder term $4^{-j} I_L(B_{\rho_j}) \leqslant 4^{-j} L^\varepsilon \int_{B_1} g$ tends to zero. Consequently

$$I_L(B_{1/2}) \leqslant 4\omega_d m^{1+\varepsilon} + \tfrac43\, 8^\varepsilon (6BN_\sigma)^{1+\varepsilon} \int_{B_1} h^{1+\varepsilon}.$$

Sending $L \to \infty$, dividing by $|B_{1/2}| = 2^{-d}\omega_d$, taking the $(1+\varepsilon)$th root, and using $8^\varepsilon \leqslant 2$, gives (4.29). □

Proof of Theorem 4.4. Let $C_0(d)$ be the constant of Lemma 4.5, and $A := C_0 \Lambda^{1/2}$, $B := C_0$. Let $C_1(d)$ be the constant of Lemma 4.6 and set $c(d) := 2(C_1 C_0)^{-1}$, so that, for $p = 2 + 2\varepsilon$, the condition $p \leqslant p_\Lambda$ implies $\varepsilon \leqslant c\Lambda^{-1/2}/2 = 1/(C_1 A)$, that is, (4.26).

(i) We may assume $p > 2$, since for $p = 2$ the estimate follows from $\fint_{B_{1/2}} w^2 \leqslant 2^d \fint_{B_1} w^2$. Apply Lemma 4.6(b) with $\sigma = 2$, $g = w^2$ and $h = |\mathbf{f}|^2$, which is allowed by Lemma 4.5(i). Taking square roots in (4.29), and using $2(1+\varepsilon) = p$, gives (4.18). The estimate for ∇u follows from (4.17).

(ii), $p \geqslant 2$. For $p = 2$, test the equation with u: $\int w^2 = -\int \mathbf{f} \cdot \nabla u \leqslant \|\mathbf{f}\|_{L^2} \|w\|_{L^2}$, so that $\|\nabla u\|_{L^2} \leqslant \|w\|_{L^2} \leqslant \|\mathbf{f}\|_{L^2}$. Let now $p > 2$. Extend w and $h = |\mathbf{f}|^2$ by zero outside B_1. They are compactly supported, $w^2 \in L^1$ and $h \in L^{1+\varepsilon}$. By Lemma 4.5(ii), (4.27) holds for every cube with $\sigma = 4$, $g = w^2$, so Lemma 4.6(a) gives $\|w\|^2_{L^p(B_1)} = \|g\|_{L^{1+\varepsilon}} \leqslant C_1 C_0 \|h\|_{L^{1+\varepsilon}} = C_1 C_0 \|\mathbf{f}\|^2_{L^p}$. The claim follows from $|\nabla u| \leqslant w$.

(ii), $p < 2$. Let $p \in [p'_\Lambda, 2)$, so that $p' \in (2, p_\Lambda]$. The transpose $\mathbf{a}^t$ also satisfies (4.1). Let $\mathbf{f}$ and u be as in the statement, and let $F \in L^{p'}(B_1; \mathbb{R}^d) \subseteq L^2(B_1; \mathbb{R}^d)$. By the Lax–Milgram theorem there is $v \in H^1_0(B_1)$ with $-\nabla \cdot \mathbf{a}^t \nabla v = \nabla \cdot F$, and, by the case already proved, $\|\nabla v\|_{L^{p'}} \leqslant C \|F\|_{L^{p'}}$. Testing the equation of u with v and the equation of v with u (both are in $H^1_0(B_1)$), we get

$$\int F \cdot \nabla u = \int \mathbf{f} \cdot \nabla v, \qquad \text{so} \qquad \left| \int F \cdot \nabla u \right| \leqslant C \|\mathbf{f}\|_{L^p} \|F\|_{L^{p'}}.$$

Since $p' > 2$, we have $L^2(B_1; \mathbb{R}^d) \cap L^{p'}(B_1; \mathbb{R}^d) = L^{p'}(B_1; \mathbb{R}^d)$, and $\nabla u \in L^2(B_1) \subseteq L^p(B_1)$. Taking the supremum over F with $\|F\|_{L^{p'}} \leqslant 1$ gives (4.19). □

Proposition 4.7 (Sharpness of the range of exponents in Theorem 4.4)**.** *Let $d \geqslant 2$, $\Lambda > 1$, $\beta := \Lambda^{-1/2}$ and $p_\star := \frac{2}{1-\beta} = 2 + \frac{2}{\Lambda^{1/2}-1}$. There exists a symmetric measurable matrix field $\mathbf{a}$ satisfying (4.1) such that the following hold.*

(i) There is a solution $u \in H^1(B_1)$ of $-\nabla \cdot \mathbf{a} \nabla u = 0$ in B_1 with $\nabla u \in L^p(B_{1/2})$ if and only if $p < p_\star$, for $p \in (0, \infty)$; in particular, $\nabla u \notin L^p(B_{1/2})$ for every $p \geqslant p_\star$.

(ii) There are $\mathbf{f} \in L^\infty(B_1; \mathbb{R}^d)$ and a solution $U \in H^1_0(B_1)$ of $-\nabla \cdot \mathbf{a} \nabla U = \nabla \cdot \mathbf{f}$ in B_1 such that $\nabla U \notin L^p(B_1)$ for every $p \geqslant p_\star$.

Consequently, let $d \geqslant 2$ and suppose that Theorem 4.4(i), respectively (ii), holds for some $c > 0$ and for every $\Lambda > 1$, with a finite constant C that is allowed to depend on d and on Λ. Then

$$c\Lambda^{-1/2} < \frac{2}{\Lambda^{1/2} - 1} \quad \text{for every } \Lambda > 1, \qquad \text{and therefore} \qquad c \leqslant 2. \tag{4.34}$$

Equivalently, if for some $\Lambda > 1$ and some $\varepsilon \geqslant \frac{2}{\Lambda^{1/2}-1}$ the estimate in Theorem 4.4(i) (respectively (ii)) held for all $p \in [2, 2+\varepsilon]$ (respectively all $p \in [p', p]$ with $p = 2 + \varepsilon$) with some finite constant, this would contradict (i) (respectively (ii)) above, whatever the dependence of the constant on Λ.

Proof. Step 1: the coefficients and the functions. In the first two coordinates use polar coordinates (r, ϑ), so that $r = |(x_1, x_2)|$, and set $e_r := (\cos\vartheta, \sin\vartheta)$, $e_\vartheta := (-\sin\vartheta, \cos\vartheta)$ (extended by zero in the other coordinates). Define

$$\mathbf{a} := \Lambda\, e_r \otimes e_r + e_\vartheta \otimes e_\vartheta + \sum_{i>2} e_i \otimes e_i$$

(and $\mathbf{a} := I$ on the axis). It is symmetric and measurable, with eigenvalues 1 and Λ, so $\mathbf{a}\xi\cdot\xi \geqslant |\xi|^2$ and $\mathbf{a}^{-1}\xi\cdot\xi \geqslant \Lambda^{-1}|\xi|^2$, that is, (4.1). Let

$$u := r^\beta \cos\vartheta, \qquad v := \Lambda^{1/2} r^\beta \sin\vartheta, \qquad \Sigma := \{x_1 = x_2 = 0\}.$$

Both depend only on (x_1, x_2), and $|u| \leqslant 1$, $|v| \leqslant \Lambda^{1/2}$ in B_1. Let R be the linear map with $Re_r = -e_\vartheta$, $Re_\vartheta = e_r$ and $Re_i = 0$ for $i > 2$; that is, $R\xi = (\xi_2, -\xi_1, 0, \dots, 0)$. It satisfies $R\xi\cdot\eta = -\xi\cdot R\eta$. On $\mathbb{R}^d\backslash\Sigma$ the functions u and v are smooth and, since $\Lambda\beta = \Lambda^{1/2}$ and $\Lambda^{1/2}\beta = 1$,

$$\begin{aligned}
\nabla u &= r^{\beta-1}\big(\beta\cos\vartheta\, e_r - \sin\vartheta\, e_\vartheta\big),\\
\mathbf{a}\nabla u &= r^{\beta-1}\big(\Lambda^{1/2}\cos\vartheta\, e_r - \sin\vartheta\, e_\vartheta\big),\\
R\nabla v &= \Lambda^{1/2} r^{\beta-1} R\big(\beta\sin\vartheta\, e_r + \cos\vartheta\, e_\vartheta\big) = r^{\beta-1}\big(\Lambda^{1/2}\cos\vartheta\, e_r - \sin\vartheta\, e_\vartheta\big) = \mathbf{a}\nabla u.
\end{aligned}$$

Moreover $|\nabla u|^2 = r^{2\beta-2}(\beta^2\cos^2\vartheta + \sin^2\vartheta)$, so

$$\beta\, r^{\beta-1} \leqslant |\nabla u| \leqslant r^{\beta-1} \qquad \text{off } \Sigma. \tag{4.35}$$

The same computation gives $|\nabla v| \leqslant \Lambda^{1/2} r^{\beta-1}$.

Step 2: $u, v \in H^1(B_1)$ and the pointwise gradient is the weak gradient. Since Σ has codimension two and $2(1-\beta) < 2$, (4.35) gives $\int_{B_1} |\nabla u|^2 \leqslant C\int_0^1 r^{2\beta-1} dr < \infty$; the same holds for v. Let $w \in \{u, v\}$, $\phi \in C_c^\infty(B_1)$ and $i \in \{1, \dots, d\}$. The set $\{r > \varepsilon\}$ has smooth boundary $\{r = \varepsilon\}$, on which the outer normal is $-e_r$, and $\mathcal{H}^{d-1}(\{r = \varepsilon\} \cap B_1) \leqslant C\varepsilon$. Integrating by parts on $\{r > \varepsilon\}$ and using $|w| \leqslant \Lambda^{1/2}$,

$$\int_{B_1} w\,\partial_i\phi = \lim_{\varepsilon\to 0}\int_{\{r>\varepsilon\}} w\,\partial_i\phi = \lim_{\varepsilon\to 0}\left(-\int_{\{r>\varepsilon\}} \partial_i w\,\phi - \int_{\{r=\varepsilon\}} w\phi\,(e_r)_i\right) = -\int_{B_1}\partial_i w\,\phi,$$

because the boundary term is bounded by $C\Lambda^{1/2}\|\phi\|_{L^\infty}\varepsilon$, and $w, \partial_i w \in L^1$. Thus the pointwise gradient is the weak gradient and $u, v \in H^1(B_1)$.

Step 3: proof of (i). Off Σ, in polar coordinates and using $\Lambda^{1/2}\beta = 1$,

$$\nabla\cdot\mathbf{a}\nabla u = \frac{1}{r}\partial_r\big(\Lambda^{1/2} r^\beta\cos\vartheta\big) + \frac{1}{r}\partial_\vartheta\big(-r^{\beta-1}\sin\vartheta\big) = \big(\Lambda^{1/2}\beta - 1\big) r^{\beta-2}\cos\vartheta = 0.$$

Let $\phi \in C_c^\infty(B_1)$. Since $\mathbf{a}\nabla u \in L^2(B_1)$, integrating by parts on $\{r > \varepsilon\}$ (outer normal $-e_r$),

$$\int_{B_1} \mathbf{a}\nabla u\cdot\nabla\phi = \lim_{\varepsilon\to 0}\int_{\{r>\varepsilon\}} \mathbf{a}\nabla u\cdot\nabla\phi = -\lim_{\varepsilon\to 0}\int_{\{r=\varepsilon\}} (\mathbf{a}\nabla u\cdot e_r)\,\phi = 0,$$

because $|\mathbf{a}\nabla u\cdot e_r| = \Lambda^{1/2}\varepsilon^{\beta-1}|\cos\vartheta|$ on $\{r = \varepsilon\}$, whose intersection with B_1 has $\mathcal{H}^{d-1}$-measure at most $C\varepsilon$, so that the boundary integral is at most $C\Lambda^{1/2}\|\phi\|_{L^\infty}\varepsilon^\beta$. Hence $u \in H^1(B_1)$ is a weak solution of $-\nabla\cdot\mathbf{a}\nabla u = 0$ in B_1. By (4.35), $\beta^p\int_{B_{1/2}} r^{p(\beta-1)} \leqslant \int_{B_{1/2}} |\nabla u|^p \leqslant \int_{B_{1/2}} r^{p(\beta-1)}$, and, since $\{r < \frac14\}\cap\{|(x_3,\dots,x_d)| < \frac14\} \subseteq B_{1/2} \subseteq \{r < \frac12\}\cap\{|(x_3,\dots,x_d)| < \frac12\}$ (for $d = 2$ there are no further coordinates),

$$c_d\int_0^{1/4} r^{p(\beta-1)+1} dr \leqslant \int_{B_{1/2}} r^{p(\beta-1)} dx \leqslant C_d\int_0^{1/2} r^{p(\beta-1)+1} dr.$$

The integral $\int_0^s r^{p(\beta-1)+1}dr$ is finite if and only if $p(\beta-1)+1>-1$, that is, $p<p_\star$. This proves (i).

Step 4: proof of (ii). Let $\chi\in C_c^\infty(B_1)$ with $\chi=1$ in B_δ for some $\delta\in(0,1)$, and set

$$U:=\chi u\in H_0^1(B_1),\qquad \mathbf{f}:=vR\nabla\chi-u\mathbf{a}\nabla\chi.$$

Then $\mathbf{f}$ is measurable, bounded (because u and v are) and supported in $\operatorname{supp}\nabla\chi$. Note that $\nabla\cdot R\nabla\chi=\partial_1\partial_2\chi-\partial_2\partial_1\chi=0$. Let $\psi\in C_c^\infty(B_1)$. Testing the equation of u with $\chi\psi$ and using $\mathbf{a}\nabla u=R\nabla v$ gives

$$\int\chi\,\mathbf{a}\nabla u\cdot\nabla\psi=-\int\psi\,\mathbf{a}\nabla u\cdot\nabla\chi=-\int\psi\,R\nabla v\cdot\nabla\chi=\int\psi\,\nabla v\cdot R\nabla\chi,$$

by the antisymmetry of R. Since $v\psi\in H^1(B_1)$ has compact support in B_1 and $R\nabla\chi$ is smooth and divergence free, $\int\nabla(v\psi)\cdot R\nabla\chi=0$, that is, $\int\psi\,\nabla v\cdot R\nabla\chi=-\int v\,\nabla\psi\cdot R\nabla\chi$. Therefore, as $\mathbf{a}$ is symmetric,

$$\int\mathbf{a}\nabla U\cdot\nabla\psi=\int\chi\,\mathbf{a}\nabla u\cdot\nabla\psi+\int u\,\mathbf{a}\nabla\chi\cdot\nabla\psi=-\int\big(vR\nabla\chi-u\mathbf{a}\nabla\chi\big)\cdot\nabla\psi=-\int\mathbf{f}\cdot\nabla\psi,$$

which is $-\nabla\cdot\mathbf{a}\nabla U=\nabla\cdot\mathbf{f}$ in B_1. Finally, $\nabla U=\nabla u$ in B_δ, so by Step 3 (with B_δ in place of $B_{1/2}$, the argument being identical) $\nabla U\notin L^p(B_1)$ for $p\geqslant p_\star$. This proves (ii).

Step 5: the consequences. Fix $\Lambda>1$ and let $\mathbf{a}$, u, U, $\mathbf{f}$ be as above; put $p_\Lambda:=2+c\Lambda^{-1/2}$. Suppose Theorem 4.4(i) holds with c and some finite constant C. Since $p_\Lambda\in[2,p_\Lambda]$, $\mathbf{f}=0$ and $u\in H^1(B_1)$, the estimate (4.18) gives $\|w\|_{\underline{L}^{p_\Lambda}(B_{1/2})}\leqslant C\|w\|_{\underline{L}^2(B_1)}<\infty$, and since $|\nabla u|\leqslant w$ we get $\nabla u\in L^{p_\Lambda}(B_{1/2})$. By (i), $p_\Lambda<p_\star$, which is the inequality in (4.34). If instead Theorem 4.4(ii) holds, then, since $p_\Lambda'\leqslant 2\leqslant p_\Lambda$ and $\mathbf{f}\in L^\infty(B_1;\mathbb{R}^d)\subseteq L^2\cap L^{p_\Lambda}$, (4.19) with $p=p_\Lambda$ gives $\nabla U\in L^{p_\Lambda}(B_1)$, and (ii) again yields $p_\Lambda<p_\star$. Multiplying by $\Lambda^{1/2}$, $c<\frac{2\Lambda^{1/2}}{\Lambda^{1/2}-1}$ for every $\Lambda>1$, and letting $\Lambda\to\infty$ gives $c\leqslant 2$. The last sentence of the statement is the same argument with $p=2+\varepsilon(\Lambda)\geqslant p_\star$ in place of p_Λ: only the finiteness of the left side of the estimate was used, not the size of C. □

The proposition shows that the length $c\Lambda^{-1/2}$ of the range $(2,p_\Lambda]$ in Theorem 4.4 has the optimal order in Λ, in the sense of (4.34); it concerns the upper endpoint p_Λ only, and we do not address the sharpness of the lower endpoint p_Λ' in (ii).

4.3 The De Giorgi-Nash Hölder estimate

In this section we give a proof of De Giorgi's celebrated result on the Hölder regularity of solutions of elliptic equations with measurable coefficients. Throughout this subsection we assume that $d\geqslant 3$, so that the Sobolev exponent $2^*=\frac{2d}{d-2}$ is finite; for $d=2$ see Proposition 4.3 and Piccinini and Spagnolo [PS72]. The matrix $\mathbf{a}$ and the ellipticity Λ are as in (4.1).

Theorem 4.8 (Interior Hölder regularity)**.** *Let $d\geqslant 3$. There exists a constant $C(d)<\infty$ such that, for every $\Lambda\geqslant 1$, there is an exponent $\alpha=\alpha(d,\Lambda)\in(0,1]$ with $\alpha\geqslant\exp\big(-C\Lambda^{\frac{d+2}{4}}\big)$ for which the following holds. If $\mathbf{a}$ satisfies (4.1) and $u\in H^1(B_1)$ satisfies*

$$-\nabla\cdot\mathbf{a}\nabla u=0\quad\text{in } B_1,$$

then u has a representative in $C^{0,\alpha}(B_{1/2})$ and

$$\|u\|_{C^{0,\alpha}(B_{1/2})} \leqslant C\Lambda^{\frac{d}{4}}\|u\|_{L^2(B_1)}. \tag{4.36}$$

We proceed by first giving De Giorgi's proof of the L^∞ upper bound for subsolutions, which is based on keeping track of the measure of the superlevel sets of the subsolution. Both proofs below use the Caccioppoli inequality in the following form, in which the gradient of the cut-off function φu_+ appears. Let $u \in H^1(B_1)$ satisfy $-\nabla\cdot\mathbf{a}\nabla u \leqslant 0$ in B_1 and let $\varphi \in W^{1,\infty}_0(B_1)$. Since $\nabla(\varphi u_+) = \varphi\nabla u_+ + u_+\nabla\varphi$, $|\nabla u_+|^2 \leqslant \nabla u_+\cdot\mathbf{a}\nabla u_+$ by (4.1), Lemma 4.1 gives

$$\int_{B_1}|\nabla(\varphi u_+)|^2 \leqslant 2\int_{B_1}\varphi^2\nabla u_+\cdot\mathbf{a}\nabla u_+ + 2\int_{B_1}u_+^2|\nabla\varphi|^2 \leqslant (8\Lambda+2)\int_{B_1}u_+^2|\nabla\varphi|^2 \leqslant 10\Lambda\int_{B_1}u_+^2|\nabla\varphi|^2. \tag{4.37}$$

Since $\mathbf{a}$ and Λ are unchanged if we add a constant to u and both sides of the inequality in Lemma 4.1 are invariant under translation and scaling of the ball, (4.37) also holds with u replaced by $u-\lambda$ for any $\lambda\in\mathbb{R}$, and with B_1 replaced by any ball $B_R(x)$ on which the inequality $-\nabla\cdot\mathbf{a}\nabla u \leqslant 0$ holds.

Proposition 4.9 (L^∞ estimate for subsolutions)**.** *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that, for every $u \in H^1(B_1)$ satisfying*

$$-\nabla\cdot\mathbf{a}\nabla u \leqslant 0 \quad \text{in } B_1, \tag{4.38}$$

we have the estimate

$$\operatorname*{ess\,sup}_{B_{1/2}} u_+ \leqslant C\Lambda^{d/4}\|u\|_{L^2(B_1)}. \tag{4.39}$$

Two proofs are given; either one suffices.

Maly's version of De Giorgi's proof. Since $u_+ \leqslant |u|$, it suffices to prove the estimate with $\|u_+\|_{L^2(B_1)}$ on the right side. By the Lebesgue differentiation theorem, it suffices to find a constant $C(d) < \infty$ such that, for every subsolution u,

$$\sup_{x\in B_{1/2}}\limsup_{k\to\infty}\|u_+\|_{\underline{L}^2(B_{2^{-k}}(x))} \leqslant C\Lambda^{d/4}\|u_+\|_{L^2(B_1)}. \tag{4.40}$$

Indeed, for $x\in B_{1/2}$ the function $\widetilde{u}(y) := u(x+y/2)$ is a subsolution in B_1 of the equation with coefficients $\widetilde{\mathbf{a}}(y) := \mathbf{a}(x+y/2)$, which satisfy (4.1) with the same Λ (note $B_{1/2}(x)\subseteq B_1$), and $\|\widetilde{u}_+\|_{L^2(B_1)} = 2^{d/2}\|u_+\|_{L^2(B_{1/2}(x))} \leqslant 2^{d/2}\|u_+\|_{L^2(B_1)}$ while $\|u_+\|_{\underline{L}^2(B_{2^{-k}}(x))} = \|\widetilde{u}_+\|_{\underline{L}^2(B_{2^{-k+1}})}$. Thus we need to prove (4.40) only for $x=0$ (with the ball radii 2^{-k} replaced by $r_k := 2^{-k-1}$, which changes the lim sup not at all).

Step 1: a reverse Hölder inequality. For each $r\in(0,1/2)$, pick $\varphi\in C_c^\infty(B_{2r})$ with $\mathbf{1}_{B_r}\leqslant\varphi\leqslant\mathbf{1}_{B_{3r/2}}$ and $|\nabla\varphi|\leqslant 4r^{-1}$, and let $\theta\in\mathbb{R}$. The function $u-\theta$ is a subsolution and $\varphi(u-\theta)_+\in H^1_0(B_{2r})$. The Sobolev inequality (Proposition 1.5 with $p=2$), rescaled to B_{2r} and written in terms of normalized norms, states that $\|v\|_{\underline{L}^{2^*}(B_{2r})}\leqslant Cr\|\nabla v\|_{\underline{L}^2(B_{2r})}$ for every $v\in H^1_0(B_{2r})$. Combining it with (4.37) we obtain

$$\begin{aligned}\|(u-\theta)_+\|_{\underline{L}^{2^*}(B_r)} \leqslant 2^{\frac{d}{2^*}}\|\varphi(u-\theta)_+\|_{\underline{L}^{2^*}(B_{2r})} \leqslant Cr\|\nabla(\varphi(u-\theta)_+)\|_{\underline{L}^2(B_{2r})} &\leqslant C\Lambda^{1/2}r\|(u-\theta)_+|\nabla\varphi|\|_{\underline{L}^2(B_{2r})}\\ &\leqslant C\Lambda^{1/2}\|(u-\theta)_+\|_{\underline{L}^2(B_{2r})}.\end{aligned}$$

Write $\chi := 2^*/2 > 1$. Since $(u-\lambda)_+ = 0$ off $\{u > \lambda\}$, while on $\{u > \lambda\}$ we have $(u-\lambda)_+ \leqslant (u-\theta)_+$ and $(u-\theta)/(\lambda-\theta) > 1$, so that $\chi > 1$ gives $(u-\lambda)_+ \leqslant (\lambda-\theta)^{1-\chi}(u-\theta)_+^{\chi}$, this inequality holds a.e. in B_1 for every $\lambda > \theta$; hence we have

$$\|(u-\lambda)_+\|_{\underline{L}^2(B_r)} \leqslant (\lambda-\theta)^{1-\chi}\|(u-\theta)_+\|^{\chi}_{\underline{L}^{2^*}(B_r)}.$$

Combining the previous two displays and raising to the power $1/\chi = 2/2^*$ yields, for every $\lambda > \theta$ and $r \in (0, 1/2)$,

$$\|(u-\lambda)_+\|^{2/2^*}_{\underline{L}^2(B_r)} \leqslant C\Lambda^{1/2}(\lambda-\theta)^{2/2^*-1}\|(u-\theta)_+\|_{\underline{L}^2(B_{2r})}. \tag{4.41}$$

Step 2: the iteration. The inequality (4.41) implies (4.40) with $x = 0$. To see this, we may assume that $\lambda_1 := \|u_+\|_{L^2(B_1)} > 0$. Let $r_k := 2^{-k-1}$ and let $\{\lambda_k\}_{k\in\mathbb{N}}$ be defined recursively by λ_1 as above and

$$\lambda_{k+1} := \lambda_k + A\|(u-\lambda_k)_+\|_{\underline{L}^2(B_{r_k})}, \quad \forall k \in \mathbb{N},$$

where $A > 1$ will be selected below. Write $\Delta_k := \lambda_{k+1} - \lambda_k \geqslant 0$, so that $\lambda_k \geqslant \lambda_1 > 0$ is nondecreasing. We claim that $\Delta_{k+1} \leqslant \frac{1}{2}\Delta_k$ for every $k \in \mathbb{N}$. If $\Delta_k = 0$, then $(u-\lambda_k)_+ = 0$ a.e. in $B_{r_k} \supseteq B_{r_{k+1}}$, so $\Delta_{k+1} = 0$ and the claim is trivial. Otherwise $\lambda_{k+1} > \lambda_k$, and we may apply (4.41) with $r = r_{k+1}$ (so that $2r = r_k$), $\lambda = \lambda_{k+1}$ and $\theta = \lambda_k$ to get

$$\begin{aligned} A^{-2/2^*}\Delta_{k+1}^{2/2^*} &= \|(u-\lambda_{k+1})_+\|^{2/2^*}_{\underline{L}^2(B_{r_{k+1}})} \\ &\leqslant C\Lambda^{1/2}\Delta_k^{2/2^*-1}\|(u-\lambda_k)_+\|_{\underline{L}^2(B_{r_k})} = C\Lambda^{1/2}A^{-1}\Delta_k^{2/2^*}. \end{aligned}$$

Multiplying by $A^{2/2^*}$ and using $1 - 2/2^* = 2/d$, we obtain $\Delta_{k+1}^{2/2^*} \leqslant C\Lambda^{1/2}A^{-2/d}\Delta_k^{2/2^*}$. Selecting $A := C_1\Lambda^{d/4}$ for a large $C_1(d)$ gives $C\Lambda^{1/2}A^{-2/d} = CC_1^{-2/d} \leqslant 2^{-2/2^*}$, so that $\Delta_{k+1} \leqslant \frac{1}{2}\Delta_k$, proving the claim.

We deduce that, for every $k \in \mathbb{N}$,

$$\lambda_k \leqslant \lambda_1 + \sum_{n=1}^{\infty}\Delta_n \leqslant \lambda_1 + 2\Delta_1 = \lambda_1 + 2A\|(u-\lambda_1)_+\|_{\underline{L}^2(B_{1/4})} \leqslant \lambda_1 + 2A\,|B_{1/4}|^{-1/2}\|u_+\|_{L^2(B_1)} \leqslant C\Lambda^{d/4}\|u_+\|_{L^2(B_1)}.$$

Moreover, $\|(u-\lambda_k)_+\|_{\underline{L}^2(B_{r_k})} = A^{-1}\Delta_k \leqslant \Delta_k \leqslant \sup_j \lambda_j$. Since $\lambda_k \geqslant 0$ implies $u_+ \leqslant (u-\lambda_k)_+ + \lambda_k$, we conclude that, for every $k \in \mathbb{N}$,

$$\|u_+\|_{\underline{L}^2(B_{r_k})} \leqslant \lambda_k + \|(u-\lambda_k)_+\|_{\underline{L}^2(B_{r_k})} \leqslant 2\sup_j \lambda_j \leqslant C\Lambda^{d/4}\|u_+\|_{L^2(B_1)}.$$

This completes the proof of (4.40) with $x = 0$. □

De Giorgi's proof. By homogeneity (cu is a subsolution for every $c > 0$, and $u_+ = 0$ if $u = 0$) we may assume that $\|u\|_{L^2(B_1)} = 1$; we will show that $u \leqslant \lambda_*$ a.e. in $B_{1/2}$ for some $\lambda_* = C_*\Lambda^{d/4}$ with $C_*(d) < \infty$, which implies the proposition. Let $\lambda \in \mathbb{R}$, $s \in (0, 1]$

and $\varphi \in W_0^{1,\infty}(B_1)$ with $\varphi = 0$ outside B_s. Applying (4.37) to $u - \lambda$, the only use of the differential inequality, we have

$$\int_{B_1} |\nabla(\varphi(u-\lambda)_+)|^2 \leqslant 10\Lambda \int_{B_1} (u-\lambda)_+^2 |\nabla\varphi|^2. \tag{4.42}$$

By the Hölder and Sobolev inequalities (Proposition 1.5 with $p = 2$), for $C(d) < \infty$,

$$\begin{aligned}\int_{B_1} (\varphi(u-\lambda)_+)^2 &\leqslant \left(\int_{B_1} (\varphi(u-\lambda)_+)^{2^*}\right)^{\frac{2}{2^*}} |\{\varphi(u-\lambda)_+ \neq 0\}|^{\frac{2^*-2}{2^*}} \\ &\leqslant C\left(\int_{B_1} |\nabla(\varphi(u-\lambda)_+)|^2\right) |\{\varphi(u-\lambda)_+ \neq 0\}|^{\frac{2^*-2}{2^*}}.\end{aligned}$$

Observing that $\frac{2^*-2}{2^*} = \frac{2}{d}$ and combining this with (4.42), we get

$$\int_{B_1} (\varphi(u-\lambda)_+)^2 \leqslant C\Lambda\left(\int_{B_1} (u-\lambda)_+^2 |\nabla\varphi|^2\right) |\{\varphi(u-\lambda)_+ \neq 0\}|^{\frac{2}{d}}.$$

Now fix $0 < r < s \leqslant 1$ and use the cutoff function

$$\varphi(x) := \left(\frac{s - |x|}{s-r}\right)_+ \wedge 1,$$

which satisfies $\varphi = 1$ in B_r, $\varphi = 0$ outside B_s and $|\nabla\varphi| \leqslant (s-r)^{-1}$. Since $\{\varphi(u-\lambda)_+ \neq 0\} \subseteq \{u > \lambda\} \cap B_s$, we obtain

$$\int_{B_r} (u-\lambda)_+^2 \leqslant \frac{C\Lambda}{(s-r)^2} |\{u > \lambda\} \cap B_s|^{\frac{2}{d}} \int_{B_s} (u-\lambda)_+^2.$$

By Chebyshev's inequality, for each $\theta < \lambda$,

$$|\{u > \lambda\} \cap B_s|^{\frac{2}{d}} \leqslant (\lambda - \theta)^{-\frac{4}{d}} \left(\int_{B_s} (u-\theta)_+^2\right)^{\frac{2}{d}}.$$

Since also $(u-\lambda)_+ \leqslant (u-\theta)_+$, these yield, for every $0 < r < s \leqslant 1$ and $\theta < \lambda$,

$$\int_{B_r} (u-\lambda)_+^2 \leqslant \frac{C\Lambda}{(s-r)^2(\lambda-\theta)^{\frac{4}{d}}} \left(\int_{B_s} (u-\theta)_+^2\right)^{1+\frac{2}{d}}. \tag{4.43}$$

We now perform the following iteration. Fix $\lambda_* > 0$ to be selected below and define, for each $n \in \mathbb{N} \cup \{0\}$,

$$r_n := \frac{1}{2}\left(1 + 2^{-n-1}\right), \quad \lambda_n := \lambda_*\left(1 - 2^{-n-1}\right), \quad \text{and} \quad A_n := \int_{B_{r_n}} (u-\lambda_n)_+^2.$$

Then $r_n \downarrow \frac{1}{2}$, $\lambda_n \uparrow \lambda_*$, $A_0 \leqslant \int_{B_1} u_+^2 \leqslant 1$, $r_n - r_{n+1} = 2^{-n-3}$ and $\lambda_{n+1} - \lambda_n = \lambda_* 2^{-n-2}$. Applying (4.43) with $r = r_{n+1}$, $s = r_n$, $\lambda = \lambda_{n+1}$ and $\theta = \lambda_n$ yields

$$A_{n+1} \leqslant C_2 \Lambda \lambda_*^{-\frac{4}{d}} b^n A_n^{1+\frac{2}{d}}, \qquad b := 2^{2+\frac{4}{d}} > 1, \tag{4.44}$$

where $C_2(d) < \infty$. Set $\gamma := \frac{2}{d}$, $\varepsilon := b^{-\frac{1}{\gamma}-\frac{1}{\gamma^2}}$ and select $\lambda_* := C_*\Lambda^{\frac{d}{4}}$ where $C_*(d) < \infty$ is so large that $C_2 C_*^{-\frac{4}{d}} \leqslant \min\{1, \varepsilon\}$; note that ε depends only on d. Then $C_2\Lambda\lambda_*^{-\frac{4}{d}} = C_2 C_*^{-\frac{4}{d}} \leqslant 1$, so that (4.44) becomes

$$A_{n+1} \leqslant b^n A_n^{1+\gamma}, \tag{4.45}$$

and, applying (4.44) with $n = 0$ and using $A_0 \leqslant 1$,

$$A_1 \leqslant C_2 C_*^{-\frac{4}{d}} \leqslant \varepsilon. \tag{4.46}$$

We claim that (4.45) and (4.46) imply that $A_n \leqslant b^{-\frac{n}{\gamma}-\frac{1}{\gamma^2}}$ for every $n \geqslant 1$. For $n = 1$ this is (4.46). If it holds for some $n \geqslant 1$, then by (4.45)

$$A_{n+1} \leqslant b^n\, b^{-(1+\gamma)(\frac{n}{\gamma}+\frac{1}{\gamma^2})} = b^{-\frac{n}{\gamma}-\frac{1}{\gamma^2}-\frac{1}{\gamma}} = b^{-\frac{n+1}{\gamma}-\frac{1}{\gamma^2}}.$$

In particular, $\lim_{n\to\infty} A_n = 0$. Since $B_{1/2} \subseteq B_{r_n}$ and $(u-\lambda_*)_+ \leqslant (u-\lambda_n)_+$, we have $\int_{B_{1/2}} (u-\lambda_*)_+^2 \leqslant A_n \to 0$, so $(u-\lambda_*)_+$ vanishes a.e. in $B_{1/2}$. That is, $\operatorname{ess\,sup}_{B_{1/2}} u \leqslant \lambda_* = C_*\Lambda^{d/4}$, which completes the proof. □

We next prove a lower bound for supersolutions which are large on a set of positive measure.

Lemma 4.10. *If $w \in H^1(B_1)$ satisfies $-\nabla\cdot\mathbf{a}\nabla w \leqslant 0$ in B_1, then w_+ satisfies the same inequality.*

Proof. Fix a nonnegative $\psi \in H_0^1(B_1) \cap L^\infty(B_1)$ and, for $\varepsilon > 0$, set $\eta_\varepsilon := \min\{w_+/\varepsilon, 1\} \in H^1(B_1)$, so that $0 \leqslant \eta_\varepsilon \leqslant 1$ and $\nabla\eta_\varepsilon = \varepsilon^{-1}\mathbf{1}_{\{0<w<\varepsilon\}}\nabla w$. Testing with $\psi\eta_\varepsilon \geqslant 0$ yields

$$0 \geqslant \int_{B_1} \nabla(\psi\eta_\varepsilon)\cdot\mathbf{a}\nabla w = \int_{B_1} \eta_\varepsilon \nabla\psi\cdot\mathbf{a}\nabla w + \varepsilon^{-1}\int_{\{0<w<\varepsilon\}} \psi\nabla w\cdot\mathbf{a}\nabla w \geqslant \int_{B_1} \eta_\varepsilon\nabla\psi\cdot\mathbf{a}\nabla w.$$

Since $\eta_\varepsilon \to \mathbf{1}_{\{w>0\}}$ pointwise as $\varepsilon\to 0$ and $\nabla w_+ = \mathbf{1}_{\{w>0\}}\nabla w$, dominated convergence gives $\int \nabla\psi\cdot\mathbf{a}\nabla w_+ \leqslant 0$. A general nonnegative $\psi \in H_0^1(B_1)$ is handled by applying this to $\min\{\psi, n\}$ and sending $n\to\infty$. □

Proposition 4.11 (Expansion of positivity)**.** *Let $d \geqslant 3$. There exists a constant $C(d) < \infty$ such that, for every $\theta \in (0, \frac{1}{2}]$ and every supersolution $u \in H^1(B_1)$ of*

$$-\nabla\cdot\mathbf{a}\nabla u \geqslant 0 \quad \text{in } B_1 \tag{4.47}$$

satisfying $u \geqslant 0$ in B_1 and

$$|\{u \geqslant 1\} \cap B_1| \geqslant (1-\theta)|B_1|, \tag{4.48}$$

we have the estimate

$$\inf_{B_{1/2}} u \geqslant \exp\Big(-C\theta^{\frac{1}{d}}\Lambda^{\frac{d+2}{4}}\Big). \tag{4.49}$$

Proof. Set $\rho := (\frac{3}{4})^{1/d} \in [\frac{3}{4}, 1)$, so that $\rho^{-d} = \frac{4}{3}$ and ρ depends only on d. Fix $\delta > 0$. The function $u + \delta$ is again a supersolution, $\{u + \delta \geqslant 1\} \supseteq \{u \geqslant 1\}$, and $\inf_{B_1}(u + \delta) \geqslant \delta > 0$. We will show that the conclusion holds for $u + \delta$ with a constant independent of δ; sending $\delta \to 0$ then yields the proposition. Thus we may assume that $\inf_{B_1} u > 0$, so that $v := -\log u \in H^1(B_1)$ (the logarithm is Lipschitz on $[\inf_{B_1} u, \infty)$) and $\nabla v = -u^{-1}\nabla u$.

Step 1: v is a subsolution. Fix a nonnegative $\psi \in H^1_0(B_1) \cap L^\infty(B_1)$. Since u is bounded below, $\psi/u \in H^1_0(B_1)$ is nonnegative and $\nabla(\psi/u) = u^{-1}\nabla\psi - \psi u^{-2}\nabla u$. Therefore

$$\begin{aligned}\int_{B_1} \nabla\psi \cdot \mathbf{a}\nabla v &= -\int_{B_1} u^{-1}\nabla\psi \cdot \mathbf{a}\nabla u \\ &= -\int_{B_1} \psi u^{-2}\nabla u \cdot \mathbf{a}\nabla u - \int_{B_1} \nabla\big(u^{-1}\psi\big) \cdot \mathbf{a}\nabla u \leqslant 0,\end{aligned}$$

where the last term is nonpositive because u is a supersolution. A general nonnegative $\psi \in H^1_0(B_1)$ is handled by truncation as in Lemma 4.10. Thus

$$-\nabla \cdot \mathbf{a}\nabla v \leqslant 0 \quad \text{in } B_1, \tag{4.50}$$

and, by Lemma 4.10, v_+ is also a subsolution.

Step 2: energy bound for v. Let $\varphi \in W^{1,\infty}_0(B_1)$ satisfy $\mathbf{1}_{B_\rho} \leqslant \varphi \leqslant \mathbf{1}_{B_{(1+\rho)/2}}$ and $\|\nabla\varphi\|_{L^\infty} \leqslant 4(1-\rho)^{-1}$. Testing the inequality for u with $\varphi^2 u^{-1} \in H^1_0(B_1)$ and using $\nabla u/u = -\nabla v$, we find

$$0 \leqslant \int_{B_1} \nabla\left(\frac{\varphi^2}{u}\right) \cdot \mathbf{a}\nabla u = -\int_{B_1} \varphi^2 \nabla v \cdot \mathbf{a}\nabla v - 2\int_{B_1} \varphi\nabla\varphi \cdot \mathbf{a}\nabla v.$$

Set $X := \int \varphi^2 \nabla v \cdot \mathbf{a}\nabla v$. Since $|\mathbf{a}e| \leqslant \Lambda^{\frac{1}{2}}(e \cdot \mathbf{a}e)^{\frac{1}{2}}$, the Cauchy-Schwarz inequality gives $X \leqslant 2\Lambda^{\frac{1}{2}}\|\nabla\varphi\|_{L^2} X^{\frac{1}{2}}$, that is, $X \leqslant 4\Lambda\|\nabla\varphi\|^2_{L^2}$. As $e \cdot \mathbf{a}e \geqslant |e|^2$ and $\varphi = 1$ on B_ρ, we obtain

$$\|\nabla v\|^2_{L^2(B_\rho)} \leqslant X \leqslant 64\Lambda(1-\rho)^{-2}|B_1| \leqslant C\Lambda. \tag{4.51}$$

Note that there is no dependence on θ here, because ρ is fixed.

Step 3: Poincaré inequality for functions vanishing on a set of positive density. Let $w := v_+$, so that $w = 0$ exactly on $\{u \geqslant 1\}$ and $|\nabla w| \leqslant |\nabla v|$. Define $E := \{w = 0\} \cap B_\rho$ and $A := \{w > 0\} \cap B_\rho = B_\rho \backslash E$. Since $A \subseteq B_1 \backslash \{u \geqslant 1\}$, we have

$$|A| \leqslant \theta|B_1| = \theta\rho^{-d}|B_\rho| = \tfrac{4}{3}\theta|B_\rho| \leqslant \tfrac{2}{3}|B_\rho|,$$

and therefore $|E| \geqslant \frac{1}{3}|B_\rho|$. We rescale B_ρ to B_1; since ρ depends only on d, all constants below depend only on d. Proposition 1.10, in the form (1.15) with density $\frac{1}{3}$, together with $(w)_E = 0$, gives

$$\big|(w)_{B_\rho}\big| = \big|(w)_{B_\rho} - (w)_E\big| \leqslant C\|\nabla w\|_{L^2(B_\rho)}.$$

By the Sobolev-Poincaré inequality (1.13) with $p = 2$ (here $d \geqslant 3$ and $2^* = \frac{2d}{d-2}$),

$$\big\|w - (w)_{B_\rho}\big\|_{L^{2^*}(B_\rho)} \leqslant C\|\nabla w\|_{L^2(B_\rho)}.$$

The triangle inequality yields $\|w\|_{L^{2^*}(B_\rho)} \leqslant C\|\nabla w\|_{L^2(B_\rho)}$. Since w vanishes on E and $\frac{1}{2} - \frac{1}{2^*} = \frac{1}{d}$, Hölder's inequality gives

$$\|w\|_{L^2(B_\rho)} = \|w\|_{L^2(A)} \leqslant |A|^{\frac{1}{d}} \|w\|_{L^{2^*}(B_\rho)} \leqslant C\theta^{\frac{1}{d}} \|\nabla v\|_{L^2(B_\rho)}. \tag{4.52}$$

The gain of the factor $\theta^{\frac{1}{d}}$ thus comes from the smallness of $|A|$ (through Hölder's inequality), not from the density of E, which is only used to control the mean.

Step 4: conclusion. Let $x \in B_{1/2}$. Then $B_{1/4}(x) \subseteq B_{3/4}$, and the function $y \mapsto v_+(x+y/4)$ is a subsolution in B_1 for the coefficient field $\mathbf{a}(x + \cdot/4)$, which satisfies the same ellipticity bounds with the same Λ. Applying Proposition 4.9 to it gives

$$\sup_{B_{1/8}(x)} v_+ \leqslant C\Lambda^{\frac{d}{4}} \|v_+\|_{L^2(B_{1/4}(x))} \leqslant C\Lambda^{\frac{d}{4}} \|v_+\|_{L^2(B_{3/4})}.$$

Covering $B_{1/2}$ by such balls and using $\frac{3}{4} \leqslant \rho$, (4.52) and (4.51), we get

$$\sup_{B_{1/2}} v \leqslant \sup_{B_{1/2}} v_+ \leqslant C\Lambda^{\frac{d}{4}} \|v_+\|_{L^2(B_\rho)} \leqslant C\theta^{\frac{1}{d}} \Lambda^{\frac{d}{4}} \|\nabla v\|_{L^2(B_\rho)} \leqslant C\theta^{\frac{1}{d}} \Lambda^{\frac{d+2}{4}}.$$

Since $\inf_{B_{1/2}} u = \exp(-\sup_{B_{1/2}} v)$, this is (4.49). □

The previous two propositions imply Theorem 4.8. Throughout, suprema and infima are essential, and the Hölder estimate is for the continuous representative, which is obtained from the bounds on the oscillation below.

Proof of Theorem 4.8. Let $d \geqslant 3$. Since the equation is invariant under translation and scaling, in the sense that $\mathbf{a}(x_0 + r\,\cdot)$ satisfies (4.1) with the same Λ, we may apply all estimates below on any ball after rescaling it to B_1. We first argue that there is $C(d) < \infty$ such that, for every solution $u \in H^1(B_1)$,

$$\operatorname*{osc}_{B_{1/2}} u \leqslant \left(1 - \exp\left(-C\Lambda^{\frac{d+2}{4}}\right)\right) \operatorname*{osc}_{B_1} u. \tag{4.53}$$

To prove (4.53), we may suppose that $\operatorname{osc}_{B_1} u < \infty$ and, after subtracting a constant and multiplying by a constant (both operations preserve solutions), that $\inf_{B_1} u = 0$ and $\sup_{B_1} u = 1$. In the case that $\left|\{u \geqslant \frac{1}{2}\} \cap B_1\right| \geqslant \frac{1}{2}|B_1|$, the function $2u \geqslant 0$ is a supersolution with $|\{2u \geqslant 1\} \cap B_1| \geqslant \frac{1}{2}|B_1|$, so Proposition 4.11 with $\theta = \frac{1}{2}$ yields $\inf_{B_{1/2}} 2u \geqslant \exp(-C\Lambda^{\frac{d+2}{4}})$. Using $\Lambda \geqslant 1$ to absorb the factor $\frac{1}{2}$ into the constant, we obtain

$$\operatorname*{osc}_{B_{1/2}} u \leqslant 1 - \inf_{B_{1/2}} u \leqslant 1 - \exp\left(-C\Lambda^{\frac{d+2}{4}}\right). \tag{4.54}$$

Otherwise we have that $\left|\{u \leqslant \frac{1}{2}\} \cap B_1\right| \geqslant \frac{1}{2}|B_1|$, in which case the function $2(1-u) \geqslant 0$ is a solution (constants are solutions) with $|\{2(1-u) \geqslant 1\} \cap B_1| \geqslant \frac{1}{2}|B_1|$, so we may again apply Proposition 4.11 with $\theta = \frac{1}{2}$, this time to $2(1-u)$ (with the same absorption of the factor $\frac{1}{2}$ into C as in the first case), to obtain

$$\operatorname*{osc}_{B_{1/2}} u = \operatorname*{osc}_{B_{1/2}} (1-u) \leqslant 1 - \inf_{B_{1/2}} (1-u) \leqslant 1 - \exp\left(-C\Lambda^{\frac{d+2}{4}}\right). \tag{4.55}$$

In either case we see that (4.53) holds.

Set $\delta := \exp(-C\Lambda^{\frac{d+2}{4}})$ with C as in (4.53), enlarged if necessary so that $\delta \leqslant \frac{1}{2}$, and $\alpha := \min\{1, \log_2 \frac{1}{1-\delta}\}$. Then $\alpha \geqslant \delta/\ln 2 \geqslant \delta$ and $1-\delta \leqslant 2^{-\alpha}$. Fix $x_0 \in B_{1/2}$, so that $B_{1/2}(x_0) \subseteq B_1$. By Proposition 4.9 applied to $\pm u$ in $B_{1/2}(x_0)$ (rescaled to B_1),

$$M := \operatorname*{osc}_{B_{1/4}(x_0)} u \leqslant 2 \sup_{B_{1/4}(x_0)} |u| \leqslant C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_{1/2}(x_0))} \leqslant C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_1)}.$$

Applying (4.53) to $u(x_0 + 2^{-j}\cdot/4)$, $j = 0, 1, 2, \dots$ successively, we get $\operatorname{osc}_{B_{2^{-j}/4}(x_0)} u \leqslant (1-\delta)^j M \leqslant 2^{-\alpha j} M$. Given $r \in (0, \frac{1}{4}]$, choose the integer $j \geqslant 0$ with $2^{-j-1}/4 < r \leqslant 2^{-j}/4$; then $2^{-j} < 8r$ and so

$$\operatorname*{osc}_{B_r(x_0)} u \leqslant \operatorname*{osc}_{B_{2^{-j}/4}(x_0)} u \leqslant (8r)^\alpha M \leqslant C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_1)} r^\alpha. \tag{4.56}$$

Now let $x, y \in B_{1/2}$ with $r := |x-y| > 0$. If $r < \frac{1}{8}$, then $y \in B_{2r}(x)$ and $2r \leqslant \frac{1}{4}$, so (4.56) with $x_0 = x$ gives $|u(x)-u(y)| \leqslant 2C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_1)} r^\alpha$. If $r \geqslant \frac{1}{8}$, then $|u(x)-u(y)| \leqslant 2\sup_{B_{1/2}} |u| \leqslant 2 \cdot 8^\alpha r^\alpha \sup_{B_{1/2}} |u| \leqslant 16 r^\alpha \sup_{B_{1/2}} |u|$. Since $\|u\|_{L^\infty(B_{1/2})} \leqslant C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_1)}$ by Proposition 4.9 applied to $\pm u$, in both cases

$$[u]_{C^{0,\alpha}(B_{1/2})} \leqslant C\Lambda^{\frac{d}{4}} \|u\|_{L^2(B_1)},$$

and adding the L^∞ bound yields (4.36). □

The following proposition shows that the power $\Lambda^{\frac{d}{4}}$ in Proposition 4.9 cannot be replaced by Λ^γ with $\gamma < \frac{d-1}{4}$. We do not know whether $\frac{d}{4}$, $\frac{d-1}{4}$ or a number in between is the optimal exponent.

Proposition 4.12 (Lower bound $\Lambda^{\frac{d-1}{4}}$ for the exponent in Proposition 4.9)**.** *Let $d \geqslant 2$. There exists $c(d) > 0$ such that, for every $\Lambda \geqslant 1$, there exist a constant matrix $\mathbf{a}$ satisfying (4.1) and a smooth function $u > 0$ on $\mathbb{R}^d$ such that $u \in H^1(B_1)$, $-\nabla \cdot \mathbf{a}\nabla u \leqslant 0$ in $\mathbb{R}^d$ and*

$$\operatorname*{ess\,sup}_{B_{1/2}} u_+ \geqslant u(0) \geqslant c\,\Lambda^{\frac{d-1}{4}} \|u\|_{L^2(B_1)}. \tag{4.57}$$

Consequently, if $d \geqslant 3$, $\gamma \in \mathbb{R}$ and $C < \infty$ are such that (4.39) holds with $C\Lambda^\gamma$ in place of $C\Lambda^{d/4}$, for every $\Lambda \geqslant 1$, every $\mathbf{a}$ satisfying (4.1) and every subsolution $u \in H^1(B_1)$, then $\gamma \geqslant \frac{d-1}{4}$.

Proof. Fix $\Lambda \geqslant 1$. Write $x = (x', x_d)$, set $s := (1 + \Lambda|x'|^2)^{1/2} \geqslant 1$, which is smooth, and define

$$\phi(x) := \tfrac{1}{2}x_d^2 - \tfrac{1}{d-1}s, \qquad u := e^\phi, \qquad \mathbf{a} := \operatorname{diag}(1, \dots, 1, \Lambda).$$

Then $\mathbf{a} \geqslant I$ and $\mathbf{a}^{-1} \geqslant \Lambda^{-1} I$, so (4.1) holds, and $u > 0$ is smooth, hence in $H^1(B_1)$.

Step 1: u is a subsolution. Since $\mathbf{a}$ is constant, $\frac{\nabla\cdot\mathbf{a}\nabla u}{u} = \Delta'\phi + |\nabla'\phi|^2 + \Lambda(\partial_{dd}\phi + (\partial_d\phi)^2)$, where ∇', Δ' act in x'. We have $\partial_d\phi = x_d$, $\partial_{dd}\phi = 1$, $\nabla'\phi = -\frac{\Lambda x'}{(d-1)s}$ and

$$\Delta'\phi = -\frac{1}{d-1}\Big(\frac{(d-1)\Lambda}{s} - \frac{\Lambda^2|x'|^2}{s^3}\Big) = -\frac{\Lambda}{s} + \frac{\Lambda^2|x'|^2}{(d-1)s^3}.$$

Therefore

$$\frac{\nabla\cdot\mathbf{a}\nabla u}{u} = \Lambda\Big(1+x_d^2-\frac{1}{s}\Big)+\frac{\Lambda^2|x'|^2}{d-1}\Big(\frac{1}{s^3}+\frac{1}{(d-1)s^2}\Big)\geqslant 0,$$

because $s\geqslant 1$. Thus $\nabla\cdot\mathbf{a}\nabla u\geqslant 0$ pointwise, and for every $\varphi\in C_c^\infty(\mathbb{R}^d)$ with $\varphi\geqslant 0$, integration by parts gives $\int\mathbf{a}\nabla u\cdot\nabla\varphi = -\int\varphi\,\nabla\cdot\mathbf{a}\nabla u\leqslant 0$; by density this holds for nonnegative $\varphi\in H_0^1(B_1)$, which is $-\nabla\cdot\mathbf{a}\nabla u\leqslant 0$ in the weak sense.

Step 2: the bounds. We have $u(0)=\exp(-\frac{1}{d-1})\geqslant e^{-1}$, and since u is continuous, $\operatorname{ess\,sup}_{B_{1/2}} u_+\geqslant u(0)$. On B_1 we have $x_d^2\leqslant 1$ and $|x'|<1$, $|x_d|<1$, and $s\geqslant\Lambda^{1/2}|x'|$, so

$$u^2=\exp\Big(x_d^2-\tfrac{2}{d-1}s\Big)\leqslant e\exp\Big(-\tfrac{2}{d-1}\Lambda^{1/2}|x'|\Big)$$

and hence, with $k:=\frac{2}{d-1}\Lambda^{1/2}$,

$$\|u\|_{L^2(B_1)}^2\leqslant 2e\int_{\mathbb{R}^{d-1}} e^{-k|x'|}\,dx' = 2e\,|S^{d-2}|\,(d-2)!\,k^{-(d-1)} = C_1(d)\,\Lambda^{-\frac{d-1}{2}},$$

where $|S^0|:=2$. Thus $\|u\|_{L^2(B_1)}\leqslant C_1^{1/2}\Lambda^{-\frac{d-1}{4}}$, and (4.57) holds with $c:=e^{-1}C_1^{-1/2}$.

Step 3: the exponent. Under the stated assumption, applying (4.39) with $C\Lambda^\gamma$ to the pair $(\mathbf{a},u)$ above and using (4.57) yields $c\Lambda^{\frac{d-1}{4}}\|u\|_{L^2(B_1)}\leqslant C\Lambda^\gamma\|u\|_{L^2(B_1)}$. Since $\|u\|_{L^2(B_1)}>0$, we get $\Lambda^{\frac{d-1}{4}-\gamma}\leqslant C/c$ for all $\Lambda\geqslant 1$, which forces $\gamma\geqslant\frac{d-1}{4}$. □

4.4 Moser iteration

We next prove a more precise version of Proposition 4.9 by using a Moser iteration. As in the previous subsection, we assume throughout that $d\geqslant 3$, so that the Sobolev inequality (with exponent $2^*=\frac{2d}{d-2}$) is available on $W_0^{1,2}$; we write $\gamma:=\frac{d}{d-2}$. Recall also that $\Lambda\geqslant 1$.

Proposition 4.13 (L^∞ estimate for subsolutions)**.** *Let $d\geqslant 3$. There exists $C(d)<\infty$ such that, for every $u\in H^1(B_1)$ satisfying*

$$-\nabla\cdot\mathbf{a}\nabla u\leqslant 0\quad \text{in } B_1, \tag{4.58}$$

and every $p\in(1,\infty)$, we have the estimate

$$\Big(\sup_{B_{1/2}} u_+\Big)^p\leqslant C\Lambda^{\frac{d}{2}}\Big(\frac{p}{p-1}\Big)^d\|u\|_{L^p(B_1)}^p. \tag{4.59}$$

Proof. Step 0. Reductions. The function u_+ is again a subsolution. Indeed, for $\phi\in C_c^\infty(B_1)$ with $\phi\geqslant 0$ and $\varepsilon>0$, the function $\phi\min\{u_+/\varepsilon,1\}$ is a nonnegative element of $H_0^1(B_1)$, and testing with it gives

$$\int_{B_1}\min\{u_+/\varepsilon,1\}\,\nabla\phi\cdot\mathbf{a}\nabla u+\frac{1}{\varepsilon}\int_{B_1\cap\{0<u<\varepsilon\}}\phi\,\nabla u\cdot\mathbf{a}\nabla u\leqslant 0.$$

The second term is nonnegative, and letting $\varepsilon\to 0$ in the first (dominated convergence) yields $\int_{B_1}\nabla\phi\cdot\mathbf{a}\nabla u_+\leqslant 0$, since $\nabla u_+=\mathbf{1}_{\{u>0\}}\nabla u$. Consequently, for every $\delta>0$, the

function $u_+ + \delta \in H^1(B_1)$ is a subsolution with $u_+ + \delta \geqslant \delta$. We will prove (4.59) with $u_+ + \delta$ in place of u and with a constant independent of δ; since $\|u_+ + \delta\|_{L^p(B_1)} \to \|u_+\|_{L^p(B_1)}$ as $\delta \to 0$ (if the latter is infinite, there is nothing to prove), the estimate for u follows.

We therefore assume from now on that $u \geqslant \delta > 0$ in B_1. By Proposition 4.9 (applied in balls contained in B_1, after translation and scaling) and a covering of $\overline{B}_{3/4}$, we have $u \in L^\infty(B_{3/4})$, say $u \leqslant M$ on $B_{3/4}$. In particular, for every $p > 1$ and every $\phi \in C_c^\infty(B_{3/4})$, the functions $\phi^2 u^{p-1}$ and $\phi u^{p/2}$ belong to $H_0^1(B_1)$, because $t \mapsto t^{p-1}$ and $t \mapsto t^{p/2}$ are Lipschitz on $[\delta, M]$.

Step 1. The test function computation. Fix $p > 1$ and $\phi \in C_c^\infty(B_{3/4})$, and write $w := u^{p/2}$. Testing the differential inequality with the nonnegative function $\phi^2 u^{p-1}$ yields

$$\begin{aligned} 0 &\geqslant \int_{B_1} \nabla\big(\phi^2 u^{p-1}\big) \cdot \mathbf{a}\nabla u \\ &= (p-1)\int_{B_1} \phi^2 u^{p-2} \nabla u \cdot \mathbf{a}\nabla u + \int_{B_1} 2\phi u^{p-1} \nabla\phi \cdot \mathbf{a}\nabla u \\ &= \frac{4(p-1)}{p^2}\int_{B_1} \phi^2 \nabla w \cdot \mathbf{a}\nabla w + \frac{4}{p}\int_{B_1} \phi w \nabla\phi \cdot \mathbf{a}\nabla w. \end{aligned}$$

Let $A := \int_{B_1} \phi^2 \nabla w \cdot \mathbf{a}\nabla w$. By (4.2), $|\mathbf{a}\nabla w| \leqslant \Lambda^{1/2}(\nabla w \cdot \mathbf{a}\nabla w)^{1/2}$, and thus, by the Cauchy-Schwarz inequality,

$$\frac{4(p-1)}{p^2} A \leqslant \frac{4}{p}\Lambda^{\frac12}\left(\int_{B_1} |\nabla\phi|^2 w^2\right)^{\frac12} A^{\frac12}.$$

Rearranging, and using $\mathbf{a} \geqslant \mathrm{I}_d$ to bound $\int \phi^2 |\nabla w|^2 \leqslant A$, we get

$$\int_{B_1} \phi^2 \big|\nabla\big(u^{\frac{p}{2}}\big)\big|^2 \leqslant \frac{\Lambda p^2}{(p-1)^2}\int_{B_1} |\nabla\phi|^2 u^p. \tag{4.60}$$

Since $|\nabla(\phi w)|^2 \leqslant 2\phi^2 |\nabla w|^2 + 2|\nabla\phi|^2 w^2$, this yields

$$\int_{B_1} \big|\nabla\big(\phi u^{\frac{p}{2}}\big)\big|^2 \leqslant \left(\frac{2\Lambda p^2}{(p-1)^2} + 2\right)\int_{B_1} |\nabla\phi|^2 u^p. \tag{4.61}$$

By the Sobolev inequality in $W_0^{1,2}(B_1)$, we get

$$\left(\int_{B_1} \phi^{2^*} u^{\frac{pd}{d-2}}\right)^{\frac{d-2}{d}} = \left(\int_{B_1} \left|\phi u^{\frac{p}{2}}\right|^{2^*}\right)^{\frac{2}{2^*}} \leqslant C\left(\frac{\Lambda p^2}{(p-1)^2} + 1\right)\int_{B_1} |\nabla\phi|^2 u^p.$$

Let $\frac12 \leqslant r < s \leqslant \frac34$ and select the cutoff function $\phi \in C_c^\infty(B_{3/4})$ so that $\mathbf{1}_{B_r} \leqslant \phi \leqslant \mathbf{1}_{B_s}$ and $\|\nabla\phi\|_{L^\infty(B_1)} \leqslant 2(s-r)^{-1}$ (if $s = \frac34$, take the support in a slightly smaller ball, which is harmless), to obtain

$$\left(\int_{B_r} u^{\frac{pd}{d-2}}\right)^{\frac{d-2}{d}} \leqslant \frac{C}{(s-r)^2}\left(\frac{\Lambda p^2}{(p-1)^2} + 1\right)\int_{B_s} u^p.$$

That is, for every r, s with $\frac{1}{2} \leqslant r < s \leqslant \frac{3}{4}$,

$$\|u\|_{L^{\frac{pd}{d-2}}(B_r)} \leqslant \left(\frac{C}{(s-r)^2} \left(\frac{\Lambda p^2}{(p-1)^2} + 1 \right) \right)^{\frac{1}{p}} \|u\|_{L^p(B_s)}. \tag{4.62}$$

Step 2. The iteration. We start from $p_0 > 1$ and let

$$p_{n+1} := \gamma\, p_n \quad \text{and} \quad r_n := \frac{1}{2} + 2^{-n-2}, \quad n \in \mathbb{N}. \tag{4.63}$$

(This differs from the choice (4.73) below only in that $r_0 = \frac{3}{4}$ rather than 1; here we need B_{r_0} to lie where u is bounded.) Then $r_0 = \frac{3}{4}$, $r_n \downarrow \frac{1}{2}$ and $r_n - r_{n+1} = 2^{-n-3}$, so $(r_n - r_{n+1})^{-2} = 64 \cdot 4^n$. Applying (4.62) with $r = r_{n+1}$, $s = r_n$ and $p = p_n$, we get

$$\|u\|_{L^{p_{n+1}}(B_{r_{n+1}})} \leqslant \left(C 4^n \left(\frac{\Lambda p_n^2}{(p_n-1)^2} + 1 \right) \right)^{\frac{1}{p_n}} \|u\|_{L^{p_n}(B_{r_n})}. \tag{4.64}$$

Iterating this inequality yields, for every $N \in \mathbb{N}$,

$$\|u\|_{L^{p_N}(B_{1/2})} \leqslant \prod_{n=0}^{N-1} \left(C 4^n \left(\frac{\Lambda p_n^2}{(p_n-1)^2} + 1 \right) \right)^{\frac{1}{p_n}} \|u\|_{L^{p_0}(B_{3/4})}.$$

Since $u \in L^\infty(B_{1/2})$ and $|B_{1/2}| < \infty$, we have $\|u\|_{L^{p_N}(B_{1/2})} \to \|u\|_{L^\infty(B_{1/2})}$ as $N \to \infty$, and so

$$\|u\|_{L^\infty(B_{1/2})} \leqslant \prod_{n=0}^{\infty} \left(C 4^n \left(\frac{\Lambda p_n^2}{(p_n-1)^2} + 1 \right) \right)^{\frac{1}{p_n}} \|u\|_{L^{p_0}(B_{3/4})}. \tag{4.65}$$

To estimate the product on the right side, we first note that, since $\Lambda \geqslant 1$ and $\frac{p}{p-1} > 1$, we have $\frac{\Lambda p^2}{(p-1)^2} + 1 \leqslant 2\Lambda \frac{p^2}{(p-1)^2}$, and that $p \mapsto \frac{p}{p-1}$ is decreasing on $(1, \infty)$, so that $\frac{p_n}{p_n - 1} \leqslant \frac{p_0}{p_0 - 1}$. We then take the logarithm of the product:

$$\begin{aligned}
\log \prod_{n=0}^{\infty} \left(C 4^n \left(\frac{\Lambda p_n^2}{(p_n-1)^2} + 1 \right) \right)^{\frac{1}{p_n}} &= \sum_{n=0}^{\infty} \frac{1}{p_n} \log \left(C 4^n \left(\frac{\Lambda p_n^2}{(p_n-1)^2} + 1 \right) \right) \\
&\leqslant \sum_{n=0}^{\infty} \frac{1}{p_n} \left(C(n+1) + \log \Lambda + \log \left(\frac{p_0^2}{(p_0-1)^2} \right) \right) \\
&\leqslant C \sum_{n=0}^{\infty} \frac{n+1}{p_n} + \left(\sum_{n=0}^{\infty} \frac{1}{p_n} \right) \left(\log \Lambda + \log \left(\frac{p_0^2}{(p_0-1)^2} \right) \right).
\end{aligned}$$

Since $\sum_{n=0}^\infty \frac{1}{p_n} = \frac{1}{p_0} \sum_{n=0}^\infty \gamma^{-n} = \frac{d}{2p_0}$ and $\sum_{n=0}^\infty \frac{n+1}{p_n} \leqslant C p_0^{-1}$, we finally obtain

$$\begin{aligned}
\|u\|_{L^\infty(B_{1/2})} &\leqslant \exp \left(C p_0^{-1} + \frac{d}{2p_0} \left(\log \Lambda + \log \left(\frac{p_0^2}{(p_0-1)^2} \right) \right) \right) \|u\|_{L^{p_0}(B_{3/4})} \\
&= C^{\frac{1}{p_0}} \left(\frac{p_0}{p_0 - 1} \right)^{\frac{d}{p_0}} \Lambda^{\frac{d}{2p_0}} \|u\|_{L^{p_0}(B_{3/4})}.
\end{aligned}$$

Raising to the power p_0 and using $\|u\|_{L^{p_0}(B_{3/4})} \leqslant \|u\|_{L^{p_0}(B_1)}$ gives (4.59) for the function $u \geqslant \delta$, with a constant independent of δ. By Step 0, this completes the proof. □

We next apply the Moser iteration method to positive supersolutions.

Proposition 4.14. *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that, for every $u \in H^1(B_1)$ satisfying $u > 0$ in B_1 and*

$$-\nabla \cdot \mathbf{a}\nabla u \geqslant 0 \quad \text{in } B_1, \tag{4.66}$$

and every $p \in (0,\infty)$, we have the estimate

$$\left(\Lambda p^2 + 1\right)^{-\frac{d}{2}} \|u^{-1}\|_{L^p(B_1)}^{-p} \leqslant C\big(\inf_{B_{1/2}} u\big)^p \tag{4.67}$$

and, for every $p, q \in (0,1)$ with $p \leqslant q$,

$$\|u\|_{L^{\frac{qd}{d-2}}(B_{1/2})}^p \leqslant C\left(\frac{\Lambda p^2}{(1-q)^2} + 1\right)^{\frac{d+2}{2}} \|u\|_{L^p(B_1)}^p. \tag{4.68}$$

Moreover, the exponent $\frac{d+2}{2}$ in (4.68) can be replaced by $\frac{d}{2}$ if $p = q\gamma^{-m}$ for some $m \in \mathbb{N}$, where $\gamma = \frac{d}{d-2}$.

Proof. We may make the qualitative assumption that $\inf_{B_1} u > 0$. Otherwise we consider $u + \delta$ in place of u, which is again a positive supersolution in $H^1(B_1)$ bounded below by δ, obtain estimates for $u + \delta$ which do not depend on δ, and then conclude by sending $\delta \to 0$. Indeed, $(u+\delta)^{-1} \leqslant u^{-1}$ implies that the left side of (4.67) for u is at most that for $u + \delta$; moreover $\inf_{B_{1/2}}(u+\delta)^p \to \inf_{B_{1/2}} u^p$, and $\|u+\delta\|_{L^p(B_1)} \to \|u\|_{L^p(B_1)}$ (finite since $u \in L^2(B_1)$ and $p < 1$), while the left side of (4.68) is monotone in u.

Step 1. The test function computation. The idea here is that we will mimic the computation from the proof of Proposition 4.13. In that argument we tested the equation with $\phi^2 u^{p-1}$ for $p > 1$, whereas here we will use $p < 1$ (although we replace p with $-p$).

Let $p \in (-1,\infty)\backslash\{0\}$, $\phi \in C_c^\infty(B_1)$, and set $w := u^{-p/2}$. Since $u \geqslant \delta$ and $1 + p > 0$, the function $t \mapsto t^{-(1+p)}$ is bounded and Lipschitz on $[\delta,\infty)$, and $t \mapsto t^{-p/2}$ is Lipschitz on $[\delta,\infty)$ with sublinear growth (recall that $|p| < 1$ if $p < 0$); therefore $\phi^2 u^{-(1+p)}$ and ϕw belong to $H_0^1(B_1)$ (no upper bound for u is needed). Testing the differential inequality with the nonnegative function $\phi^2 u^{-(1+p)}$ yields

$$\begin{aligned}
0 &\leqslant \int_{B_1} \nabla\left(\phi^2 u^{-(1+p)}\right) \cdot \mathbf{a}\nabla u \\
&= -(1+p)\int_{B_1} \phi^2 u^{-p-2} \nabla u \cdot \mathbf{a}\nabla u + \int_{B_1} 2\phi u^{-p-1} \nabla\phi \cdot \mathbf{a}\nabla u \\
&= -\frac{4(p+1)}{p^2}\int_{B_1} \phi^2 \nabla w \cdot \mathbf{a}\nabla w - \frac{4}{p}\int_{B_1} \phi w \nabla\phi \cdot \mathbf{a}\nabla w.
\end{aligned}$$

As in the previous proof, with $A := \int_{B_1} \phi^2 \nabla w \cdot \mathbf{a}\nabla w$, (4.2) and the Cauchy-Schwarz inequality give $\frac{4(1+p)}{p^2} A \leqslant \frac{4}{|p|}\Lambda^{\frac{1}{2}}\big(\int_{B_1} |\nabla\phi|^2 w^2\big)^{\frac{1}{2}} A^{\frac{1}{2}}$, and hence

$$\int_{B_1} \phi^2 \left|\nabla\left(u^{-\frac{p}{2}}\right)\right|^2 \leqslant \frac{\Lambda p^2}{(1+p)^2}\int_{B_1} |\nabla\phi|^2 u^{-p}. \tag{4.69}$$

This yields

$$\int_{B_1} \left|\nabla\left(\phi u^{-\frac{p}{2}}\right)\right|^2 \leqslant \left(\frac{2\Lambda p^2}{(1+p)^2} + 2\right) \int_{B_1} |\nabla\phi|^2 u^{-p}. \tag{4.70}$$

The Sobolev inequality gives

$$\left(\int_{B_1} \phi^{2^*} u^{-\frac{pd}{d-2}}\right)^{\frac{d-2}{d}} = \left(\int_{B_1} \left|\phi u^{-\frac{p}{2}}\right|^{2^*}\right)^{\frac{2}{2^*}} \leqslant C\left(\frac{\Lambda p^2}{(1+p)^2} + 1\right) \int_{B_1} |\nabla\phi|^2 u^{-p}.$$

Splitting into the two cases (depending on the sign of p; in the case $p < 0$ we replace p by $-p \in (0,1)$), fixing $0 < r < s \leqslant 1$ and taking $\phi \in C_c^\infty(B_1)$ such that $\mathbf{1}_{B_r} \leqslant \phi \leqslant \mathbf{1}_{B_s}$ and $\|\nabla\phi\|_{L^\infty(B_1)} \leqslant 2(s-r)^{-1}$ (for $s = 1$ take the support in a slightly smaller ball and let it increase to B_1), we get

$$\left\|u^{-1}\right\|_{L^{\frac{pd}{d-2}}(B_r)} \leqslant C^{\frac{1}{p}} \left(\frac{\Lambda p^2}{(1+p)^2} + 1\right)^{\frac{1}{p}} (s-r)^{-\frac{2}{p}} \left\|u^{-1}\right\|_{L^p(B_s)} \quad \text{if } p \in (0,\infty), \tag{4.71}$$

and

$$\|u\|_{L^{\frac{pd}{d-2}}(B_r)} \leqslant C^{\frac{1}{p}} \left(\frac{\Lambda p^2}{(1-p)^2} + 1\right)^{\frac{1}{p}} (s-r)^{-\frac{2}{p}} \|u\|_{L^p(B_s)} \quad \text{if } p \in (0,1). \tag{4.72}$$

We now need to perform two separate iterations. Let p_n and r_n be given by

$$p_{n+1} := \gamma\, p_n \quad \text{and} \quad r_n := \frac{1}{2}\left(1 + 2^{-n}\right), \quad n \in \mathbb{N}, \tag{4.73}$$

so that $r_n \downarrow \frac{1}{2}$ and $(r_n - r_{n+1})^{-2} = 16 \cdot 4^n$.

Step 2. The iteration of (4.71). We take $p_0 \in (0,\infty)$ and define $\{p_n\}$ and $\{r_n\}$ as in (4.73). Applying (4.71) with $r = r_{n+1}$, $s = r_n$, $p = p_n$, and iterating (recall that $u^{-1} \leqslant \delta^{-1}$ is bounded, so that $\|u^{-1}\|_{L^{p_N}(B_{1/2})} \to \|u^{-1}\|_{L^\infty(B_{1/2})}$), we get

$$\left\|u^{-1}\right\|_{L^\infty(B_{1/2})} \leqslant \prod_{n=0}^{\infty} \left(C4^n \left(\frac{\Lambda p_n^2}{(1+p_n)^2} + 1\right)\right)^{\frac{1}{p_n}} \left\|u^{-1}\right\|_{L^{p_0}(B_1)}. \tag{4.74}$$

Since $p_n \leqslant \gamma^n p_0$, we have $\frac{\Lambda p_n^2}{(1+p_n)^2} + 1 \leqslant \gamma^{2n}(\Lambda p_0^2 + 1)$, and we estimate the product by

$$\begin{aligned} \log \prod_{n=0}^{\infty} \left(C4^n \left(\frac{\Lambda p_n^2}{(1+p_n)^2} + 1\right)\right)^{\frac{1}{p_n}} &\leqslant \sum_{n=0}^{\infty} \frac{1}{p_n} \log\left(C4^n \gamma^{2n}\left(\Lambda p_0^2 + 1\right)\right) \\ &\leqslant \sum_{n=0}^{\infty} \frac{1}{p_n} \log\left(C^{n+1}\left(\Lambda p_0^2 + 1\right)\right). \end{aligned}$$

Since $\sum_n p_n^{-1} = \frac{d}{2p_0}$ and $\sum_n (n+1)p_n^{-1} \leqslant C p_0^{-1}$, we get

$$\prod_{n=0}^{\infty} \left(C4^n \left(\frac{\Lambda p_n^2}{(1+p_n)^2} + 1\right)\right)^{\frac{1}{p_n}} \leqslant \exp\left(C p_0^{-1}\right)\left(\Lambda p_0^2 + 1\right)^{\frac{d}{2p_0}}. \tag{4.75}$$

Combining this with the estimate (4.74) and using $\|u^{-1}\|_{L^\infty(B_{1/2})}^{p_0} = (\inf_{B_{1/2}} u)^{-p_0}$, we get

$$\big(\inf_{B_{1/2}} u\big)^{-p_0} \leqslant C\big(\Lambda p_0^2 + 1\big)^{\frac{d}{2}} \|u^{-1}\|_{L^{p_0}(B_1)}^{p_0}. \tag{4.76}$$

This is (4.67).

Step 3. The iteration of (4.72). Fix $0 < p \leqslant q < 1$ and set

$$X := \frac{\Lambda p^2}{(1-q)^2} + 1.$$

Let $m \in \mathbb{N} \cup \{0\}$ be the largest integer such that $p' := q\gamma^{-m} \geqslant p$; then $p' \in [p, \gamma p)$.

Case $p < p'$. Define $p_0 := p$ and $p_n := q\gamma^{n-1-m}$ for $n \in \{1, \dots, m+1\}$, so that $p_1 = p'$, $p_{n+1} = \gamma p_n$ for $1 \leqslant n \leqslant m$, and $p_{m+1} = q$. All p_n belong to $(0, q]$, so that $p_n < 1$ and $(1-p_n)^{-2} \leqslant (1-q)^{-2}$. With r_n as in (4.73), we first apply (4.72) with $r = r_1$, $s = r_0$ and exponent $p_0 = p$, and then Hölder's inequality (using $p_1 = p' < \gamma p_0$ and $|B_1| < \infty$):

$$\|u\|_{L^{p_1}(B_{r_1})} \leqslant |B_1|^{\frac{1}{p_1} - \frac{1}{\gamma p_0}} \|u\|_{L^{\gamma p_0}(B_{r_1})} \leqslant C^{\frac{1}{p_0}} \left(\frac{\Lambda p_0^2}{(1-p_0)^2} + 1 \right)^{\frac{1}{p_0}} \|u\|_{L^{p_0}(B_{r_0})} \leqslant \big(CX\big)^{\frac{1}{p_0}} \|u\|_{L^{p_0}(B_{r_0})}.$$

(Here $|B_1|^{\frac{1}{p_1} - \frac{1}{\gamma p_0}} \leqslant \max\{1, |B_1|\}^{\frac{1}{p_0}}$ and we used $16 \leqslant C$.) For $n \in \{1, \dots, m+1\}$ we apply (4.72) with $r = r_{n+1}$, $s = r_n$ and exponent p_n, and obtain

$$\|u\|_{L^{p_{n+1}}(B_{r_{n+1}})} \leqslant \left(C 4^n \left(\frac{\Lambda p_n^2}{(1-p_n)^2} + 1 \right) \right)^{\frac{1}{p_n}} \|u\|_{L^{p_n}(B_{r_n})},$$

where $p_{m+2} := \gamma q = \frac{qd}{d-2}$. Since $p_n = p'\gamma^{n-1} \leqslant \gamma^n p$ for $n \geqslant 1$, we have

$$\frac{\Lambda p_n^2}{(1-p_n)^2} + 1 \leqslant \gamma^{2n} \frac{\Lambda p^2}{(1-q)^2} + 1 \leqslant \gamma^{2n} X.$$

Iterating, and using $r_{m+2} > \frac{1}{2}$, we get

$$\|u\|_{L^{\frac{qd}{d-2}}(B_{1/2})} \leqslant \prod_{n=0}^{m+1} \big(C^{n+1} X\big)^{\frac{1}{p_n}} \|u\|_{L^p(B_1)}.$$

Now $\sum_{n=0}^{m+1} p_n^{-1} \leqslant p^{-1} + \frac{1}{p'} \sum_{j \geqslant 0} \gamma^{-j} = p^{-1} + \frac{d}{2p'} \leqslant \frac{d+2}{2p}$ because $p' \geqslant p$, and likewise $\sum_{n=0}^{m+1} (n+1) p_n^{-1} \leqslant C p^{-1}$. Taking logarithms as in Step 2 yields

$$\|u\|_{L^{\frac{qd}{d-2}}(B_{1/2})} \leqslant \exp\big(Cp^{-1}\big) X^{\frac{d+2}{2p}} \|u\|_{L^p(B_1)},$$

which is (4.68).

Case $p = p'$, i.e. $p = q\gamma^{-m}$. Here we omit the first step: set $p_n := p\gamma^n$ for $n \in \{0, \dots, m\}$, so that $p_m = q$, and apply (4.72) with $r = r_{n+1}$, $s = r_n$ and exponent p_n for $n = 0, \dots, m$. As above, $\frac{\Lambda p_n^2}{(1-p_n)^2} + 1 \leqslant \gamma^{2n} X$, and now $\sum_n p_n^{-1} \leqslant \frac{d}{2p}$ and $\sum_n (n+1) p_n^{-1} \leqslant C p^{-1}$, so that

$$\|u\|_{L^{\frac{qd}{d-2}}(B_{1/2})} \leqslant \exp\big(Cp^{-1}\big) X^{\frac{d}{2p}} \|u\|_{L^p(B_1)}.$$

Raising to the power p gives (4.68) with exponent $\frac{d}{2}$ in this case, and in the previous case with exponent $\frac{d+2}{2}$. □

Proposition 4.15 (Expansion of positivity for small θ)**.** *Let $d \geqslant 3$. There exist $c(d) \in (0, \frac{1}{4}]$ and $C(d) < \infty$ such that, for every $\Lambda \geqslant 1$, every $\mathbf{a}$ satisfying (4.1), every $\theta \in (0, c]$ and every $u \in H^1(B_1)$ with $u \geqslant 0$ in B_1,*

$$-\nabla \cdot \mathbf{a}\nabla u \geqslant 0 \quad \text{in } B_1 \qquad \text{and} \qquad |\{u \geqslant 1\} \cap B_1| \geqslant (1-\theta)|B_1|,$$

we have

$$\operatorname*{ess\,inf}_{B_{1/2}} u \geqslant \exp\big(-C\Lambda^{1/2}\big). \tag{4.77}$$

This improves the bound $\exp(-C\Lambda^{\frac{d+2}{4}})$ of Proposition 4.11 for small θ. We do not claim that the dependence on Λ in (4.77) is optimal.

Proof. Let $C_0(d) \geqslant 1$ be the constant in (4.67) and define

$$C_* := \max\{1,\, 2^{\frac{d}{2}} C_0 |B_1|\}, \qquad c := \frac{1}{4C_*}, \qquad C := \ln(8C_*).$$

Set $p := \Lambda^{-1/2} \in (0,1]$, so that $\Lambda p^2 + 1 = 2$, and $\lambda_0 := \operatorname{ess\,inf}_{B_{1/2}} u \geqslant 0$. Fix $\varepsilon > 0$. The function $u + \varepsilon \in H^1(B_1)$ is a positive supersolution, and $\operatorname{ess\,inf}_{B_{1/2}}(u+\varepsilon) = \lambda_0 + \varepsilon$, so (4.67) gives

$$2^{-\frac{d}{2}} \Big(\int_{B_1} (u+\varepsilon)^{-p} \Big)^{-1} \leqslant C_0 (\lambda_0 + \varepsilon)^p.$$

On $\{u \geqslant 1\} \cap B_1$ we have $(u+\varepsilon)^{-p} \leqslant 1$, and on the rest of B_1, which has measure at most $\theta|B_1|$, we have $(u+\varepsilon)^{-p} \leqslant \varepsilon^{-p}$. Hence $\int_{B_1} (u+\varepsilon)^{-p} \leqslant |B_1|(1 + \theta\varepsilon^{-p})$ and, by the previous display,

$$\varepsilon^p \leqslant C_* (\lambda_0 + \varepsilon)^p (\varepsilon^p + \theta) \qquad \text{for every } \varepsilon > 0. \tag{4.78}$$

Let $\varepsilon_0 := (8C_*)^{-1/p}$, so that $\varepsilon_0^p = \frac{1}{8C_*}$. Suppose that $\lambda_0 \leqslant \varepsilon_0$. Then $(\lambda_0 + \varepsilon_0)^p \leqslant (2\varepsilon_0)^p \leqslant 2\varepsilon_0^p$ since $p \leqslant 1$, and (4.78) with $\varepsilon = \varepsilon_0$ gives, after dividing by ε_0^p and using $\theta \leqslant c = \frac{1}{4C_*}$,

$$1 \leqslant 2C_* \big(\varepsilon_0^p + \theta\big) \leqslant \frac{1}{4} + \frac{1}{2} < 1,$$

a contradiction. Hence $\lambda_0 > \varepsilon_0 = (8C_*)^{-\Lambda^{1/2}} = \exp(-C\Lambda^{1/2})$. □

In order to link the two estimates in Proposition 4.14 together we need the following *crossover estimate*. It does not use the Sobolev inequality, and holds in every dimension.

Proposition 4.16 (Crossover estimate)**.** *There exist $C(d) < \infty$ and $c(d) \in (0, \frac{1}{2}]$ such that, for every $u \in H^1(B_1)$ satisfying $u > 0$ in B_1 and*

$$-\nabla \cdot \mathbf{a}\nabla u \geqslant 0 \quad \text{in } B_1, \tag{4.79}$$

we have

$$\big\| u^{c\Lambda^{-\frac{1}{2}}} \big\|_{\underline{L}^1(B_{1/2})} \big\| u^{-c\Lambda^{-\frac{1}{2}}} \big\|_{\underline{L}^1(B_{1/2})} \leqslant C. \tag{4.80}$$

The proof uses the logarithm of the supersolution, similar to the first paragraph of the proof of Proposition 4.11, together with the John–Nirenberg inequality (Proposition 1.17).

Proof. Step 0. Reduction. Assume first that $\inf_{B_1} u > 0$. Then $u^{-1} \in H^1(B_1) \cap L^\infty(B_1)$ and $v := -\log u \in H^1(B_1)$. The general case is treated in Step 3.

Step 1. The logarithm is in BMO. Let φ be Lipschitz with compact support in B_1. The function $\varphi^2 u^{-1}$ belongs to $H^1_0(B_1)$ and is nonnegative, so it is an admissible test function. Since $\nabla u/u = -\nabla v$, we obtain

$$0 \leqslant \int_{B_1} \nabla\Big(\frac{\varphi^2}{u}\Big) \cdot \mathbf{a}\nabla u = -\int_{B_1} \varphi^2 \nabla v \cdot \mathbf{a}\nabla v - 2\int_{B_1} \varphi \nabla\varphi \cdot \mathbf{a}\nabla v.$$

Let $X := \int_{B_1} \varphi^2 \nabla v \cdot \mathbf{a}\nabla v$. By (4.2) and the Cauchy-Schwarz inequality, $X \leqslant 2\Lambda^{\frac12}\big(\int_{B_1} |\nabla\varphi|^2\big)^{\frac12} X^{\frac12}$, that is, $X \leqslant 4\Lambda \int_{B_1} |\nabla\varphi|^2$. Since $|\nabla v|^2 \leqslant \nabla v \cdot \mathbf{a}\nabla v$, we get

$$\int_{B_1} \varphi^2 |\nabla v|^2 \leqslant 4\Lambda \int_{B_1} |\nabla\varphi|^2. \tag{4.81}$$

Let $B_r(x)$ be a ball with $\overline{B_r(x)} \subseteq B_{1/2}$. Then $|x| + 2r < 1$, so we may take $\varphi := \min\{1, (2r - |\cdot - x|)_+/r\}$ in (4.81), which has compact support in B_1, equals one in $B_r(x)$ and satisfies $|\nabla\varphi| \leqslant r^{-1}$. We get

$$\int_{B_r(x)} |\nabla v|^2 \leqslant \frac{4\Lambda |B_{2r}|}{r^2} = C\Lambda r^{d-2}. \tag{4.82}$$

By the Poincaré inequality in $B_r(x)$ and Jensen's inequality, we deduce that

$$\fint_{B_r(x)} \big|v - (v)_{B_r(x)}\big| \leqslant \left(\fint_{B_r(x)} \big|v - (v)_{B_r(x)}\big|^2\right)^{\frac12} \leqslant C_1 \Lambda^{\frac12}.$$

That is, $\|v\|_{\mathrm{BMO}(B_{1/2})} \leqslant C_1\Lambda^{\frac12}$ with $C_1(d) < \infty$.

Step 2. The exponential integrability. Let $C_0(d)$ be the constant in Proposition 1.17 (applied in the ball $B = B_{1/2}$, for which $v \in \mathrm{BMO}(B_{1/2})$ by Step 1), enlarged if necessary so that $C_0 \geqslant 1$, and set $a_0 := (C_0 C_1 \Lambda^{\frac12})^{-1}$. Since $\|v\|_{\mathrm{BMO}(B_{1/2})} \leqslant C_1\Lambda^{\frac12}$, (1.31) yields

$$\frac{|\{x \in B_{1/2} : |v - (v)_{B_{1/2}}| > t\}|}{|B_{1/2}|} \leqslant C_0 e^{-a_0 t} \qquad \text{for every } t > 0.$$

Therefore, for every $a \in (0, a_0)$, by the layer-cake formula,

$$\fint_{B_{1/2}} e^{a|v-(v)_{B_{1/2}}|} = 1 + a\int_0^\infty e^{at}\, \frac{|\{|v - (v)_{B_{1/2}}| > t\}|}{|B_{1/2}|}\, dt \leqslant 1 + \frac{C_0 a}{a_0 - a}.$$

Define $c := \min\big\{\frac12, (C_0C_1(1 + C_0))^{-1}\big\}$ and $a := c\Lambda^{-\frac12} \leqslant a_0/(1 + C_0)$. The right side above is nondecreasing in a and equals 2 at $a = a_0/(1 + C_0)$, so

$$\fint_{B_{1/2}} \exp\big(c\Lambda^{-\frac12}\big|v - (v)_{B_{1/2}}\big|\big) \leqslant 2. \tag{4.83}$$

Now set $w := \exp\big(-c\Lambda^{-\frac{1}{2}}(v-(v)_{B_{1/2}})\big) = e^{-c\Lambda^{-1/2}(\log u)_{B_{1/2}}}\, u^{c\Lambda^{-1/2}}$. Then w and w^{-1} are both bounded pointwise by the exponential in (4.83), so $\|w\|_{\underline{L}^1(B_{1/2})} \leqslant 2$ and $\|w^{-1}\|_{\underline{L}^1(B_{1/2})} \leqslant 2$. The constants $e^{\mp c\Lambda^{-1/2}(\log u)_{B_{1/2}}}$ cancel in the product, and we obtain

$$\|u^{c\Lambda^{-\frac{1}{2}}}\|_{\underline{L}^1(B_{1/2})}\|u^{-c\Lambda^{-\frac{1}{2}}}\|_{\underline{L}^1(B_{1/2})} = \|w\|_{\underline{L}^1(B_{1/2})}\|w^{-1}\|_{\underline{L}^1(B_{1/2})} \leqslant 4. \tag{4.84}$$

Step 3. The general case. For $u > 0$ in B_1, and $\delta > 0$, the function $u+\delta$ is a positive supersolution in $H^1(B_1)$ with $\inf_{B_1}(u+\delta) \geqslant \delta > 0$. Write $\gamma := c\Lambda^{-\frac{1}{2}} \leqslant 1$. By the previous steps, $\|(u+\delta)^{\gamma}\|_{\underline{L}^1(B_{1/2})}\|(u+\delta)^{-\gamma}\|_{\underline{L}^1(B_{1/2})} \leqslant 4$. As $\delta \downarrow 0$, the first factor converges to $\|u^{\gamma}\|_{\underline{L}^1(B_{1/2})} < \infty$ (dominated convergence, since $u \in L^2(B_1)$ and $\gamma \leqslant 1$), and the second increases to $\|u^{-\gamma}\|_{\underline{L}^1(B_{1/2})}$ (monotone convergence). This proves (4.80) with $C = 4$. □

By using the crossover estimate (4.80) to bridge the gap between (4.67) and (4.68), we obtain the weak Harnack inequality for positive supersolutions.

Proposition 4.17 (Weak Harnack inequality)**.** *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that, for every $u \in H^1(B_1)$ satisfying $u > 0$ in B_1 and*

$$-\nabla \cdot \mathbf{a}\nabla u \geqslant 0 \quad \text{in } B_1, \tag{4.85}$$

and every $q \in (0,1)$, we have the estimate

$$\|u\|_{L^{\frac{qd}{d-2}}(B_{1/4})} \leqslant \left(\frac{C}{(1-q)^C}\right)^{\Lambda^{\frac{1}{2}}} \inf_{B_{1/4}} u. \tag{4.86}$$

Proof. Write $\chi := \frac{d}{d-2}$, let $c \in (0, \frac{1}{2}]$ be the constant in Proposition 4.16, and set $g := c\Lambda^{-\frac{1}{2}} \in (0, \frac{1}{2}]$.

Step 1. The case $q \in [\frac{1}{2}, 1)$. Let $\bar{u}(y) := u(y/2)$ for $y \in B_1$. Then $\bar{u} \in H^1(B_1)$ is a positive supersolution of $-\nabla\cdot\bar{\mathbf{a}}\nabla\bar{u} \geqslant 0$ in B_1, with $\bar{\mathbf{a}}(y) := \mathbf{a}(y/2)$, which satisfies (4.1) with the same Λ. Moreover, $\|\bar{u}\|^r_{L^r(B_\rho)} = 2^d\|u\|^r_{L^r(B_{\rho/2})}$ and $\inf_{B_\rho}\bar{u} = \inf_{B_{\rho/2}} u$. (The dilation is used since Proposition 4.14 has its input on B_1 and Proposition 4.16 its output on $B_{1/2}$.)

We apply (4.68) to $\bar{u}$ with $p = g \leqslant \frac{1}{2} \leqslant q$. Since $\Lambda g^2 = c^2 \leqslant 1$, we have $\big(\frac{\Lambda g^2}{(1-q)^2}+1\big)^{\frac{d+2}{2}} \leqslant C(1-q)^{-(d+2)}$, and thus

$$\|u\|^g_{L^{q\chi}(B_{1/4})} \leqslant C(1-q)^{-(d+2)}\|u\|^g_{L^g(B_{1/2})} = C(1-q)^{-(d+2)}\int_{B_{1/2}} u^g \leqslant C(1-q)^{-(d+2)}\|u^g\|_{\underline{L}^1(B_{1/2})},$$

where we used $|B_{1/2}| < 1$. Next, (4.80) gives

$$\|u^g\|_{\underline{L}^1(B_{1/2})} \leqslant C\|u^{-g}\|^{-1}_{\underline{L}^1(B_{1/2})}.$$

Finally, (4.67) applied to $\bar{u}$ with $p = g$, for which $\Lambda g^2 + 1 \leqslant 2$, gives

$$\|u^{-g}\|^{-1}_{\underline{L}^1(B_{1/2})} = |B_{1/2}|\Big(\int_{B_{1/2}} u^{-g}\Big)^{-1} \leqslant C\|\bar{u}^{-1}\|^{-g}_{L^g(B_1)} \leqslant C\big(\inf_{B_{1/2}}\bar{u}\big)^g = C\big(\inf_{B_{1/4}} u\big)^g.$$

Chaining the three displays we obtain $\|u\|^g_{L^{q\chi}(B_{1/4})} \leqslant C(1-q)^{-(d+2)}(\inf_{B_{1/4}} u)^g$. Taking the power $1/g = \Lambda^{\frac12}/c$ yields (4.86), with the constant $C(d)$ in the exponent of $(1-q)^{-1}$ equal to $(d+2)/c$, and with $C \geqslant 1$.

Step 2. The case $q \in (0, \frac12)$. Since $|B_{1/4}| < 1$, for $0 < r \leqslant s$ we have $\|f\|_{L^r(B_{1/4})} \leqslant \|f\|_{L^s(B_{1/4})}$ (Hölder's inequality). Hence, by Step 1 with $q = \frac12$ and constants C_1 (prefactor) and C_1 (exponent of $(1-q)^{-1}$), the left side of (4.86) is at most $\big(C_1 2^{C_1}\big)^{\Lambda^{\frac12}} \inf_{B_{1/4}} u$. Since $(1-q)^{-C} \geqslant 1$, (4.86) follows after enlarging C so that $C \geqslant C_1 2^{C_1}$. □

Even for the Laplacian, the estimate (4.86) is false for $q = 1$. Indeed, let $G(x) := |x|^{2-d}$ and, for $k \in \mathbb{N}$, consider the positive supersolutions $\min\{G, k\} \in H^1(B_1)$ of $-\Delta u \geqslant 0$ (here $\mathbf{a} = \mathrm{I}_d$). Then $\inf_{B_{1/4}} \min\{G,k\} = 4^{d-2}$ for every $k \geqslant 4^{d-2}$, but $\|\min\{G,k\}\|_{L^{\frac{d}{d-2}}(B_{1/4})} \to \infty$ as $k \to \infty$ since $G^{\frac{d}{d-2}} = |x|^{-d}$ is not integrable at the origin.

Combining Propositions 4.13 and 4.17, we obtain Moser's Harnack inequality.

Theorem 4.18 (Harnack inequality)**.** *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that, for every $u \in H^1(B_1)$ satisfying $u > 0$ in B_1 and*

$$-\nabla \cdot \mathbf{a}\nabla u = 0 \quad \text{in } B_1, \tag{4.87}$$

we have

$$\sup_{B_{1/2}} u \leqslant \exp\big(C\Lambda^{\frac12}\big) \inf_{B_{1/2}} u. \tag{4.88}$$

Proof. Let $q := \frac{d-1}{d}$ and $p := \frac{qd}{d-2} = \frac{d-1}{d-2} > 1$, so that $\frac{p}{p-1} = d-1$. The function u is continuous by Theorem 4.8.

Step 1. A local Harnack inequality. Fix $x_0 \in B_{1/2}$. Then $B_{1/2}(x_0) \subseteq B_1$, and $\widetilde{u}(y) := u(x_0 + y/2)$ is a positive solution in B_1 of an equation with coefficient field $\mathbf{a}(x_0 + \cdot/2)$, which satisfies (4.1) with the same Λ. Proposition 4.17 (with the above q) gives

$$\|\widetilde{u}\|_{L^p(B_{1/4})} \leqslant e^{C\Lambda^{\frac12}} \inf_{B_{1/4}} \widetilde{u}.$$

Applying (4.59) to $z \mapsto \widetilde{u}(z/4)$, $z \in B_1$, we get, since $p > 1$ and $\widetilde{u} > 0$,

$$\sup_{B_{1/8}} \widetilde{u} \leqslant C\Lambda^{\frac{d}{2p}} \|\widetilde{u}\|_{L^p(B_{1/4})}.$$

Since $\Lambda \geqslant 1$ implies $\log\Lambda \leqslant 2\Lambda^{\frac12}$, we have $\Lambda^{\frac{d}{2p}} \leqslant e^{C\Lambda^{1/2}}$. Combining the last two displays and using $\inf_{B_{1/4}} \widetilde{u} \leqslant \inf_{B_{1/8}} \widetilde{u}$, we obtain $\sup_{B_{1/8}} \widetilde{u} \leqslant e^{C\Lambda^{1/2}} \inf_{B_{1/8}} \widetilde{u}$, that is,

$$\sup_{B_{1/16}(x_0)} u \leqslant e^{C\Lambda^{\frac12}} \inf_{B_{1/16}(x_0)} u \qquad \text{for every } x_0 \in B_{1/2}. \tag{4.89}$$

Step 2. Chaining. Let $y, z \in B_{1/2}$ and put $x_i := y + \frac{i}{17}(z-y)$ for $i \in \{0, \dots, 17\}$. By convexity $x_i \in B_{1/2}$, and $|x_{i+1} - x_i| < \frac{1}{17} < \frac{1}{16}$, so $x_{i+1} \in B_{1/16}(x_i)$. By (4.89), $u(x_i) \leqslant e^{C\Lambda^{1/2}} u(x_{i+1})$ for each i, and therefore $u(y) \leqslant e^{17C\Lambda^{1/2}} u(z)$. Taking the supremum in y and the infimum in z gives (4.88). □

Remark 4.19 (The case $d = 2$)**.** The Harnack inequality (4.88) also holds for $d = 2$, with the constant $C = C(3)$, by adding a variable. Let $u \in H^1(B_1)$ be a positive solution of $-\nabla \cdot \mathbf{a}\nabla u = 0$ in $B_1 \subseteq \mathbb{R}^2$, and put $U(x_1, x_2, x_3) := u(x_1, x_2)$ and $\tilde{\mathbf{a}} := \mathbf{a} \oplus 1$. Then $\tilde{\mathbf{a}}$ is measurable and satisfies (4.1) with the same Λ: for $e = (e', e_3)$ we have $e \cdot \tilde{\mathbf{a}} e = e' \cdot \mathbf{a} e' + e_3^2 \geqslant |e|^2$ and $e \cdot \tilde{\mathbf{a}}^{-1} e = e' \cdot \mathbf{a}^{-1} e' + e_3^2 \geqslant \Lambda^{-1}|e|^2$, since $\Lambda \geqslant 1$. Moreover $U \in H^1(B_1^3)$, since $B_1^3 \subseteq B_1^2 \times (-1, 1)$, and U solves $-\nabla \cdot \tilde{\mathbf{a}} \nabla U = 0$ in B_1^3: for $\phi \in C_c^\infty(B_1^3)$ and every x_3, the function $\phi(\cdot, x_3)$ belongs to $C_c^\infty(B_1^2)$, so by Fubini's theorem

$$\int_{B_1^3} \tilde{\mathbf{a}} \nabla U \cdot \nabla \phi \, dx = \int_{-1}^{1} \Big(\int_{B_1^2} \mathbf{a} \nabla u \cdot \nabla_{x'} \phi(\cdot, x_3) \, dx' \Big) dx_3 + \int_{B_1^3} 0 \cdot \partial_3 \phi \, dx = 0.$$

Hence U is a weak solution. Applying the case $d = 3$ to $U > 0$, and noting that $U(x', x_3) = u(x')$ and that the projection of $B_{1/2}^3$ onto $\mathbb{R}^2$ is $B_{1/2}^2$, gives $\sup_{B_{1/2}} u \leqslant \exp(C\Lambda^{\frac{1}{2}}) \inf_{B_{1/2}} u$ (the function $u = U(\cdot, 0)$ is continuous, so sup and inf may be read as essential supremum and infimum).

Proposition 4.20 (Sharpness of the Harnack inequality)**.** *Let $d \geqslant 2$ and $\Lambda \geqslant 1$. There exist a constant matrix $\mathbf{a}$ satisfying (4.1) and a function $u \in C^\infty(\mathbb{R}^d)$ with $u > 0$ in B_1 and $-\nabla \cdot \mathbf{a}\nabla u = 0$ in $\mathbb{R}^d$ such that*

$$\sup_{B_{1/2}} u \geqslant \exp\big(\Lambda^{\frac{1}{2}}\big) \inf_{B_{1/2}} u. \tag{4.90}$$

Consequently, if $C_H \in [1, \infty)$ is such that $\sup_{B_{1/2}} v \leqslant C_H \inf_{B_{1/2}} v$ for every positive solution $v \in H^1(B_1)$ of an equation with coefficients satisfying (4.1) with constant Λ, then $\log C_H \geqslant \Lambda^{\frac{1}{2}}$. In particular, in (4.88) (and in Remark 4.19) the constant satisfies $C \geqslant 1$ for every $\Lambda \geqslant 1$, so the exponent $\Lambda^{\frac{1}{2}}$ cannot be replaced by Λ^β with $\beta < \frac{1}{2}$: a bound $\sup_{B_{1/2}} v \leqslant \exp(C\Lambda^\beta) \inf_{B_{1/2}} v$ with C independent of Λ would force $C\Lambda^\beta \geqslant \Lambda^{\frac{1}{2}}$ for all $\Lambda \geqslant 1$, which is impossible.

Proof. Write $k := \Lambda^{\frac{1}{2}}$ and set

$$u(x) := \exp(kx_1)\cos(x_2), \qquad \mathbf{a} := e_1 \otimes e_1 + \Lambda\, e_2 \otimes e_2 + \sum_{j=3}^{d} e_j \otimes e_j$$

(the sum is empty if $d = 2$). The matrix $\mathbf{a}$ is constant and symmetric with eigenvalues $1, \Lambda, 1, \dots, 1$, so $e \cdot \mathbf{a} e \geqslant |e|^2$ and, since $\Lambda \geqslant 1$, $e \cdot \mathbf{a}^{-1} e \geqslant \Lambda^{-1}|e|^2$; that is, (4.1) holds. The function u is smooth, and $\partial_{11} u = \Lambda u$, $\partial_{22} u = -u$, $\partial_{jj} u = 0$ for $j \geqslant 3$, so that $\nabla \cdot \mathbf{a}\nabla u = \partial_{11} u + \Lambda \partial_{22} u = 0$. Since u is smooth, it belongs to $H^1(B_1)$ and, integrating by parts, is a weak solution. Since $B_1 \subseteq \{|x_2| < 1\}$ and $\cos > 0$ on $(-1, 1)$, we have $u > 0$ in B_1.

For $t \in (0, \frac{1}{2})$ the points $\pm t e_1$ lie in $B_{1/2}$ and $u(\pm t e_1) = e^{\pm kt}$. Therefore $\sup_{B_{1/2}} u \geqslant e^{kt}$ and $\inf_{B_{1/2}} u \leqslant e^{-kt}$, so $\sup_{B_{1/2}} u \geqslant e^{2kt} \inf_{B_{1/2}} u$. Letting $t \uparrow \frac{1}{2}$ gives (4.90).

For the consequence, $u \in H^1(B_1)$ is an admissible v, so $e^k \inf_{B_{1/2}} u \leqslant \sup_{B_{1/2}} u \leqslant C_H \inf_{B_{1/2}} u$, and $\inf_{B_{1/2}} u > 0$ by continuity of u and $u > 0$ on $\overline{B_{1/2}}$. Hence $C_H \geqslant e^k$. Finally, if $\sup_{B_{1/2}} v \leqslant \exp(C\Lambda^{\frac{1}{2}}) \inf_{B_{1/2}} v$ holds for all such v, then $C_H = e^{C\Lambda^{1/2}}$ is admissible and so $C\Lambda^{\frac{1}{2}} \geqslant \Lambda^{\frac{1}{2}}$, that is, $C \geqslant 1$. □

Using Theorem 4.18, we can improve the statement of Theorem 4.8.

Theorem 4.21 (Interior Hölder regularity)**.** *Let $d \geqslant 3$. There exist a constant $C(d) < \infty$ and an exponent $\alpha \in \big[\exp\big(-C\Lambda^{\frac{1}{2}}\big), 1\big]$ such that, for any $u \in H^1(B_1)$ satisfying*

$$-\nabla \cdot \mathbf{a}\nabla u = 0 \quad \text{in } B_1,$$

and any $p \in (1, \infty)$, we have

$$\|u\|_{C^{0,\alpha}(B_{1/2})} \leqslant C\Lambda^{\frac{d}{2p}} \Big(\frac{p}{p-1}\Big)^{d/p} \|u\|_{L^p(B_1)}. \tag{4.91}$$

Proof. We may assume that $\|u\|_{L^p(B_1)} < \infty$. By Theorem 4.8, u is continuous in B_1. Set $C_H := \max\{3, \exp(C\Lambda^{1/2})\}$, where C is the constant in (4.88), so that (4.88) holds with C_H in place of $\exp(C\Lambda^{1/2})$.

Step 1. Oscillation decay. Let $x \in B_{3/4}$, so that $B_{1/4}(x) \subseteq B_1$, and let $0 < r \leqslant \frac{1}{4}$. Set $M := \sup_{B_r(x)} u$, $m := \inf_{B_r(x)} u$, $M' := \sup_{B_{r/2}(x)} u$ and $m' := \inf_{B_{r/2}(x)} u$. The functions $M - u$ and $u - m$ are nonnegative solutions in $B_r(x)$; applying (4.88) (after translation and dilation of $B_r(x)$ to B_1) to $M - u + \varepsilon$ and $u - m + \varepsilon$ and sending $\varepsilon \to 0$, we get

$$M - m' \leqslant C_H(M - M') \quad \text{and} \quad M' - m \leqslant C_H(m' - m).$$

Adding these inequalities and rearranging yields

$$\operatorname*{osc}_{B_{r/2}(x)} u \leqslant \sigma \operatorname*{osc}_{B_r(x)} u, \qquad \sigma := \frac{C_H - 1}{C_H + 1} \in \big[\tfrac{1}{2}, 1\big).$$

Define $\alpha := \log_2 \frac{C_H+1}{C_H-1}$, so that $\sigma = 2^{-\alpha}$. Since $C_H \geqslant 3$, we have $\alpha \leqslant 1$, and since $\log(1+t) \geqslant \frac{t}{1+t}$,

$$\alpha = \frac{1}{\log 2} \log\Big(1 + \frac{2}{C_H - 1}\Big) \geqslant \frac{2}{(C_H + 1)\log 2} \geqslant \frac{1}{C_H} \geqslant \exp\big(-C\Lambda^{\frac{1}{2}}\big)$$

(with a larger C). Iterating, for every $n \in \mathbb{N} \cup \{0\}$ we have $\operatorname{osc}_{B_{2^{-n}/4}(x)} u \leqslant 2^{-\alpha n} \operatorname{osc}_{B_{1/4}(x)} u$. For $\rho \in (0, \frac{1}{4}]$, choose $n \in \mathbb{N} \cup \{0\}$ with $2^{-n-1} < 4\rho \leqslant 2^{-n}$; then $2^{-\alpha n} \leqslant 2^{\alpha}(4\rho)^{\alpha} \leqslant 2(4\rho)^{\alpha}$, and so

$$\operatorname*{osc}_{B_\rho(x)} u \leqslant 2(4\rho)^{\alpha} \operatorname*{osc}_{B_{1/4}(x)} u \qquad \text{for all } x \in B_{3/4},\ \rho \in \big(0, \tfrac{1}{4}\big]. \tag{4.92}$$

Step 2. The L^∞ bound. Let $z \in B_{3/4}$. Then $B_{1/4}(z) \subseteq B_1$, and both $\pm u(z + \cdot/4)$ are subsolutions in B_1 (for the rescaled coefficients). Applying (4.59) to each of them, and using $\|u(z + \cdot/4)\|^p_{L^p(B_1)} = 4^d \|u\|^p_{L^p(B_{1/4}(z))}$, we get

$$\sup_{B_{1/8}(z)} |u| \leqslant C\Lambda^{\frac{d}{2p}} \Big(\frac{p}{p-1}\Big)^{d/p} \|u\|_{L^p(B_1)},$$

where we used $p > 1$ to bound $4^{d/p} \leqslant 4^d$. Since the balls $B_{1/8}(z)$, $z \in B_{3/4}$, cover $B_{3/4}$,

$$\|u\|_{L^\infty(B_{3/4})} \leqslant C\Lambda^{\frac{d}{2p}} \Big(\frac{p}{p-1}\Big)^{d/p} \|u\|_{L^p(B_1)}. \tag{4.93}$$

Step 3. Conclusion. Let $x, y \in B_{1/2}$ with $x \neq y$. If $|x-y| < \frac{1}{4}$, then, for every $\rho \in (|x-y|, \frac{1}{4}]$, $y \in B_\rho(x)$, and (4.92) gives $|u(x)-u(y)| \leqslant 2(4\rho)^\alpha \cdot 2\|u\|_{L^\infty(B_{3/4})}$. Letting $\rho \downarrow |x-y|$ we get $|u(x)-u(y)| \leqslant 16|x-y|^\alpha \|u\|_{L^\infty(B_{3/4})}$. If $|x-y| \geqslant \frac{1}{4}$, then $|u(x)-u(y)| \leqslant 2\|u\|_{L^\infty(B_{3/4})} \leqslant 2(4|x-y|)^\alpha \|u\|_{L^\infty(B_{3/4})}$. In either case $[u]_{C^{0,\alpha}(B_{1/2})} \leqslant 16\|u\|_{L^\infty(B_{3/4})}$, and also $\|u\|_{L^\infty(B_{1/2})} \leqslant \|u\|_{L^\infty(B_{3/4})}$. Combining with (4.93) yields (4.91). □

Proposition 4.22 (Lower bound for the density estimate)**.** *Let $d \geqslant 3$ and $C_\star := 2|B_1^{d-1}|/|B_1^d|$. For every $\Lambda \geqslant 1$ and $\theta \in (0, \frac{1}{2}]$ there exist a constant matrix $\mathbf{a}$ satisfying (4.1) and a function $u \in C^\infty(\mathbb{R}^d)$ with $u > 0$ in B_1 and $-\nabla \cdot \mathbf{a}\nabla u = 0$ in $\mathbb{R}^d$, such that*

$$\big|\{u \geqslant 1\} \cap B_1\big| \geqslant (1-\theta)|B_1| \qquad \text{and} \qquad \inf_{B_{1/2}} u \leqslant 4\exp\big(-\theta\Lambda^{\frac{1}{2}}/C_\star\big). \tag{4.94}$$

Consequently, if $C(d)$ is a constant for which the conclusion (4.49) of Proposition 4.11 holds (for all $\Lambda \geqslant 1$, $\theta \in (0, \frac{1}{2}]$ and constant matrices satisfying (4.1)), then

$$C\theta^{\frac{1}{d}}\Lambda^{\frac{d+2}{4}} \geqslant \frac{\theta\Lambda^{\frac{1}{2}}}{C_\star} - \log 4. \tag{4.95}$$

Proof. Write $k := \Lambda^{\frac{1}{2}}$, let $\lambda := \theta/C_\star \in (0, \frac{1}{2C_\star}]$, and set

$$u(x) := \frac{\cosh(kx_1)\cos(x_2)}{\cosh(k\lambda)\cos(1)}, \qquad \mathbf{a} := e_1 \otimes e_1 + \Lambda\, e_2 \otimes e_2 + \sum_{j=3}^{d} e_j \otimes e_j.$$

As in the proof of Proposition 4.20, $\mathbf{a}$ satisfies (4.1), u is smooth, and $\partial_{11}u = \Lambda u$, $\partial_{22}u = -u$, $\partial_{jj}u = 0$ for $j \geqslant 3$, so $\nabla \cdot \mathbf{a}\nabla u = 0$. Also $u > 0$ in B_1 since $\cos(x_2) \geqslant \cos(1) > 0$ for $|x_2| < 1$.

If $x \in B_1$ and $|x_1| \geqslant \lambda$, then $\cosh(kx_1) \geqslant \cosh(k\lambda)$ and $\cos(x_2) \geqslant \cos(1)$, so $u(x) \geqslant 1$. The complement in B_1 of $\{|x_1| \geqslant \lambda\}$ is contained in $\{|x_1| < \lambda\} \cap B_1$, whose measure is at most $2\lambda|B_1^{d-1}| = C_\star\lambda|B_1| = \theta|B_1|$. This gives the first part of (4.94). For the second, $0 \in B_{1/2}$, $\cosh t \geqslant \frac{1}{2}e^t$ and $\cos(1) > \frac{1}{2}$ (as $1 < \frac{\pi}{3}$), so

$$\inf_{B_{1/2}} u \leqslant u(0) = \big(\cosh(k\lambda)\cos(1)\big)^{-1} \leqslant 4e^{-k\lambda} = 4\exp\big(-\theta\Lambda^{\frac{1}{2}}/C_\star\big).$$

Finally, $u \in H^1(B_1)$ is a nonnegative solution, hence a supersolution, satisfying the hypotheses of Proposition 4.11 at level θ, so (4.49) gives $\exp(-C\theta^{1/d}\Lambda^{(d+2)/4}) \leqslant \inf_{B_{1/2}} u \leqslant 4\exp(-\theta\Lambda^{1/2}/C_\star)$. Taking logarithms gives (4.95). □

4.5 The sharp exponent in two dimensions

In this subsection we prove the monotonicity formula of Piccinini and Spagnolo [PS72] in two dimensions. Combined with Morrey's Dirichlet growth theorem it gives the Hölder exponent $\Lambda^{-1/2}$ for symmetric coefficient fields, and Meyers' example [Mey63] shows that this exponent cannot be improved. Throughout, a *solution* of $-\nabla \cdot \mathbf{a}\nabla u = 0$ in B_1 is a function $u \in H^1(B_1)$ with $\int_{B_1} \nabla\varphi \cdot \mathbf{a}\nabla u = 0$ for every $\varphi \in C_c^\infty(B_1)$. Since $\mathbf{a}\nabla u \in L^2(B_1; \mathbb{R}^d)$, this identity extends by mollification to every $\varphi \in H^1(B_1)$ with compact support in B_1, in particular to products ψu with ψ Lipschitz and compactly supported in B_1.

If $\mathbf{a}$ is symmetric, then (4.1) is equivalent to the eigenvalue bounds $|\xi|^2 \leqslant \xi\cdot\mathbf{a}(x)\xi \leqslant \Lambda|\xi|^2$ for all $\xi \in \mathbb{R}^d$ and $x \in B_1$: indeed, for symmetric $\mathbf{a}$ the second inequality in (4.1) says that the eigenvalues of $\mathbf{a}^{-1}$ are at least Λ^{-1}.

Lemma 4.23 (Wirtinger inequality)**.** *Let $w : \mathbb{R} \to \mathbb{R}$ be 2π-periodic and absolutely continuous with $w' \in L^2(0, 2\pi)$, and let $\bar{w} := \frac{1}{2\pi}\int_0^{2\pi} w\,d\theta$. Then*

$$\int_0^{2\pi} (w - \bar{w})^2\,d\theta \leqslant \int_0^{2\pi} (w')^2\,d\theta.$$

Proof. Let $c_n := \frac{1}{2\pi}\int_0^{2\pi} w(\theta)e^{-in\theta}\,d\theta$, $n \in \mathbb{Z}$. Integrating by parts (which is legitimate for absolutely continuous periodic functions, the boundary terms cancelling by periodicity) shows that the Fourier coefficients of $w' \in L^2$ are inc_n. Parseval's identity gives $\int_0^{2\pi}(w - \bar{w})^2 = 2\pi\sum_{n\neq 0}|c_n|^2 \leqslant 2\pi\sum_{n\neq 0} n^2|c_n|^2 = \int_0^{2\pi}(w')^2$, since $c_0 = \bar{w}$. □

Theorem 4.24 (Piccinini and Spagnolo [PS72])**.** *Suppose $d = 2$, and let $\mathbf{a} : B_1 \to \mathbb{R}^{2\times 2}$ be measurable, symmetric and satisfy (4.1). Let $u \in H^1(B_1)$ be a solution of*

$$-\nabla\cdot\mathbf{a}\nabla u = 0 \quad \text{in } B_1.$$

Then

$$r \mapsto r^{-2/\sqrt{\Lambda}} \int_{B_r} \nabla u\cdot\mathbf{a}\nabla u \quad \text{is nondecreasing on } (0,1). \tag{4.96}$$

Proof. We write $\sigma := \mathbf{a}\nabla u \in L^2(B_1;\mathbb{R}^2)$, $\beta := 2\Lambda^{-1/2}$ and

$$\phi(r) := \int_{B_r} \nabla u\cdot\mathbf{a}\nabla u\,dx, \qquad r \in (0,1).$$

Step 1: polar coordinates. Let $e_\rho(\theta) := (\cos\theta, \sin\theta)$, $e_\theta(\theta) := (-\sin\theta, \cos\theta)$, $T(\rho,\theta) := \rho e_\rho(\theta)$ and $\Phi := u \circ T$ on $(0,1)\times\mathbb{R}$. The map T is a smooth local diffeomorphism onto $B_1\backslash\{0\}$ and 2π-periodic in θ, so by the chain rule for Sobolev functions $\Phi \in H^1_{\mathrm{loc}}((0,1)\times\mathbb{R})$, with

$$\partial_\rho\Phi = (\nabla u\cdot e_\rho)\circ T, \qquad \partial_\theta\Phi = \rho\,(\nabla u\cdot e_\theta)\circ T.$$

We use the polar integration formula $\int_{B_1} f\,dx = \int_0^1\int_0^{2\pi} f(T(\rho,\theta))\rho\,d\theta\,d\rho$ for $f \in L^1(B_1)$. By the characterization of Sobolev functions by absolute continuity on lines (ACL) and Fubini's theorem, for a.e. $\rho \in (0,1)$ the function $\theta \mapsto \Phi(\rho,\theta)$ coincides a.e. with an absolutely continuous function, which is then 2π-periodic by continuity, whose derivative is $\partial_\theta\Phi(\rho,\cdot)$; we always use this representative for such ρ. Moreover, by Fubini and $\nabla u, \sigma \in L^2$, for a.e. ρ all the functions $\Phi, \partial_\rho\Phi, \partial_\theta\Phi, \sigma\circ T$ are in $L^2(0,2\pi)$ as functions of θ.

Write $u_\rho := \partial_\rho\Phi$, $\rho^{-1}u_\theta := \rho^{-1}\partial_\theta\Phi$ (so $\nabla u\circ T = u_\rho e_\rho + \rho^{-1}u_\theta e_\theta$) and let $P := J^*\mathbf{a}\circ T\,J$ with $J = (e_\rho \mid e_\theta)$, that is $p_{11} = e_\rho\cdot\mathbf{a}e_\rho$, $p_{12} = p_{21} = e_\rho\cdot\mathbf{a}e_\theta$, $p_{22} = e_\theta\cdot\mathbf{a}e_\theta$ (symmetry of $\mathbf{a}$ is used here). Since J is orthogonal, $|\xi|^2 \leqslant \xi\cdot P\xi \leqslant \Lambda|\xi|^2$ for all $\xi \in \mathbb{R}^2$; in particular $1 \leqslant p_{11} \leqslant \Lambda$. We have, at $x = T(\rho,\theta)$,

$$\sigma\cdot e_\rho = p_{11}u_\rho + p_{12}\rho^{-1}u_\theta, \tag{4.97}$$

$$\nabla u\cdot\mathbf{a}\nabla u = p_{11}u_\rho^2 + 2p_{12}\rho^{-1}u_\rho u_\theta + p_{22}\rho^{-2}u_\theta^2. \tag{4.98}$$

Finally, taking $\xi = (-p_{12}/p_{11}, 1)$ in the ellipticity of P gives $1 \leqslant |\xi|^2 \leqslant \xi \cdot P\xi = p_{22} - p_{12}^2/p_{11}$, hence

$$1 + \frac{p_{12}^2}{p_{11}} \leqslant p_{22}. \tag{4.99}$$

Step 2: the flux identity. Define, for a.e. $\rho \in (0,1)$,

$$F(\rho) := \int_0^{2\pi} (\sigma \cdot e_\rho)\rho \, d\theta, \quad G(\rho) := \int_0^{2\pi} \Phi\,(\sigma \cdot e_\rho)\rho \, d\theta, \quad e(\rho) := \int_0^{2\pi} (\nabla u \cdot \mathbf{a}\nabla u)\rho \, d\theta.$$

These belong to $L^1(0,1)$: $\sigma \cdot e_\rho$ and $\nabla u \cdot \mathbf{a}\nabla u$ are in L^2 resp. $L^1(B_1)$, and $u \in L^2(B_1)$, so $u\,\sigma \cdot \frac{x}{|x|} \in L^1(B_1)$ by Cauchy–Schwarz. By the polar integration formula, $\phi(r) = \int_0^r e(\rho)\,d\rho$; hence ϕ is absolutely continuous on $[0,1)$ with $\phi(0) = 0$ and $\phi' = e$ a.e. Let $g \in L^\infty(0,1)$ have compact support in $(0,1)$ and put $\eta(\rho) := -\int_\rho^1 g$. Then η is Lipschitz, vanishes near 1 and is constant near 0, so $\psi(x) := \eta(|x|)$ is Lipschitz with compact support in B_1, and $\nabla\psi = g(|x|)\frac{x}{|x|}$ a.e. Testing the equation with ψ and, using the extension noted above, with ψu, and integrating in polar coordinates, gives

$$0 = \int_0^1 g\,F\,d\rho, \qquad 0 = \int_0^1 g\,G\,d\rho + \int_0^1 \eta\,e\,d\rho.$$

Since ϕ is absolutely continuous, η Lipschitz, $\phi(0) = 0$ and $\eta = 0$ near 1, we have $\int_0^1 \eta\phi' = -\int_0^1 g\phi$. Because g is arbitrary, $F = 0$ and $G = \phi$ a.e. in $(0,1)$. Consequently there is a null set $N \subseteq (0,1)$ such that, for every $r \in (0,1)\backslash N$, the properties of Step 1 hold, $\phi'(r) = e(r)$, and for *every* $k \in \mathbb{R}$

$$\phi(r) = G(r) - kF(r) = \int_0^{2\pi} \big(\Phi(r,\theta) - k\big)(\sigma \cdot e_\rho)\,r\,d\theta. \tag{4.100}$$

Step 3: the differential inequality. Fix $r \in (0,1)\backslash N$ and set $w := \Phi(r,\cdot)$, $A := u_\rho(r,\cdot)$, $B := r^{-1}w'$ and $k := \frac{1}{2\pi}\int_0^{2\pi} w\,d\theta$. By (4.100) and (4.97), and writing $Y := p_{11}^{1/2}A + p_{12}p_{11}^{-1/2}B$,

$$\phi(r) = r\int_0^{2\pi} (w-k)(p_{11}A + p_{12}B)\,d\theta = \int_0^{2\pi} \big(p_{11}^{1/2}(w-k)\big)\,(rY)\,d\theta.$$

(All integrands are integrable, since w is bounded and A, B, p_{ij} are in L^2, resp. L^∞.) Young's inequality $ab \leqslant \frac{1}{2}(\lambda^{-1}a^2 + \lambda b^2)$ with $\lambda = \Lambda^{1/2}$ gives

$$\phi(r) \leqslant \frac{1}{2}\Lambda^{-1/2}\int_0^{2\pi} p_{11}(w-k)^2\,d\theta + \frac{1}{2}\Lambda^{1/2}r^2\int_0^{2\pi} Y^2\,d\theta.$$

By $p_{11} \leqslant \Lambda$ and Lemma 4.23, the first term is at most $\frac{1}{2}\Lambda^{1/2}\int_0^{2\pi}(w')^2\,d\theta = \frac{1}{2}\Lambda^{1/2}r^2\int_0^{2\pi} B^2\,d\theta$. Therefore, using (4.99) and (4.98),

$$\begin{aligned}\phi(r) &\leqslant \frac{1}{2}\Lambda^{1/2}r^2\int_0^{2\pi}\big(B^2 + Y^2\big)\,d\theta = \frac{1}{2}\Lambda^{1/2}r^2\int_0^{2\pi}\Big(p_{11}A^2 + 2p_{12}AB + \big(1 + \tfrac{p_{12}^2}{p_{11}}\big)B^2\Big)d\theta \\ &\leqslant \frac{1}{2}\Lambda^{1/2}r^2\int_0^{2\pi}\big(p_{11}A^2 + 2p_{12}AB + p_{22}B^2\big)d\theta = \frac{1}{2}\Lambda^{1/2}\,r\,e(r) = \frac{1}{2}\Lambda^{1/2}\,r\,\phi'(r).\end{aligned}$$

That is, $r\phi'(r) \geqslant \beta\phi(r)$ for a.e. $r \in (0,1)$.

Step 4: conclusion. The function $\chi(r) := r^{-\beta}\phi(r)$ is absolutely continuous on every compact subinterval of $(0,1)$, being the product of such a function and a smooth one, and $\chi'(r) = r^{-\beta-1}\big(r\phi'(r) - \beta\phi(r)\big) \geqslant 0$ for a.e. r. Hence $\chi(b) - \chi(a) = \int_a^b \chi' \geqslant 0$ for $0 < a < b < 1$, which is (4.96). □

We next state Morrey's Dirichlet growth theorem in the form needed. It follows from the Campanato characterization of Hölder spaces, Lemma 1.19, and the Poincaré inequality.

Lemma 4.25 (Morrey's Dirichlet growth theorem)**.** *Let $d \geqslant 2$. There is $C(d) < \infty$ such that the following holds. Let $\alpha \in (0,1]$, $K \in [0,\infty)$ and $u \in H^1(B_1)$ satisfy*

$$\int_{B_r(x)} |\nabla u|^2 \leqslant K\, r^{d-2+2\alpha} \quad \text{for all } x \in B_{1/2},\ r \in (0, 1/2]. \tag{4.101}$$

Then u has a representative in $C^{0,\alpha}(B_{1/2})$ and $[u]_{C^{0,\alpha}(B_{1/2})} \leqslant C\alpha^{-1}K^{1/2}$. In particular, this holds if (4.101) is assumed for all balls $B_r(x) \subseteq B_1$.

Proof. Let $x \in B_{1/2}$ and $r \in (0, 1/4)$; then $B_r(x) \subseteq B_1$. By the Poincaré inequality in the form $\fint_B |u - (u)_B| \leqslant Cr \fint_B |\nabla u|$ for balls B of radius r, which follows from Theorem 1.9 with $p = 1$ (where $\gamma_1 = 1$, $p^* = \frac{d}{d-1}$, the convex set is $B_r(x)$ of diameter $2r$) together with $\|\cdot\|_{\underline{L}^1} \leqslant \|\cdot\|_{\underline{L}^{p^*}}$ (Jensen), and Cauchy–Schwarz,

$$r^{-\alpha}\|u-(u)_{B_r(x)}\|_{\underline{L}^1(B_r(x))} \leqslant Cr^{1-\alpha}\Big(\frac{1}{|B_1|r^d}\int_{B_r(x)}|\nabla u|^2\Big)^{1/2} \leqslant C|B_1|^{-1/2}K^{1/2}r^{1-\alpha-\frac{d}{2}+\frac{d-2+2\alpha}{2}} = C|B_1|^{-1/2}K^{1/2}.$$

The supremum over x, r of the left side is therefore at most $CK^{1/2}$, and Lemma 1.19 gives the claim (its proof shows that u agrees a.e. with a continuous function, which is the representative in the statement). □

Proposition 4.26 (Meyers' example)**.** *Let $d \geqslant 2$, $\Lambda \geqslant 1$, and let*

$$\alpha(d,\Lambda) := \frac{1}{2}\left(-(d-2) + \sqrt{(d-2)^2 + \frac{4(d-1)}{\Lambda}}\right) \tag{4.102}$$

be the positive root of $\alpha^2 + (d-2)\alpha - \frac{d-1}{\Lambda} = 0$. Write $\alpha := \alpha(d,\Lambda)$ and define

$$\mathbf{a}(x) := I + (\Lambda - 1)\frac{x \otimes x}{|x|^2} \quad (x \neq 0), \qquad \mathbf{a}(0) := I, \qquad u(x) := |x|^{\alpha-1}x_1 \quad (x \neq 0), \qquad u(0) := 0.$$

Then:

(i) $\alpha \in (0,1]$, *with $\alpha = 1$ if and only if $\Lambda = 1$, and $\alpha(2,\Lambda) = \Lambda^{-1/2}$;*

(ii) $\mathbf{a}$ *is measurable and symmetric and satisfies (4.1);*

(iii) $u \in H^1(B_1) \cap C(B_1)$ *and $-\nabla\cdot\mathbf{a}\nabla u = 0$ in B_1 in the weak sense, that is $\int_{B_1} \nabla\varphi\cdot\mathbf{a}\nabla u = 0$ for all $\varphi \in C_c^\infty(B_1)$;*

(iv) $u(te_1) = t^\alpha$ *for* $t \in (0,1)$. *In particular* $u \notin C^{0,\alpha'}(B_{1/2})$ *for every* $\alpha' > \alpha$ *(there is no representative with finite* α'*-Hölder seminorm on* $B_{1/2}$*).*

Proof. *(i)* Since $4(d-1)/\Lambda > 0$, the square root exceeds $d-2$ and $\alpha > 0$. The polynomial $f(s) := s^2 + (d-2)s - \frac{d-1}{\Lambda}$ is strictly increasing on $[0,\infty)$ with $f(0) < 0$ and $f(1) = (d-1)(1-\Lambda^{-1}) \geqslant 0$, with equality iff $\Lambda = 1$; hence $\alpha \in (0,1]$ and $\alpha = 1$ iff $\Lambda = 1$. For $d = 2$, $f(s) = s^2 - \Lambda^{-1}$.

(ii) $\mathbf{a}$ is smooth away from 0, hence measurable, and symmetric. With $\pi(x) := x \otimes x/|x|^2$ an orthogonal projection, $\xi \cdot \mathbf{a}\xi = |\xi|^2 + (\Lambda-1)|\pi\xi|^2 \in [|\xi|^2, \Lambda|\xi|^2]$. Since $(I+s\pi)(I+t\pi) = I + (s+t+st)\pi$ and $s = \Lambda - 1$, $t = \Lambda^{-1} - 1$ give $s+t+st = 0$, we have $\mathbf{a}^{-1} = I + (\Lambda^{-1}-1)\pi$ and $e \cdot \mathbf{a}^{-1}e = |e|^2 + (\Lambda^{-1}-1)|\pi e|^2 \geqslant \Lambda^{-1}|e|^2$. At $x = 0$ both hold trivially.

(iii) For $x \neq 0$, u is smooth and

$$\nabla u = |x|^{\alpha-1}e_1 + (\alpha-1)|x|^{\alpha-3}x_1 x, \qquad x \cdot \nabla u = \alpha|x|^{\alpha-1}x_1.$$

Since $|x_1||x| \leqslant |x|^2$, $|\nabla u| \leqslant (1 + |\alpha - 1|)|x|^{\alpha-1} \leqslant 2|x|^{\alpha-1}$, and $|u| \leqslant |x|^\alpha$; thus u is continuous on B_1 and, in polar coordinates, $\int_{B_1} |\nabla u|^2 \leqslant 4\,|\partial B_1| \int_0^1 \rho^{2\alpha-2+d-1} d\rho < \infty$ because $2\alpha + d - 2 > 0$. We check that the pointwise gradient is the weak gradient on all of B_1. Let $\eta_\varepsilon(x) := \eta(|x|/\varepsilon)$ with η smooth, $\eta = 0$ on $[0,1]$, $\eta = 1$ on $[2,\infty)$, so $|\nabla \eta_\varepsilon| \leqslant C/\varepsilon$. For $\psi \in C_c^\infty(B_1)$, integration by parts on $B_1 \backslash \{0\}$ (where $u\psi\eta_\varepsilon$ is smooth and compactly supported) gives $\int u\, \partial_i \psi\, \eta_\varepsilon = -\int \partial_i u\, \psi \eta_\varepsilon - \int u\psi\, \partial_i \eta_\varepsilon$, and $|\int u\psi \partial_i \eta_\varepsilon| \leqslant C\|\psi\|_\infty \varepsilon^{-1} \int_{B_{2\varepsilon}} |x|^\alpha \leqslant C\varepsilon^{\alpha+d-1} \to 0$. Letting $\varepsilon \to 0$ (dominated convergence) shows that the pointwise gradient is the weak gradient, so $u \in H^1(B_1)$.

Next, for $x \neq 0$, using $\mathbf{a}\xi = \xi + (\Lambda-1)\pi\xi$ and the formulas above,

$$\sigma := \mathbf{a}\nabla u = |x|^{\alpha-1}e_1 + (\Lambda\alpha - 1)|x|^{\alpha-3}x_1 x, \qquad \frac{x}{|x|} \cdot \sigma = \Lambda\alpha|x|^{\alpha-2}x_1 = \Lambda\alpha \frac{u}{|x|}.$$

Since $\nabla \cdot (|x|^{\alpha-1}e_1) = (\alpha-1)|x|^{\alpha-3}x_1$ and $\nabla \cdot (|x|^{\alpha-3}x_1 x) = (\alpha - 2 + d)|x|^{\alpha-3}x_1$,

$$\nabla \cdot \sigma = |x|^{\alpha-3}x_1\big(\alpha - 1 + (\Lambda\alpha - 1)(\alpha + d - 2)\big) = \Lambda\,|x|^{\alpha-3}x_1\, f(\alpha) = 0 \quad \text{in } B_1\backslash\{0\}.$$

Let $\varphi \in C_c^\infty(B_1)$ and $\varepsilon \in (0,1)$. The divergence theorem on $B_1 \backslash B_\varepsilon$, whose outer normal on ∂B_ε is $-x/|x|$, gives

$$\int_{B_1\backslash B_\varepsilon} \nabla\varphi \cdot \sigma = \int_{B_1\backslash B_\varepsilon} \nabla \cdot (\varphi\sigma) = -\int_{\partial B_\varepsilon} \varphi\, \sigma \cdot \tfrac{x}{|x|}\, d\sigma,$$

and the right side is bounded by $\|\varphi\|_\infty \Lambda\alpha\, \varepsilon^{\alpha-1}\, |\partial B_1| \varepsilon^{d-1} = C\varepsilon^{\alpha+d-2} \to 0$, as $\alpha > 0$, $d \geqslant 2$. Since $\sigma \in L^2 \subseteq L^1$, the left side tends to $\int_{B_1} \nabla\varphi \cdot \sigma$. Thus $\int_{B_1} \nabla\varphi \cdot \mathbf{a}\nabla u = 0$.

(iv) $u(te_1) = t^{\alpha-1}t = t^\alpha$ and $u(0) = 0$. Since u is continuous, any representative in $C^{0,\alpha'}(B_{1/2})$ equals u everywhere on $B_{1/2}$; then $t^\alpha = |u(te_1) - u(0)| \leqslant [u]_{C^{0,\alpha'}} t^{\alpha'}$ for all $t \in (0, 1/2)$, which is impossible as $t \to 0$ if $\alpha' > \alpha$. □

Proposition 4.27 (Sharp Hölder exponent in $d = 2$)**.** *Let* $d = 2$, $\Lambda \geqslant 1$ *and* $\alpha_\Lambda := \Lambda^{-1/2}$.

(i) Let $\mathbf{a}$ *be measurable, symmetric and satisfy* (4.1)*, and let* $u \in H^1(B_1)$ *solve* $-\nabla\cdot\mathbf{a}\nabla u = 0$ *in* B_1. *Then for all* $x_0 \in B_{1/2}$ *and* $r \in (0, 1/2]$,

$$\int_{B_r(x_0)} |\nabla u|^2 \leqslant \Lambda(2r)^{2\alpha_\Lambda} \int_{B_1} |\nabla u|^2, \tag{4.103}$$

and u has a representative in $C^{0,\alpha_\Lambda}(B_{1/2})$ with $[u]_{C^{0,\alpha_\Lambda}(B_{1/2})} \leqslant C\Lambda\|\nabla u\|_{L^2(B_1)}$, where C is an absolute constant.

(ii) If $\Lambda > 1$ and $\alpha' \in (\alpha_\Lambda, 1]$, there exist $\mathbf{a}$ and u as in (i) (namely those of Proposition 4.26 with $d = 2$) such that $u \notin C^{0,\alpha'}(B_{1/2})$.

Proof. (i) Fix $x_0 \in B_{1/2}$, so $B_{1/2}(x_0) \subseteq B_1$, and set $v(y) := u(x_0 + \frac{y}{2})$, $\mathbf{a}_v(y) := \mathbf{a}(x_0 + \frac{y}{2})$ for $y \in B_1$. The change of variables shows that $v \in H^1(B_1)$ solves $-\nabla \cdot \mathbf{a}_v \nabla v = 0$, and $\mathbf{a}_v$ is measurable, symmetric and satisfies (4.1). Moreover, since $d = 2$ and $\nabla v(y) = \frac{1}{2}\nabla u(x_0 + \frac{y}{2})$, $\int_{B_s} \nabla v \cdot \mathbf{a}_v \nabla v = \int_{B_{s/2}(x_0)} \nabla u \cdot \mathbf{a} \nabla u$ for $s \in (0, 1]$. Let $\beta := 2\alpha_\Lambda$. By Theorem 4.24, for $0 < s' \leqslant s < 1$, $s'^{-\beta} \int_{B_{s'}} \nabla v \cdot \mathbf{a}_v \nabla v \leqslant s^{-\beta} \int_{B_s} \nabla v \cdot \mathbf{a}_v \nabla v$; letting $s \uparrow 1$ (monotone convergence) and putting $s' = 2r$ for $r < 1/2$ yields

$$\int_{B_r(x_0)} \nabla u \cdot \mathbf{a} \nabla u \leqslant (2r)^\beta \int_{B_{1/2}(x_0)} \nabla u \cdot \mathbf{a} \nabla u \leqslant (2r)^\beta \int_{B_1} \nabla u \cdot \mathbf{a} \nabla u.$$

Since $|\nabla u|^2 \leqslant \nabla u \cdot \mathbf{a} \nabla u \leqslant \Lambda |\nabla u|^2$, this is (4.103). It says that (4.101) holds with $d = 2$, $\alpha = \alpha_\Lambda \in (0, 1]$ and $K = \Lambda 2^{2\alpha_\Lambda} \|\nabla u\|^2_{L^2(B_1)} \leqslant 4\Lambda \|\nabla u\|^2_{L^2(B_1)}$, and Lemma 4.25 gives $[u]_{C^{0,\alpha_\Lambda}(B_{1/2})} \leqslant C\alpha_\Lambda^{-1}(4\Lambda)^{1/2}\|\nabla u\|_{L^2(B_1)} = 2C\Lambda\|\nabla u\|_{L^2(B_1)}$.

(ii) By Proposition 4.26(i)–(iv) with $d = 2$, the Meyers pair is admissible in (i) and $u \notin C^{0,\alpha'}(B_{1/2})$ for $\alpha' > \alpha(2, \Lambda) = \alpha_\Lambda$. □

Remark 4.28. Symmetric coefficient fields satisfying (4.1) form a subclass of those allowed in this chapter. Hence, for $d = 2$, Proposition 4.27 shows that no exponent larger than $\Lambda^{-1/2}$ is valid in Proposition 4.3, whose exponent $(C\Lambda)^{-1/2}$ is therefore optimal up to the constant C; for symmetric $\mathbf{a}$ the exponent $\Lambda^{-1/2}$ itself is valid and optimal. (The proof of Theorem 4.24 uses the symmetry of $\mathbf{a}$; we do not treat the sharp constant for nonsymmetric $\mathbf{a}$.) For $\Lambda = 1$ the Meyers pair is $\mathbf{a} = I$, $u = x_1$, and $\alpha(d, 1) = 1$ in every dimension.

Remark 4.29 (Cited, not proved here, and its resolution)**.** Statements (a) and (b) below are recorded for context only. They are not proved here; statement (c) is proved in the next subsection.

(a) For fixed $d \geqslant 3$, an elementary expansion gives $\alpha(d, \Lambda) \sim \frac{d-1}{d-2}\Lambda^{-1}$ as $\Lambda \to \infty$; for $d = 2$ instead $\alpha(2, \Lambda) = \Lambda^{-1/2}$.

(b) For $d \geqslant 2$, $\mathbf{a}$ symmetric with $I \leqslant \mathbf{a} \leqslant \Lambda I$ and $u \in H^1(B_1)$ a solution of $-\nabla \cdot \mathbf{a} \nabla u = 0$, one may ask whether

$$r \mapsto r^{-\beta} \int_{B_r} \nabla u \cdot \mathbf{a} \nabla u \ \text{ is nondecreasing on } (0, 1), \qquad \beta := \Big((d-2)^2 + \frac{4(d-1)}{\Lambda}\Big)^{1/2}. \tag{4.104}$$

For $d = 2$ we have $\beta = 2\Lambda^{-1/2}$, and the answer is yes by Theorem 4.24. This monotonicity formula, extended to $d \geqslant 3$, is sometimes posed as a question (a natural conjecture, suggested by Meyers' example), with the natural exponent $\alpha(d, \Lambda) = \frac{1}{2}(\beta - (d-2))$ of Proposition 4.26, which equals the Meyers exponent. By Morrey's theorem in the form of Lemma 4.25, (4.104) would imply Hölder regularity with exponent $\alpha(d, \Lambda)$.

(c) The conjectured exponent in (b) is not correct for $d \geqslant 3$ once Λ is large; this is proved below, see Proposition 4.30 and Corollary 4.32. Indeed, $\alpha(d,\Lambda) = \frac{2(d-1)}{\Lambda(\beta+d-2)} \geqslant \Lambda^{-1}$ for $\Lambda \geqslant 1$, since $\beta \leqslant d$ in (4.104). By Proposition 4.30 there are symmetric $\mathbf{a}$ and a solution u with $u \notin C^{0,\alpha'}(B_{1/2})$ for every $\alpha' > \alpha_\gamma$, where $\alpha_\gamma \leqslant \exp(-\Lambda^{1/2}/40) < \Lambda^{-1}$ once $\Lambda \geqslant 3 \cdot 10^5$. If (4.104) held on all balls (after translation), Lemma 4.25 would give $u \in C^{0,\alpha(d,\Lambda)}$ with $\alpha(d,\Lambda) > \alpha_\gamma$, a contradiction. This is due to Guerra [Gue26].

4.6 Sharpness of the Hölder exponent and of the Harnack ratio for $d \geqslant 3$

In dimension at least three, the exponential dependence on $\Lambda^{1/2}$ is necessary for the reciprocal of the Hölder exponent (the corresponding statement for the Harnack constant is Proposition 4.20). This is due to Guerra, who constructed a homogeneous solution from an eigenfunction of a weighted Laplacian [Gue26, Theorem 1.1 and Lemma 2.1]. What is new here is that the explicit hyperbolic-tangent profile below replaces the eigenvalue problem, and that the construction is formulated directly in Euclidean coordinates, so that all differentiations and estimates are elementary. The Harnack half of the statement is recorded for a single coefficient field, with the exact ratio (4.110); Proposition 4.20 is stronger. The angular profile is nearly constant on each hemisphere and changes sign in a narrow band around the equator. Its derivative at the poles is exponentially small. Cancellation of the equation at the poles determines the homogeneity exponent, while an off-diagonal coefficient cancels the remaining error. All differentiations below take place in $\mathbb{R}^d$.

Throughout this subsection $\alpha = \alpha_\gamma$ denotes the exponent (4.106), not the function $\alpha(d,\Lambda)$ of Proposition 4.26. We use the Euclidean length $|x| = (\sum_{i=1}^d x_i^2)^{1/2}$ and the round ball $B_R = \{x : |x| < R\}$. For vectors v, w, the matrix $v \otimes w$ acts by $(v \otimes w)\xi = v(w \cdot \xi)$.

Proposition 4.30 (Sharpness of the Hölder exponent and of the Harnack ratio)**.** *Let $d \geqslant 3$. For every $\gamma \geqslant 1$, set*

$$b = 400\gamma^2, \qquad K = 80\gamma, \qquad \Lambda = K^2, \qquad \mu = \frac{2(d-1)\gamma}{\sinh(2\gamma)}\,. \tag{4.105}$$

Define

$$\alpha = \frac{\sqrt{(d-2)^2 + 4\mu/b} - (d-2)}{2} = \frac{2\mu}{b\big(\sqrt{(d-2)^2 + 4\mu/b} + d - 2\big)}\,. \tag{4.106}$$

There is a bounded measurable symmetric matrix field $\mathbf{a}_\gamma$ on B_1 satisfying, for every $x \in B_1$ and $\xi \in \mathbb{R}^d$,

$$\xi \cdot \mathbf{a}_\gamma(x)\xi \geqslant |\xi|^2, \qquad \xi \cdot \mathbf{a}_\gamma^{-1}(x)\xi \geqslant \Lambda^{-1}|\xi|^2\,. \tag{4.107}$$

The function

$$u_\gamma(0) = 0, \qquad u_\gamma(x) = |x|^\alpha \frac{\tanh(\gamma x_d/|x|)}{\tanh\gamma} \quad (x \neq 0) \tag{4.108}$$

is positively α-homogeneous, belongs to $H^1(B_1) \cap C^{0,\alpha}(B_1)$, and solves $-\nabla \cdot \mathbf{a}_\gamma \nabla u_\gamma = 0$ weakly in B_1. Moreover,

$$0 < \alpha \leqslant e^{-2\gamma} = \exp\big(-\Lambda^{1/2}/40\big)\,. \tag{4.109}$$

Moreover, $|u_\gamma(x)| \leqslant |x|^\alpha$ *and* $[u_\gamma]_{C^{0,\alpha}(B_1)} \leqslant 5\gamma$. *For every* $\rho \in (0,1]$ *and* $\alpha' > \alpha$, *no representative of* u_γ *belongs to* $C^{0,\alpha'}(B_\rho)$. *The weak formulation holds with test functions in* $H^1_0(B_1)$, *where* $\mathbf{a}_\gamma(x) = K\,A\big(B(x)/(20\gamma), \nabla u_\gamma(x)\big)$ *for* $x \neq 0$ *and* $\mathbf{a}_\gamma(0) = 2\mathrm{I}_d$ *(notation of the proof). The positive weak solution* $v_\gamma = 1 + u_\gamma$ *satisfies, with* sup *and* inf *taken (pointwise, for the continuous function* v_γ, *equal to the essential ones) over the open ball* $B_{1/2}$, *where the supremum is not attained,*

$$\frac{\sup_{B_{1/2}} v_\gamma}{\inf_{B_{1/2}} v_\gamma} = \frac{1 + 2^{-\alpha}}{1 - 2^{-\alpha}} \geqslant \frac{2}{\alpha \log 2} \geqslant \exp\big(\Lambda^{1/2}/40\big)\,. \tag{4.110}$$

We first record a symmetrization which preserves the flux of a single solution. Its proof uses only quadratic inequalities.

Lemma 4.31 (Symmetrization of a flux)**.** *Let* $k \geqslant 1$ *and let an invertible matrix* C *satisfy*

$$\xi \cdot C\xi \geqslant k^{-1}|\xi|^2, \qquad \xi \cdot C^{-1}\xi \geqslant k^{-1}|\xi|^2 \quad (\xi \in \mathbb{R}^d)\,. \tag{4.111}$$

For every $p \in \mathbb{R}^d$, *there is an explicitly defined symmetric matrix* $A = A(C,p)$ *such that*

$$Ap = Cp, \qquad \frac{1}{2k}|\xi|^2 \leqslant \xi \cdot A\xi \leqslant 2k|\xi|^2 \quad (\xi \in \mathbb{R}^d)\,. \tag{4.112}$$

The choice $A(C,p)$ *is measurable in its arguments.*

Proof. For $p = 0$, take $A = k^{-1}\mathrm{I}_d$. Otherwise set

$$e = \frac{p}{|p|}, \qquad \sigma = e \cdot Ce, \qquad z = Ce - \sigma e, \qquad A = \sigma \mathrm{I}_d + e \otimes z + z \otimes e + \frac{2}{\sigma} z \otimes z\,. \tag{4.113}$$

Then $e \cdot z = 0$, $Ae = Ce$, and $Ap = Cp$. Applying (4.111) to e and to Ce gives

$$\sigma \geqslant k^{-1}, \qquad |Ce|^2 \leqslant k\sigma, \qquad \sigma + \frac{|z|^2}{\sigma} = \frac{|Ce|^2}{\sigma} \leqslant k\,. \tag{4.114}$$

Write $\xi = se + w$ with $e \cdot w = 0$. Completing a square yields

$$\begin{aligned} \xi \cdot A\xi &= \sigma s^2 + \sigma|w|^2 + 2s(z \cdot w) + \frac{2}{\sigma}(z \cdot w)^2 \\ &= \frac{\sigma}{2}s^2 + \sigma|w|^2 + \frac{2}{\sigma}\Big(z \cdot w + \frac{\sigma}{2}s\Big)^2 \geqslant \frac{\sigma}{2}|\xi|^2\,. \end{aligned}$$

For the upper bound, use $2s(z \cdot w) \leqslant \sigma|w|^2 + \sigma^{-1}|z|^2 s^2$ and $(z \cdot w)^2 \leqslant |z|^2|w|^2$. Thus

$$\xi \cdot A\xi \leqslant \left(\sigma + \frac{|z|^2}{\sigma}\right)s^2 + 2\left(\sigma + \frac{|z|^2}{\sigma}\right)|w|^2 \leqslant 2k|\xi|^2\,.$$

The formula is continuous wherever $p \neq 0$ and $\sigma > 0$; its specified value on $\{p = 0\}$ makes it measurable. □

Proof of Proposition 4.30. Step 1: the profile and the exponent. Put $\beta = \alpha + d - 2$ and define, for $-1 \leqslant t \leqslant 1$,

$$\varphi(t) = \frac{\tanh(\gamma t)}{\tanh \gamma}, \qquad \varphi'(t) = \frac{\gamma}{\tanh\gamma \cosh^2(\gamma t)}, \qquad \varphi''(t) = -2\gamma \tanh(\gamma t)\varphi'(t)\,. \tag{4.115}$$

In particular, $\varphi(\pm 1) = \pm 1$, $|\varphi| \leqslant 1$, and $\varphi' > 0$. The choice (4.106) is the positive root of

$$b\alpha\beta = \mu = (d-1)\varphi'(1)\,. \tag{4.116}$$

Since $\beta \geqslant d-2$ and $^{d-1}\!/_{d-2} \leqslant 2$,

$$\alpha = \frac{\mu}{b\beta} \leqslant \frac{1}{100\gamma \sinh(2\gamma)} \leqslant \frac{3}{100\gamma} e^{-2\gamma} \leqslant e^{-2\gamma}\,. \tag{4.117}$$

Here $e^{4\gamma} \geqslant 1 + 4\gamma \geqslant 5$ gives $\sinh(2\gamma) = {}^1\!/_2 e^{2\gamma}(1 - e^{-4\gamma}) \geqslant {}^1\!/_3 e^{2\gamma}$ and $\coth(2\gamma) = \frac{e^{4\gamma}+1}{e^{4\gamma}-1} \leqslant {}^3\!/_2$. In particular, $0 < \alpha < 1$.

Step 2: the matrix in ambient coordinates. The scalar function

$$F(t) = t - \frac{\sinh(2\gamma t)}{\sinh(2\gamma)} \tag{4.118}$$

is odd and vanishes at both endpoints. For $|t| \leqslant 1$,

$$|F'(t)| \leqslant 1 + 2\gamma \coth(2\gamma) \leqslant 4\gamma\,.$$

Indeed, $\coth(2\gamma) \leqslant {}^3\!/_2$ by $e^{4\gamma} \geqslant 5$. Integrating F' from the nearest endpoint gives

$$|F(t)| \leqslant 4\gamma(1 - |t|) \leqslant 4\gamma(1 - t^2)\,. \tag{4.119}$$

For $|t| < 1$, set

$$h(t) = \frac{1}{\beta}\left(2\gamma \tanh(\gamma t) + (d-1)\frac{F(t)}{1-t^2}\right), \qquad h(-1) = h(1) = 0\,. \tag{4.120}$$

The endpoint values are a convention: h jumps at $t = \pm 1$ (since $F(t)/(1-t^2) \to \mp F'(\pm 1)/2 \neq 0$), but the vector multiplied by h below satisfies $|hq| \leqslant 10\gamma\sqrt{1-t^2}$, so it vanishes there and B below is continuous across the poles. Since $d \geqslant 3$, we have

$$|h(t)| \leqslant \frac{2 + 4(d-1)}{d-2}\gamma \leqslant 10\gamma\,. \tag{4.121}$$

For $x \neq 0$, introduce the following functions of x:

$$r = |x|, \qquad \omega = \frac{x}{r}, \qquad t = \frac{x_d}{r}, \qquad q = e_d - t\omega, \qquad Y = h(t)q\,. \tag{4.122}$$

These are ambient vectors, and $\omega \cdot q = 0$, $|q|^2 = 1 - t^2$. Define

$$\begin{aligned} B(x) &= \mathrm{I}_d + (b-1)\omega \otimes \omega + \omega \otimes Y \\ &= \mathrm{I}_d + (b-1)\frac{x \otimes x}{|x|^2} + h\left(\frac{x_d}{|x|}\right)\frac{x}{|x|} \otimes \left(e_d - \frac{x_d x}{|x|^2}\right), \end{aligned} \tag{4.123}$$

and set $B(0) = \mathrm{I}_d$. This is an explicit bounded measurable matrix field, invariant under positive dilations away from the origin.

To verify ellipticity, write $\xi = s\omega + w$ with $\omega \cdot w = 0$. Then

$$B\xi = (bs + Y \cdot w)\omega + w, \qquad B^{-1}\xi = b^{-1}(s - Y \cdot w)\omega + w\,.$$

In particular, B is invertible and, for $x \neq 0$,

$$B^{-1} = \mathrm{I}_d + (b^{-1} - 1)\omega \otimes \omega - b^{-1}\omega \otimes Y\,. \tag{4.124}$$

Since $|Y|^2 \leqslant 100\gamma^2 = b/4$, the inequalities

$$|s(Y \cdot w)| \leqslant \frac{b}{2}s^2 + \frac{|Y|^2}{2b}|w|^2, \qquad b^{-1}|s(Y \cdot w)| \leqslant \frac{1}{2b}s^2 + \frac{|Y|^2}{2b}|w|^2$$

imply

$$\xi \cdot B\xi \geqslant \frac{1}{2}(bs^2 + |w|^2), \qquad \xi \cdot B^{-1}\xi \geqslant \frac{1}{2}(b^{-1}s^2 + |w|^2)\,. \tag{4.125}$$

Consequently, $C = B/(20\gamma) = B/\sqrt{b}$ satisfies (4.111) with $k = 40\gamma$, including at $x = 0$.

Step 3: the equation. Write $u = r^\alpha \varphi(t)$ away from the origin. Coordinate differentiation gives

$$\partial_i r = \frac{x_i}{r}, \qquad \partial_i t = \frac{\delta_{id}}{r} - \frac{x_d x_i}{r^3}, \qquad \nabla u = r^{\alpha-1}\big(\alpha\varphi(t)\omega + \varphi'(t)q\big)\,. \tag{4.126}$$

Define the scalar expression

$$L(t) = (1 - t^2)\varphi''(t) - (d-1)t\varphi'(t)\,. \tag{4.127}$$

The hyperbolic identity $\tanh s \cosh^2 s = \tfrac{1}{2}\sinh(2s)$ gives

$$\mu\varphi(t) = (d-1)\varphi'(t)\frac{\sinh(2\gamma t)}{\sinh(2\gamma)}\,. \tag{4.128}$$

Substituting (4.115) and (4.128) shows, for $|t| < 1$,

$$\beta h(t)(1 - t^2)\varphi'(t) = -L(t) - \mu\varphi(t)\,. \tag{4.129}$$

This also holds at $t = \pm 1$: there $L(1) = -\mu$, $L(-1) = \mu$, and $\varphi(\pm 1) = \pm 1$. This is the reason for (4.116).

The action of B on the displayed gradient, followed by (4.116) and (4.129), gives

$$\begin{aligned} B\nabla u &= r^{\alpha-1}\big(\big[b\alpha\varphi + h(t)(1-t^2)\varphi'\big]\omega + \varphi' q\big) \\ &= r^{\alpha-1}\big(Q(t)\omega + \varphi'(t)q\big), \qquad Q(t) = -\frac{L(t)}{\beta}\,. \end{aligned} \tag{4.130}$$

The final expression is smooth on $\mathbb{R}^d\backslash\{0\}$, including on the axis, because Q and φ are real analytic. Thus we need no derivatives of h to compute its divergence.

For completeness, if f, g are smooth functions of one real variable (here they extend smoothly to a neighborhood of $[-1,1]$, since tanh and sinh are entire, so (4.131) may be used at $|t| = 1$), the two coordinate identities in (4.126) give

$$\operatorname{div}\big(r^{\alpha-2} f(t) x + r^{\alpha-1} g(t) e_d\big) = r^{\alpha-2}\big(\beta f(t) + (\alpha-1) t g(t) + (1-t^2) g'(t)\big)\,. \tag{4.131}$$

Indeed, $\sum_i x_i \partial_i r = r$ and $\sum_i x_i \partial_i t = 0$, so differentiation of the first summand gives $\beta r^{\alpha-2} f(t)$. Differentiation of the second summand in the dth coordinate gives $r^{\alpha-2}((\alpha-1)tg + (1-t^2)g')$. In (4.130), take $f = Q - t\varphi'$ and $g = \varphi'$. Since $\alpha - 1 - \beta = -(d-1)$, we obtain

$$\operatorname{div}(B\nabla u) = r^{\alpha-2}\big(\beta Q + (1-t^2)\varphi'' - (d-1) t \varphi'\big) = r^{\alpha-2}(-L + L) = 0\,. \tag{4.132}$$

Equivalently, the coefficient of x in this smooth ambient flux is

$$f(t) = \frac{(1-\alpha) t \varphi'(t) - (1-t^2)\varphi''(t)}{\beta}\,.$$

Step 4: the origin. Since $\tanh \gamma \geqslant 1/2$, we have $|\varphi'| \leqslant 2\gamma$ and hence

$$|u(x)| \leqslant r^{\alpha}, \qquad |\nabla u(x)| \leqslant 3\gamma r^{\alpha-1}\,. \tag{4.133}$$

The bound on $\tanh \gamma$ follows from $e^{2\gamma} \geqslant 3$. If s_{d-1} denotes the area of the unit sphere, polar integration gives

$$\int_{B_1} |u|^2 \leqslant \frac{s_{d-1}}{2\alpha + d}, \qquad \int_{B_1} |\nabla u|^2 \leqslant \frac{9\gamma^2 s_{d-1}}{2\alpha + d - 2} < \infty\,.$$

To identify this classical gradient with the weak gradient, integrate $u\partial_i\zeta$ by parts on $B_1 \backslash \overline{B_\varepsilon}$ for $\zeta \in C_c^\infty(B_1)$. The inner boundary term has absolute value at most $s_{d-1}\|\zeta\|_\infty \varepsilon^{\alpha+d-1}$, which tends to zero. The volume integrals converge by the preceding bounds. Thus $u \in H^1(B_1)$.

Now set $J = B\nabla u$, using the smooth expression (4.130) away from zero. Choose $R < 1$ such that ζ vanishes near ∂B_R. The divergence theorem on $B_R \backslash \overline{B_\varepsilon}$, whose inner outward normal is $-\omega$, gives

$$\int_{B_R \backslash \overline{B_\varepsilon}} J \cdot \nabla\zeta = -\int_{\partial B_\varepsilon} \zeta J \cdot \omega\,.$$

By (4.133) and boundedness of B, its absolute value is bounded by

$$3\gamma \|B\|_\infty s_{d-1} \|\zeta\|_\infty \varepsilon^{\alpha+d-2} \longrightarrow 0\,.$$

Since $J \in L^2(B_1)$, passing to the limit proves $\int_{B_1} B\nabla u \cdot \nabla\zeta = 0$. Density extends this identity to every $\zeta \in H_0^1(B_1)$. The two integrability and removability conditions used here are $2\alpha + d - 2 > 0$ and $\alpha + d - 2 > 0$; both hold for $d \geqslant 3$.

Step 5: symmetry and Hölder regularity. Apply Lemma 4.31 pointwise to $C = B/(20\gamma)$ and $p = \nabla u$. It gives a bounded measurable symmetric field A with

$$A\nabla u = C\nabla u, \qquad K^{-1}|\xi|^2 \leqslant \xi \cdot A\xi \leqslant K|\xi|^2, \qquad K = 80\gamma\,.$$

Define $\mathbf{a}_\gamma = KA$. Then $\mathbf{a}_\gamma \nabla u = 4B\nabla u$ and $|\xi|^2 \leqslant \xi \cdot \mathbf{a}_\gamma \xi \leqslant K^2|\xi|^2$. For a symmetric positive matrix the last upper bound implies $\xi \cdot \mathbf{a}_\gamma^{-1}\xi \geqslant K^{-2}|\xi|^2$. One may see this without diagonalizing: with $y = \mathbf{a}_\gamma^{-1}\xi$, so that $y \cdot \mathbf{a}_\gamma y = \xi \cdot \mathbf{a}_\gamma^{-1}\xi$, Cauchy–Schwarz for the positive form $\mathbf{a}_\gamma$ gives

$$|\xi|^4 = (y \cdot \mathbf{a}_\gamma \xi)^2 \leqslant (y \cdot \mathbf{a}_\gamma y)(\xi \cdot \mathbf{a}_\gamma \xi) \leqslant K^2(\xi \cdot \mathbf{a}_\gamma^{-1}\xi)|\xi|^2 .$$

All denominators in (4.113) are positive. In fact $\nabla u \neq 0$ away from zero: for $|t| < 1$, its orthogonal component $\varphi' q$ is nonzero, and at the poles its radial component is $\pm\alpha$. Thus (4.123), (4.126), and (4.113) explicitly specify the entire coefficient field. Its value at the origin from the lemma is $\mathbf{a}_\gamma(0) = 2\mathrm{I}_d$.

We also verify the asserted Hölder regularity directly. For $x, y \in B_1$, write $\delta = |x - y|$ and $r_0 = \min\{|x|, |y|\}$. If $\delta \geqslant r_0/2$, then the two radii are at most 2δ and 3δ, respectively, and $|u(x) - u(y)| \leqslant 5\delta^\alpha$. If $0 < \delta < r_0/2$, the segment joining them has distance at least $r_0/2 > \delta$ from the origin. Integrating (4.133) along that segment gives $|u(x) - u(y)| \leqslant 3\gamma\delta^\alpha$. Therefore $[u]_{C^{0,\alpha}(B_1)} \leqslant 5\gamma$.

The formula gives $u(sx) = s^\alpha u(x)$ for $s > 0$. In particular, $u(se_d) = s^\alpha$ and $u(0) = 0$. A $C^{0,\alpha'}$ bound at zero for $\alpha' > \alpha$ would force $s^{\alpha-\alpha'}$ to remain bounded as $s \downarrow 0$, which is impossible. A continuous representative equal to u almost everywhere equals u everywhere, since a nonempty open set has positive Lebesgue measure. This proves the assertion about all representatives.

Step 6: a positive solution. Since $|u(x)| \leqslant |x|^\alpha < 1$ on B_1, the function $v = 1 + u$ is positive there. For every $0 < R \leqslant 1$, the limits along $\pm se_d$ as $s \uparrow R$ show

$$\sup_{B_R} u = R^\alpha, \qquad \inf_{B_R} u = -R^\alpha . \tag{4.134}$$

Hence the ratio in (4.110) is exact. To bound it from below, put $z = \frac{1}{2}\alpha \log 2$. Then $\frac{1+2^{-\alpha}}{1-2^{-\alpha}} = \coth z \geqslant z^{-1}$. Indeed, the function $z\cosh z - \sinh z$ vanishes at zero and has derivative $z \sinh z \geqslant 0$. Finally (4.109) and $\frac{2}{\log 2} > 1$ give the last bound in (4.110). □

The relation between Harnack constants and Hölder exponents is the one used in Step 1 of the proof of Theorem 4.21: a Harnack constant $H \geqslant 3$ valid for the class, with the same bound after translation and dilation, gives the contraction $\operatorname{osc}_{B_{r/2}} u \leqslant \frac{H-1}{H+1} \operatorname{osc}_{B_r} u$ and hence the exponent $\beta_H = \log_2 \frac{H+1}{H-1} \in [H^{-1}, 1]$. Conversely, for the solution u_γ of Proposition 4.30, (4.134) gives $\operatorname{osc}_{B_R} u_\gamma = 2R^\alpha$, so the contraction can hold only if $2^{-\alpha} \leqslant \frac{H-1}{H+1}$, that is, $H \geqslant \frac{1+2^{-\alpha}}{1-2^{-\alpha}}$. This is the first inequality in (4.110) read backwards.

Corollary 4.32. *Let $d \geqslant 3$ and $\Lambda \geqslant 6400$. Suppose $\alpha_* \in (0, 1]$ is such that every weak solution $u \in H^1(B_1)$ of $-\nabla \cdot \mathbf{a}\nabla u = 0$ in B_1, with $\mathbf{a}$ measurable, symmetric and satisfying (4.1), has a representative in $C^{0,\alpha_*}(B_{1/2})$. Then $\alpha_* \leqslant \exp(-\Lambda^{1/2}/40)$. If $C_H \in [1, \infty)$ is such that $\sup_{B_{1/2}} v \leqslant C_H \inf_{B_{1/2}} v$ for every such positive weak solution v, then $C_H \geqslant \exp(\Lambda^{1/2}/40)$. In particular, for $\Lambda \geqslant 6400$ and symmetric $\mathbf{a}$ the exponent of Theorem 4.21 can be taken in $[\exp(-C\Lambda^{1/2}), \exp(-\Lambda^{1/2}/40)]$ with $C = C(d)$, and no better.*

Proof. Take $\gamma = \Lambda^{1/2}/80 \geqslant 1$ in Proposition 4.30. The function u_γ is a weak solution which, by the assertion about representatives in Proposition 4.30, is not in $C^{0,\alpha'}(B_{1/2})$ for any

$\alpha' > \alpha$; hence $\alpha_* \leqslant \alpha \leqslant e^{-2\gamma}$. The Harnack statement follows from (4.110) applied to v_γ. □

Symmetric fields form a subclass of the coefficient fields allowed in (4.1), so these obstructions also apply to the full class of possibly nonsymmetric coefficients. Combined with Theorem 4.21 and Theorem 4.18 (both stated for $d \geqslant 3$), this proves sharpness of the exponential orders as $\Lambda \to \infty$. The threshold 6400 is harmless for this conclusion; it cannot simply be omitted from the asserted counterexample bound, since for $\Lambda = 1$ the symmetric coefficient is I_d and harmonic functions have the uniform exponent 1.

The Harnack half of Corollary 4.32 is weaker than Proposition 4.20, which gives the lower bound $\exp(\Lambda^{1/2})$ already for a constant matrix, in every dimension $d \geqslant 2$ and for every $\Lambda \geqslant 1$. The point of the homogeneous construction is that a single coefficient field yields the Hölder obstruction, and the exact Harnack ratio (4.110) in terms of the Hölder exponent α. The example of Proposition 4.20 is smooth and so gives no Hölder obstruction.

If symmetry is unnecessary, one may omit the symmetrization (the first half of Step 5) and take $\mathbf{a} = 2B$: by (4.125), $\xi \cdot 2B\xi \geqslant bs^2 + |w|^2 \geqslant |\xi|^2$ and $\xi \cdot (2B)^{-1}\xi \geqslant \frac{1}{4}(b^{-1}s^2 + |w|^2) \geqslant \frac{1}{4b}|\xi|^2$, so $2B$ is a bounded measurable (nonsymmetric) field satisfying (4.1) with $\Lambda = 4b = 1600\gamma^2$. Since $2B\nabla u_\gamma = \frac{1}{2}\mathbf{a}_\gamma \nabla u_\gamma$, the same u_γ is a weak solution, and all other conclusions of Proposition 4.30 hold verbatim. Thus for $\Lambda \geqslant 1600$ and $\gamma = \Lambda^{1/2}/40$ one gets $\alpha \leqslant e^{-2\gamma} = \exp(-\Lambda^{1/2}/20)$ and Harnack ratio $\geqslant \exp(\Lambda^{1/2}/20)$ for the nonsymmetric class; the threshold and the constant are better by factors 4 and 2.

4.7 Green function decay estimates

This subsection has two parts. In the first, we bound, far from the source, the solution of a Dirichlet problem whose source is the divergence of a vector field supported in B_1. A duality argument reduces this bound to an oscillation decay estimate for $\mathbf{a}^t$-harmonic functions, so that different regularity inputs give different decay rates: the De Giorgi–Nash estimate (Theorem 4.8) gives the rate $r^{2-\alpha-d}$, and for constant coefficients the gradient bound for harmonic functions gives the rate r^{1-d}, which is the rate of decay of the gradient of the fundamental solution of the Laplace equation. In the second part, for $d \geqslant 3$, we define the Green function of $-\nabla \cdot \mathbf{a}\nabla$ in a ball, with pole at the origin, as the kernel of the map which sends a source g to the value at the origin of the solution of the adjoint Dirichlet problem. We prove that it exists and is unique, and that it is comparable to $|x|^{2-d}$ in $B_{R/2}$, where B_R is the ball, up to factors $\exp(\pm C\Lambda^{1/2})$.

Throughout this subsection we fix a radius $R > 0$ and assume that $\mathbf{a} \in L^\infty(B_R; \mathbb{R}^{d\times d})$ satisfies (4.1) for almost every $x \in B_R$. The transpose $\mathbf{a}^t$ satisfies (4.1) with the same Λ, because $e \cdot \mathbf{a}^t e = e \cdot \mathbf{a} e$ and $e \cdot (\mathbf{a}^t)^{-1} e = e \cdot \mathbf{a}^{-1} e$; in particular, (4.2) and $|\mathbf{b}e| \leqslant \Lambda|e|$ hold for $\mathbf{b} \in \{\mathbf{a}, \mathbf{a}^t\}$. Solutions are understood in the weak sense: if $\mathbf{b} \in \{\mathbf{a}, \mathbf{a}^t\}$, $U \subseteq B_R$ is open, $g \in L^2(U)$ and $\mathbf{f} \in L^2(U; \mathbb{R}^d)$, then $u \in H^1(U)$ is a solution of $-\nabla \cdot \mathbf{b}\nabla u = g + \nabla \cdot \mathbf{f}$ in U if

$$\int_U \nabla\varphi \cdot \mathbf{b}\nabla u = \int_U \big(g\varphi - \mathbf{f} \cdot \nabla\varphi\big) \qquad \forall \varphi \in H^1_0(U)\,. \tag{4.135}$$

Since $\mathbf{b}$ is bounded and $C^\infty_c(U)$ is dense in $H^1_0(U)$, it is equivalent to require (4.135) for every $\varphi \in C^\infty_c(U)$. If $g = 0$ and $\mathbf{f} = 0$, we say that u is $\mathbf{b}$-*harmonic* in U. We write $A_r := B_r \backslash B_{r/2}$ for the dyadic annuli, and $\operatorname{ess\,osc}_S u := \operatorname{ess\,sup}_S u - \operatorname{ess\,inf}_S u$.

We use two Poincaré inequalities. Let $d \geqslant 2$. There exists $C_1(d) < \infty$ such that, for every $s > 0$,

$$\|v\|_{L^2(B_s)} \leqslant C_1 s \|\nabla v\|_{L^2(B_s)} \qquad \forall v \in H^1_0(B_s)\,, \tag{4.136}$$

$$\|w - (w)_{B_s}\|_{L^2(B_s)} \leqslant C_1 s \|\nabla w\|_{L^2(B_s)} \qquad \forall w \in H^1(B_s)\,. \tag{4.137}$$

For (4.136), let $p := \frac{2d}{d+2} \in [1, d)$, so that $p^* = 2$ and $\frac{1}{p} - \frac{1}{2} = \frac{1}{d}$. Since $H^1_0(B_s) \subseteq W^{1,p}_0(B_s)$, Proposition 1.5 and Hölder's inequality give

$$\|v\|_{L^2(B_s)} \leqslant C\gamma_p \|\nabla v\|_{L^p(B_s)} \leqslant C\gamma_p |B_s|^{1/d} \|\nabla v\|_{L^2(B_s)}\,.$$

For (4.137), apply Proposition 1.10 with $E = B_1$ and $\theta = \frac{1}{2}$ to the function $x \mapsto w(sx)$.

We also record the Caccioppoli inequality in the form in which we use it; the argument is that of Lemma 4.1. Let $\mathbf{b} \in \{\mathbf{a}, \mathbf{a}^t\}$, let $U \subseteq B_R$ be open, $u \in H^1(U)$ and $k \in \mathbb{R}$, and let η be a bounded Lipschitz function on $\mathbb{R}^d$ such that

$$\eta^2(u-k) \in H^1_0(U) \qquad \text{and} \qquad \int_U \nabla\big(\eta^2(u-k)\big) \cdot \mathbf{b}\nabla u = 0\,.$$

For example, this holds if u is $\mathbf{b}$-harmonic in U and η has compact support in U. Then

$$\int_U \eta^2\, \nabla u \cdot \mathbf{b}\nabla u \leqslant 4\Lambda \int_U (u-k)^2 |\nabla \eta|^2\,. \tag{4.138}$$

Indeed, $\nabla(\eta^2(u-k)) = \eta^2 \nabla u + 2\eta(u-k)\nabla\eta$, so (4.2) for $\mathbf{b}$ and Young's inequality give

$$\int_U \eta^2\, \nabla u \cdot \mathbf{b}\nabla u = -2\int_U \eta\,(u-k)\,\nabla\eta \cdot \mathbf{b}\nabla u \leqslant \frac{1}{2}\int_U \eta^2\, \nabla u \cdot \mathbf{b}\nabla u + 2\Lambda \int_U (u-k)^2 |\nabla\eta|^2\,,$$

and the first term on the right is finite.

The Dirichlet problems we need are solved by the Lax–Milgram theorem, which we recall: if H is a real Hilbert space and B is a bilinear form on H with $|B(v,\varphi)| \leqslant M\|v\|_H\|\varphi\|_H$ and $B(v,v) \geqslant c\|v\|_H^2$ for some $M < \infty$ and $c > 0$, then for every bounded linear functional ℓ on H there is exactly one $v \in H$ such that $B(v,\varphi) = \ell(\varphi)$ for every $\varphi \in H$.

Lemma 4.33 (Dirichlet problem)**.** *Let $d \geqslant 2$ and $\mathbf{b} \in \{\mathbf{a}, \mathbf{a}^t\}$. For every $g \in L^2(B_R)$ and $\mathbf{f} \in L^2(B_R; \mathbb{R}^d)$, there is exactly one $v \in H^1_0(B_R)$ which is a solution of $-\nabla \cdot \mathbf{b}\nabla v = g + \nabla \cdot \mathbf{f}$ in B_R. If $\mathbf{f} = 0$, then*

$$\|\nabla v\|_{L^2(B_R)} \leqslant C_1 R \|g\|_{L^2(B_R)} \qquad \text{and} \qquad \|v\|_{L^2(B_R)} \leqslant C_1^2 R^2 \|g\|_{L^2(B_R)}\,, \tag{4.139}$$

where $C_1(d)$ is the constant in (4.136).

Proof. In this proof, all norms are taken on B_R. By (4.136), the bilinear form $\int_{B_R} \nabla v \cdot \nabla w$ is an inner product on $H^1_0(B_R)$ whose norm $\|\nabla v\|_{L^2}$ is equivalent to the H^1 norm, so $H^1_0(B_R)$ is a Hilbert space with this inner product. The bilinear form $B(v,\varphi) := \int_{B_R} \nabla\varphi \cdot \mathbf{b}\nabla v$ satisfies $|B(v,\varphi)| \leqslant \Lambda \|\nabla v\|_{L^2}\|\nabla\varphi\|_{L^2}$ and, by (4.1), $B(v,v) \geqslant \|\nabla v\|_{L^2}^2$. By (4.136), the linear

functional $\ell(\varphi) := \int_{B_R}(g\varphi - \mathbf{f}\cdot\nabla\varphi)$ satisfies $|\ell(\varphi)| \leqslant (C_1R\|g\|_{L^2} + \|\mathbf{f}\|_{L^2})\|\nabla\varphi\|_{L^2}$. The Lax–Milgram theorem gives exactly one $v \in H^1_0(B_R)$ with $B(v,\varphi) = \ell(\varphi)$ for every $\varphi \in H^1_0(B_R)$, which is the first statement. If $\mathbf{f} = 0$, testing with $\varphi = v$ and using (4.136) gives

$$\|\nabla v\|_{L^2}^2 \leqslant \int_{B_R} gv \leqslant \|g\|_{L^2}\|v\|_{L^2} \leqslant C_1R\|g\|_{L^2}\|\nabla v\|_{L^2}\,,$$

which gives the first bound of (4.139); the second follows from the first and (4.136). □

Lemma 4.34 (Weak maximum principle)**.** *Let $d \geqslant 2$ and $\mathbf{b} \in \{\mathbf{a}, \mathbf{a}^t\}$, and let $v \in H^1_0(B_R)$ satisfy $\int_{B_R}\nabla\varphi\cdot\mathbf{b}\nabla v \geqslant 0$ for every $\varphi \in H^1_0(B_R)$ with $\varphi \geqslant 0$. Then $v \geqslant 0$ almost everywhere in B_R. In particular, if $g \in L^2(B_R)$ and $g \geqslant 0$, then the solution $v \in H^1_0(B_R)$ of $-\nabla\cdot\mathbf{b}\nabla v = g$ in B_R satisfies $v \geqslant 0$.*

Proof. The negative part $v_- := \max\{-v, 0\}$ belongs to $H^1_0(B_R)$ and $\nabla v_- = -\mathbf{1}_{\{v<0\}}\nabla v$. Testing with $\varphi = v_- \geqslant 0$ and using (4.1), we get

$$0 \leqslant \int_{B_R}\nabla v_-\cdot\mathbf{b}\nabla v = -\int_{B_R}\nabla v_-\cdot\mathbf{b}\nabla v_- \leqslant -\|\nabla v_-\|^2_{L^2(B_R)}\,.$$

Hence $\nabla v_- = 0$, and $v_- = 0$ by (4.136). □

We turn to solutions with a divergence-form source supported in B_1. The following lemma converts an oscillation decay estimate for $\mathbf{a}^t$-harmonic functions into a decay estimate for such solutions.

Lemma 4.35. *Let $d \geqslant 2$ and $R \geqslant 10$. Fix $\beta \in (0,1]$ and $C_\star < \infty$, and suppose that, for every $\rho \in [4, R]$ and every $w \in H^1(B_\rho)$ which is $\mathbf{a}^t$-harmonic in B_ρ,*

$$\operatorname*{ess\,osc}_{B_2} w \leqslant C_\star\rho^{-\beta}\|w - (w)_{B_\rho}\|_{\underline{L}^2(B_\rho)}\,. \tag{4.140}$$

Let $\mathbf{f} \in L^2(B_1;\mathbb{R}^d)$, extended by zero to B_R, and let $u \in H^1_0(B_R)$ satisfy

$$\int_{B_R}\nabla\varphi\cdot\mathbf{a}\nabla u = -\int_{B_1}\mathbf{f}\cdot\nabla\varphi \qquad \forall\varphi \in H^1_0(B_R)\,, \tag{4.141}$$

that is, $-\nabla\cdot\mathbf{a}\nabla u = \nabla\cdot\mathbf{f}$ in B_R. Then, for every $r \in [10, R]$,

$$\|u - (u)_{A_r}\|_{\underline{L}^2(A_r)} \leqslant CC_\star\Lambda^{1/2}\,r^{2-\beta-d}\|\mathbf{f}\|_{L^2(B_1)}\,, \tag{4.142}$$

where $C(d) < \infty$.

Proof. Fix $r \in [10, R]$ and let $g := (u - (u)_{A_r})\mathbf{1}_{A_r} \in L^2(B_R)$. Then $\int_{B_R} g = 0$, $g = 0$ in $B_{r/2}$, and

$$\int_{B_R} ug = \int_{A_r}\big(u - (u)_{A_r}\big)g = \|g\|^2_{L^2(B_R)}\,.$$

Let $v \in H^1_0(B_R)$ be the solution, given by Lemma 4.33, of the adjoint Dirichlet problem

$$-\nabla\cdot\mathbf{a}^t\nabla v = g \quad \text{in } B_R\,. \tag{4.143}$$

Step 1. Energy estimate. We test (4.143) with v. Since g has zero mean and vanishes outside $B_r \subseteq B_R$, the inequality (4.137) in B_r gives

$$\|\nabla v\|_{L^2(B_R)}^2 \leqslant \int_{B_R} \nabla v \cdot \mathbf{a}^t \nabla v = \int_{B_r} g\big(v - (v)_{B_r}\big) \leqslant C_1 r \, \|g\|_{L^2(B_R)} \|\nabla v\|_{L^2(B_r)} \,,$$

and therefore

$$\|\nabla v\|_{L^2(B_R)} \leqslant C_1 r \|g\|_{L^2(B_R)} \,. \tag{4.144}$$

Step 2. Duality. Testing (4.143) with u and (4.141) with v, and using the identity $\nabla u \cdot \mathbf{a}^t \nabla v = \nabla v \cdot \mathbf{a} \nabla u$, we obtain

$$\|g\|_{L^2(B_R)}^2 = \int_{B_R} ug = \int_{B_R} \nabla v \cdot \mathbf{a} \nabla u = -\int_{B_1} \mathbf{f} \cdot \nabla v \leqslant \|\mathbf{f}\|_{L^2(B_1)} \|\nabla v\|_{L^2(B_1)} \,. \tag{4.145}$$

Step 3. Estimate of ∇v in B_1. If $\varphi \in H_0^1(B_{r/2})$, then its extension by zero belongs to $H_0^1(B_R)$ and vanishes where $g \neq 0$. Hence v is $\mathbf{a}^t$-harmonic in $B_{r/2}$, where $\frac{r}{2} \in [5, \frac{R}{2}]$. By (4.140) with $\rho = \frac{r}{2}$ and $w = v$, the essential oscillation of v on B_2 is finite; let $c := \frac{1}{2}(\operatorname{ess\,sup}_{B_2} v + \operatorname{ess\,inf}_{B_2} v)$, so that $|v - c| \leqslant \frac{1}{2} \operatorname{ess\,osc}_{B_2} v$ almost everywhere in B_2. The cutoff $\eta(x) := \min\{1, (3 - 2|x|)_+\}$ equals 1 in B_1, has compact support in $B_{r/2}$ and satisfies $|\nabla \eta| \leqslant 2\mathbf{1}_{B_2}$. Thus (4.138), with $U = B_{r/2}$, $\mathbf{b} = \mathbf{a}^t$ and $k = c$, gives

$$\|\nabla v\|_{L^2(B_1)}^2 \leqslant \int_{B_{r/2}} \eta^2 \, \nabla v \cdot \mathbf{a}^t \nabla v \leqslant 16\Lambda \int_{B_2} (v - c)^2 \leqslant 4\Lambda |B_2| \Big(\operatorname*{ess\,osc}_{B_2} v \Big)^2 .$$

Next, (4.140) with $\rho = \frac{r}{2}$, the inequality (4.137) in $B_{r/2}$ and (4.144) give, since $\beta \leqslant 1$,

$$\begin{aligned} \operatorname*{ess\,osc}_{B_2} v &\leqslant C_\star \Big(\frac{r}{2}\Big)^{-\beta} |B_{r/2}|^{-1/2} \|v - (v)_{B_{r/2}}\|_{L^2(B_{r/2})} \\ &\leqslant 2C_\star r^{-\beta} |B_1|^{-1/2} \Big(\frac{r}{2}\Big)^{-d/2} C_1 \frac{r}{2} \, \|\nabla v\|_{L^2(B_R)} \leqslant 2^{d/2} |B_1|^{-1/2} C_1^2 C_\star \, r^{2-\beta-d/2} \|g\|_{L^2(B_R)} \,. \end{aligned}$$

Combining the two previous displays and using $|B_2| = 2^d |B_1|$, we obtain

$$\|\nabla v\|_{L^2(B_1)} \leqslant 2^{d+1} C_1^2 C_\star \Lambda^{1/2} \, r^{2-\beta-d/2} \|g\|_{L^2(B_R)} \,. \tag{4.146}$$

Step 4. Conclusion. By (4.145) and (4.146), $\|g\|_{L^2(B_R)} \leqslant 2^{d+1} C_1^2 C_\star \Lambda^{1/2} r^{2-\beta-d/2} \|\mathbf{f}\|_{L^2(B_1)}$. Since $|A_r| = (1 - 2^{-d})|B_1| r^d \geqslant \frac{1}{2}|B_1| r^d$ and $\|u - (u)_{A_r}\|_{\underline{L}^2(A_r)} = |A_r|^{-1/2} \|g\|_{L^2(B_R)}$, this gives (4.142) with $C := 2^{d+3/2} |B_1|^{-1/2} C_1^2$. □

Corollary 4.36 (Measurable coefficients)**.** *Let $d \geqslant 3$, and let $\alpha = \alpha(d, \Lambda) \in (0, 1]$ be the exponent of Theorem 4.8, so that $\alpha \geqslant \exp(-C\Lambda^{(d+2)/4})$. Let $R \geqslant 10$, let $\mathbf{f} \in L^2(B_1; \mathbb{R}^d)$ and let $u \in H_0^1(B_R)$ satisfy (4.141). Then, for every $r \in [10, R]$,*

$$\|u - (u)_{A_r}\|_{\underline{L}^2(A_r)} \leqslant C\Lambda^{1/2 + d/4} \, r^{2-\alpha-d} \|\mathbf{f}\|_{L^2(B_1)} \,, \tag{4.147}$$

where $C(d) < \infty$.

Proof. Let $C_D(d)$ be the constant in (4.36). We show that (4.140) holds with $\beta = \alpha$ and $C_\star = 4|B_1|^{1/2} C_D \Lambda^{d/4}$; the corollary then follows from Lemma 4.35.

Let $\rho \in [4, R]$ and let $w \in H^1(B_\rho)$ be $\mathbf{a}^t$-harmonic in B_ρ. By the change of variables $x = \rho y$, the function $w_\rho(y) := w(\rho y) - (w)_{B_\rho}$ belongs to $H^1(B_1)$ and is a solution in B_1 of the equation with coefficient field $\mathbf{a}^t(\rho\,\cdot)$, which satisfies (4.1). Indeed, if $\phi \in C_c^\infty(B_1)$, then $\psi(x) := \phi(x/\rho)$ belongs to $C_c^\infty(B_\rho)$, and, since $\nabla w_\rho(y) = \rho \nabla w(\rho y)$, $\nabla \phi(y) = \rho \nabla \psi(\rho y)$ and $dy = \rho^{-d} dx$, $\int_{B_1} \nabla \phi \cdot \mathbf{a}^t(\rho y) \nabla w_\rho \, dy = \rho^{2-d} \int_{B_\rho} \nabla \psi \cdot \mathbf{a}^t \nabla w \, dx = 0$. By Theorem 4.8, w_ρ has a representative $\widetilde{w}_\rho \in C^{0,\alpha}(B_{1/2})$ with

$$[\widetilde{w}_\rho]_{C^{0,\alpha}(B_{1/2})} \leqslant C_D \Lambda^{d/4} \|w_\rho\|_{L^2(B_1)} = C_D \Lambda^{d/4} |B_1|^{1/2} \|w - (w)_{B_\rho}\|_{\underline{L}^2(B_\rho)} .$$

Since $\rho \geqslant 4$, we have $x/\rho \in B_{1/2}$ and $|x/\rho - x'/\rho| < 4/\rho$ for $x, x' \in B_2$. Therefore the representative $x \mapsto \widetilde{w}_\rho(x/\rho) + (w)_{B_\rho}$ of w in B_2 gives, using $4^\alpha \leqslant 4$,

$$\operatorname*{ess\,osc}_{B_2} w \leqslant \Big(\frac{4}{\rho}\Big)^\alpha [\widetilde{w}_\rho]_{C^{0,\alpha}(B_{1/2})} \leqslant 4|B_1|^{1/2} C_D \Lambda^{d/4} \rho^{-\alpha} \|w - (w)_{B_\rho}\|_{\underline{L}^2(B_\rho)} .$$

This is (4.140) with the claimed β and $C_\star$. □

When the coefficients are constant, the $\mathbf{a}^t$-harmonic functions are harmonic functions after a linear change of variables, and they satisfy (4.140) with $\beta = 1$.

Corollary 4.37 (Constant coefficients)**.** *Let $d \geqslant 2$ and $R \geqslant 10$, and suppose that $\mathbf{a}(x) = \mathbf{a}_0$ for almost every $x \in B_R$, where $\mathbf{a}_0 \in \mathbb{R}^{d\times d}$ satisfies (4.1). Let $\mathbf{f} \in L^2(B_1; \mathbb{R}^d)$ and let $u \in H_0^1(B_R)$ satisfy (4.141). Then, for every $r \in [10, R]$,*

$$\|u - (u)_{A_r}\|_{\underline{L}^2(A_r)} \leqslant C \Lambda^{\frac{d+4}{4}} r^{1-d} \|\mathbf{f}\|_{L^2(B_1)} , \tag{4.148}$$

where $C(d) < \infty$.

Proof. We show that (4.140) holds with $\beta = 1$ and $C_\star = C\Lambda^{(d+2)/4}$; the corollary then follows from Lemma 4.35. Let $\mathbf{s} := \frac{1}{2}(\mathbf{a}_0 + \mathbf{a}_0^t)$, $\mathbf{k} := \frac{1}{2}(\mathbf{a}_0^t - \mathbf{a}_0)$, and let $T := \mathbf{s}^{1/2}$ be the symmetric positive definite square root of $\mathbf{s}$. By (4.1), $|e|^2 \leqslant e \cdot \mathbf{s} e = e \cdot \mathbf{a}_0 e \leqslant \Lambda |e|^2$, so the eigenvalues of T lie in $[1, \Lambda^{1/2}]$; hence $|Te| \leqslant \Lambda^{1/2}|e|$, $|T^{-1}e| \leqslant |e|$ and $\det T \geqslant 1$. Fix $\rho \in [4, R]$ and let $w \in H^1(B_\rho)$ be $\mathbf{a}^t$-harmonic in B_ρ.

Step 1. Reduction to the Laplace equation. For $\varphi \in C_c^\infty(B_\rho)$, the definition of the weak derivatives of w, applied with the test functions $\partial_i \varphi$, gives

$$\int_{B_\rho} \nabla\varphi \cdot \mathbf{k} \nabla w = \sum_{i,j=1}^d \mathbf{k}_{ij} \int_{B_\rho} \partial_i\varphi \, \partial_j w = -\int_{B_\rho} w \sum_{i,j=1}^d \mathbf{k}_{ij} \, \partial_j \partial_i \varphi = 0 ,$$

because $\mathbf{k}$ is antisymmetric. Since $\mathbf{a}_0^t = \mathbf{s} + \mathbf{k}$, it follows that $\int_{B_\rho} \nabla\varphi \cdot \mathbf{s} \nabla w = 0$ for every $\varphi \in C_c^\infty(B_\rho)$. Let $E := T^{-1}(B_\rho)$ and $h(y) := w(Ty)$ for $y \in E$. Then $h \in H^1(E)$ and $\nabla h(y) = T(\nabla w)(Ty)$. If $\varphi \in C_c^\infty(E)$, then $\psi := \varphi \circ T^{-1} \in C_c^\infty(B_\rho)$ and $\nabla\varphi(y) = T(\nabla\psi)(Ty)$, so the change of variables $x = Ty$ gives

$$\int_E \nabla\varphi \cdot \nabla h = \int_E (\nabla\psi)(Ty) \cdot T^2 (\nabla w)(Ty) \, dy = (\det T)^{-1} \int_{B_\rho} \nabla\psi \cdot \mathbf{s} \nabla w = 0 .$$

Thus h is harmonic in E.

Step 2. Gradient bound. Let $\sigma := \rho/(2\Lambda^{1/2})$ and $y_0 \in T^{-1}(B_2)$. For $y \in B_\sigma(y_0)$ we have $|Ty| \leqslant |Ty_0| + \Lambda^{1/2}|y - y_0| < 2 + \frac{\rho}{2} \leqslant \rho$, so $B_\sigma(y_0) \subseteq E$. Proposition 2.1, applied with $m = 1$ to $z \mapsto h(y_0 + z)$ in B_σ, shows that h has a real analytic representative in $B_\sigma(y_0)$ and that, with $C(d)$ the constant of (2.1) and for every $c \in \mathbb{R}$,

$$|\nabla h(y_0)| \leqslant \frac{C}{\sigma}\|h - (h)_{B_\sigma(y_0)}\|_{\underline{L}^2(B_\sigma(y_0))} \leqslant \frac{C}{\sigma}\|h - c\|_{\underline{L}^2(B_\sigma(y_0))}\,.$$

Two such representatives, in overlapping balls, are continuous and agree almost everywhere on the overlap, hence agree on it; therefore h has a representative which is C^1 in an open set containing $T^{-1}(B_2)$, and the previous display holds for it at every $y_0 \in T^{-1}(B_2)$. By the change of variables $x = Ty$, $\det T \geqslant 1$ and $B_\sigma(y_0) \subseteq E$,

$$\|h - c\|^2_{\underline{L}^2(B_\sigma(y_0))} \leqslant \frac{1}{|B_\sigma|\det T}\int_{B_\rho}|w - c|^2 \leqslant \frac{|B_\rho|}{|B_\sigma|}\|w - c\|^2_{\underline{L}^2(B_\rho)} = \left(2\Lambda^{1/2}\right)^d\|w - c\|^2_{\underline{L}^2(B_\rho)}\,.$$

Taking $c = (w)_{B_\rho}$, we obtain, for every $y_0 \in T^{-1}(B_2)$,

$$|\nabla h(y_0)| \leqslant 2^{1+d/2}C\Lambda^{\frac{d+2}{4}}\rho^{-1}\|w - (w)_{B_\rho}\|_{\underline{L}^2(B_\rho)}\,.$$

Step 3. Conclusion. Let h be the representative of Step 2. Then $\widetilde{w} := h \circ T^{-1}$ is a representative of w which is C^1 in an open set containing B_2, and $|\nabla\widetilde{w}(x)| = |T^{-1}\nabla h(T^{-1}x)| \leqslant |\nabla h(T^{-1}x)|$ for $x \in B_2$. Since B_2 is convex with diameter 4,

$$\operatorname*{ess\,osc}_{B_2} w \leqslant \sup_{B_2}\widetilde{w} - \inf_{B_2}\widetilde{w} \leqslant 4\sup_{B_2}|\nabla\widetilde{w}| \leqslant 2^{3+d/2}C\Lambda^{\frac{d+2}{4}}\rho^{-1}\|w - (w)_{B_\rho}\|_{\underline{L}^2(B_\rho)}\,.$$

This is (4.140) with $\beta = 1$ and $C_\star = 2^{3+d/2}C\Lambda^{(d+2)/4}$. □

We next estimate the Green function itself, and *for the rest of this subsection we assume that* $d \geqslant 3$. Formally, the Green function $G = G(\cdot, 0)$ of $-\nabla\cdot\mathbf{a}\nabla$ in B_R with pole at the origin solves $-\nabla\cdot\mathbf{a}\nabla G = \delta_0$ in B_R with $G = 0$ on ∂B_R. If $v \in H^1_0(B_R)$ solves the adjoint problem $-\nabla\cdot\mathbf{a}^t\nabla v = g$, then testing each equation with the other solution gives, at least heuristically, $\int_{B_R} Gg = v(0)$. We take this identity as the definition of G. It involves only the Dirichlet problem of Lemma 4.33, and it makes sense for sources g which vanish near the origin: then v is $\mathbf{a}^t$-harmonic near the origin, and hence continuous there by Theorem 4.8.

For $g \in L^2(B_R)$, we denote by $v_g \in H^1_0(B_R)$ the solution of $-\nabla\cdot\mathbf{a}^t\nabla v_g = g$ in B_R given by Lemma 4.33. By uniqueness, the map $g \mapsto v_g$ is linear.

Lemma 4.38 (Value at the pole)**.** *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that the following holds for every $\varepsilon \in (0, R]$ and every $g \in L^2(B_R)$ with $g = 0$ almost everywhere in B_ε. The function v_g has a representative which is continuous in $B_{\varepsilon/2}$; in particular, the limit*

$$v_g(0) := \lim_{\rho\to 0}\fint_{B_\rho} v_g$$

exists. Moreover, with $K_\varepsilon := C\Lambda^{d/4}\varepsilon^{-d/2}R^2$,

$$|v_g(0)| \leqslant K_\varepsilon\|g\|_{L^2(B_R)} \qquad \text{and} \qquad \left|\fint_{B_\rho} v_g\right| \leqslant K_\varepsilon\|g\|_{L^2(B_R)} \quad \forall\rho \in \left(0, \tfrac{\varepsilon}{2}\right]. \tag{4.149}$$

Proof. If $\varphi \in H^1_0(B_\varepsilon)$, its extension by zero belongs to $H^1_0(B_R)$ and vanishes where $g \neq 0$, so $\int_{B_\varepsilon} \nabla\varphi \cdot \mathbf{a}^t \nabla v_g = \int_{B_R} g\varphi = 0$. Hence v_g is $\mathbf{a}^t$-harmonic in B_ε, and $w(y) := v_g(\varepsilon y)$ is a solution in B_1 of the equation with coefficient field $\mathbf{a}^t(\varepsilon\,\cdot)$, which satisfies (4.1). By Theorem 4.8, with $C_D(d)$ the constant in (4.36), w has a representative $\widetilde{w}$ which is continuous in $B_{1/2}$ and satisfies

$$\sup_{B_{1/2}} |\widetilde{w}| \leqslant C_D \Lambda^{d/4} \|w\|_{L^2(B_1)} = C_D \Lambda^{d/4} \varepsilon^{-d/2} \|v_g\|_{L^2(B_\varepsilon)} .$$

Thus $\widetilde{v}(x) := \widetilde{w}(x/\varepsilon)$ is a representative of v_g which is continuous in $B_{\varepsilon/2}$, and, by (4.139),

$$\sup_{B_{\varepsilon/2}} |\widetilde{v}| \leqslant C_D \Lambda^{d/4} \varepsilon^{-d/2} C_1^2 R^2 \|g\|_{L^2(B_R)} .$$

For $\rho \in (0, \frac{\varepsilon}{2}]$, the average $⨍_{B_\rho} v_g = ⨍_{B_\rho} \widetilde{v}$ is bounded in absolute value by the right side, and it converges to $\widetilde{v}(0)$ as $\rho \to 0$, by continuity. This proves the lemma with $C := C_D C_1^2$. □

Definition 4.39 (Green function)**.** Let $d \geqslant 3$. A *Green function of* $-\nabla \cdot \mathbf{a}\nabla$ *in* B_R *with pole at the origin* is a measurable function $G : B_R \to \mathbb{R}$ such that $G \in L^2(B_R \backslash B_\varepsilon)$ for every $\varepsilon \in (0, R)$ and

$$\int_{B_R} G\, g = v_g(0) \tag{4.150}$$

for every $\varepsilon \in (0, R)$ and every $g \in L^2(B_R)$ with $g = 0$ almost everywhere in B_ε.

Both sides of (4.150) are defined: the integrand on the left vanishes in B_ε, and the right side is defined by Lemma 4.38. The equation $-\nabla \cdot \mathbf{a}\nabla G = \delta_0$ is recovered in the weak form (4.152) below. In the proof of the next proposition we use the approximations $G_\rho \in H^1_0(B_R)$ which solve $-\nabla \cdot \mathbf{a}\nabla G_\rho = |B_\rho|^{-1} \mathbf{1}_{B_\rho}$, together with Stampacchia's truncation argument.

Proposition 4.40 (Existence and basic properties of the Green function)**.** *Let $d \geqslant 3$. There exists a Green function G of $-\nabla \cdot \mathbf{a}\nabla$ in B_R with pole at the origin, and it is unique up to a set of measure zero. Moreover, there exists $C_0(d) < \infty$ such that the following statements hold.*

(i) $G \geqslant 0$ *almost everywhere in* B_R.

(ii) For every measurable set $E \subseteq B_R$,

$$\int_E G \leqslant C_0 |E|^{2/d} . \tag{4.151}$$

In particular, $G \in L^1(B_R)$ *and* $|\{G > t\}| \leqslant (C_0/t)^{\frac{d}{d-2}}$ *for every* $t > 0$.

(iii) For every $\varepsilon \in (0, R)$, *we have* $G \in H^1(B_R \backslash \overline{B}_\varepsilon)$ *and* G *is* $\mathbf{a}$*-harmonic in* $B_R \backslash \overline{B}_\varepsilon$. *Moreover,* G *has a representative which is continuous in* $B_R \backslash \{0\}$.

(iv) For every $\varepsilon \in (0,R)$ and every Lipschitz function φ with compact support in B_R which is constant in B_ε,

$$\int_{B_R\setminus B_\varepsilon} \nabla\varphi\cdot\mathbf{a}\nabla G = \varphi(0)\,. \tag{4.152}$$

The constant C_0 in (4.151) depends neither on Λ nor on R.

Proof. For $\varepsilon\in(0,R)$, let $H_\varepsilon := \{g\in L^2(B_R) : g=0 \text{ a.e. in } B_\varepsilon\}$. This is a closed subspace of $L^2(B_R)$, which we identify with $L^2(B_R\setminus B_\varepsilon)$.

Step 1. Uniqueness. Let G and G' be Green functions, and let $\varepsilon\in(0,R)$. The function $g := (G-G')\mathbf{1}_{B_R\setminus B_\varepsilon}$ belongs to H_ε, and subtracting (4.150) for G' from (4.150) for G gives $\int_{B_R\setminus B_\varepsilon}(G-G')^2 = 0$. Hence $G=G'$ almost everywhere in $B_R\setminus B_\varepsilon$. Letting $\varepsilon\to 0$ along a sequence, we get $G=G'$ almost everywhere in B_R.

Step 2. Existence. Fix $\varepsilon\in(0,R)$. By Lemma 4.38 and the linearity of $g\mapsto v_g$, the map $g\mapsto v_g(0)$ is a linear functional on the Hilbert space H_ε of norm at most K_ε. By the Riesz representation theorem, there is $G^{(\varepsilon)}\in H_\varepsilon$ with $\int_{B_R} G^{(\varepsilon)}g = v_g(0)$ for every $g\in H_\varepsilon$. If $0<\varepsilon'<\varepsilon$, then $H_\varepsilon\subseteq H_{\varepsilon'}$, and both $G^{(\varepsilon)}$ and $G^{(\varepsilon')}\mathbf{1}_{B_R\setminus B_\varepsilon}$ represent the same functional on H_ε; hence $G^{(\varepsilon')}=G^{(\varepsilon)}$ almost everywhere in $B_R\setminus B_\varepsilon$. Let $\varepsilon_n := 2^{-n}R$ for $n\geqslant 0$, and define $G(x) := G^{(\varepsilon_n)}(x)$ for $\varepsilon_n\leqslant|x|<\varepsilon_{n-1}$ and $n\geqslant 1$, and $G(0):=0$. Then $G=G^{(\varepsilon_n)}$ almost everywhere in $B_R\setminus B_{\varepsilon_n}$, for every $n\geqslant 1$. Given $\varepsilon\in(0,R)$, choose n with $\varepsilon_n\leqslant\varepsilon$. Then $G=G^{(\varepsilon_n)}=G^{(\varepsilon)}$ almost everywhere in $B_R\setminus B_\varepsilon$, so $G\in L^2(B_R\setminus B_\varepsilon)$ and $\int_{B_R} Gg = \int_{B_R} G^{(\varepsilon)}g = v_g(0)$ for every $g\in H_\varepsilon$. Thus G is a Green function.

Step 3. Proof of (i). Let $\varepsilon\in(0,R)$ and $g := G_-\mathbf{1}_{B_R\setminus B_\varepsilon}\in H_\varepsilon$, where $G_- := \max\{-G,0\}$. Since $g\geqslant 0$, Lemma 4.34 gives $v_g\geqslant 0$, and hence $v_g(0)=\lim_{\rho\to 0}\fint_{B_\rho} v_g\geqslant 0$. By (4.150),

$$-\int_{B_R\setminus B_\varepsilon} G_-^2 = \int_{B_R} Gg = v_g(0)\geqslant 0\,,$$

so $G_-=0$ almost everywhere in $B_R\setminus B_\varepsilon$. Letting $\varepsilon\to 0$ along a sequence proves (i).

Step 4. Approximate Green functions. For $\rho\in(0,R)$, let $G_\rho\in H_0^1(B_R)$ be the solution, given by Lemma 4.33, of

$$-\nabla\cdot\mathbf{a}\nabla G_\rho = |B_\rho|^{-1}\mathbf{1}_{B_\rho}\quad\text{in } B_R\,. \tag{4.153}$$

By Lemma 4.34, $G_\rho\geqslant 0$. For every $g\in L^2(B_R)$, testing the equation of v_g with G_ρ and (4.153) with v_g, and using $\nabla G_\rho\cdot\mathbf{a}^t\nabla v_g = \nabla v_g\cdot\mathbf{a}\nabla G_\rho$, we obtain

$$\int_{B_R} G_\rho\, g = \int_{B_R}\nabla G_\rho\cdot\mathbf{a}^t\nabla v_g = \int_{B_R}\nabla v_g\cdot\mathbf{a}\nabla G_\rho = \fint_{B_\rho} v_g\,. \tag{4.154}$$

Now let $\varepsilon\in(0,R)$ and $g\in H_\varepsilon$. By Lemma 4.38 and (4.150), the right side of (4.154) is bounded in absolute value by $K_\varepsilon\|g\|_{L^2(B_R)}$ for $\rho\in(0,\frac{\varepsilon}{2}]$, and it converges to $v_g(0)=\int_{B_R} Gg$ as $\rho\to 0$. Taking $g=G_\rho\mathbf{1}_{B_R\setminus B_\varepsilon}$ in the first statement, we conclude that, for every $\varepsilon\in(0,R)$,

$$\|G_\rho\|_{L^2(B_R\setminus B_\varepsilon)}\leqslant K_\varepsilon\quad\forall\rho\in\left(0,\tfrac{\varepsilon}{2}\right]\qquad\text{and}\qquad\lim_{\rho\to 0}\int_{B_R} G_\rho\, g = \int_{B_R} G\, g\quad\forall g\in H_\varepsilon\,. \tag{4.155}$$

Step 5. Proof of (ii). Fix $\rho \in (0,R)$ and $t>0$, and let $w_t := \min\{G_\rho, t\} = G_\rho - (G_\rho - t)_+$. Then $w_t \in H^1_0(B_R)$ and $\nabla w_t = \mathbf{1}_{\{G_\rho<t\}}\nabla G_\rho$ almost everywhere, since $\nabla G_\rho = 0$ almost everywhere on $\{G_\rho = t\}$. Testing (4.153) with w_t and using (4.1) and $w_t \leqslant t$, we get

$$\|\nabla w_t\|^2_{L^2(B_R)} \leqslant \int_{\{G_\rho<t\}} \nabla G_\rho \cdot \mathbf{a}\nabla G_\rho = \int_{B_R} \nabla w_t \cdot \mathbf{a}\nabla G_\rho = \fint_{B_\rho} w_t \leqslant t\,.$$

Let $C_S := C\gamma_2$ be the constant of Proposition 1.5 with $p=2$, and recall $2^* = \frac{2d}{d-2}$. Since $w_t = t$ on $\{G_\rho \geqslant t\}$, Proposition 1.5 in $U = B_R$ gives

$$t\,\big|\{G_\rho \geqslant t\}\big|^{1/2^*} \leqslant \|w_t\|_{L^{2^*}(B_R)} \leqslant C_S\|\nabla w_t\|_{L^2(B_R)} \leqslant C_S t^{1/2}\,,$$

that is, $|\{G_\rho \geqslant t\}| \leqslant C_S^{2^*} t^{-\frac{d}{d-2}}$ for every $t>0$. Let $E \subseteq B_R$ be measurable with $|E|>0$, and let $t_0 := C_S^2|E|^{-\frac{d-2}{d}}$. By the layer-cake formula,

$$\begin{aligned}\int_E G_\rho = \int_0^\infty \big|E\cap\{G_\rho>t\}\big|\,dt &\leqslant \int_0^{t_0} |E|\,dt + \int_{t_0}^\infty C_S^{2^*} t^{-\frac{d}{d-2}}\,dt \\ &= t_0|E| + \frac{d-2}{2} C_S^{2^*} t_0^{-\frac{2}{d-2}} = \frac{d}{2} C_S^2 |E|^{2/d}\,.\end{aligned}$$

Let $C_0 := \frac{d}{2}C_S^2$. We have shown that $\int_E G_\rho \leqslant C_0|E|^{2/d}$ for every measurable $E \subseteq B_R$ and every $\rho \in (0,R)$. If $E \subseteq B_R\backslash B_\varepsilon$ for some $\varepsilon \in (0,R)$, then $\mathbf{1}_E \in H_\varepsilon$ and (4.155) gives $\int_E G = \lim_{\rho\to 0}\int_E G_\rho \leqslant C_0|E|^{2/d}$. For a general measurable $E \subseteq B_R$, we apply this to $E\backslash B_\varepsilon$ and let $\varepsilon \to 0$; since $G \geqslant 0$, the monotone convergence theorem gives (4.151). Taking $E = B_R$ shows that $G \in L^1(B_R)$. Taking $E = \{G>t\}$ gives $t|E| \leqslant \int_E G \leqslant C_0|E|^{2/d}$, hence $|E|^{\frac{d-2}{d}} \leqslant C_0/t$ whenever $|E|>0$, which is the last statement of (ii).

Step 6. Proof of (iii). Fix $\varepsilon \in (0,R)$, let $U := B_R\backslash\overline{B}_\varepsilon$, and let $\rho \in (0, \frac{\varepsilon}{4}]$. The Lipschitz function $\eta(x) := \min\{1, (2|x|/\varepsilon - 1)_+\}$ vanishes in $B_{\varepsilon/2}$, equals 1 outside B_ε, and satisfies $|\nabla\eta| \leqslant 2/\varepsilon$. The function $\eta^2 G_\rho$ belongs to $H^1_0(B_R)$ and vanishes in B_ρ, so testing (4.153) with it gives $\int_{B_R}\nabla(\eta^2 G_\rho)\cdot\mathbf{a}\nabla G_\rho = 0$. Hence (4.138), with U replaced by B_R and $k=0$, and (4.155) with $\varepsilon/2$ in place of ε give

$$\|\nabla G_\rho\|^2_{L^2(U)} \leqslant \int_{B_R} \eta^2\,\nabla G_\rho\cdot\mathbf{a}\nabla G_\rho \leqslant \frac{16\Lambda}{\varepsilon^2}\int_{B_R\backslash B_{\varepsilon/2}} G_\rho^2 \leqslant M^2\,, \qquad M := \frac{4\Lambda^{1/2}}{\varepsilon} K_{\varepsilon/2}\,.$$

Let $\Psi \in C^\infty_c(U;\mathbb{R}^d)$. Then $\nabla\cdot\Psi$, extended by zero, belongs to H_ε, so (4.155) and the definition of the weak gradient of G_ρ give

$$\int_U G\,\nabla\cdot\Psi = \lim_{\rho\to0}\int_U G_\rho\,\nabla\cdot\Psi = -\lim_{\rho\to0}\int_U \nabla G_\rho\cdot\Psi\,, \qquad \text{hence} \qquad \left|\int_U G\,\nabla\cdot\Psi\right| \leqslant M\|\Psi\|_{L^2(U)}\,.$$

Since $C^\infty_c(U;\mathbb{R}^d)$ is dense in $L^2(U;\mathbb{R}^d)$, the functional $\Psi \mapsto -\int_U G\,\nabla\cdot\Psi$ extends to a bounded linear functional on $L^2(U;\mathbb{R}^d)$ of norm at most M, and the Riesz representation theorem gives $F \in L^2(U;\mathbb{R}^d)$ with $\|F\|_{L^2(U)} \leqslant M$ and $\int_U F\cdot\Psi = -\int_U G\,\nabla\cdot\Psi$ for every $\Psi \in C^\infty_c(U;\mathbb{R}^d)$. Thus F is the weak gradient of G in U, and $G \in H^1(U)$, since $G \in L^2(U)$. We have also shown that $\int_U \nabla G_\rho\cdot\Psi \to \int_U \nabla G\cdot\Psi$ for every $\Psi \in C^\infty_c(U;\mathbb{R}^d)$. Since $\|\nabla G_\rho\|_{L^2(U)} \leqslant$

M for $\rho \leqslant \frac{\varepsilon}{4}$ and $\|\nabla G\|_{L^2(U)} \leqslant M$, approximating a general $\Psi \in L^2(U;\mathbb{R}^d)$ in L^2 by elements of $C_c^\infty(U;\mathbb{R}^d)$ gives

$$\lim_{\rho\to 0} \int_U \nabla G_\rho \cdot \Psi = \int_U \nabla G \cdot \Psi \qquad \forall \Psi \in L^2(U;\mathbb{R}^d)\,. \tag{4.156}$$

Now let $\varphi \in H_0^1(U)$. Its extension by zero belongs to $H_0^1(B_R)$ and vanishes in B_ε, so testing (4.153) with it gives $\int_U \nabla\varphi \cdot \mathbf{a}\nabla G_\rho = 0$ for $\rho < \varepsilon$. Since $\nabla\varphi \cdot \mathbf{a}\nabla G_\rho = (\mathbf{a}^t\nabla\varphi)\cdot\nabla G_\rho$ and $\mathbf{a}^t\nabla\varphi \in L^2(U;\mathbb{R}^d)$, (4.156) gives $\int_U \nabla\varphi\cdot\mathbf{a}\nabla G = 0$. Hence G is $\mathbf{a}$-harmonic in U.

Finally, let $y \in B_R\backslash\{0\}$ and $s := \frac{1}{3}\min\{|y|, R-|y|\}$. Then $B_{2s}(y) \subseteq B_R\backslash\overline{B}_s$, so $G \in H^1(B_{2s}(y))$ is $\mathbf{a}$-harmonic in $B_{2s}(y)$, by what we have just proved with $\varepsilon = s$. Theorem 4.8, applied to $z \mapsto G(y+2sz)$ in B_1, gives a representative of G which is continuous in $B_s(y)$. Two continuous functions which agree almost everywhere on an open set agree everywhere on it. Hence these representatives agree on the overlaps of the balls $B_s(y)$, $y \in B_R\backslash\{0\}$, and define a continuous function on $B_R\backslash\{0\}$. This function agrees with G almost everywhere, because $B_R\backslash\{0\}$ is a countable union of compact sets, each of which is covered by finitely many of the balls $B_s(y)$.

Step 7. Proof of (iv). Let ε and φ be as in (iv). Then $\varphi \in H_0^1(B_R)$, and $\nabla\varphi = 0$ almost everywhere in B_ε. For $\rho \in (0,\varepsilon)$, testing (4.153) with φ gives

$$\int_{B_R\backslash B_\varepsilon} \nabla\varphi\cdot\mathbf{a}\nabla G_\rho = \fint_{B_\rho} \varphi = \varphi(0)\,.$$

By (4.156), with $\varepsilon/2$ in place of ε and $\Psi := \mathbf{1}_{B_R\backslash B_\varepsilon}\mathbf{a}^t\nabla\varphi$, the left side converges to $\int_{B_R\backslash B_\varepsilon}\nabla\varphi\cdot\mathbf{a}\nabla G$ as $\rho\to 0$. This is (4.152). □

The Harnack inequality (Theorem 4.18) is stated in balls. Since G is $\mathbf{a}$-harmonic only away from the pole, we need a version in annuli, which follows by chaining balls.

Lemma 4.41 (Harnack inequality in annuli)**.** *Let $d \geqslant 3$. There exists $C(d) < \infty$ such that the following holds for every $r \in (0, \frac{R}{2}]$. Let $V := B_{2r}\backslash\overline{B}_{r/8}$ and $D_r := B_{3r/2}\backslash B_{r/4}$. If $u \in H^1(V)$ is $\mathbf{a}$-harmonic in V and $u \geqslant 0$ almost everywhere in V, then*

$$\operatorname*{ess\,sup}_{D_r} u \leqslant \exp\bigl(C\Lambda^{1/2}\bigr) \operatorname*{ess\,inf}_{D_r} u\,. \tag{4.157}$$

Proof. Let $H := \exp(C_H\Lambda^{1/2})$, where $C_H(d)$ is the constant in (4.88), and let $s := \frac{r}{16}$. For $y \in \overline{D}_r$, write $M(y) := \operatorname{ess\,sup}_{B_s(y)} u$ and $m(y) := \operatorname{ess\,inf}_{B_s(y)} u$.

Step 1. One ball. Let $y \in \overline{D}_r$. If $z \in B_{2s}(y)$, then $\frac{r}{8} = \frac{r}{4} - 2s < |z| < \frac{3r}{2} + 2s < 2r$, so $B_{2s}(y) \subseteq V$. We claim that $M(y) \leqslant Hm(y)$. Let $\delta > 0$. The function $\widetilde{u}(z) := u(y+2sz) + \delta$ belongs to $H^1(B_1)$, is a solution in B_1 of the equation with coefficient field $\mathbf{a}(y+2s\,\cdot)$, which satisfies (4.1), and $\widetilde{u} \geqslant \delta$ almost everywhere in B_1. Therefore the continuous representative of $\widetilde{u}$ given by Theorem 4.8 (applied in balls contained in B_1) is positive in B_1, and Theorem 4.18 gives $\operatorname{ess\,sup}_{B_{1/2}} \widetilde{u} \leqslant H \operatorname{ess\,inf}_{B_{1/2}} \widetilde{u}$; here we used that the supremum and the infimum of a continuous function over the open ball $B_{1/2}$ coincide with its essential supremum and infimum. That is, $M(y) + \delta \leqslant H(m(y)+\delta)$. Letting $\delta \to 0$ proves the claim.

Step 2. Chains of balls. Let $y, z \in \overline{D}_r$ and $N := 80$. We claim that

$$M(y) \leqslant H^{N+1} m(z)\,.$$

Consider the path which goes from y to $ry/|y|$ along a radial segment, then from $ry/|y|$ to $rz/|z|$ along an arc of a great circle of the sphere ∂B_r, and then from $rz/|z|$ to z along a radial segment. It lies in $\overline{D}_r$, and its length is at most $\frac{3r}{4} + \pi r + \frac{3r}{4} < 5r = Ns$. Hence there are points $y_0 = y, y_1, \dots, y_N = z$ on this path with $|y_i - y_{i+1}| < s$ for every i. The set $B_s(y_i) \cap B_s(y_{i+1})$ is open and contains y_{i+1}, so it has positive measure. Since $u \geqslant m(y_i)$ and $u \leqslant M(y_{i+1})$ almost everywhere on it, we get $m(y_i) \leqslant M(y_{i+1}) \leqslant H m(y_{i+1})$, by Step 1. Combining these inequalities for $i = 0, \dots, N-1$ with $M(y_0) \leqslant H m(y_0)$ proves the claim.

Step 3. Conclusion. Since $\overline{D}_r$ is compact, there are finitely many points $y^1, \dots, y^K \in \overline{D}_r$ such that the balls $B_s(y^k)$ cover $\overline{D}_r$. Almost everywhere in D_r, we have $u \leqslant \max_k M(y^k)$ and $u \geqslant \min_l m(y^l)$. By Step 2, $M(y^k) \leqslant H^{81} m(y^l)$ for all k and l. This gives (4.157) with $C := 81 C_H$. □

We now prove the two-sided bound for the Green function. The upper bound combines the averaged bound (4.151), which does not depend on Λ, with the Harnack inequality in annuli. The lower bound follows from the flux identity (4.152): the total flux of $\mathbf{a}\nabla G$ through an annulus equals 1, while the Caccioppoli inequality bounds this flux by the size of G in a slightly larger annulus.

Proposition 4.42 (Green function estimates, $d \geqslant 3$)**.** *Let $d \geqslant 3$ and let G be the Green function of $-\nabla \cdot \mathbf{a}\nabla$ in B_R with pole at the origin. There exists $C(d) < \infty$ such that, for every $r \in (0, \frac{R}{2}]$,*

$$\operatorname*{ess\,sup}_{A_r} G \leqslant \exp\bigl(C\Lambda^{1/2}\bigr)\, r^{2-d} \tag{4.158}$$

and

$$\operatorname*{ess\,inf}_{A_r} G \geqslant \exp\bigl(-C\Lambda^{1/2}\bigr)\, r^{2-d}\,. \tag{4.159}$$

Consequently, the representative of G which is continuous in $B_R \backslash \{0\}$ satisfies

$$\exp\bigl(-C\Lambda^{1/2}\bigr)|x|^{2-d} \leqslant G(x) \leqslant \exp\bigl(C\Lambda^{1/2}\bigr)|x|^{2-d} \qquad \forall x \in \overline{B}_{R/2} \backslash \{0\}\,. \tag{4.160}$$

Proof. Let $r \in (0, \frac{R}{2}]$ and let $D_r := B_{3r/2} \backslash B_{r/4} \supseteq A_r$ and $V := B_{2r} \backslash \overline{B}_{r/8}$ be as in Lemma 4.41. Since $V \subseteq B_R \backslash \overline{B}_{r/8}$, parts (i) and (iii) of Proposition 4.40 (with $\varepsilon = r/8$) show that $G \in H^1(V)$, $G \geqslant 0$, and G is $\mathbf{a}$-harmonic in V; indeed, functions in $H^1_0(V)$, extended by zero, belong to $H^1_0(B_R \backslash \overline{B}_{r/8})$. Therefore Lemma 4.41 gives, with $C_2(d)$ its constant,

$$\operatorname*{ess\,sup}_{D_r} G \leqslant \exp\bigl(C_2\Lambda^{1/2}\bigr) \operatorname*{ess\,inf}_{D_r} G\,. \tag{4.161}$$

Throughout the proof we use that, since $\Lambda \geqslant 1$, every constant $K(d) > 0$ satisfies $K \leqslant \exp(\log^+ K\, \Lambda^{1/2})$ and $K^{-1} \geqslant \exp(-\log^+ K\, \Lambda^{1/2})$ (here $\log^+ K = \max\{\log K, 0\}$), and that $\Lambda \leqslant \exp(2\Lambda^{1/2})$.

Step 1. Upper bound. Since $|A_r| = (1 - 2^{-d})|B_1| r^d \geqslant \frac{1}{2}|B_1| r^d$, (4.151) with $E = A_r$ gives

$$\fint_{A_r} G \leqslant C_0 |A_r|^{-\frac{d-2}{d}} \leqslant C_0 \Bigl(\frac{|B_1|}{2}\Bigr)^{-\frac{d-2}{d}} r^{2-d}\,. \tag{4.162}$$

Since $A_r \subseteq D_r$ and the essential infimum of G on A_r is at most its average there, (4.161) and (4.162) give

$$\operatorname*{ess\,sup}_{D_r} G \leqslant \exp\bigl(C_2\Lambda^{1/2}\bigr) \operatorname*{ess\,inf}_{A_r} G \leqslant \exp\bigl(C_2\Lambda^{1/2}\bigr) \fint_{A_r} G \leqslant \exp\bigl(C\Lambda^{1/2}\bigr) r^{2-d}\,.$$

Step 2. Lower bound. The Lipschitz function $\varphi(x) := \min\{1, (2 - 2|x|/r)_+\}$ equals 1 in $B_{r/2}$, has compact support $\overline{B}_r \subseteq B_R$, and satisfies $|\nabla\varphi| \leqslant 2/r$, with $\nabla\varphi = 0$ almost everywhere outside A_r. By (4.152) with $\varepsilon = r/2$ and by (4.2),

$$1 = \int_{A_r} \nabla\varphi \cdot \mathbf{a}\nabla G \leqslant \frac{2\Lambda^{1/2}}{r} \int_{A_r} \bigl(\nabla G \cdot \mathbf{a}\nabla G\bigr)^{1/2} \leqslant \frac{2\Lambda^{1/2}}{r} |A_r|^{1/2} \biggl(\int_{A_r} \nabla G \cdot \mathbf{a}\nabla G\biggr)^{1/2}.$$

The Lipschitz function $\eta(x) := \min\{(4|x|/r - 1)_+, 1, (3 - 2|x|/r)_+\}$ equals 1 in A_r, vanishes outside D_r, satisfies $|\nabla\eta| \leqslant 4/r$, and has compact support in $B_R\backslash\overline{B}_{r/8}$, where G is $\mathbf{a}$-harmonic. Hence (4.138), with $U = B_R\backslash\overline{B}_{r/8}$ and $k = 0$, gives

$$\int_{A_r} \nabla G \cdot \mathbf{a}\nabla G \leqslant \int_U \eta^2\, \nabla G \cdot \mathbf{a}\nabla G \leqslant \frac{64\Lambda}{r^2} \int_{D_r} G^2 \leqslant \frac{64\Lambda}{r^2} |D_r| \Bigl(\operatorname*{ess\,sup}_{D_r} G\Bigr)^2.$$

Combining the two previous displays and using $|A_r|, |D_r| \leqslant |B_{3r/2}|$, we obtain

$$1 \leqslant \frac{16\Lambda}{r^2} |B_{3r/2}| \operatorname*{ess\,sup}_{D_r} G = 16\Bigl(\frac{3}{2}\Bigr)^d |B_1|\, \Lambda\, r^{d-2} \operatorname*{ess\,sup}_{D_r} G\,.$$

By (4.161) and $\Lambda \leqslant \exp(2\Lambda^{1/2})$, it follows that

$$\begin{aligned}\operatorname*{ess\,inf}_{D_r} G \geqslant \exp\bigl(-C_2\Lambda^{1/2}\bigr) \operatorname*{ess\,sup}_{D_r} G &\geqslant \Bigl(16\Bigl(\frac{3}{2}\Bigr)^d |B_1|\Bigr)^{-1} \Lambda^{-1} \exp\bigl(-C_2\Lambda^{1/2}\bigr) r^{2-d} \\ &\geqslant \exp\bigl(-C\Lambda^{1/2}\bigr) r^{2-d}\,.\end{aligned}$$

Step 3. Conclusion. Steps 1 and 2 give, for every $r \in (0, \frac{R}{2}]$,

$$\exp\bigl(-C\Lambda^{1/2}\bigr) r^{2-d} \leqslant \operatorname*{ess\,inf}_{D_r} G \leqslant \operatorname*{ess\,sup}_{D_r} G \leqslant \exp\bigl(C\Lambda^{1/2}\bigr) r^{2-d}\,.$$

Since $A_r \subseteq D_r$, this implies (4.158) and (4.159). Let $x \in \overline{B}_{R/2}\backslash\{0\}$ and $r := |x|$. Then x belongs to the open set $O := B_{3r/2}\backslash\overline{B}_{r/4} \subseteq D_r$. If a function is continuous on an open set, its value at each point of the set lies between its essential infimum and its essential supremum on the set. Hence the continuous representative of G satisfies $\operatorname{ess\,inf}_{D_r} G \leqslant \operatorname{ess\,inf}_O G \leqslant G(x) \leqslant \operatorname{ess\,sup}_O G \leqslant \operatorname{ess\,sup}_{D_r} G$, and the previous display gives (4.160). □

The derivation of (4.162) from (4.151) applies to every $r \in (0, R]$: the averages of G on the annuli A_r are bounded by Cr^{2-d} with C independent of Λ. The factors $\exp(\pm C\Lambda^{1/2})$ in Proposition 4.42 come from the Harnack inequality, which is used to pass from averages to pointwise values. We have assumed $d \geqslant 3$ in the construction of the Green function, and the case $d = 2$ is not treated here.

5. Parabolic equations

5.1 Function spaces for parabolic equations

Throughout this subsection, $I \subseteq \mathbb{R}$ is an interval (open, half-open or closed) with interior I°, and $U \subseteq \mathbb{R}^d$ is open. Points of space-time are written (t, x) with the time variable first, and all integrals over $I \times U$ are with respect to Lebesgue measure $dx\,dt$. The *parabolic cylinder* of radius $r > 0$ centered at (t_0, x_0) is

$$Q_r(t_0, x_0) := (t_0 - r^2, t_0] \times B_r(x_0)\,,$$

with volume $|Q_r| = r^{d+2}|Q_1|$, and its normalized norms are

$$\|f\|_{\underline{L}^p(Q_r(t_0,x_0))} := \left(\frac{1}{|Q_r|}\int_{Q_r(t_0,x_0)} |f|^p\right)^{1/p}, \qquad p \in (0,\infty)\,.$$

The class of equations below is invariant under the parabolic scaling $(t, x) \mapsto (r^2 t, rx)$ and under translations in (t, x): if u solves the equation with coefficient field $\mathbf{a}$ and $r > 0$, then $u(r^2\,\cdot, r\,\cdot)$ solves it with the coefficient field $\mathbf{a}(r^2\,\cdot, r\,\cdot)$, which satisfies the same bounds.

Coefficients. The coefficient field is a Lebesgue measurable map $\mathbf{a} : I \times U \to \mathbb{R}^{d\times d}$ (jointly measurable in (t, x), which is what Fubini's theorem requires), and there are constants $0 < \lambda \leqslant \Lambda < \infty$ such that, for almost every $(t, x) \in I \times U$,

$$e \cdot \mathbf{a}(t,x)e \geqslant \lambda|e|^2 \quad \text{and} \quad e \cdot \mathbf{a}^{-1}(t,x)e \geqslant \Lambda^{-1}|e|^2\,, \quad \forall e \in \mathbb{R}^d. \tag{5.1}$$

(The first inequality shows that $\mathbf{a}(t, x)$ is invertible.) The ratio Λ/λ is called the *parabolicity contrast ratio*. The matrix $\mathbf{a}(t, x)$ may not be symmetric, and we write

$$\mathbf{s}(t,x) := \tfrac{1}{2}\big(\mathbf{a}(t,x) + \mathbf{a}(t,x)^\top\big)$$

for its symmetric part, so that $e\cdot\mathbf{s}e = e\cdot\mathbf{a}e$. Note that (5.1) is the parabolic version of (4.1): it implies that

$$|\mathbf{a}(t,x)e| \leqslant \Lambda^{1/2}(e\cdot\mathbf{s}(t,x)e)^{1/2} \quad \text{and} \quad |\mathbf{a}(t,x)^\top e| \leqslant \Lambda^{1/2}(e\cdot\mathbf{s}(t,x)e)^{1/2}\,, \quad \forall e \in \mathbb{R}^d, \tag{5.2}$$

which is how we will often use it. Indeed, applying the second inequality of (5.1) to $f = \mathbf{a}e$ gives $\Lambda^{-1}|\mathbf{a}e|^2 \leqslant f\cdot\mathbf{a}^{-1}f = \mathbf{a}e\cdot e = e\cdot\mathbf{s}e$; and since $e\cdot(\mathbf{a}^\top)^{-1}e = e\cdot\mathbf{a}^{-1}e$, the matrix $\mathbf{a}^\top$ satisfies (5.1) with the same constants and the same symmetric part. In particular $|\mathbf{a}e| \leqslant \Lambda|e|$, so $\mathbf{a} \in L^\infty(I \times U; \mathbb{R}^{d\times d})$ and $\mathbf{a}\nabla u \in L^2$ whenever $\nabla u \in L^2$.

The energy space. We write $L^2_{\mathrm{c}}(U)$ for the functions in $L^2(U)$ vanishing outside a compact subset of U, and $C^1_{\mathrm{c}}(U)$ for the C^1 functions with compact support in U.

Definition 5.1 (The space $H^1_{\mathrm{par}}(I \times U)$)**.** The space $H^1_{\mathrm{par}}(I \times U)$ consists of the (jointly) measurable functions $u : I \times U \to \mathbb{R}$, with two such functions identified if $u(t,\cdot) = v(t,\cdot)$ almost everywhere in U *for every* $t \in I$ (by Remark 5.2(a) this is equivalent to $u = v$ a.e. in $I \times U$, so elements of $H^1_{\mathrm{par}}(I \times U)$ are a.e.-classes in $L^2(I \times U)$ admitting a representative satisfying (ii)), for which there is a measurable $\nabla u : I \times U \to \mathbb{R}^d$ such that:

(i) u and ∇u belong to $L^2(I \times U)$, and, for almost every $t \in I$, $u(t,\cdot) \in H^1(U)$ with weak gradient $\nabla u(t,\cdot)$;

(ii) for every $t \in I$, $u(t,\cdot) \in L^2_{\mathrm{loc}}(U)$, and for every $\psi \in L^2_{\mathrm{c}}(U)$ the function

$$t \mapsto \int_U u(t,x)\psi(x)\,dx \quad \text{is continuous on } I. \tag{5.3}$$

Thus $H^1_{\mathrm{par}}(I \times U)$ is a subspace of $L^2(I; H^1(U))$, singled out by the requirement that $t \mapsto u(t,\cdot)$ be weakly continuous into $L^2_{\mathrm{loc}}(U)$. This makes the slice $u(t,\cdot)$ meaningful for *every* $t \in I$, including an endpoint belonging to I such as $t = 0$ for $I = (-1, 0]$. The following two remarks show that this is consistent.

Remark 5.2 (Slices are well-defined and locally bounded)**.** *(a)* If two functions u, v as in Definition 5.1 satisfy $u(t,\cdot) = v(t,\cdot)$ a.e. for almost every t, then for each $\psi \in L^2_{\mathrm{c}}(U)$ the continuous function $t \mapsto \int (u - v)(t,\cdot)\psi$ vanishes for a.e. t, hence for every t; thus $u(t,\cdot) = v(t,\cdot)$ a.e. for every t. So an element of $H^1_{\mathrm{par}}(I \times U)$ is determined by its a.e. class in $I \times U$, and the every-time slices are canonical.

(b) For every compact interval $J \subseteq I$ and compact $K \subseteq U$, $\sup_{t \in J} \|u(t,\cdot)\|_{L^2(K)} < \infty$. Indeed, the functionals $\psi \mapsto \int u(t,\cdot)\psi$ are bounded on the Hilbert space $L^2(K)$ and, for each fixed ψ, bounded in $t \in J$ by (5.3); the uniform boundedness principle gives the claim.

Remark 5.3 (On the definition)**.** For a weak solution $u \in L^2(I; H^1(U))$ of the equation below, the representative in (ii) exists automatically (the time derivative $\partial_t u \in L^2(I; H^{-1})$ makes u continuous into L^2_{loc} after modification on a null set of times; see Lions and Magenes). We do not prove this here: we build (ii) into the definition, so that only elements with a genuine time trace are considered. Continuity into $L^2(U)$ up to ∂U cannot be expected for elements of $L^2(I; H^1(U))$ without boundary conditions, which is why only $L^2_{\mathrm{loc}}(U)$ appears.

Weak solutions. We use test functions with compact support in space, and in time either compact support in I° or (when a slice is involved) cutoffs that need not vanish at an endpoint of I belonging to I. Write $C^1_{\mathrm{c}}(I^\circ \times U)$ for the C^1 functions with compact support in the open set $I^\circ \times U$, and $C^1_{\mathrm{c}}(I \times U)$ for the functions $\varphi \in C^1(I \times \mathbb{R}^d)$ (one-sided derivative at an endpoint in I) vanishing outside a compact subset of $I \times U$. For example, $\varphi \in C^1_{\mathrm{c}}((-1, 0] \times B_1)$ vanishes for t near -1 and for x near ∂B_1, but not necessarily near $t = 0$.

Definition 5.4 (Weak solutions, sub- and supersolutions)**.** Let $\mathbf{a}$ be as in (5.1) and $u \in H^1_{\mathrm{par}}(I \times U)$. We say that u is a *weak solution*, *subsolution*, respectively *supersolution* of

$$\partial_t u - \nabla \cdot \mathbf{a}\nabla u = 0 \quad \text{in } I \times U \tag{5.4}$$

if

$$\int_{I \times U} \big(-u\,\partial_t\varphi + \mathbf{a}\nabla u \cdot \nabla\varphi\big) \;=\; 0\,, \quad \leqslant 0\,, \quad \text{respectively} \quad \geqslant 0\,, \tag{5.5}$$

for every $\varphi \in C^1_{\mathrm{c}}(I^\circ \times U)$ (in the case of sub- and supersolutions, for every non-negative such φ).

The integrand is integrable because $u, \nabla u \in L^2$ and $\mathbf{a}$ is bounded. The (in)equalities (5.5) are the weak form of $\partial_t u - \nabla \cdot \mathbf{a}\nabla u = 0, \leqslant 0, \geqslant 0$ (integrate $(\partial_t u - \nabla \cdot \mathbf{a}\nabla u)\varphi$ by parts for smooth u).

The next lemma converts (5.5) into a statement about slices, valid at *every* time and including endpoints of I.

Lemma 5.5 (Weak formulation at every time)**.** *Let $u \in H^1_{\mathrm{par}}(I \times U)$ be a weak solution of (5.4). Then for every $\zeta \in C^1_{\mathrm{c}}(U)$ and every $s < t$ in I,*

$$\int_U u(t,x)\zeta(x)\,dx - \int_U u(s,x)\zeta(x)\,dx = -\int_s^t\!\!\int_U \mathbf{a}\nabla u \cdot \nabla\zeta\,dx\,d\tau\,. \tag{5.6}$$

Proof. Set $m(\tau) := \int_U u(\tau,\cdot)\zeta$ and $g(\tau) := \int_U \mathbf{a}(\tau,\cdot)\nabla u(\tau,\cdot)\cdot\nabla\zeta$. By (5.3), m is continuous on I, and by Cauchy–Schwarz, $|g(\tau)| \leqslant \Lambda\|\nabla u(\tau,\cdot)\|_{L^2(U)}\|\nabla\zeta\|_{L^2}$, so $g \in L^2(I) \subseteq L^1_{\mathrm{loc}}(I)$. Let $\theta \in C^1_{\mathrm{c}}(I^\circ)$. The function $\varphi(\tau,x) = \theta(\tau)\zeta(x)$ is admissible in (5.5); by Fubini, $\int_I(-m\theta' + g\theta)\,d\tau = 0$. Fix $\tau_0 \in I^\circ$ and let $G(\tau) := \int_{\tau_0}^{\tau} g$, which is locally absolutely continuous on I with $\int G\theta' = -\int g\theta$. Hence $\int_I (m+G)\theta'\,d\tau = 0$ for all θ, so $m+G$ is a.e. equal to a constant (du Bois-Reymond lemma). Both m and G are continuous on I, so $m+G$ is constant on all of I, endpoints included. This is (5.6). □

The energy identity. The following chain rule is used in every energy argument of this chapter. In the smooth case it is the identity $\frac{d}{dt}\int \frac12\varphi^2u^2 = \int u^2\varphi\partial_t\varphi + \int\varphi^2 u\,\partial_t u$ combined with the equation; the point is that it holds for elements of H^1_{par}, with slices at every time.

Lemma 5.6 (Energy identity)**.** *Let $u \in H^1_{\mathrm{par}}(I\times U)$ be a weak solution of (5.4) and let $\varphi \in C^1_{\mathrm{c}}(I\times U)$. Then the function*

$$E(t) := \frac12\int_U \varphi(t,x)^2 u(t,x)^2\,dx\,, \qquad t \in I,$$

is finite and locally absolutely continuous on I, and for all $s < t$ in I,

$$E(t) - E(s) = \int_s^t\!\!\int_U u^2\varphi\,\partial_t\varphi\,dx\,d\tau - \int_s^t\!\!\int_U \mathbf{a}\nabla u\cdot\nabla(\varphi^2 u)\,dx\,d\tau\,. \tag{5.7}$$

In particular $E' = \int u^2\varphi\partial_t\varphi - \int \mathbf{a}\nabla u\cdot\nabla(\varphi^2u)$ almost everywhere.

Proof. Let $Z \subseteq I\times U$ be a compact set outside which $\varphi = 0$, $K \subseteq U$ its (compact) projection onto space, $J \subseteq I$ a compact interval containing s,t, and $\delta_0 > 0$ so small that $K_1 := \{x : \operatorname{dist}(x,K) \leqslant 2\delta_0\} \subseteq U$. Write $F := \mathbf{a}\nabla u \in L^2(I\times U;\mathbb{R}^d)$. Finiteness of E follows from Remark 5.2(b): $E(t) \leqslant \frac12\|\varphi\|_\infty^2 \sup_J \|u(t,\cdot)\|^2_{L^2(K)} < \infty$ for every compact interval $J \subseteq I$. (Continuity of E is *not* a consequence of the weak continuity (5.3), which only gives lower semicontinuity; it is a consequence of the identity.) So it suffices to prove (5.7): once it holds for all $s < t$ in I, the right side is the increment of the integral of an L^1_{loc} function of the upper limit, so E is continuous and locally absolutely continuous, with the stated derivative almost everywhere.

Step 1: mollified equation. Let ρ_δ be a standard mollifier supported in B_δ, $0 < \delta < \delta_0$, and set $u_\delta(\tau,x) := \int \rho_\delta(x-y)u(\tau,y)\,dy$ for $x \in K_0 := \{\operatorname{dist}(x,K) \leqslant \delta_0\}$ (so that $B_\delta(x) \subseteq$

$K_1 \subseteq U$) and $\tau \in J$; this is smooth in x for each τ, and, by Remark 5.2(b), $|u_\delta| \leqslant C_\delta$ on $J \times K_0$. Applying Lemma 5.5 with $\zeta(y) = \rho_\delta(x-y)$ (so $\nabla_y \zeta(y) = -(\nabla\rho_\delta)(x-y)$) gives, for *every* $x \in K$ and $\tau_1 < \tau_2$ in J,

$$u_\delta(\tau_2,x) - u_\delta(\tau_1,x) = \int_{\tau_1}^{\tau_2} D_\delta(\tau,x)\,d\tau\,, \qquad D_\delta(\tau,x) := \int F(\tau,y)\cdot(\nabla\rho_\delta)(x-y)\,dy = \nabla\cdot F_\delta(\tau,x)\,,$$

where $F_\delta := F * \rho_\delta$ (convolution in x). Since $|D_\delta(\tau,x)| \leqslant \|F(\tau)\|_{L^2(K_1)}\|\nabla\rho_\delta\|_{L^2}$, we have $D_\delta \in L^2(J\times K)$. Thus, for each $x \in K$, $\tau \mapsto u_\delta(\tau,x)$ is absolutely continuous with derivative $D_\delta(\cdot,x)$ (a.e.), and so is $\tau \mapsto \frac12\varphi^2 u_\delta^2$, with derivative $\varphi\partial_t\varphi\, u_\delta^2 + \varphi^2 u_\delta D_\delta$. Integrating over $[s,t]\times K$ (everything is integrable: u_δ and φ are bounded, $D_\delta \in L^2$) and using Fubini,

$$\frac12\int \varphi^2 u_\delta^2(t) - \frac12\int \varphi^2 u_\delta^2(s) = \int_s^t\!\!\int \varphi\partial_t\varphi\, u_\delta^2 + \int_s^t\!\!\int \varphi^2 u_\delta\,\nabla\cdot F_\delta\,.$$

The function $\varphi^2 u_\delta(\tau,\cdot)$, defined on K_0 and extended by 0, is C^1 with support in K (since $\varphi = 0$ outside K), so we may integrate by parts in x:

$$\frac12\int \varphi^2 u_\delta^2(t) - \frac12\int \varphi^2 u_\delta^2(s) = \int_s^t\!\!\int \varphi\partial_t\varphi\, u_\delta^2 - \int_s^t\!\!\int \nabla(\varphi^2 u_\delta)\cdot F_\delta\,. \tag{5.8}$$

Step 2: $\delta \to 0$. For every τ, $u(\tau,\cdot) \in L^2(K_1)$, so $u_\delta(\tau,\cdot) \to u(\tau,\cdot)$ in $L^2(K)$; this gives the convergence of the two terms on the left side of (5.8), at the fixed times s and t. Moreover, since $\|u_\delta(\tau)\|_{L^2(K)} \leqslant \|u(\tau)\|_{L^2(K_1)}$, which is bounded on J, dominated convergence gives $u_\delta \to u$ in $L^2(J\times K)$. For almost every τ, $u(\tau,\cdot) \in H^1(U)$, so $\nabla u_\delta(\tau,\cdot) = (\nabla u)(\tau,\cdot) * \rho_\delta \to \nabla u(\tau,\cdot)$ in $L^2(K)$, with $\|\nabla u_\delta(\tau)\|_{L^2(K)} \leqslant \|\nabla u(\tau)\|_{L^2(K_1)}$; likewise $F_\delta(\tau) \to F(\tau)$ in $L^2(K)$. By dominated convergence, $\nabla u_\delta \to \nabla u$ and $F_\delta \to F$ in $L^2(J\times K)$. Since $\nabla(\varphi^2 u_\delta) = \varphi^2\nabla u_\delta + 2\varphi u_\delta\nabla\varphi$ and $\varphi, \nabla\varphi, \partial_t\varphi$ are bounded, the integrands on the right of (5.8) converge in $L^1(J\times K)$ (products of L^2-convergent sequences) to those of (5.7). This proves (5.7). □

5.2 Parabolic Caccioppoli inequality

We begin by generalizing the Caccioppoli inequality (Lemma 4.1) to solutions of parabolic equations. The proof uses only the upper bound in (5.2); the constant λ plays no role.

Lemma 5.7 (Caccioppoli inequality, parabolic version)**.** *Let* $\mathbf{a}$ *satisfy* (5.1) *in* $(-1,0]\times B_1$. *For every weak solution* $u \in H^1_{\mathrm{par}}((-1,0]\times B_1)$ *of*

$$\partial_t u - \nabla\cdot\mathbf{a}\nabla u = 0 \quad \text{in } (-1,0]\times B_1, \tag{5.9}$$

every cutoff function $\varphi \in C^1_{\mathrm{c}}((-1,0]\times B_1)$ *and every* $t_1 \in (-1,0]$, *writing* $Q_{t_1} := (-1,t_1]\times B_1$, *we have*

$$\int_{Q_{t_1}} \varphi^2\nabla u\cdot\mathbf{s}\nabla u + \int_{B_1}(\varphi^2 u^2)(t_1,\cdot) \leqslant \int_{Q_{t_1}} u^2\big(2|\varphi||\partial_t\varphi| + 4\Lambda|\nabla\varphi|^2\big)\,. \tag{5.10}$$

In particular, with $t_1 = 0$, *and since the right side of* (5.10) *is non-decreasing in* t_1, *each of the two terms on the left side (the second one at every time* t_1*) is bounded by the right side with* $t_1 = 0$.

Proof. Fix t_1 and choose $t_0 \in (-1, t_1)$ such that $\varphi \equiv 0$ on $(-1, t_0] \times B_1$ (possible since $\operatorname{supp}\varphi$ is a compact subset of $(-1, 0] \times B_1$, so t_0 may be taken below t_1 and below the minimum of the time projection of the support); then $E(t_0) = 0$ in the notation of Lemma 5.6. This lemma (applied with $I = (-1, 0]$, $U = B_1$, $s = t_0$, $t = t_1$) and the fact that $\varphi = 0$ and $\nabla(\varphi^2 u) = 0$ on $(-1, t_0] \times B_1$ give the identity

$$\int_{Q_{t_1}} \mathbf{a}\nabla u \cdot \nabla(\varphi^2 u) = \int_{Q_{t_1}} u^2 \varphi\, \partial_t \varphi - \frac{1}{2} \int_{B_1} (u^2 \varphi^2)(t_1, \cdot)\,. \tag{5.11}$$

Expanding the left side, all terms being integrable,

$$\int_{Q_{t_1}} \varphi^2 \nabla u \cdot \mathbf{a}\nabla u = \int_{Q_{t_1}} \mathbf{a}\nabla u \cdot \nabla(\varphi^2 u) - 2 \int_{Q_{t_1}} \varphi u\, \mathbf{a}\nabla u \cdot \nabla\varphi\,.$$

By the first inequality in (5.2) and $2xy \leqslant \frac{1}{2}x^2 + 2y^2$ (with $x = |\varphi|(\nabla u \cdot \mathbf{s}\nabla u)^{1/2}$ and $y = \Lambda^{1/2}|u||\nabla\varphi|$),

$$2|\varphi u\, \mathbf{a}\nabla u \cdot \nabla\varphi| \leqslant 2|\varphi||u||\nabla\varphi|\, \Lambda^{1/2} (\nabla u \cdot \mathbf{s}\nabla u)^{1/2} \leqslant \frac{1}{2}\varphi^2 \nabla u \cdot \mathbf{s}\nabla u + 2\Lambda u^2 |\nabla\varphi|^2\,.$$

Since $\nabla u \cdot \mathbf{a}\nabla u = \nabla u \cdot \mathbf{s}\nabla u$ and $\int_{Q_{t_1}} \varphi^2 \nabla u \cdot \mathbf{s}\nabla u < \infty$, we may reabsorb the second term. Substituting (5.11),

$$\frac{1}{2} \int_{Q_{t_1}} \varphi^2 \nabla u \cdot \mathbf{s}\nabla u + \frac{1}{2} \int_{B_1} (\varphi^2 u^2)(t_1, \cdot) \leqslant \int_{Q_{t_1}} u^2 \varphi\, \partial_t \varphi + 2\Lambda \int_{Q_{t_1}} u^2 |\nabla\varphi|^2\,.$$

Multiplying by two and using $\varphi\partial_t\varphi \leqslant |\varphi||\partial_t\varphi|$ yields (5.10). □

Remark 5.8. *(i)* The inequality is unchanged if $(-1, 0] \times B_1$ is replaced by any cylinder $(t_a, t_b] \times B_r(x_0)$, with the same proof. *(ii)* For subsolutions and supersolutions one expects the same inequality for $(u - k)_\pm$ (compare Lemma 4.1). This needs a chain rule for truncations in the class H^1_{par}, which we do not prove here since it is not used in the sequel.

The following lemma is a companion of the Caccioppoli inequality which controls the gradient at each fixed time. It is not used in the rest of the chapter, and we keep the proof minimal. It is not vacuous: $u = e^{-\mu t} w(x)$ with w an eigenfunction satisfies its hypotheses. It requires a time-independent coefficient field and, as stated, regularity of the time derivative, which holds on subcylinders away from the initial time by a standard difference-quotient argument that we do not reproduce; on the whole cylinder it fails in general, since the initial datum is only in L^2.

Lemma 5.9 (Time slice estimate)**.** *Assume that $\mathbf{a}(t, x) = \mathbf{a}(x)$ is independent of the time variable and satisfies (5.1). Let $u \in H^1_{\mathrm{par}}((-1, 0] \times B_1)$ be a weak solution of (5.9) whose distributional time derivative $v := \partial_t u$ belongs to $H^1_{\mathrm{par}}((-1, 0] \times B_1)$. Let $\varphi \in C^1_{\mathrm{c}}((-1, 0] \times B_1)$ with $|\varphi| \leqslant 1$, set*

$$M := 4\|\partial_t\varphi\|_{L^\infty} + 16\Lambda\|\nabla\varphi\|^2_{L^\infty}\,, \qquad G := \int_{\{\varphi \neq 0\}} \nabla u \cdot \mathbf{s}\nabla u\,,$$

the latter being finite by Lemma 5.7 applied with a cutoff equal to 1 *on* $\operatorname{supp}\varphi$*. Then, for almost every* $t \in (-1,0]$*, with* $\nabla u(t,\cdot)$ *denoting the a.e.-defined slice of the gradient (which lies in* $L^2(B_1;\mathbb{R}^d)$ *for a.e.* t *by (i), and, since* $\partial_t \nabla u = \nabla v \in L^2$*, agrees for a.e.* t *with the absolutely continuous* L^2*-valued representative),*

$$\int_{B_1} \varphi^4 \, \nabla u \cdot \mathbf{s} \nabla u \, (t,\cdot) \leqslant 7 \, \frac{\Lambda}{\lambda} \, M \, G \,. \tag{5.12}$$

Proof. Write $A := \nabla u \cdot \mathbf{s} \nabla u$ and $B := \nabla v \cdot \mathbf{s} \nabla v$; recall $|\nabla u|^2 \leqslant \lambda^{-1} A$. Throughout, integrals are over $(-1,0] \times B_1$ unless indicated, and $\|\cdot\|_\infty$ denotes sup norms.

Step 1: v is a weak solution. Since $\mathbf{a}$ does not depend on t, for $\chi \in C_c^2((-1,0)\times B_1)$ we can test (5.5) with $\partial_t \chi$. Because $\partial_t \nabla u = \nabla v$ in the sense of distributions (test with ∇ of smooth functions, and use $u(t) \in H^1$), and both are in L^2, we have $\int \nabla v \cdot H = -\int \nabla u \cdot \partial_t H$ for every bounded H which is C^1 in t with compact support in $(-1,0)\times B_1$ (approximate H by mollification in (t,x): $H \in L^2$ and $\partial_t H = \mathbf{a}^\top \nabla \partial_t \chi \in L^2$ suffice, and $\partial_t H$ is $\mathbf{a}^\top \nabla \partial_t \chi$ since $\mathbf{a}$ is independent of t). Taking $H = \mathbf{a}^\top \nabla \chi$ gives

$$\int \mathbf{a} \nabla v \cdot \nabla \chi = -\int \mathbf{a} \nabla u \cdot \nabla \partial_t \chi = -\int u \, \partial_t^2 \chi = \int v \, \partial_t \chi \,,$$

i.e. $\int (-v \partial_t \chi + \mathbf{a} \nabla v \cdot \nabla \chi) = 0$; by density (uniform convergence of χ, $\partial_t \chi$, $\nabla \chi$ under $C^2 \to C^1$ approximation) this extends to $\chi \in C_c^1((-1,0)\times B_1)$. So v is a weak solution and Lemma 5.7 applies to v.

Step 2: L^2 bound of v. Set $X := \int \varphi^2 v^2 < \infty$. Testing the weak form of the equation of u with $\varphi^2 v \in L^2(I; H_0^1(B_1))$ (approximating $\varphi^2 v$ in $L^2(H^1)$ by smooth compactly supported functions, after a cutoff in time near $t=0$, and using $-\int u \, \partial_t \psi = \int v \psi$) gives $X = -\int \mathbf{a} \nabla u \cdot \nabla(\varphi^2 v)$, hence

$$X = -\int \varphi^2 \, \mathbf{a} \nabla u \cdot \nabla v - 2 \int \varphi v \, \mathbf{a} \nabla u \cdot \nabla \varphi \,.$$

By (5.2), $|\mathbf{a} \nabla u \cdot \nabla v| = |\nabla u \cdot \mathbf{a}^\top \nabla v| \leqslant \Lambda^{1/2} |\nabla u| B^{1/2} \leqslant (\Lambda/\lambda)^{1/2} A^{1/2} B^{1/2}$, and $|\mathbf{a} \nabla u \cdot \nabla \varphi| \leqslant \Lambda^{1/2} A^{1/2} |\nabla \varphi|$. Using $2xy \leqslant \frac{1}{2} x^2 + 2y^2$ on the second term (with $x = |\varphi||v|$), and $\nabla \varphi = 0$ a.e. on $\{\varphi = 0\}$ (the set $\{\varphi = 0, \nabla \varphi \neq 0\}$ is locally a C^1 graph, hence null; so $\int \varphi^2 A |\nabla \varphi|^2$-type terms are bounded by $\|\nabla \varphi\|_\infty^2 G$),

$$X \leqslant \kappa \int \varphi^2 A^{1/2} B^{1/2} + \frac{1}{2} X + 2\Lambda \|\nabla \varphi\|_\infty^2 G \,, \qquad \kappa := (\Lambda/\lambda)^{1/2} \,.$$

Now $\varphi^2 A^{1/2} B^{1/2} \leqslant \frac{\theta}{2} \varphi^4 B + \frac{1}{2\theta} A \mathbf{1}_{\{\varphi \neq 0\}}$ for $\theta > 0$, and Lemma 5.7 for v with the cutoff φ^2 gives (as $2\varphi^2 |\partial_t \varphi^2| = 4|\varphi|^3 |\partial_t \varphi|$ and $4\Lambda |\nabla \varphi^2|^2 = 16 \Lambda \varphi^2 |\nabla \varphi|^2$, and $|\varphi| \leqslant 1$)

$$\int \varphi^4 B \leqslant \int v^2 \varphi^2 \big(4|\varphi||\partial_t \varphi| + 16 \Lambda |\nabla \varphi|^2 \big) \leqslant M \, X \,.$$

Choosing $\theta = 1/(2\kappa M)$ (if $M = 0$ then $\varphi \equiv 0$ and there is nothing to prove) and absorbing, we get $\frac{1}{2} X \leqslant \frac{1}{4} X + \kappa^2 M G + 2\Lambda \|\nabla \varphi\|_\infty^2 G$, that is

$$X \leqslant 4 \frac{\Lambda}{\lambda} \, M G + 8 \Lambda \|\nabla \varphi\|_\infty^2 G \,. \tag{5.13}$$

Step 3: the slice. Let $\mathcal{E}(t) := \int_{B_1} \varphi^4 A(t,\cdot)$. Since $\partial_t \nabla u = \nabla v \in L^2$, the map $t \mapsto \nabla u(t,\cdot) \in L^2(B_1;\mathbb{R}^d)$ has an absolutely continuous representative, so $\mathcal{E}$ is absolutely continuous on compact subintervals (of the times where φ is supported), with, since $\mathbf{s}$ is symmetric and independent of t,

$$\mathcal{E}' = \int 4\varphi^3 \partial_t\varphi\, A + 2\int \varphi^4 \nabla u \cdot \mathbf{s}\nabla v \leqslant \int \big(4|\varphi|^3|\partial_t\varphi| + \varepsilon^{-1}\varphi^4\big)A + \varepsilon\int \varphi^4 B\,,$$

for any $\varepsilon > 0$, by Cauchy–Schwarz for the form $\mathbf{s}$ and $2xy \leqslant \varepsilon^{-1}x^2 + \varepsilon y^2$. Choose t_0 with $\varphi \equiv 0$ on $(-1,t_0]\times B_1$, so $\mathcal{E}(t_0) = 0$, and integrate over (t_0,t). Using $|\varphi| \leqslant 1$ and the bound on $\int \varphi^4 B$ from Step 2, for a.e. t,

$$\mathcal{E}(t) \leqslant \varepsilon M X + \big(4\|\partial_t\varphi\|_\infty + \varepsilon^{-1}\big)G\,.$$

Take $\varepsilon = 1/M$ and use $4\|\partial_t\varphi\|_\infty \leqslant M$, $8\Lambda\|\nabla\varphi\|_\infty^2 \leqslant M/2$, $\Lambda/\lambda \geqslant 1$ and (5.13): $\mathcal{E}(t) \leqslant X + 2MG \leqslant (4\Lambda/\lambda + \frac{5}{2})MG \leqslant 7(\Lambda/\lambda)MG$. □

5.3 The Nash-Aronson estimate

In this subsection we allow the coefficients $\mathbf{a}(t,x)$ to depend on a time variable $t \in \mathbb{R}$. We assume that (5.1) holds with $\lambda = 1$ for every $(t,x) \in \mathbb{R}^{1+d}$; neither symmetry nor independence of time is assumed. For $s < t$ and $f \in L^2(\mathbb{R}^d)$ we write $S(t,s)f := u(t,\cdot)$, where $u \in H^1_{\mathrm{par}}([s,T]\times\mathbb{R}^d)$ is the weak solution (Definition 5.4) of the Cauchy problem with $u(s,\cdot) = f$. The following standard facts about this evolution are used without proof: existence and uniqueness, positivity, conservation of mass, L^2 contraction and the energy identity (including its weighted version for bounded Lipschitz weights), and duality $\int g\, S(t,s)f = \int f\, S^*(s,t)g$, where S^* is the backward evolution of $\mathbf{a}^{\mathsf{t}}$. Supports of L^1 functions are closed essential supports.

The *Nash-Aronson estimates* state that the kernel $P(t,x;s,y)$ of $S(t,s)$ satisfies, for constants $c, C, \alpha, \beta > 0$ depending only on (d,Λ) and all $s < t$, $x, y \in \mathbb{R}^d$,

$$c(t-s)^{-d/2}\exp\!\left(-\beta\frac{|x-y|^2}{t-s}\right) \leqslant P(t,x;s,y) \leqslant C(t-s)^{-d/2}\exp\!\left(-\alpha\frac{|x-y|^2}{t-s}\right).$$

We prove them in the kernel-free form of Proposition 5.10 and Theorem 5.14. The upper bound is easier, and is closely related to Propositions 4.9 and 4.13; the lower bound is more subtle, and is analogous to Proposition 4.17. The exponent α can be any number smaller than $1/(4\Lambda)$, as follows from (5.14); obtaining a good bound on β is open.

Proposition 5.10 (Nash-Aronson upper bound)**.** *There is $C(d) < \infty$ such that, if $f, g \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$ have supports at distance at least $r \geqslant 0$, then, for $s < t$,*

$$\left|\int_{\mathbb{R}^d} g\, S(t,s)f\right| \leqslant C(t-s)^{-d/2}\exp\!\left(-\frac{r^2}{4\Lambda(t-s)} + \sqrt{\frac{d+2}{2}}\,\frac{r}{\sqrt{\Lambda(t-s)}}\right)\|f\|_{L^1}\|g\|_{L^1}\,. \tag{5.14}$$

Proof. Set $u(t) = S(t,s)f$. Positivity and mass conservation, applied also to $|f|$, give $\|u(t)\|_1 \leqslant \|f\|_1$. The energy identity and Nash's inequality (Proposition 1.22) give

$$\partial_h \int |u|^2 = -2\int \nabla u \cdot \mathbf{a}\nabla u \leqslant -2\int |\nabla u|^2 \leqslant -c\left(\int |u|^2\right)^{1+\frac{2}{d}}\|f\|_1^{-\frac{4}{d}}.$$

Integrating, we get

$$\|u(s+h)\|_2^2 \leqslant \min\{\|f\|_2^2,\, Ch^{-d/2}\|f\|_1^2\} \qquad (h>0). \tag{5.15}$$

Let K be the closed essential support of f, $\rho(x)=\operatorname{dist}(x,K)$, $\rho_R=\min\{\rho,R\}$, and $\psi=\exp(\rho_R^2/(4A(h+\sigma)))$, where $A>\Lambda$ and $R,\sigma>0$. The weighted energy identity (first with a spatial cutoff, then by approximation) gives, with $Y=\int\psi^2u^2$,

$$Y' \leqslant 2\int u^2(\psi\partial_h\psi+\Lambda|\nabla\psi|^2) \leqslant -\frac{2(1-\Lambda/A)}{h+\sigma}\int\frac{\rho_R^2}{4A(h+\sigma)}\psi^2u^2\,.$$

Put $\kappa=1+\frac{d}{2}$ and $m=\kappa/(2(1-\Lambda/A))$. On $\{\rho_R^2<4A(h+\sigma)m\}$ we have $\psi^2\leqslant e^{2m}$; splitting the last integral yields

$$\frac{d}{dh}\big((h+\sigma)^\kappa Y\big) \leqslant \kappa e^{2m}(h+\sigma)^{\kappa-1}\|u(s+h)\|_2^2\,.$$

Since $Y(0)=\|f\|_2^2$, integration on $(0,\sigma)$ using L^2 contraction and on (σ,h) using (5.15) gives

$$(h+\sigma)^\kappa Y(h) \leqslant Ce^{2m}\big(\sigma^\kappa\|f\|_2^2+h\|f\|_1^2\big).$$

Letting $R\uparrow\infty$ and $\sigma\downarrow 0$, we obtain

$$\int_{\mathbb{R}^d}\exp\!\left(\frac{\operatorname{dist}(x,K)^2}{2Ah}\right)|S(s+h,s)f(x)|^2\,dx \leqslant C\exp\!\left(\frac{d+2}{2}\frac{A}{A-\Lambda}\right)h^{-d/2}\|f\|_1^2\,. \tag{5.16}$$

Apply this estimate to f on $(s,(s+t)/2)$ and to g under the backward adjoint evolution on $((s+t)/2,t)$. If K_f,K_g are their supports, then $\operatorname{dist}(z,K_f)^2+\operatorname{dist}(z,K_g)^2\geqslant r^2/2$. Duality and Cauchy–Schwarz give, with $h=(t-s)/2$,

$$\left|\int g\,S(t,s)f\right| \leqslant Ch^{-d/2}\exp\!\left(-\frac{r^2}{8Ah}+\kappa\frac{A}{A-\Lambda}\right)\|f\|_1\|g\|_1.$$

For $r^2/(8h)>\kappa\Lambda$, minimizing over $A>\Lambda$ gives $-r^2/(8\Lambda h)+2\sqrt{\kappa r^2/(8\Lambda h)}$; in the remaining range take $A=2\Lambda$ and enlarge $C(d)$. Substituting $2h=t-s$ proves (5.14). □

The weighted estimate also gives a lower bound on the mass that remains near the initial support. We record it together with its L^2 consequence; symmetry is not required.

Proposition 5.11 (On-diagonal lower bound)**.** *There is $c(d)>0$ such that, if $f\geqslant 0$, $\int f=1$, $f\in L^1\cap L^2$, and* $\operatorname{supp} f\subseteq B_\varepsilon(y)$ *with $\varepsilon^2\leqslant\Lambda(t-s)$, then*

$$\int_{B_R(y)} S(t,s)f \geqslant \tfrac12 \qquad \text{with } R=C(d)\sqrt{\Lambda(t-s)\log(e\Lambda)}\geqslant\varepsilon, \tag{5.17}$$

and consequently

$$\int_{\mathbb{R}^d}|S(t,s)f|^2 \geqslant c\big(\Lambda\log(e\Lambda)(t-s)\big)^{-d/2}. \tag{5.18}$$

Proof. Write $h = t - s$ and $\rho = \operatorname{dist}(\cdot, \operatorname{supp} f)$. Since $\varepsilon^2 \leqslant \Lambda h$, the inclusion $\operatorname{supp} f \subseteq B_\varepsilon(y)$ gives $\int e^{-\rho^2/(8\Lambda h)} \leqslant C(d)(\Lambda h)^{d/2}$. On $\{\rho > R\}$, split $e^{-\rho^2/(4\Lambda h)}$ into two equal factors and bound one by $e^{-R^2/(8\Lambda h)}$. Thus (5.16) with $A = 2\Lambda$ and Cauchy–Schwarz give

$$\int_{\{\rho>R\}} S(t,s)f \leqslant C\Lambda^{d/4} \exp\bigl(-R^2/(16\Lambda h)\bigr).$$

Take $R_0 = C(d)\sqrt{\Lambda h \log(e\Lambda)}$ so that the right side is at most $\frac{1}{2}$. Mass conservation and $\operatorname{supp} f \subseteq B_\varepsilon(y)$ imply $\int_{B_{R_0+\varepsilon}(y)} S(t,s)f \geqslant \frac{1}{2}$, which is (5.17) with $R = R_0 + \varepsilon$ (enlarging $C(d)$). Cauchy–Schwarz on this ball yields (5.18). □

The local estimate needed for the lower bound is the parabolic weak Harnack inequality. Its proof uses only cylinders contained in the domain of the supersolution.

Proposition 5.12 (Parabolic weak Harnack inequality)**.** *If $u \in H^1_{\mathrm{par}}((-1,0] \times B_1)$ is a nonnegative supersolution of $\partial_t u - \nabla \cdot \mathbf{a}\nabla u \geqslant 0$, then, for every $q \in (0,1)$,*

$$\|u\|_{\underline{L}^{q(d+2)/d}((-3/4,-1/2)\times B_{1/2})} \leqslant C(d,\Lambda,q) \operatorname*{ess\,inf}_{(-1/4,0]\times B_{1/2}} u\,. \tag{5.19}$$

Proof. Set $v = u+\varepsilon$, $\varepsilon > 0$, and $\chi = 1+2/d$. The nonlinear test functions used below (powers of v, $\log v$) are justified for solutions as in Lemma 5.6, since $v \geqslant \varepsilon$, and for supersolutions by first regularizing in time with Steklov averages; this standard extension of Lemma 5.6 is not proved here. Testing the supersolution inequality with $\eta^2 v^{p-1}$ for $0 < p < 1$, and with $\eta^2 v^{-p-1}$ for $p > 0$, gives the energy estimates for v^p toward earlier times and v^{-p} toward later times. In either case the gradient term is a positive multiple of $\int \eta^2 |\nabla v^{\pm p/2}|^2$; the mixed term is absorbed using (5.2). For a cutoff vanishing at the appropriate time endpoint, the result has the form

$$\operatorname*{ess\,sup}_\tau \int \eta^2 v^{\pm p}(\tau) + \iint |\nabla(\eta v^{\pm p/2})|^2 \leqslant C(d,\Lambda,p) \iint v^{\pm p}(|\nabla\eta|^2 + |\eta\partial_t\eta|).$$

The parabolic Sobolev inequality

$$\iint |w|^{2\chi} \leqslant C(d)\bigl(\operatorname*{ess\,sup}_t \int |w(t)|^2\bigr)^{2/d} \iint |\nabla w|^2$$

then gives, on nested cylinders $Q' \subset Q$ with a gap at the cutoff time endpoint,

$$\|v^{\pm 1}\|_{\underline{L}^{\chi p}(Q')} \leqslant \bigl(C(d,\Lambda,p)(\sigma-\sigma')^{-C(d)}\bigr)^{1/p} \|v^{\pm 1}\|_{\underline{L}^{p}(Q)}\,. \tag{5.20}$$

Here Q' and Q are members of a nested family with parameters $\sigma' < \sigma$; for the plus sign $p < 1$, and for the minus sign $p > 0$. The constants inside the parentheses stay bounded as $p \downarrow 0$. Successive applications give the positive-power estimate up to every exponent smaller than χ, and the negative-power estimate up to L^∞.

To cross between the two estimates, choose a fixed radial cutoff ζ supported in $B_{7/8}$ and equal to one on $B_{3/4}$, put $w = \log v$, $d\nu = \zeta^2 dx / \int \zeta^2$ and $m(t) = \int w\, d\nu$. Testing with ζ^2/v gives

$$m'(t) \geqslant c \int |\nabla w|^2\, d\nu - C\Lambda \geqslant c \int |w - m(t)|^2\, d\nu - C\Lambda\,.$$

The last inequality is the weighted Poincaré inequality for this radial cutoff. Set $t_0 = -\frac{3}{8}$ and $M(t) = m(t) + C\Lambda t$, with the constant large enough that $M'(t) \geqslant c\int |w - m(t)|^2\, d\nu$. Put $k = M(t_0)$. Chebyshev's inequality gives, for $\ell \geqslant 2C\Lambda$,

$$\int_{-7/8}^{-3/8} \nu(\{w > k + \ell\})\, dt \leqslant C \int_{-7/8}^{-3/8} \frac{M'(t)}{(\ell - C\Lambda + M(t_0) - M(t))^2}\, dt \leqslant \frac{C}{\ell}\,.$$

The same calculation after t_0 controls $\{w < k - \ell\}$; for smaller ℓ use the volume bound. Since $\zeta = 1$ on $B_{3/4}$, this gives

$$\big|\{(t,x) \in (-7/8, -3/8) \times B_{3/4} : \log v(t,x) > k + \ell\}\big| + \big|\{(t,x) \in (-3/8, 0) \times B_{3/4} : \log v(t,x) < k - \ell\}\big| \leqslant \frac{C(d,\Lambda)}{\ell}\,. \tag{5.21}$$

One preliminary application of (5.20) supplies the finite $L^{q\chi}$ norm needed below. Apply the Bombieri–Giusti lemma (Lemma 1.12; see also [BG72, Mos71]) to $e^{-k}v$ on cylinders nested between $(-\frac{7}{8}, -\frac{3}{8}) \times B_{3/4}$ and $Q^- = (-\frac{3}{4}, -\frac{1}{2}) \times B_{1/2}$, using (5.20) and the first tail bound in (5.21). It gives $\|v\|_{\underline{L}^{q\chi}(Q^-)} \leqslant Ce^k$. Apply the same lemma to e^k/v (with $\alpha_0 = 1$) between $(-\frac{3}{8}, 0) \times B_{3/4}$ and the intermediate cylinder $(-\frac{5}{16}, 0) \times B_{5/8}$, using the second tail bound, and then iterate (5.20) from this cylinder down to $Q^+ = (-\frac{1}{4}, 0) \times B_{1/2}$ to obtain $\|v^{-1}\|_{L^\infty(Q^+)} \leqslant Ce^{-k}$. Multiplication proves (5.19) for v. The constants do not depend on ε; let $\varepsilon \downarrow 0$. □

Corollary 5.13 (Parabolic Harnack inequality)**.** *There is $H(d,\Lambda) \geqslant 2$ such that, if $u \geqslant 0$ is a solution in a cylinder $(\tau - \frac{3}{2}\theta, \tau + \frac{5}{4}\theta] \times B_{\sqrt{2\theta}}(\frac{z+z'}{2})$ and $|z - z'| \leqslant \sqrt{\theta}/2$, then*

$$\operatorname*{ess\,sup}_{B_{\sqrt{\theta}/4}(z)} u(\tau, \cdot) \leqslant H \operatorname*{ess\,inf}_{B_{\sqrt{\theta}/4}(z')} u(\tau + \theta, \cdot). \tag{5.22}$$

Proof. For positive solutions the power test with truncated powers $p > 1$ gives the local maximum estimate, as in Proposition 4.13; choose $q = \frac{d+1}{d+2}$ so that $q\chi > 1$. Combining it with Proposition 5.12 on nested cylinders gives $\operatorname{ess\,sup}_{Q_*^-} u \leqslant H \operatorname{ess\,inf}_{Q^+} u$, where $Q_*^- = (-\frac{11}{16}, -\frac{1}{2}) \times B_{3/8}$ is a subcylinder of Q^- with a gap at the bottom. Rescale with $r^2 = 2\theta$ and top time $\tau + \frac{5}{4}\theta$ mapped to 0: then (τ, z) and $(\tau + \theta, z')$ correspond to times $-\frac{5}{8}$ and $-\frac{1}{8}$ and to points with $|y| \leqslant \frac{1}{2\sqrt{2}} < \frac{1}{2}$. By (5.3) a bound $u \leqslant M$ or $u \geqslant m$ a.e. on a space-time cylinder passes to every slice at interior times, since $\int u(\tau)\psi = \lim \int u(\tau_n)\psi$ for $\psi \geqslant 0$. □

Theorem 5.14 (Nash-Aronson lower bound)**.** *There are $c(d,\Lambda), \beta(d,\Lambda) > 0$ such that, whenever $s < t$, $f, g \in L^1 \cap L^2$ are nonnegative, $\operatorname{supp} f \subseteq B_{\sqrt{t-s}/8}(y)$ and $\operatorname{supp} g \subseteq B_{\sqrt{t-s}/8}(x)$, one has*

$$\int_{\mathbb{R}^d} g\, S(t,s) f \geqslant c(t-s)^{-d/2} \exp\left(-\beta \frac{|x-y|^2}{t-s}\right) \|f\|_1 \|g\|_1\,. \tag{5.23}$$

Proof. By homogeneity assume $\|f\|_1 = 1$, and put $h = t - s$, $u(\tau) = S(\tau, s)f$. Proposition 5.11, at time $s + h/4$ with $\varepsilon = \sqrt{h}/8$, gives $R_0 = C(d)\sqrt{\Lambda \log(e\Lambda)} \geqslant 1$ such that

$$\int_{B_{R_0\sqrt{h}}(y)} u(s + h/4, z)\, dz \geqslant \tfrac{1}{2}.$$

Covering the ball by balls of radius $\sqrt{\theta}/4$ (with θ as below), we find a center z with $\operatorname{ess\,sup}_{B_{\sqrt{\theta}/4}(z)} u(s+h/4,\cdot) \geqslant c(d) R_0^{-d} h^{-d/2}$. Fix $x' \in B_{\sqrt{h}/8}(x)$ and divide the segment from z to x' and the interval $[s+h/4, t]$ into

$$N = \bigl\lceil C(d)\bigl(1 + R_0^2 + |x-y|^2/h\bigr)\bigr\rceil$$

equal parts. Choose $C(d)$ large enough that each spatial step is at most half the square root of its time step θ and the first enlarged cylinder stays above time s. The last step needs the solution on a cylinder that extends a quarter step beyond time t, while u is only given on $[s,t]$: extend $\mathbf{a}$ by the identity matrix for times after t, let $\widetilde{u}$ be the solution of the Cauchy problem with the same datum on $[s, 2t-s]$, and note that $\widetilde{u} = u$ on $[s,t]$ by uniqueness, so it suffices to argue with $\widetilde{u}$. Apply Corollary 5.13 successively to $u+\varepsilon$ and let $\varepsilon \downarrow 0$ to get, for a.e. $x' \in B_{\sqrt{h}/8}(x)$,

$$u(t,x') \geqslant H^{-N}\, c(d) R_0^{-d} h^{-d/2} \geqslant c(d) R_0^{-d} H^{-C(d)(1+R_0^2)} h^{-d/2} \exp\!\left(-C(d)\log H \,\frac{|x-y|^2}{h}\right).$$

Integrating against g proves the assertion. One may take $c = c(d) R_0^{-d} H^{-C(d)(1+R_0^2)}$ and $\beta = C(d)\log H$. □

Remark 5.15. The lower bound uses a local Harnack step, which contains the logarithmic crossover in Proposition 5.12. No symmetry is needed. The dependence of c and β on Λ here is not claimed to be sharp. For symmetric, time-independent coefficients only, the identity $\int P(t,z;0,y)^2\,dz = P(2t,y;0,y)$ also turns Proposition 5.11 into an on-diagonal estimate at points of continuity of the kernel.

The upper bound at $r=0$ makes $S(t,s) : L^1 \to L^\infty$ bounded. Consequently it has a nonnegative measurable kernel $P(t,x;s,y)$ for each $s<t$. By testing the two bilinear bounds against approximate identities and absorbing the linear term of (5.14) into half the quadratic term, one obtains, for almost every (x,y),

$$\begin{aligned} c(d,\Lambda)(t-s)^{-d/2}\exp\!\left(-\beta(d,\Lambda)\frac{|x-y|^2}{t-s}\right) &\leqslant P(t,x;s,y) \\ &\leqslant C(d)(t-s)^{-d/2}\exp\!\left(-\frac{|x-y|^2}{8\Lambda(t-s)}\right). \end{aligned} \tag{5.24}$$

A jointly continuous representative, when available from parabolic interior regularity, satisfies the same bounds everywhere.

Notes and references

Nash's decay argument is from [Nas58]; the weighted upper bound follows the method of [Dav87]. The two-sided Gaussian bounds are due to Aronson [Aro68]; the logarithmic argument and the parabolic Harnack inequality go back to [Mos64, Mos67], the route through the logarithm to [Mos71], and the crossing lemma to [BG72]. See also [FS86] for the kernel method.

Inputs used here without proof: (1) the Cauchy evolution on $\mathbb{R}^d$ with positivity, mass conservation, L^2 contraction, the global and weighted energy identities, and duality; (2) the parabolic Sobolev inequality, for $w(t,\cdot) \in H^1_0$; (3) the weighted Poincaré inequality for the radial cutoff ζ^2; (4) the Moser bookkeeping for powers of both signs and the parabolic local maximum estimate; (5) the $L^1 \to L^\infty$ kernel (Dunford–Pettis) and Lebesgue differentiation in $\mathbb{R}^{2d}$; (6) the chain rule for powers, the logarithm and truncated powers, for solutions as in Lemma 5.6 and for supersolutions via Steklov averages.

References

[ACF84] H. W. Alt, L. A. Caffarelli, and A. Friedman. Variational problems with two phases and their free boundaries. *Trans. Amer. Math. Soc.*, 282(2):431–461, 1984.

[Aro68] D. G. Aronson. Non-negative solutions of linear parabolic equations. *Ann. Scuola Norm. Sup. Pisa (3)*, 22:607–694, 1968.

[BG72] E. Bombieri and E. Giusti. Harnack's inequality for elliptic differential equations on minimal surfaces. *Invent. Math.*, 15:24–46, 1972.

[Dav87] E. B. Davies. Explicit constants for Gaussian upper bounds on heat kernels. *Amer. J. Math.*, 109(2):319–333, 1987.

[DLS09] Camillo De Lellis and Jr. Székelyhidi, László. The Euler equations as a differential inclusion. *Ann. of Math. (2)*, 170(3):1417–1436, 2009.

[Eva98] L. C. Evans. *Partial differential equations*, volume 19 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 1998.

[FH76] S. Friedland and W. K. Hayman. Eigenvalue inequalities for the Dirichlet problem on spheres and the growth of subharmonic functions. *Comment. Math. Helv.*, 51(2):133–161, 1976.

[FS86] E. B. Fabes and D. W. Stroock. A new proof of Moser's parabolic Harnack inequality using the old ideas of Nash. *Arch. Rational Mech. Anal.*, 96(4):327–338, 1986.

[Geh73] F. W. Gehring. The L^p-integrability of the partial derivatives of a quasiconformal mapping. *Acta Math.*, 130:265–277, 1973.

[GT01] D. Gilbarg and N. S. Trudinger. *Elliptic partial differential equations of second order*. Classics in Mathematics. Springer-Verlag, Berlin, 2001.

[Gue26] André Guerra. On the sharp Hölder exponent in the De Giorgi–Nash–Moser theory, 2026. arXiv:2606.28244.

[Mey63] N. G. Meyers. An L^p-estimate for the gradient of solutions of second order elliptic divergence equations. *Ann. Scuola Norm. Sup. Pisa (3)*, 17:189–206, 1963.

[Mos64] J. Moser. A Harnack inequality for parabolic differential equations. *Comm. Pure Appl. Math.*, 17:101–134, 1964.

[Mos67] J. Moser. Correction to: "A Harnack inequality for parabolic differential equations". *Comm. Pure Appl. Math.*, 20:231–236, 1967.

[Mos71] J. Moser. On a pointwise estimate for parabolic differential equations. *Comm. Pure Appl. Math.*, 24:727–740, 1971.

[Nas58] J. Nash. Continuity of solutions of parabolic and elliptic equations. *Amer. J. Math.*, 80:931–954, 1958.

[PS72] L. C. Piccinini and S. Spagnolo. On the Hölder continuity of solutions of second order elliptic equations in two variables. *Ann. Scuola Norm. Sup. Pisa (3)*, 26:391–402, 1972.